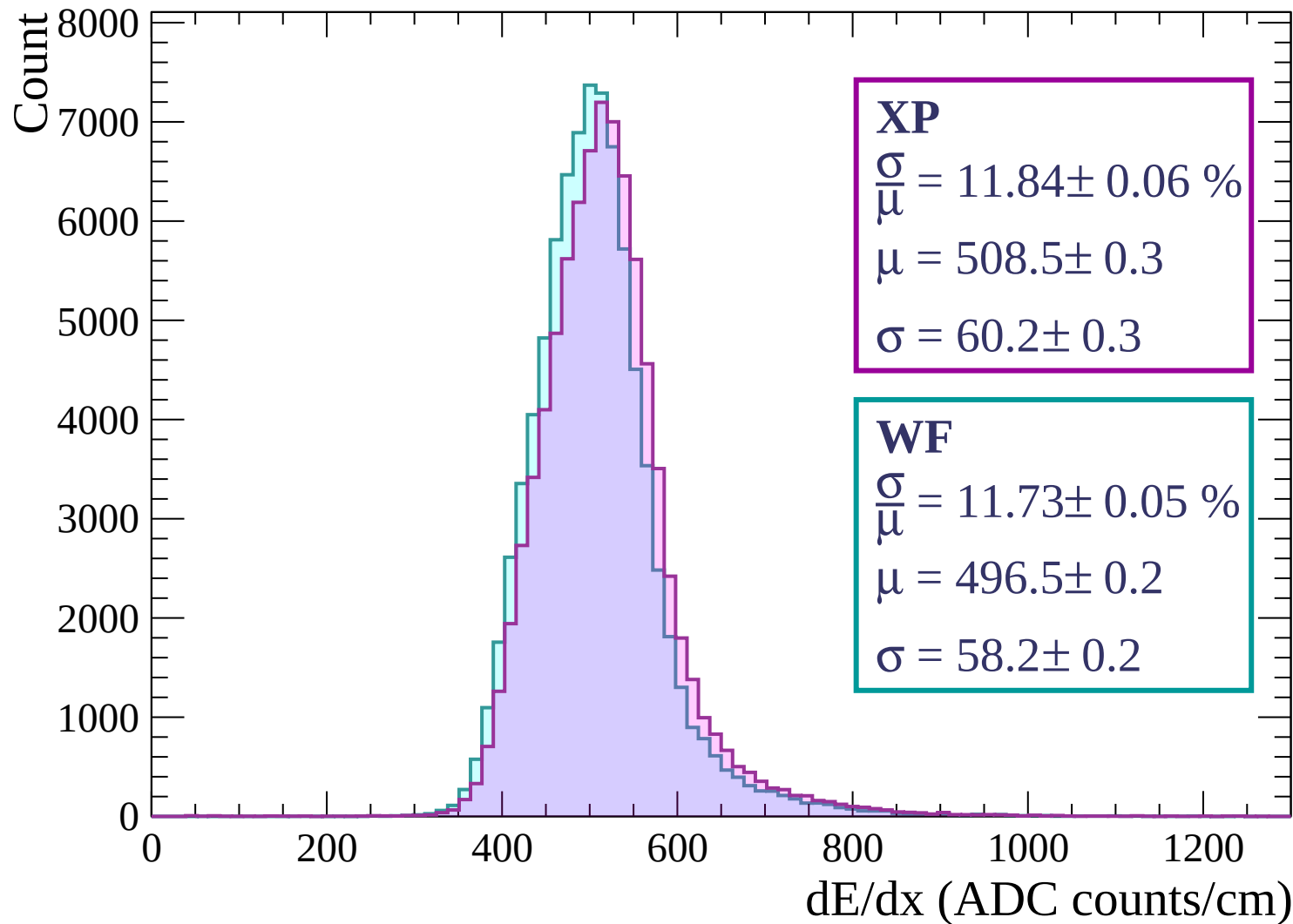
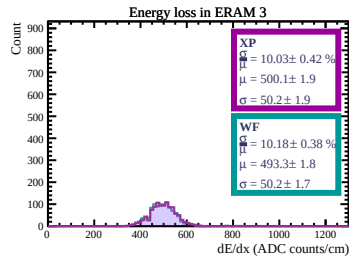
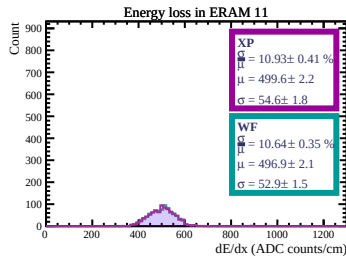
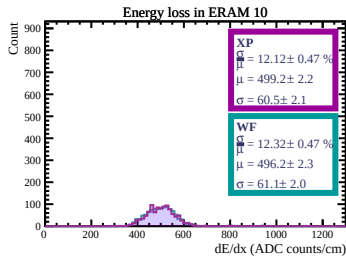
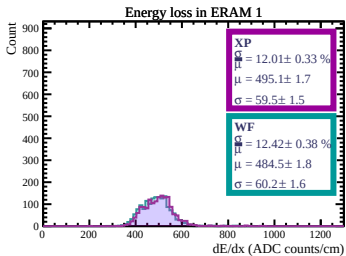
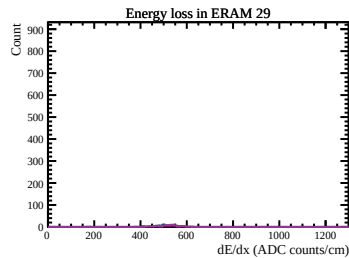
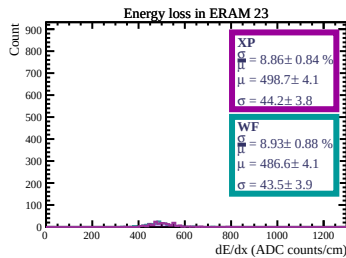
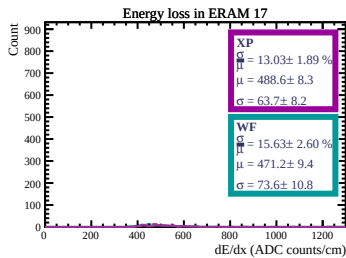
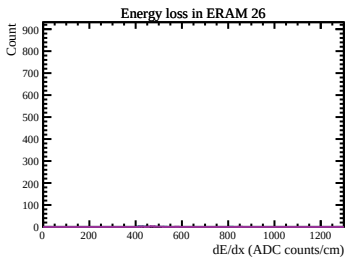
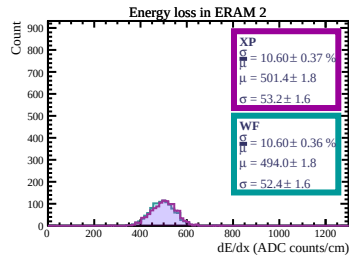
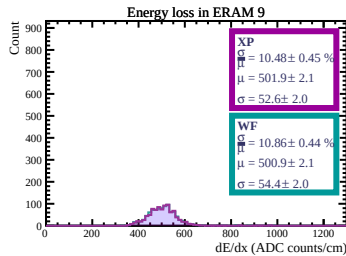
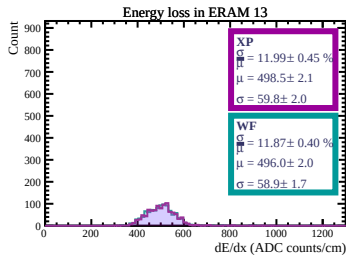
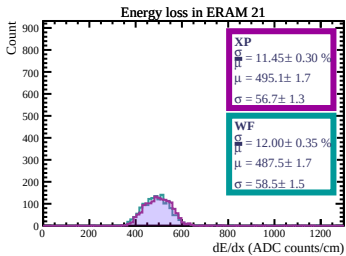
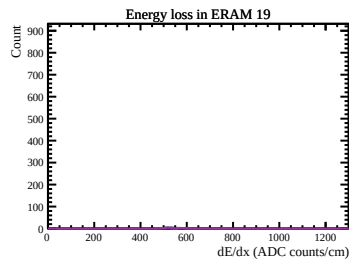
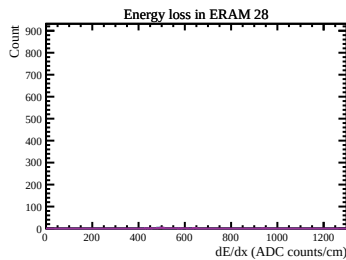
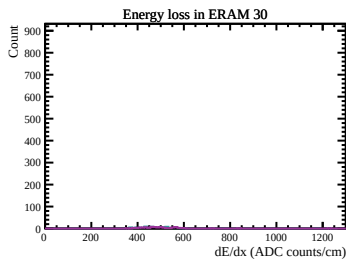
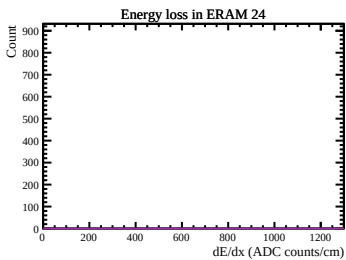
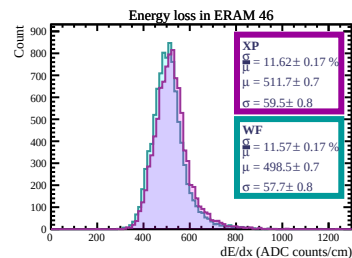
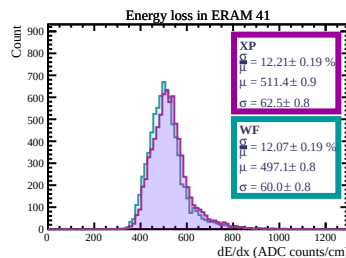
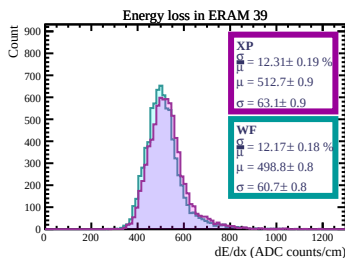
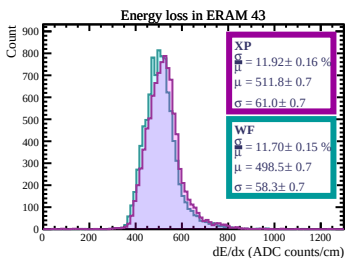
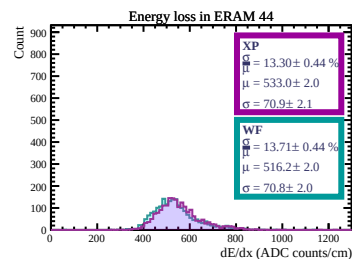
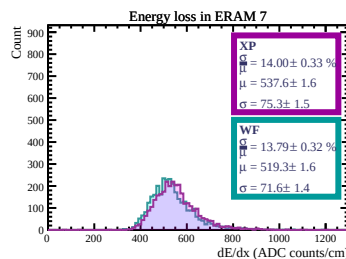
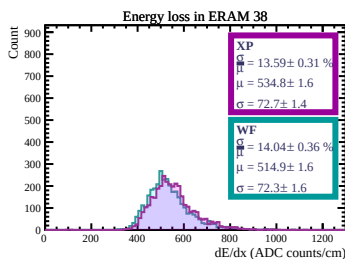
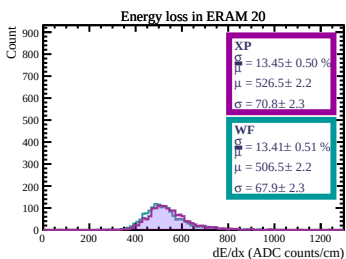
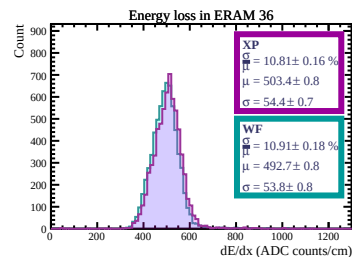
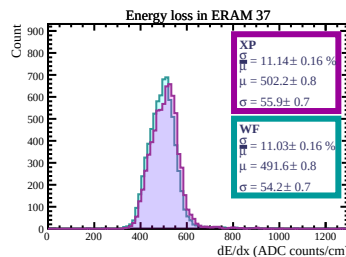
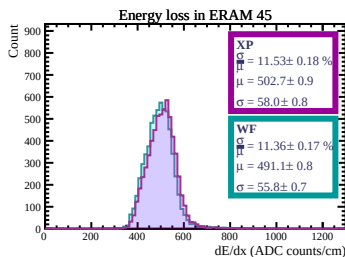
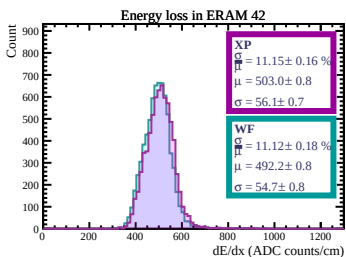
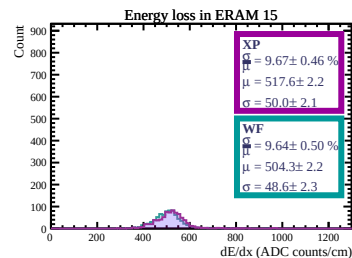
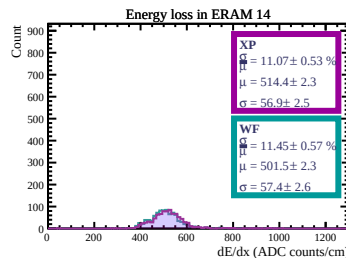
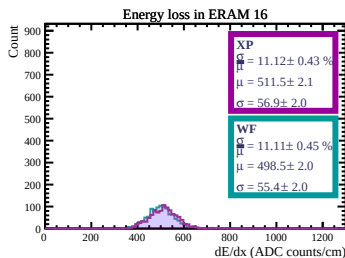
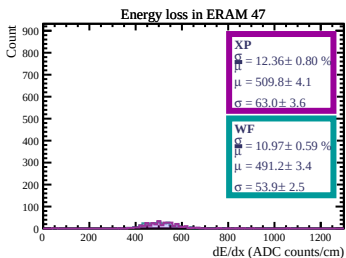
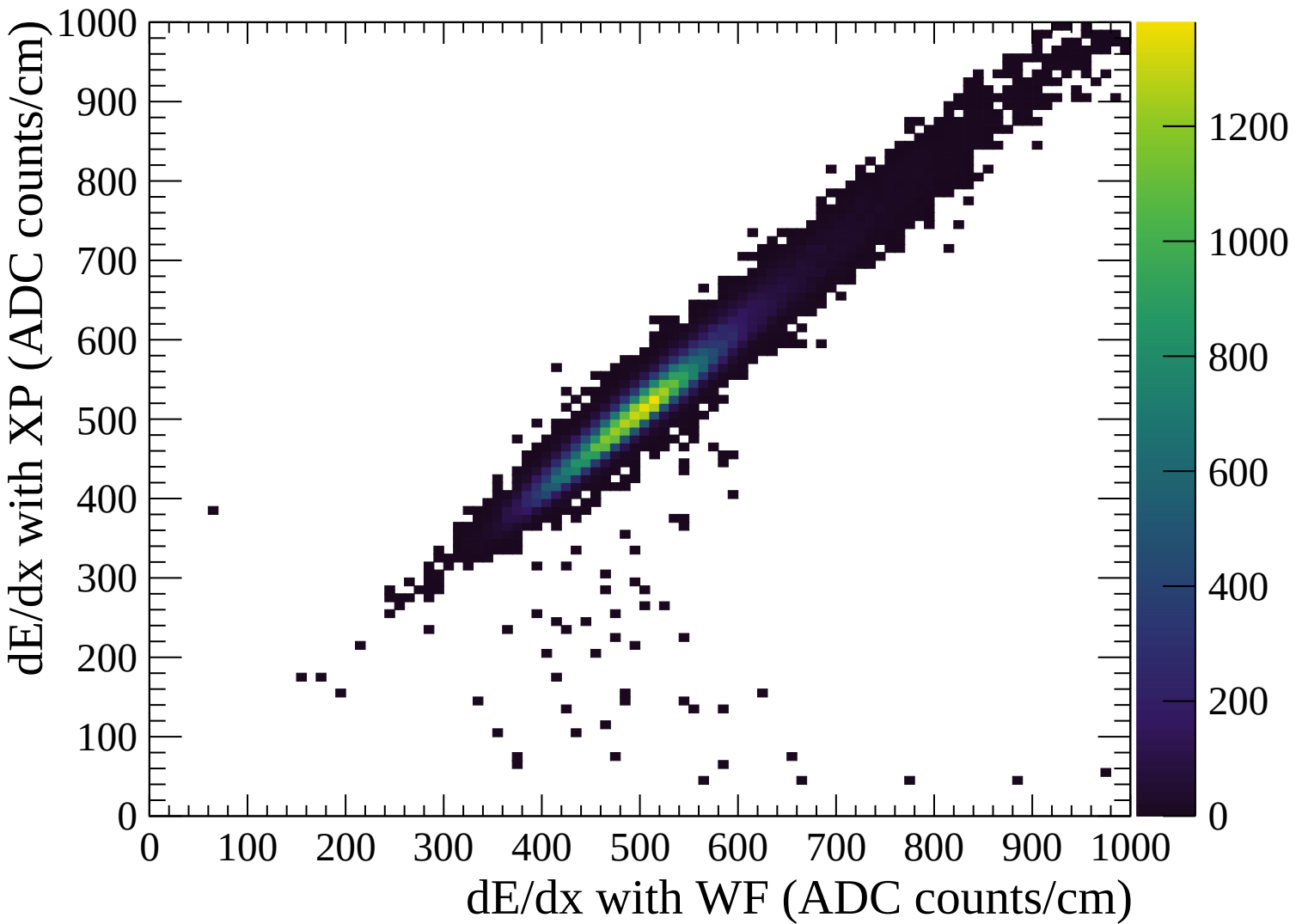
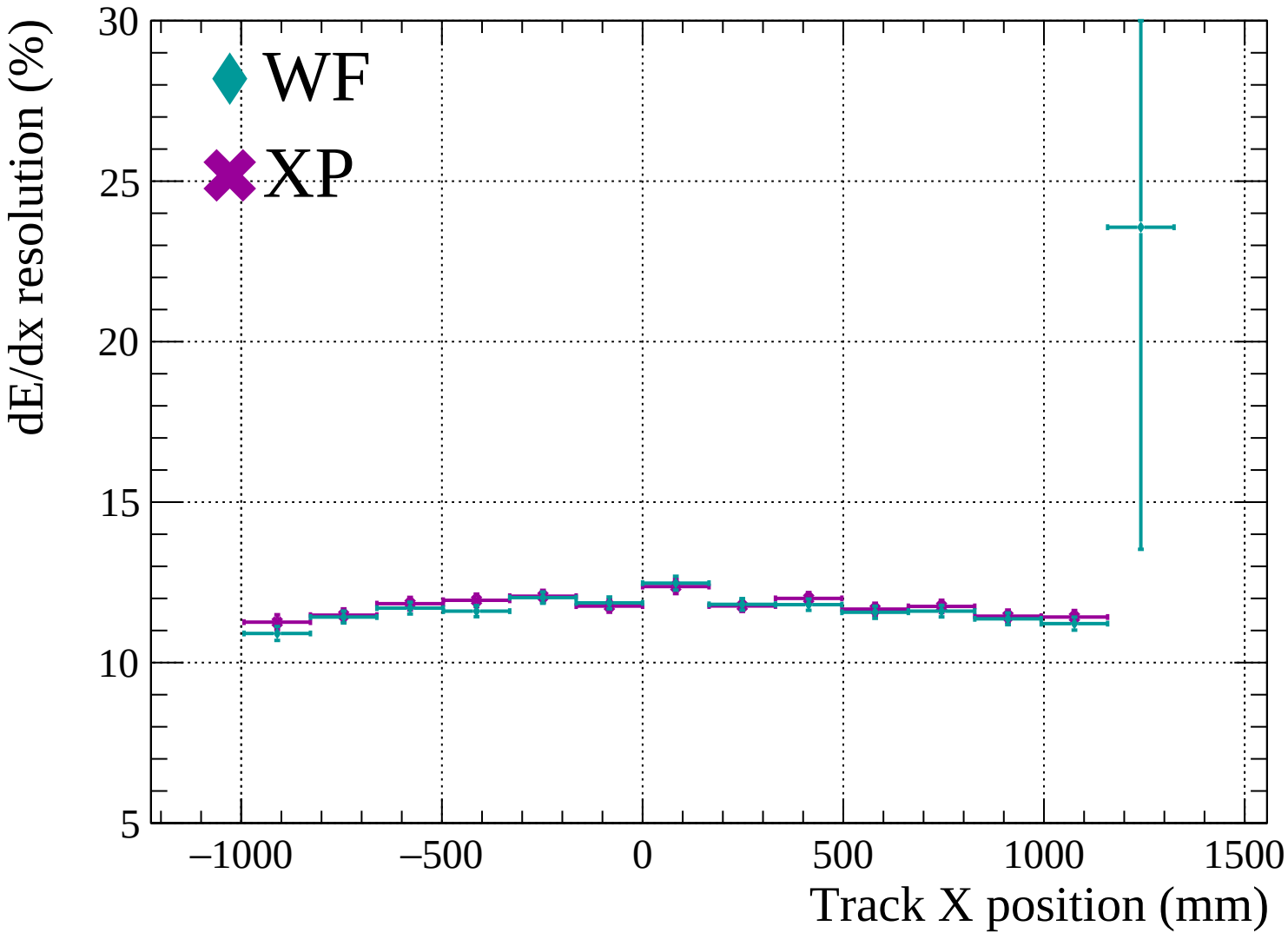


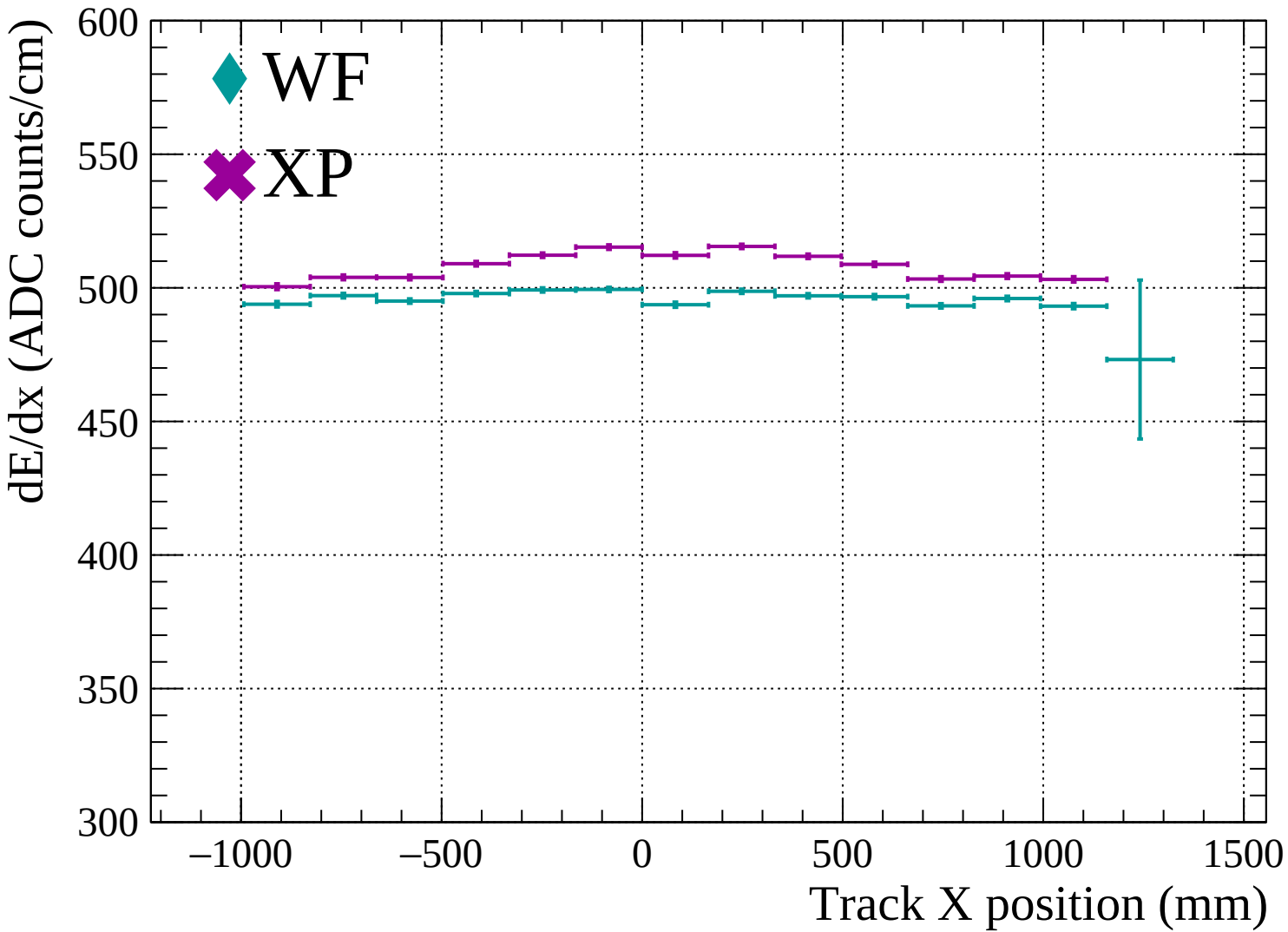


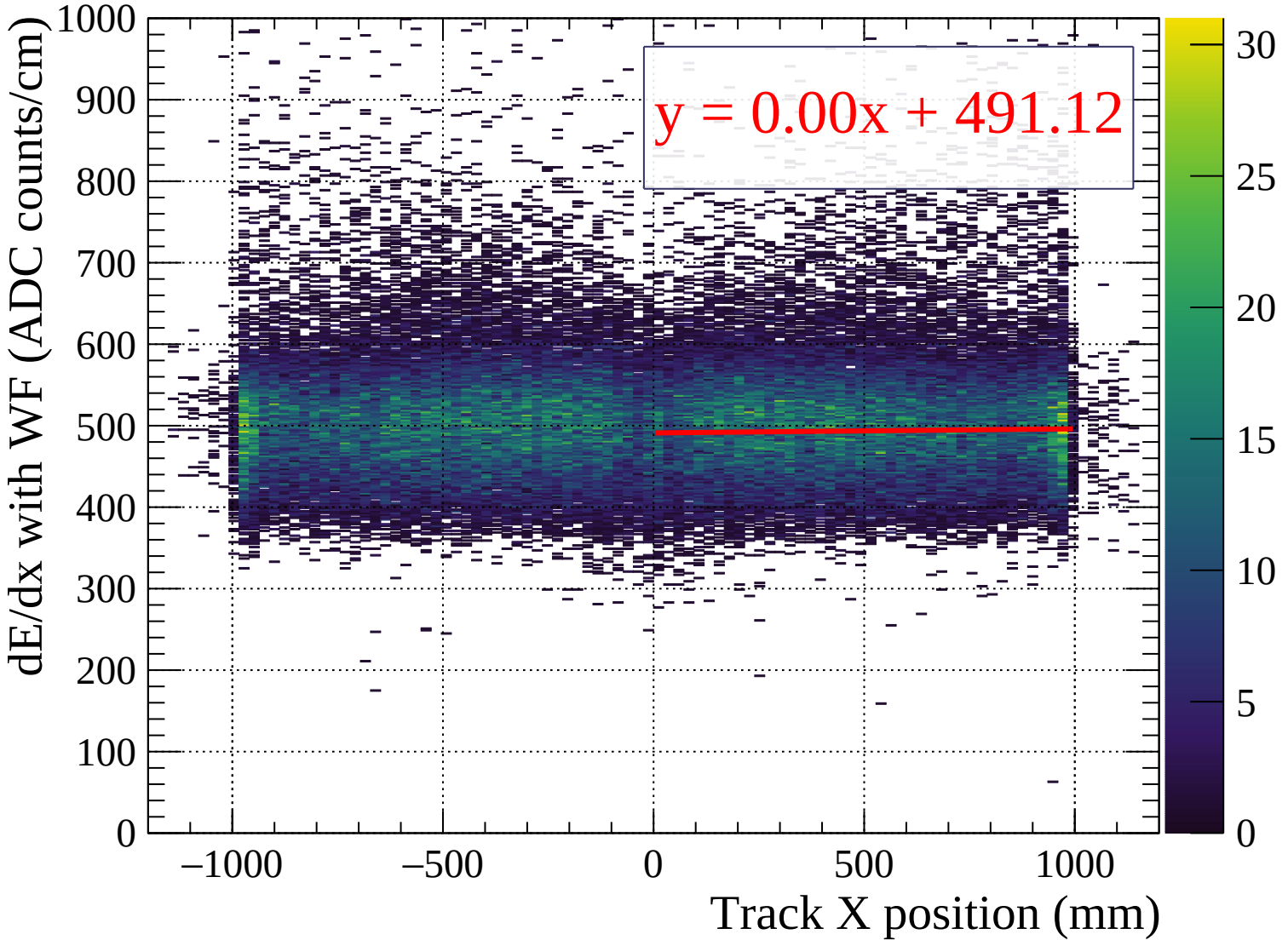


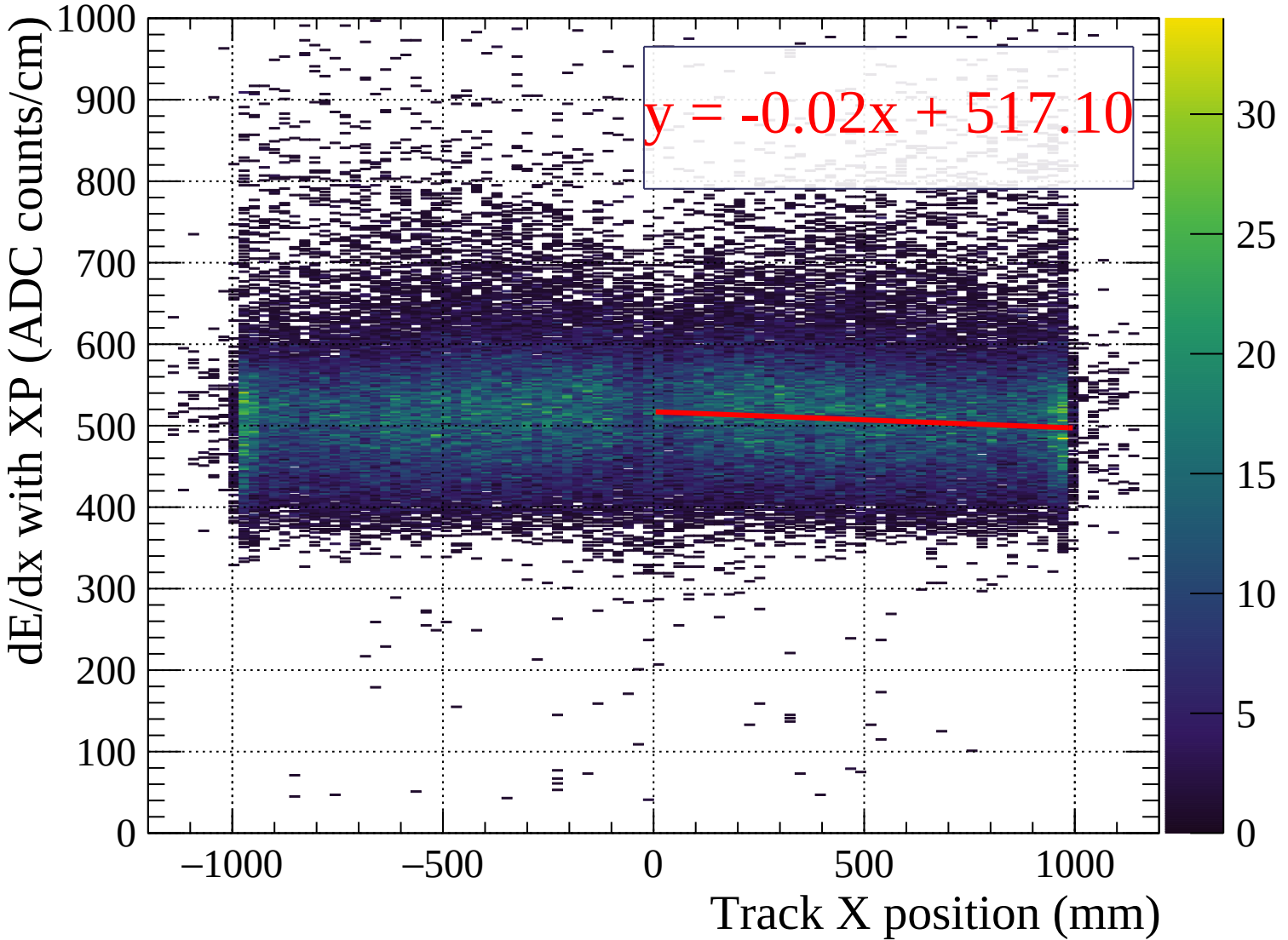


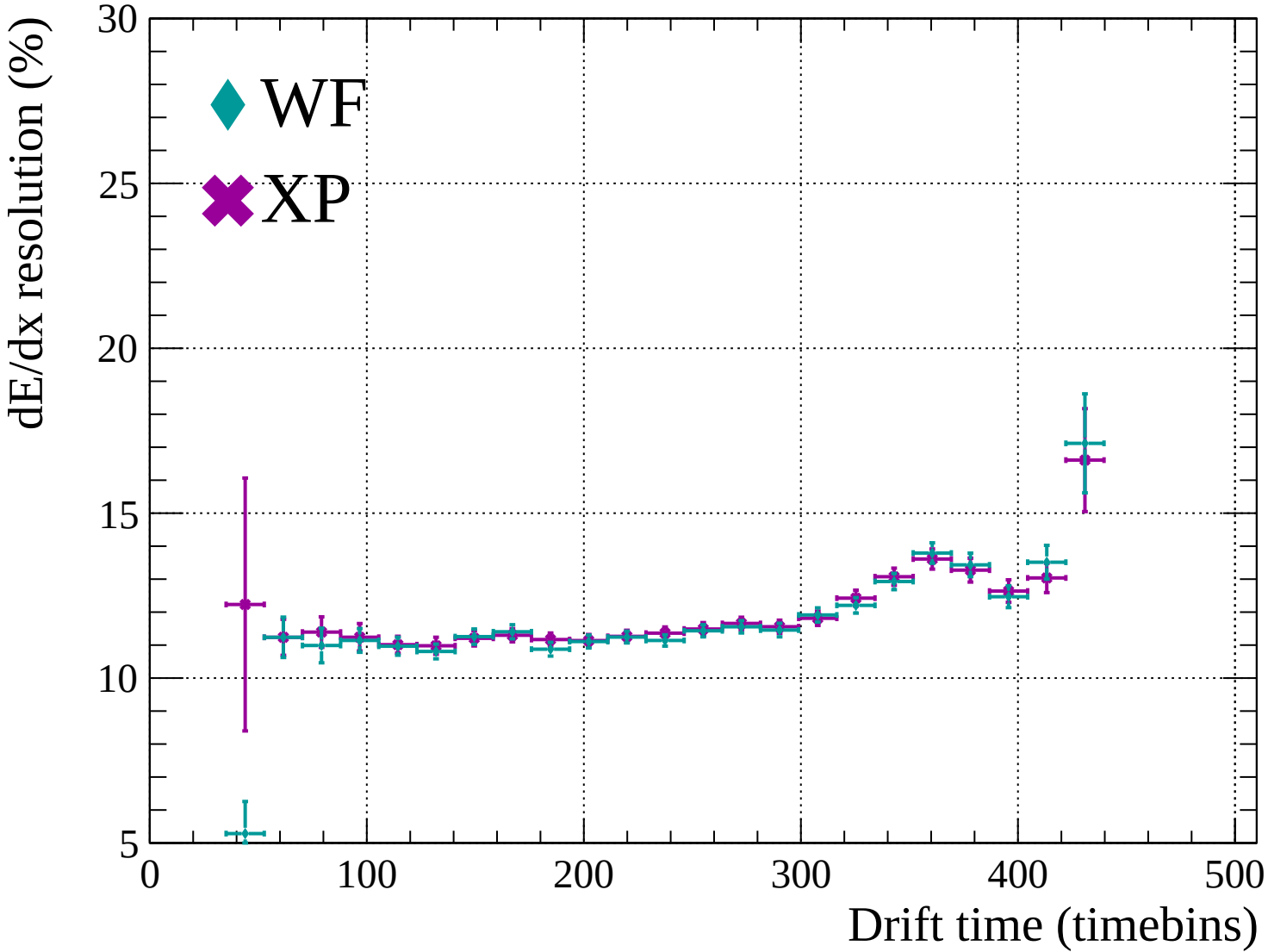


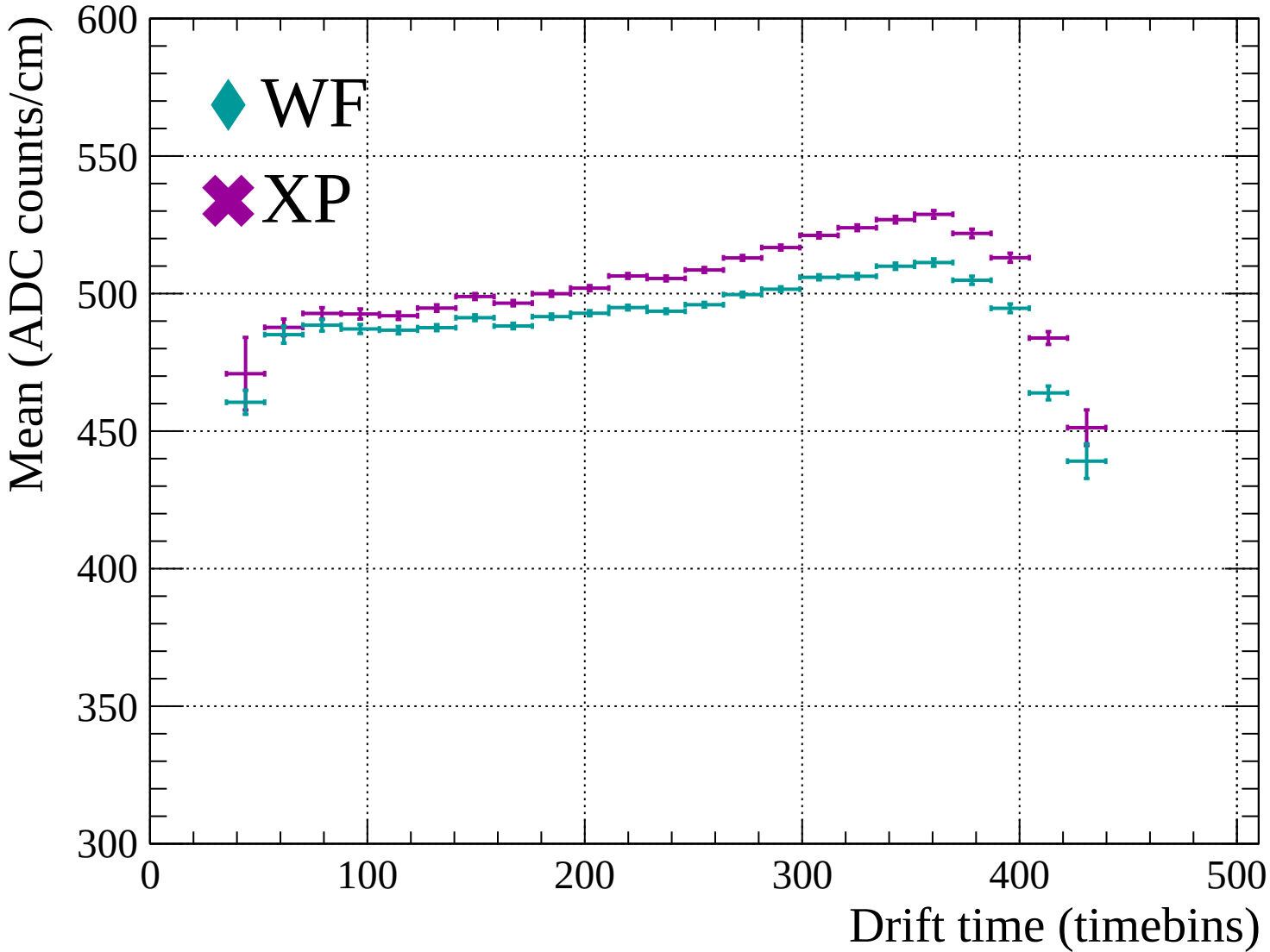


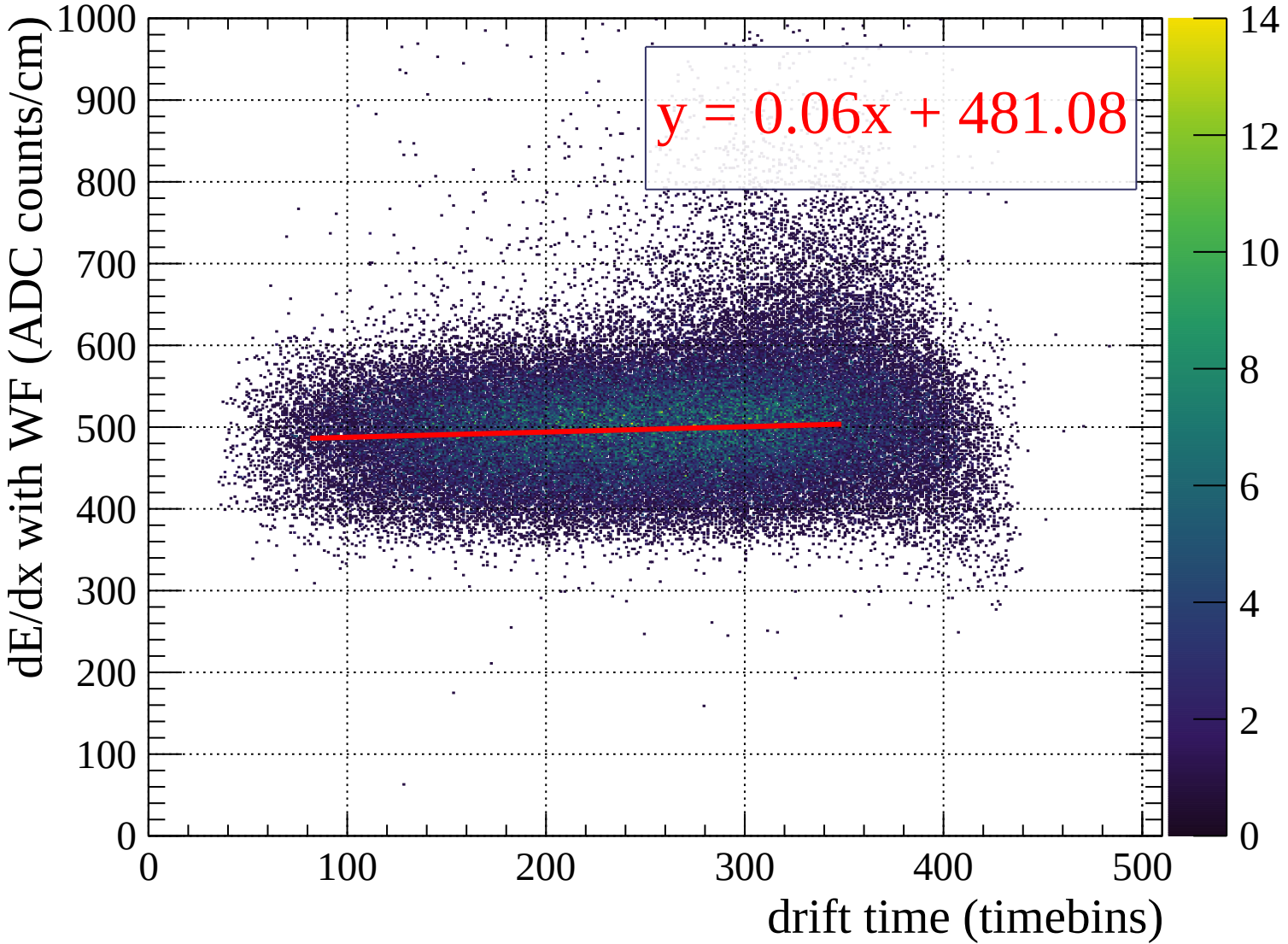


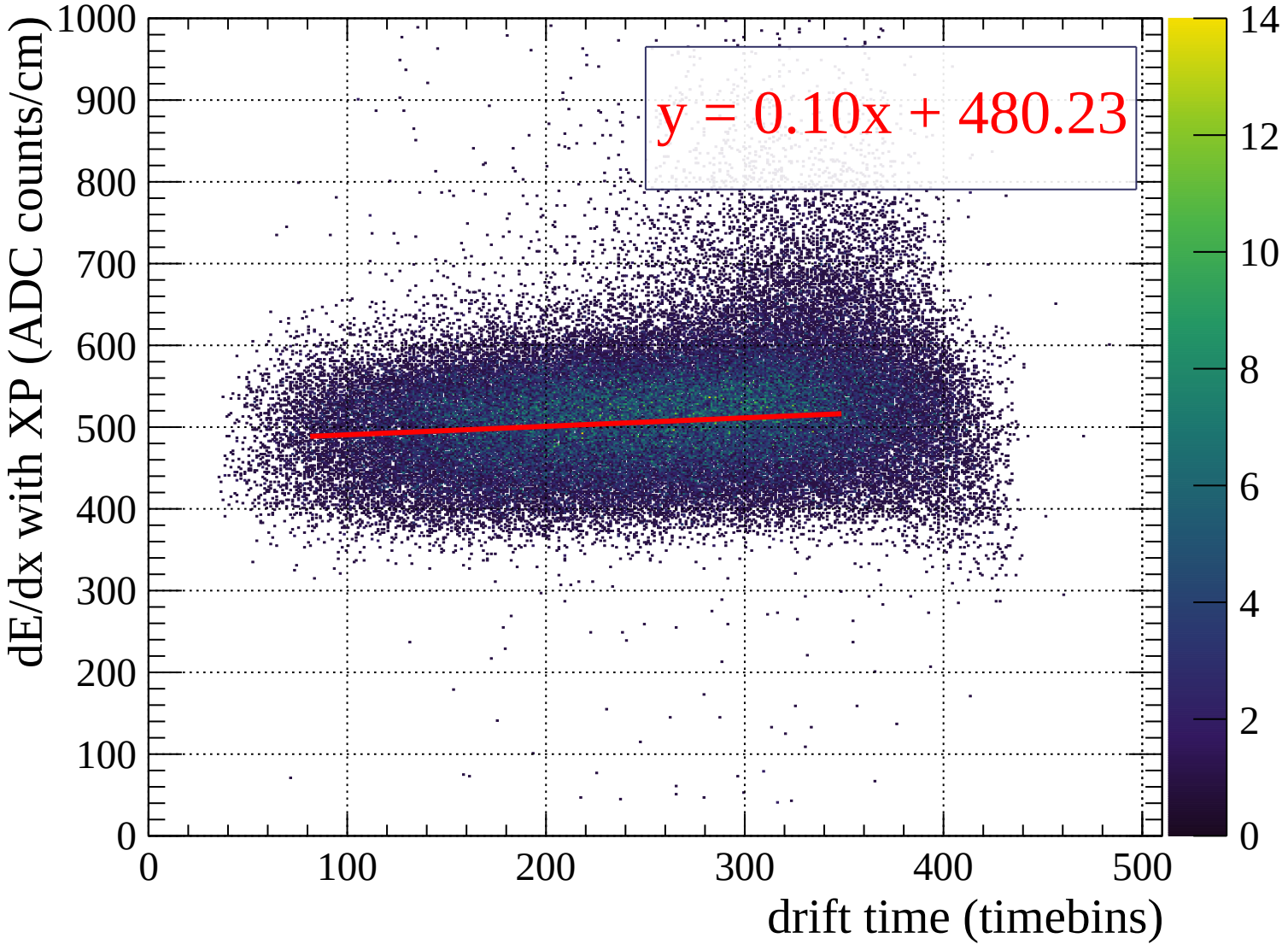


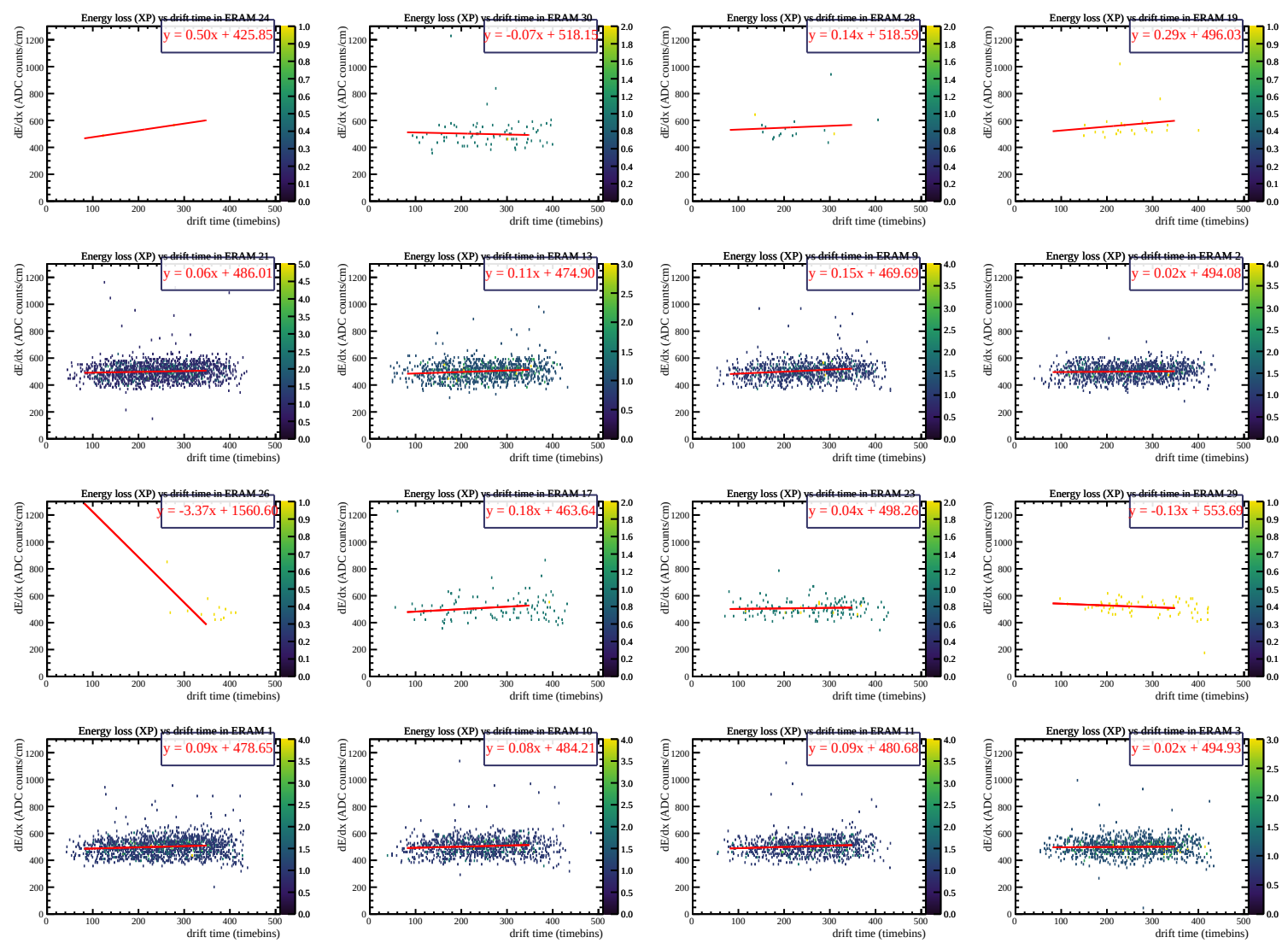


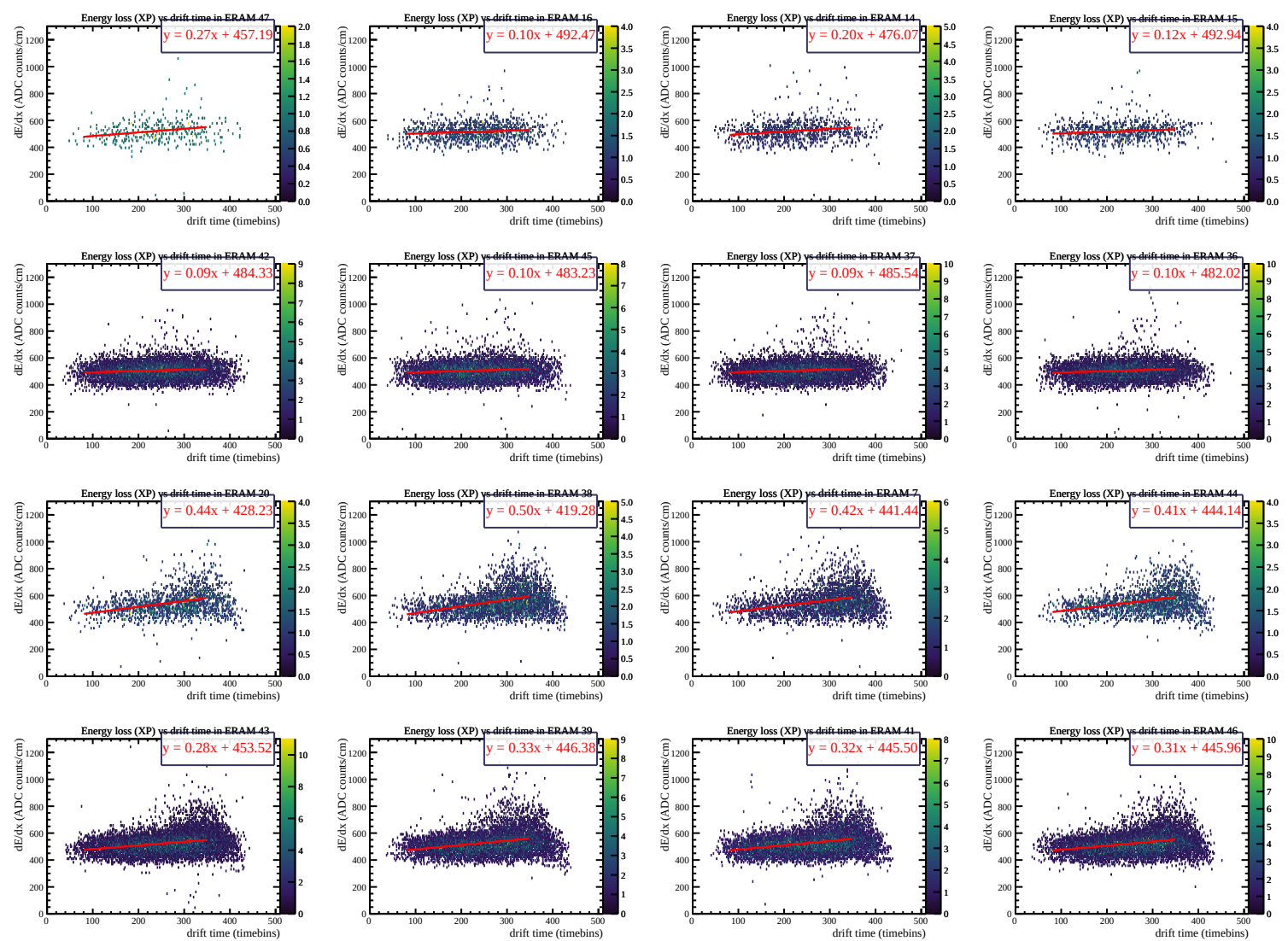


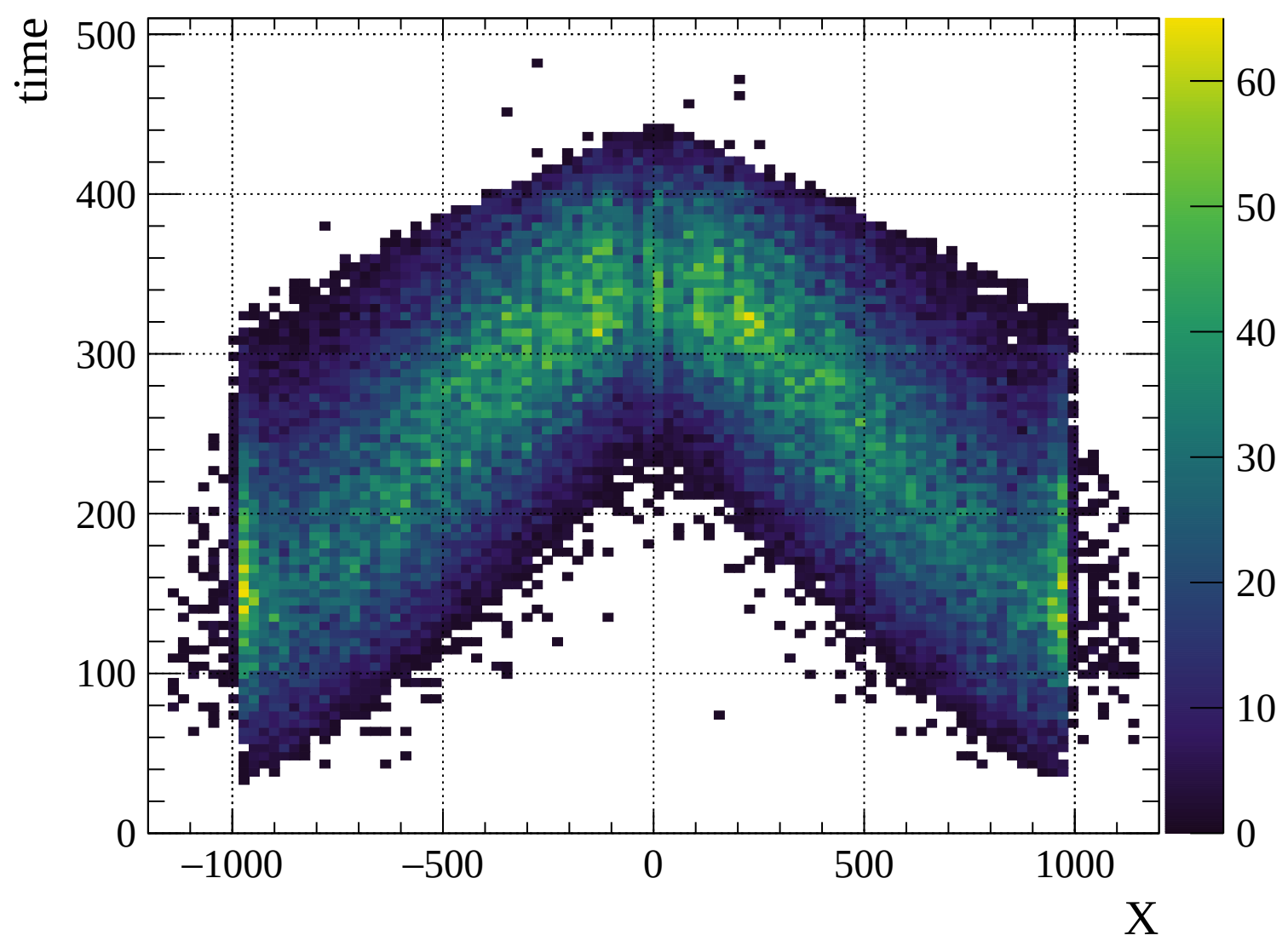


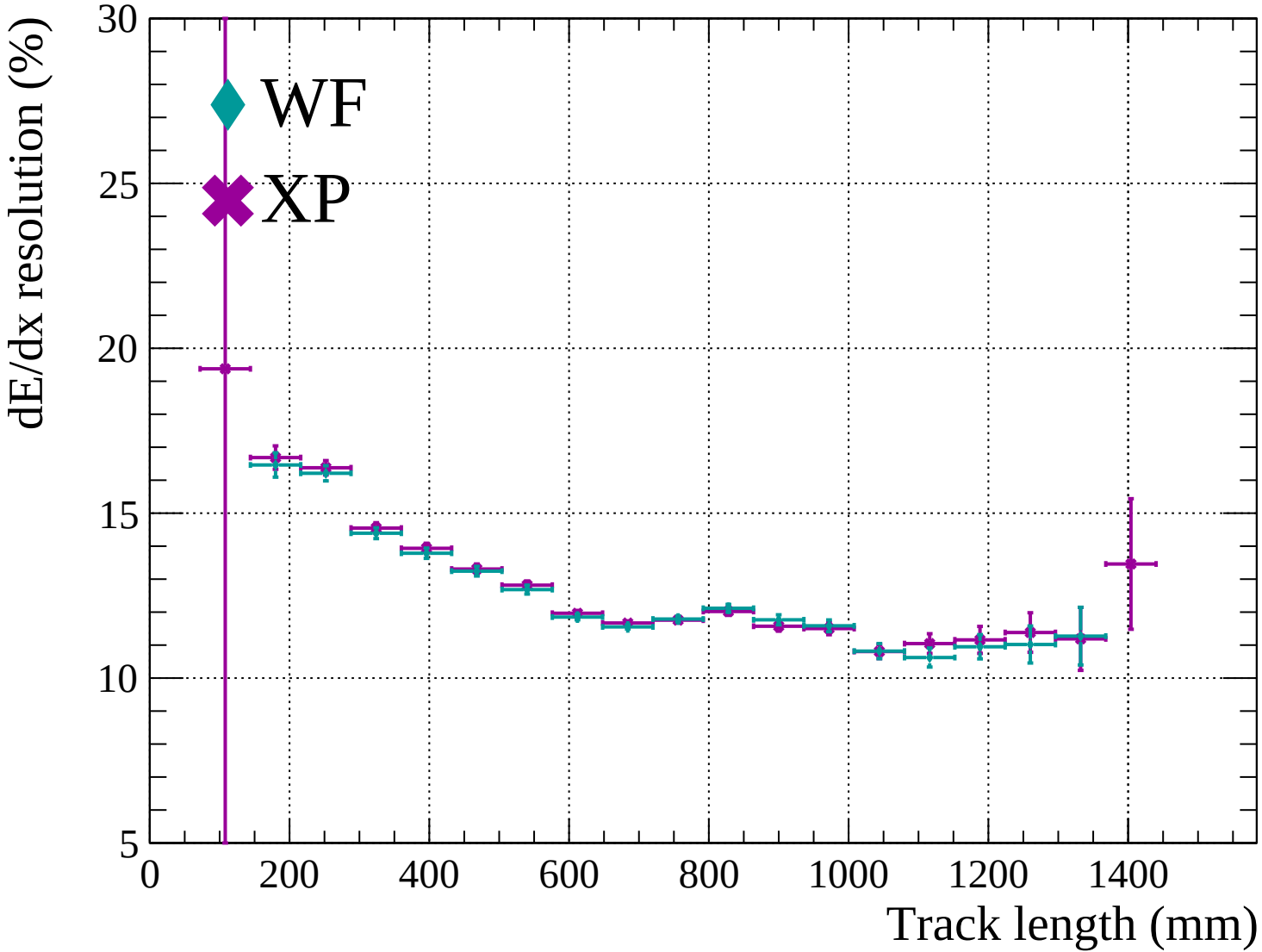


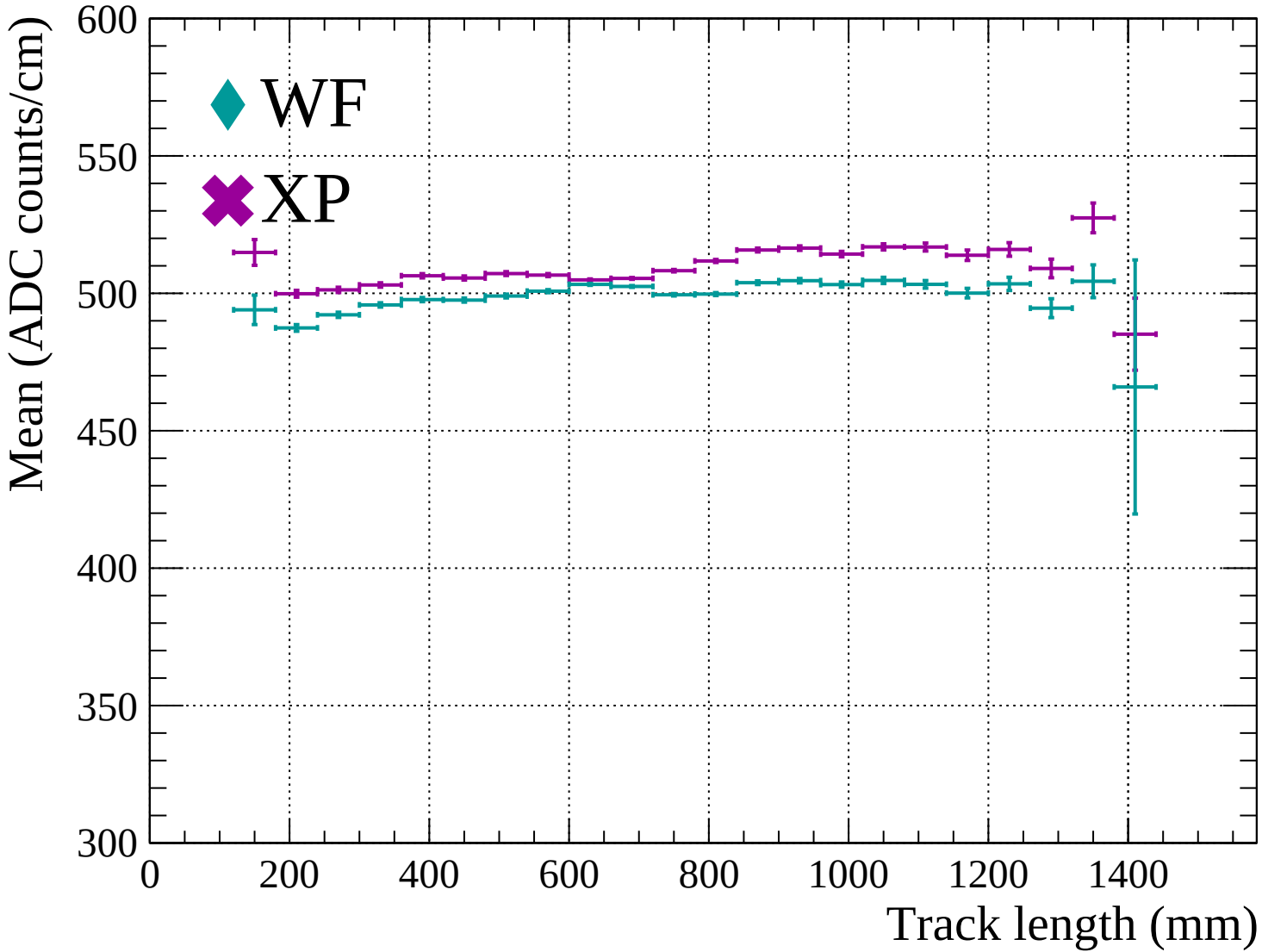


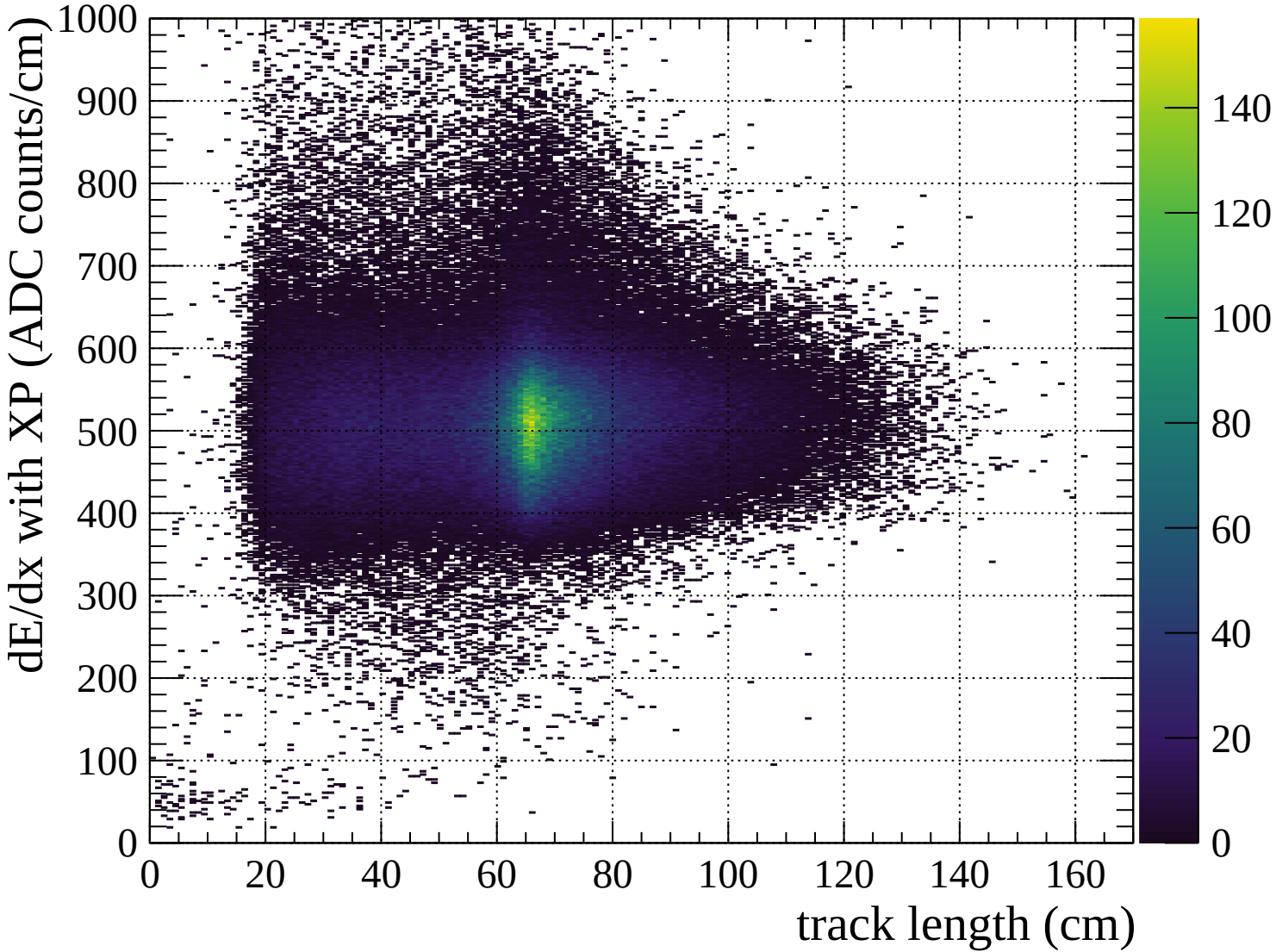


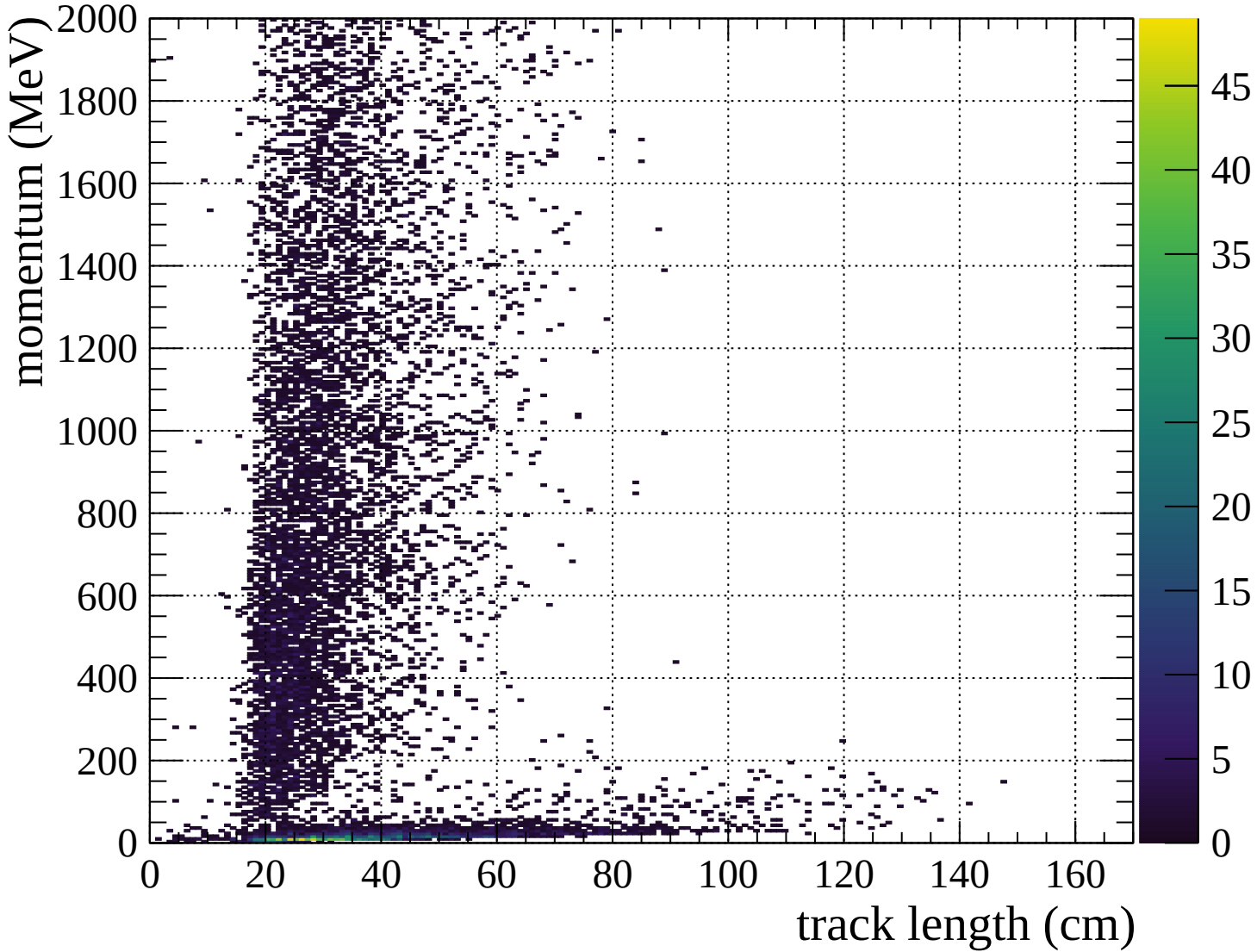


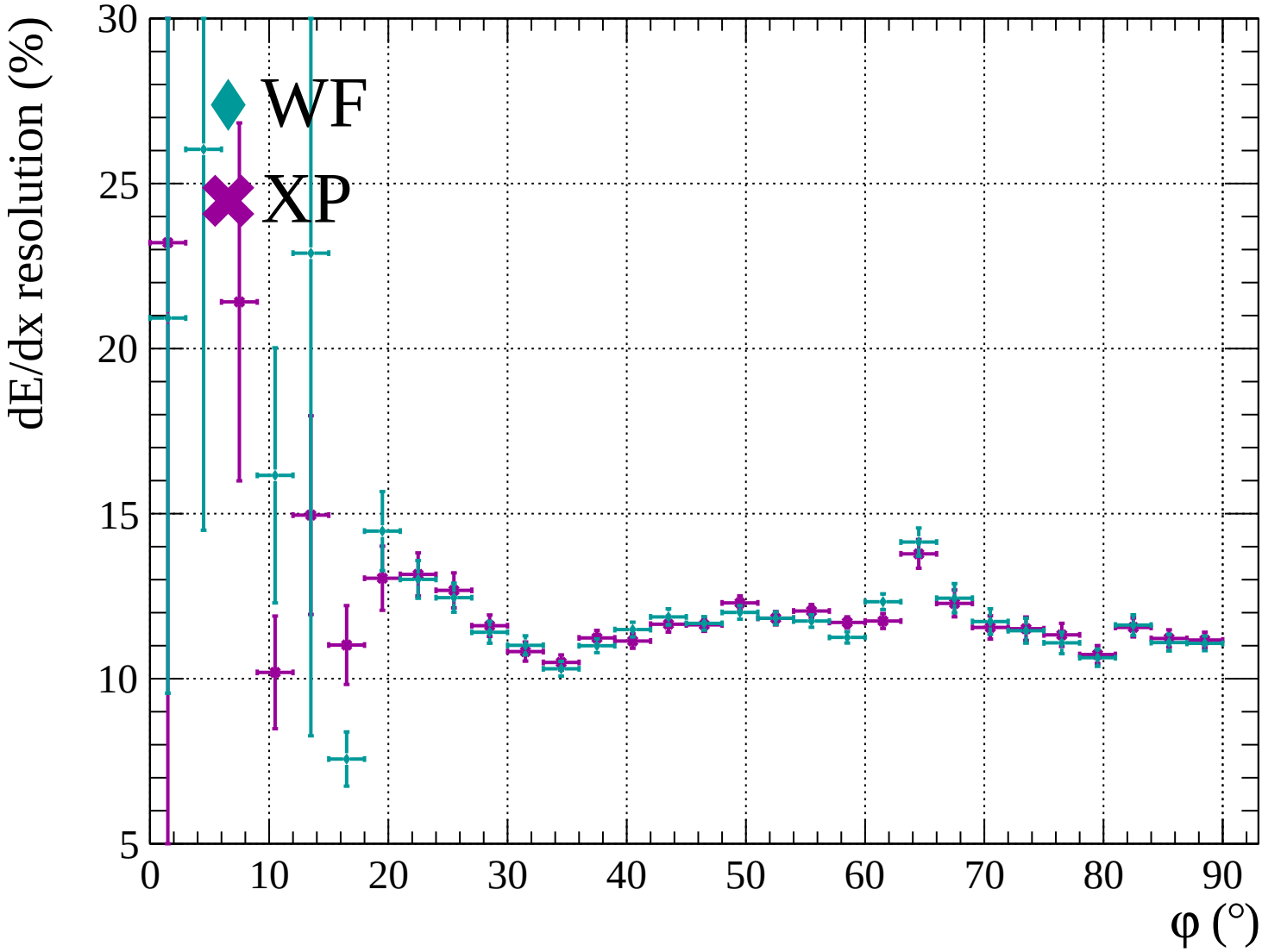


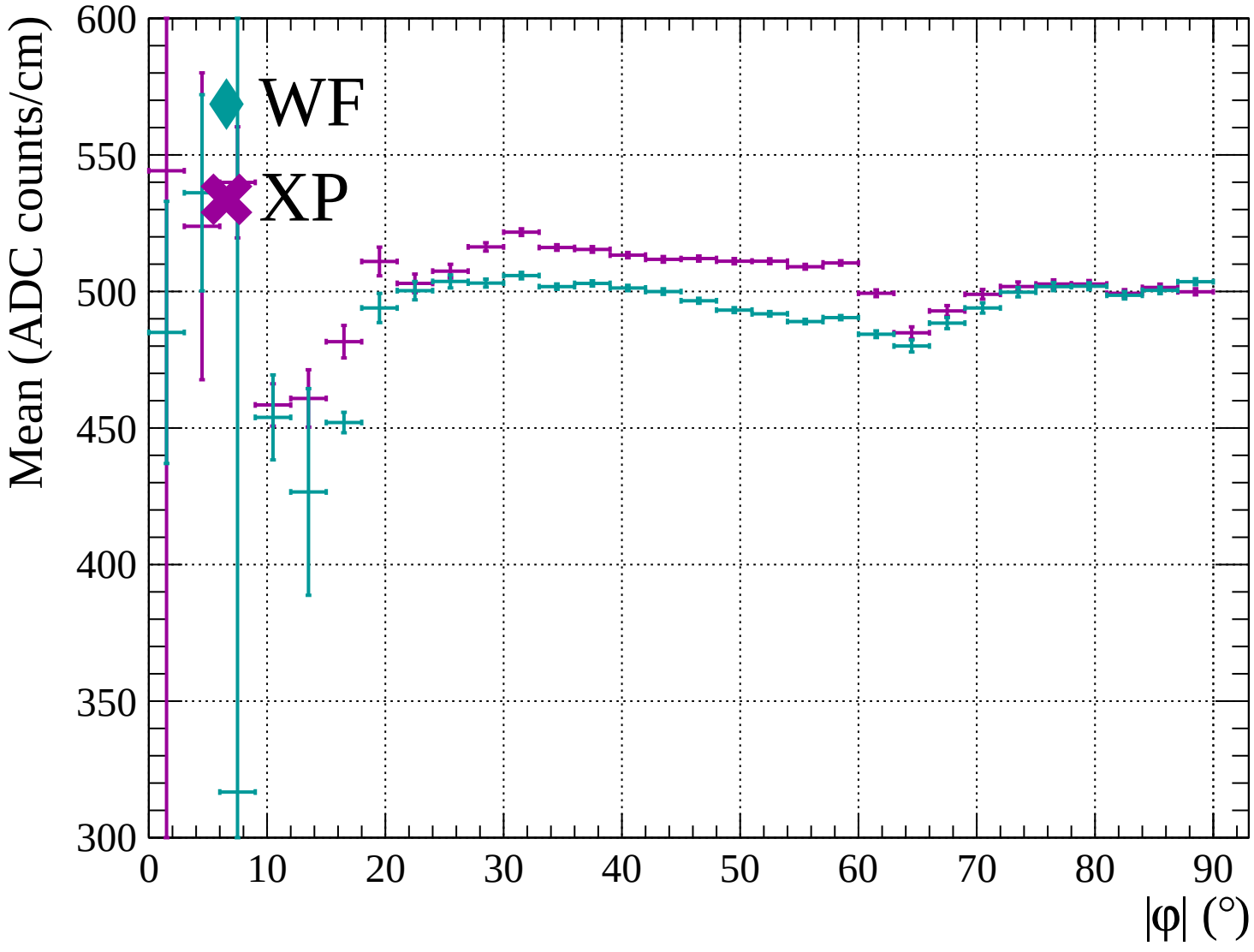


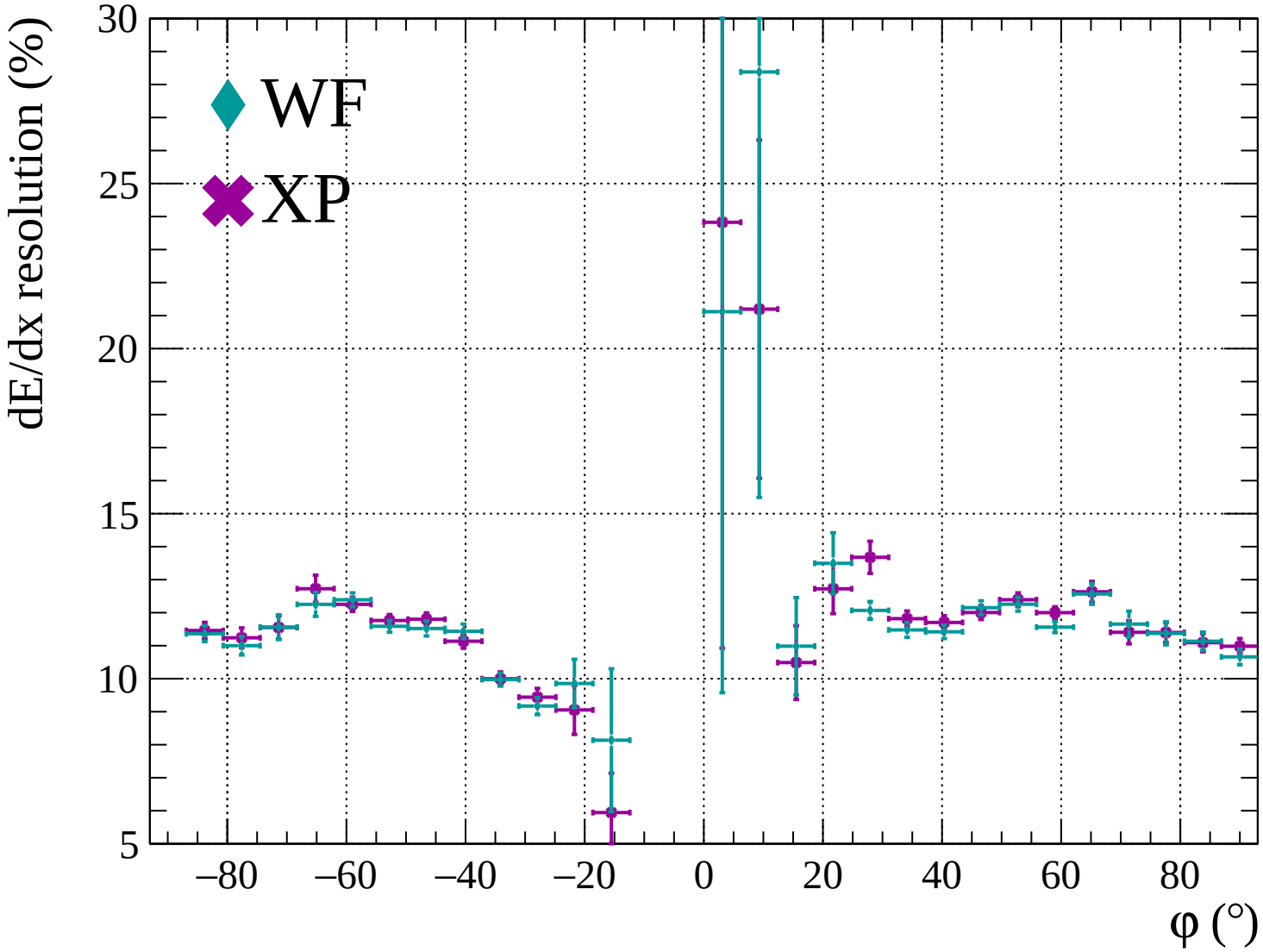


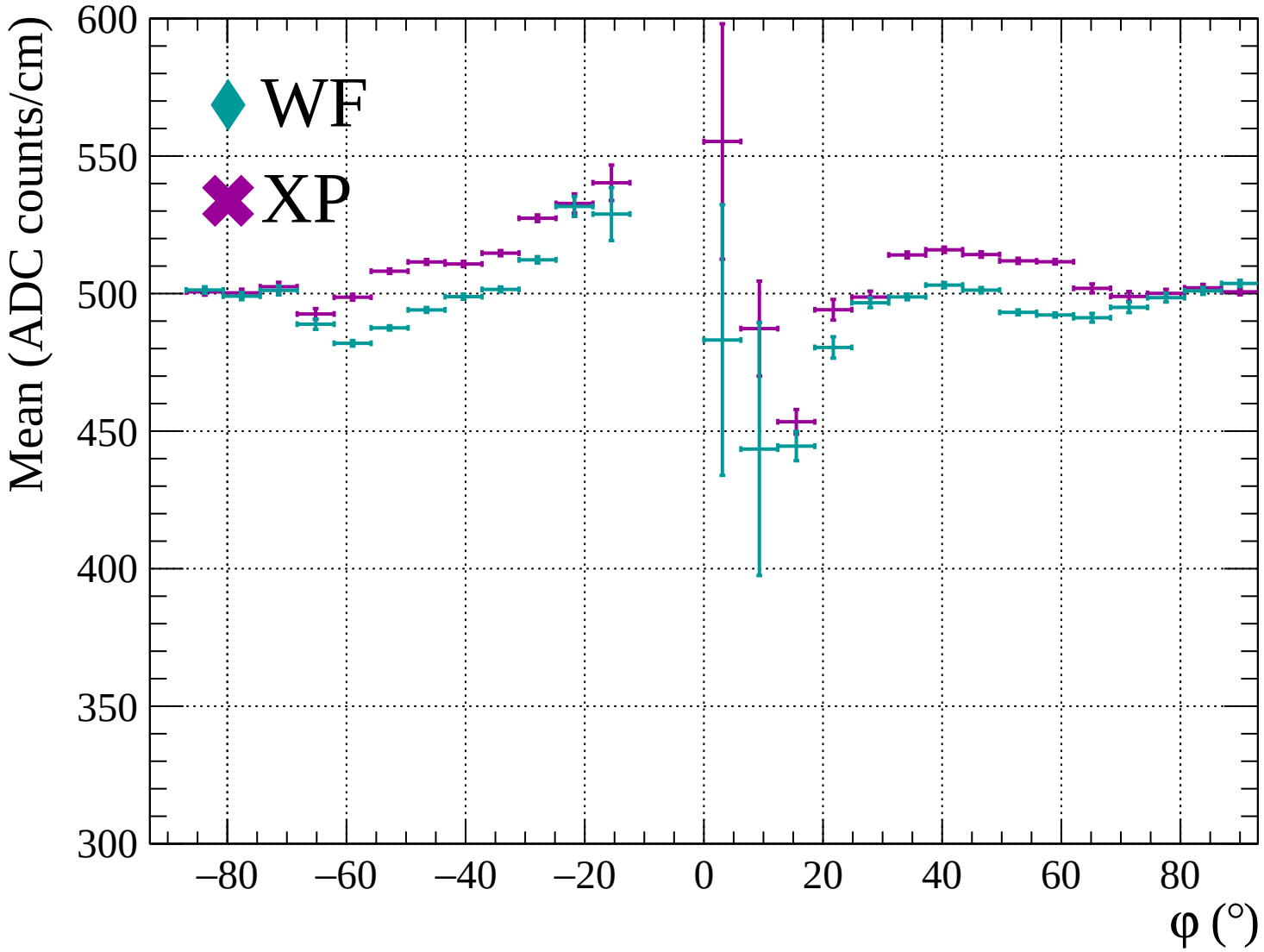


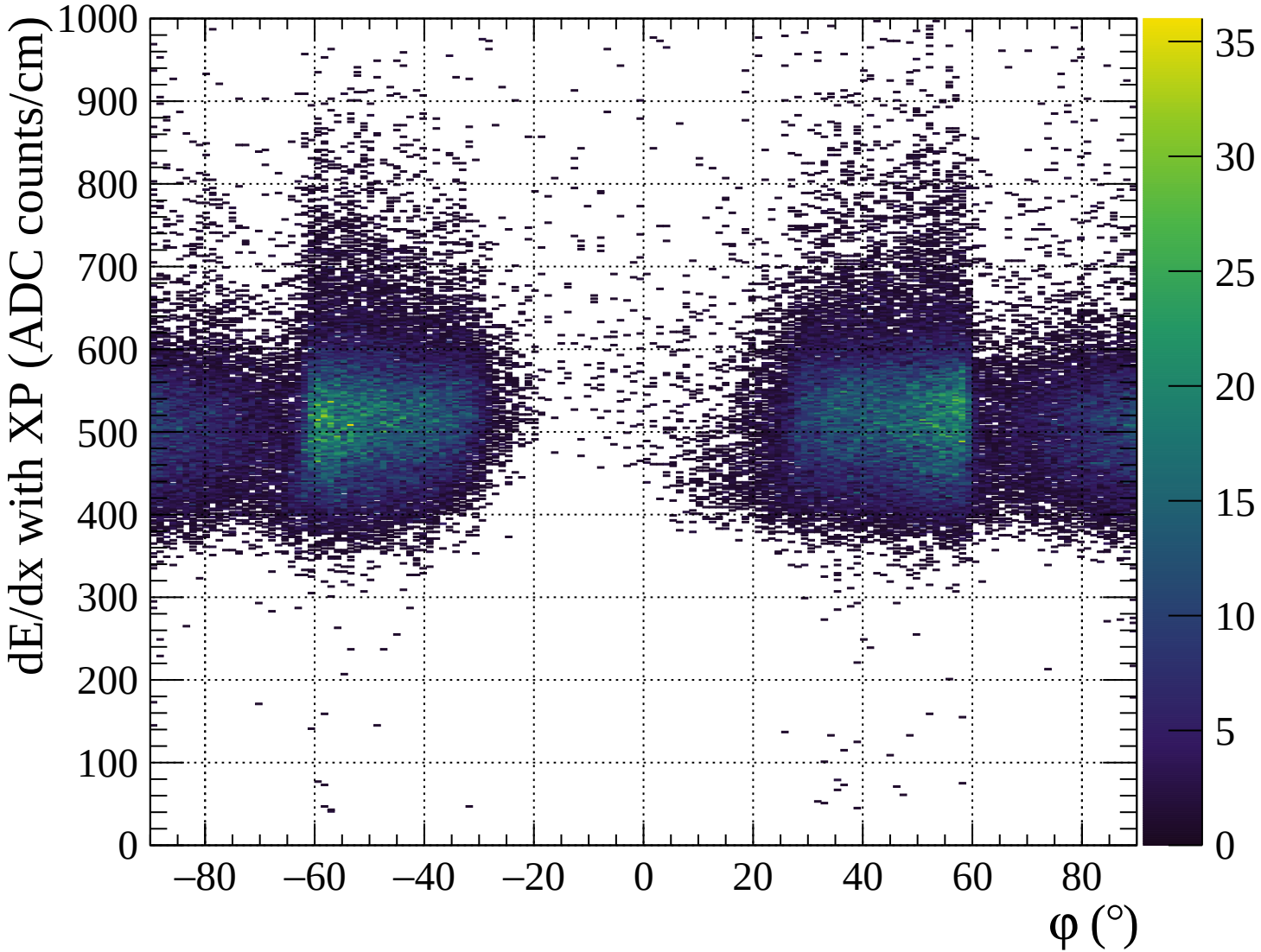


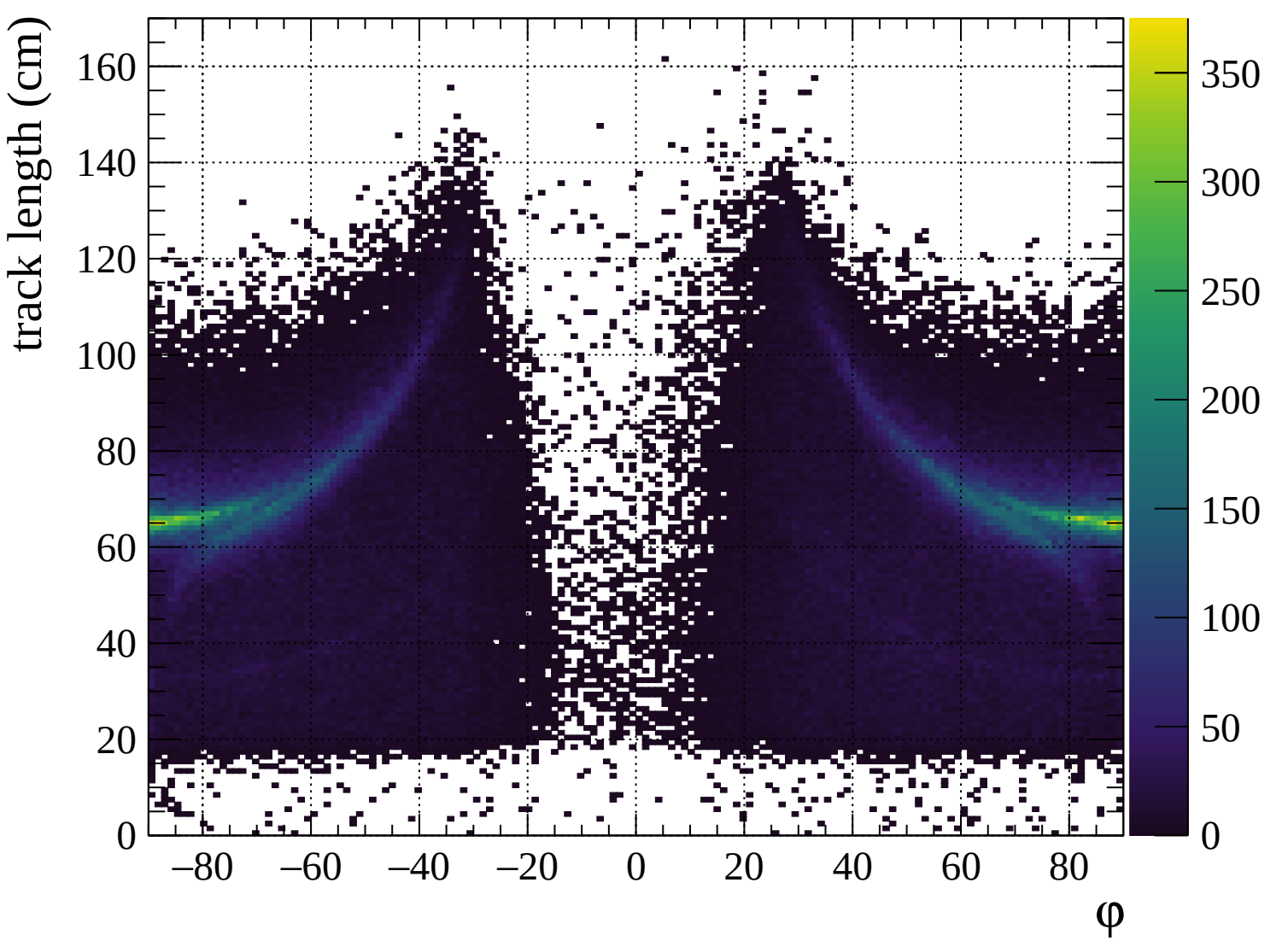


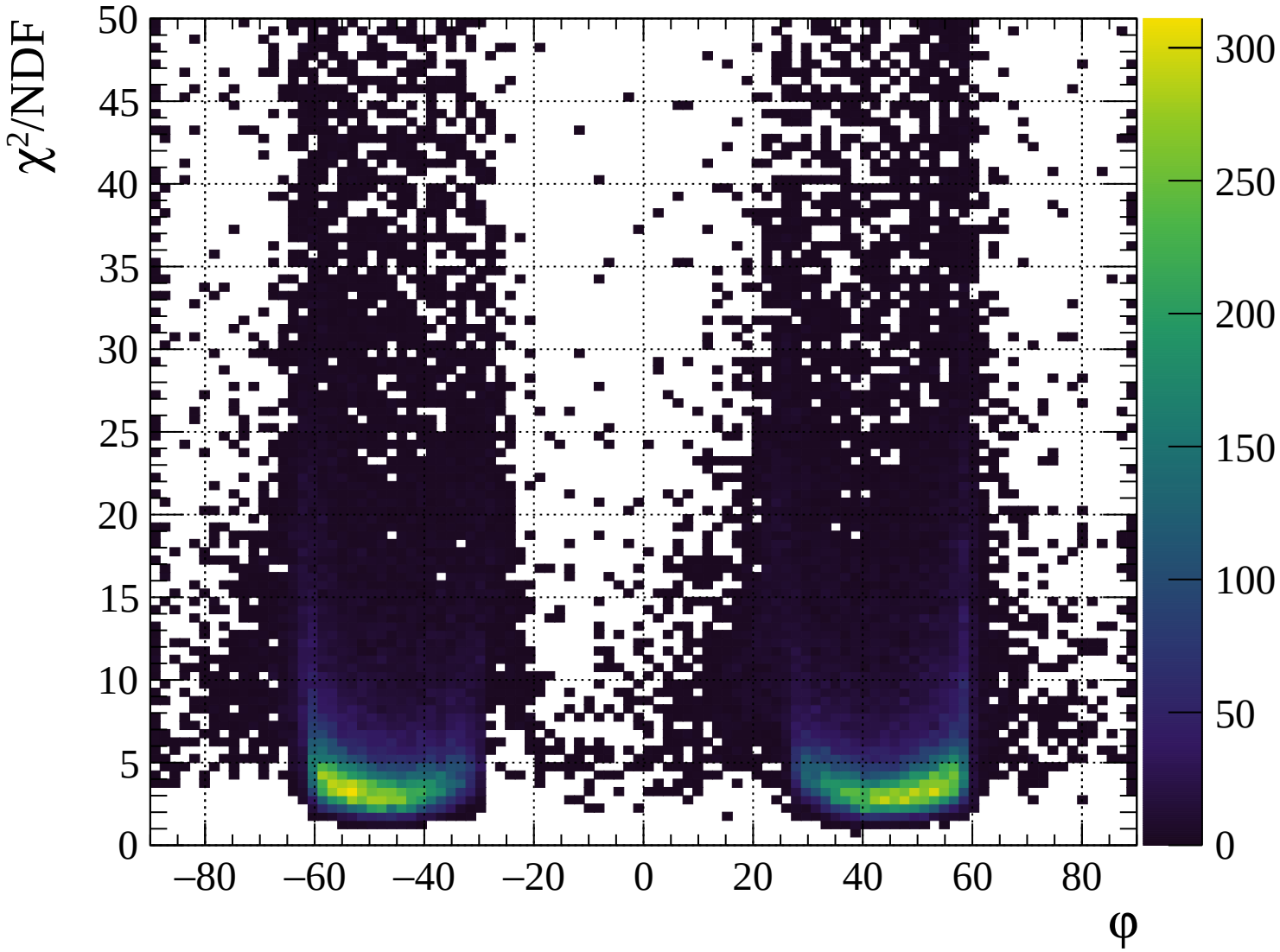


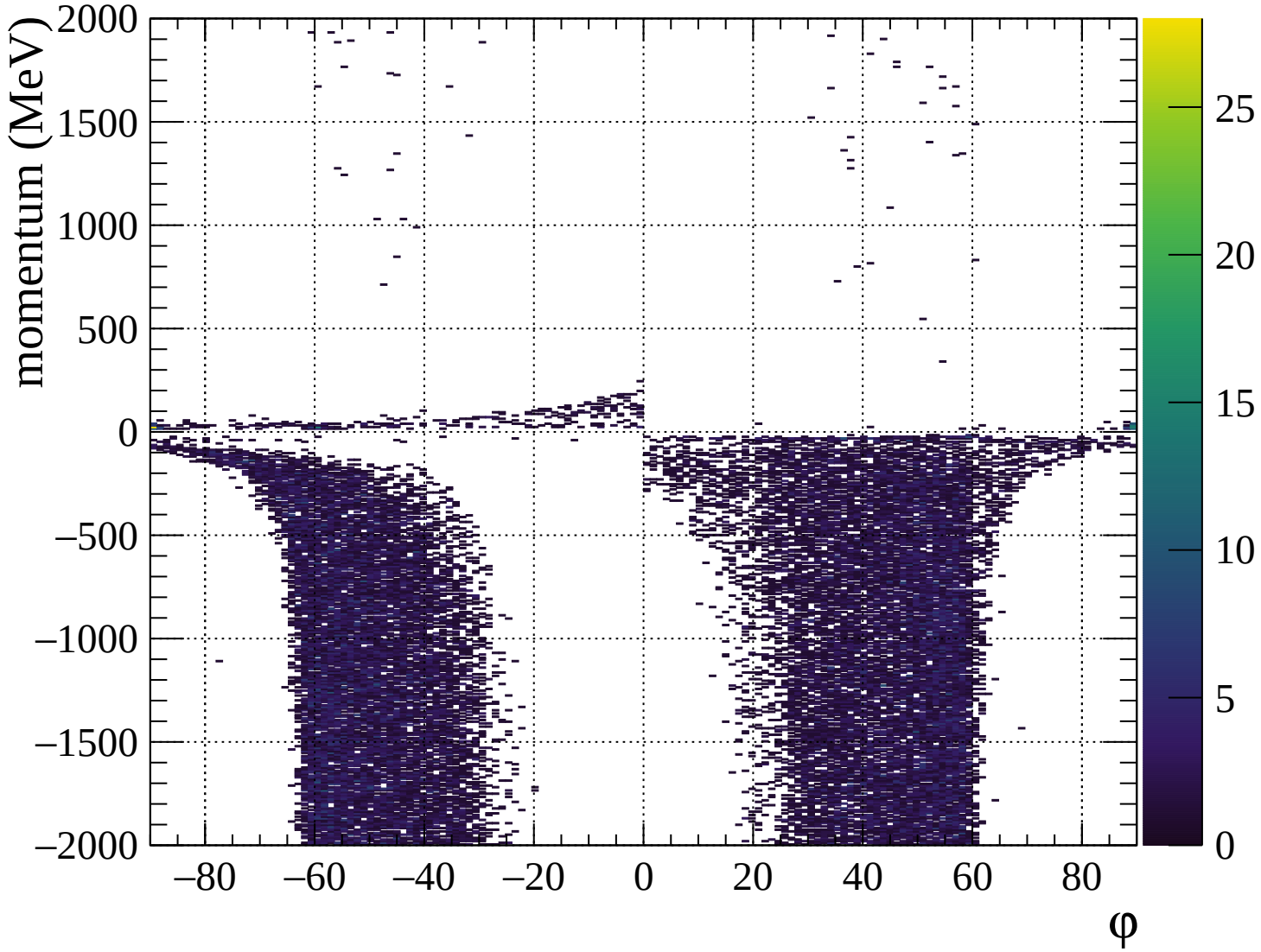


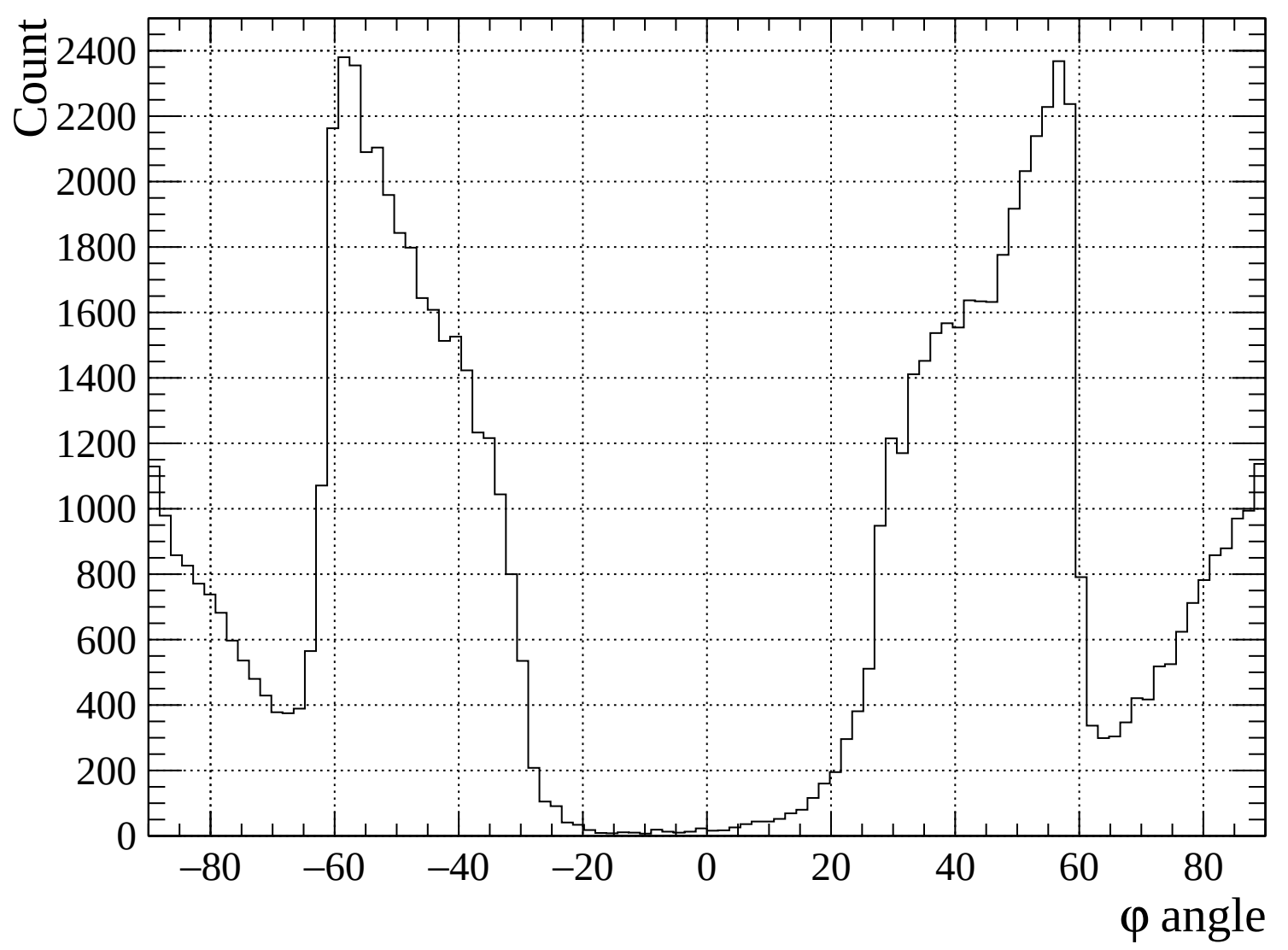


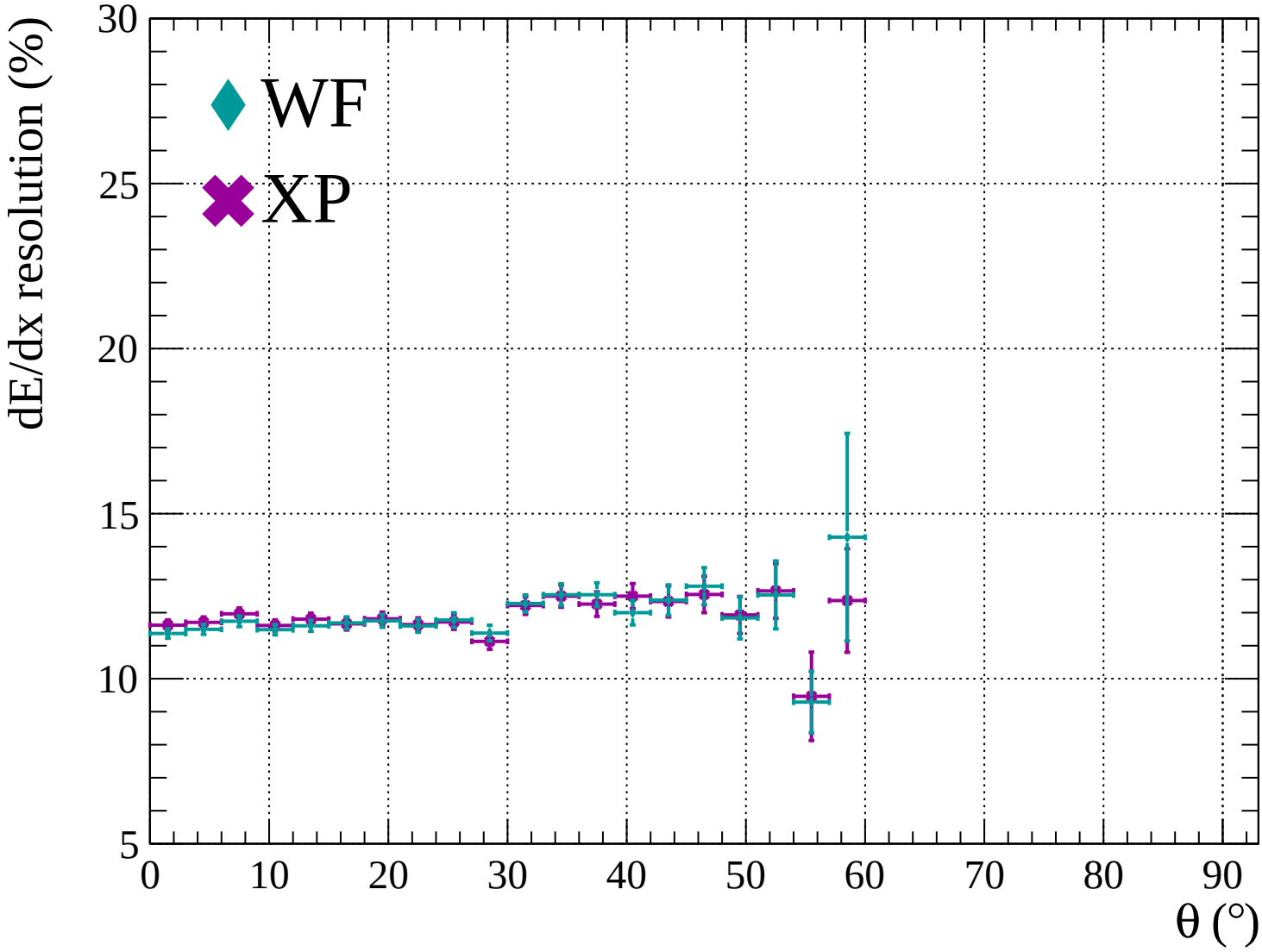


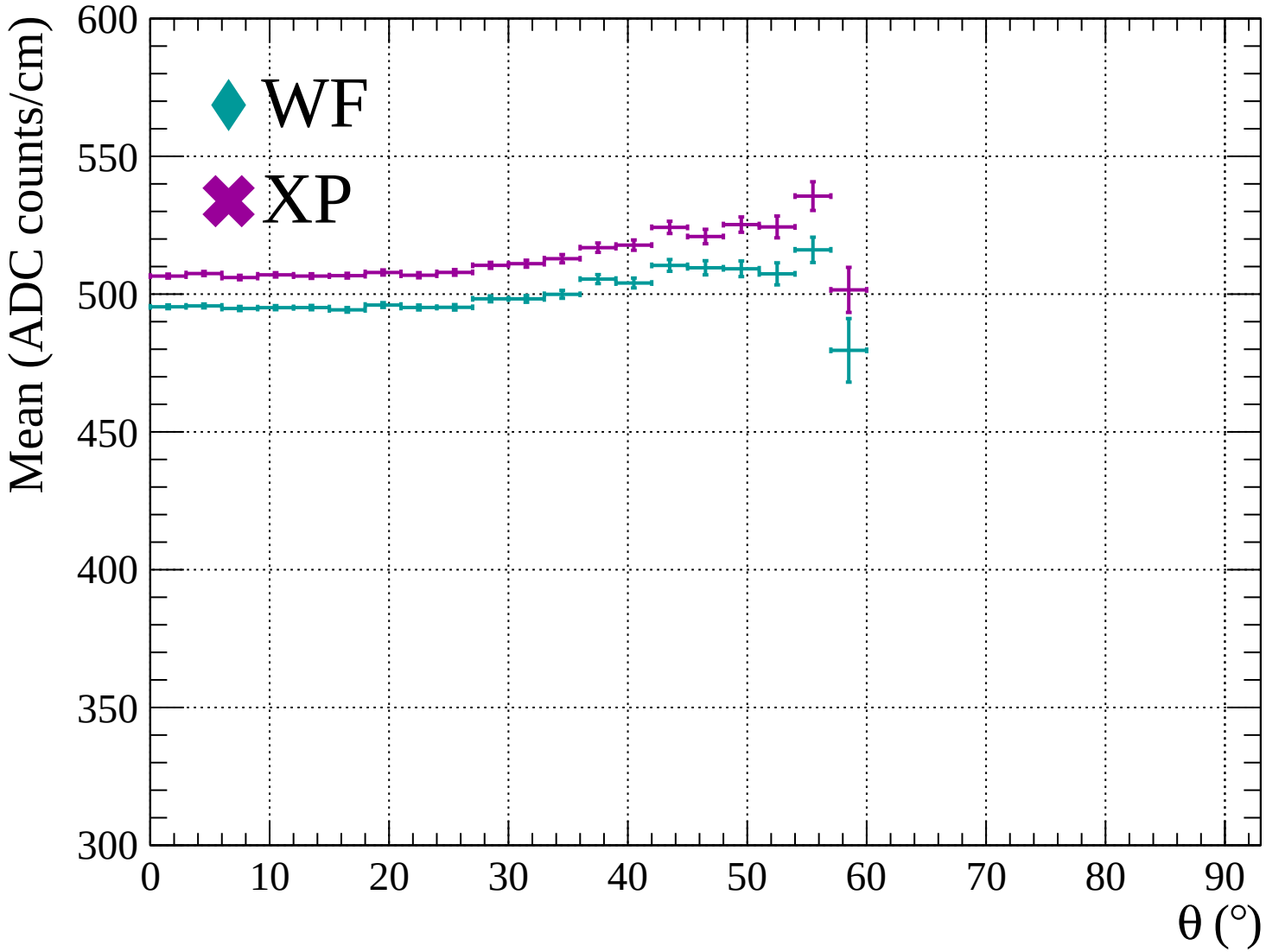


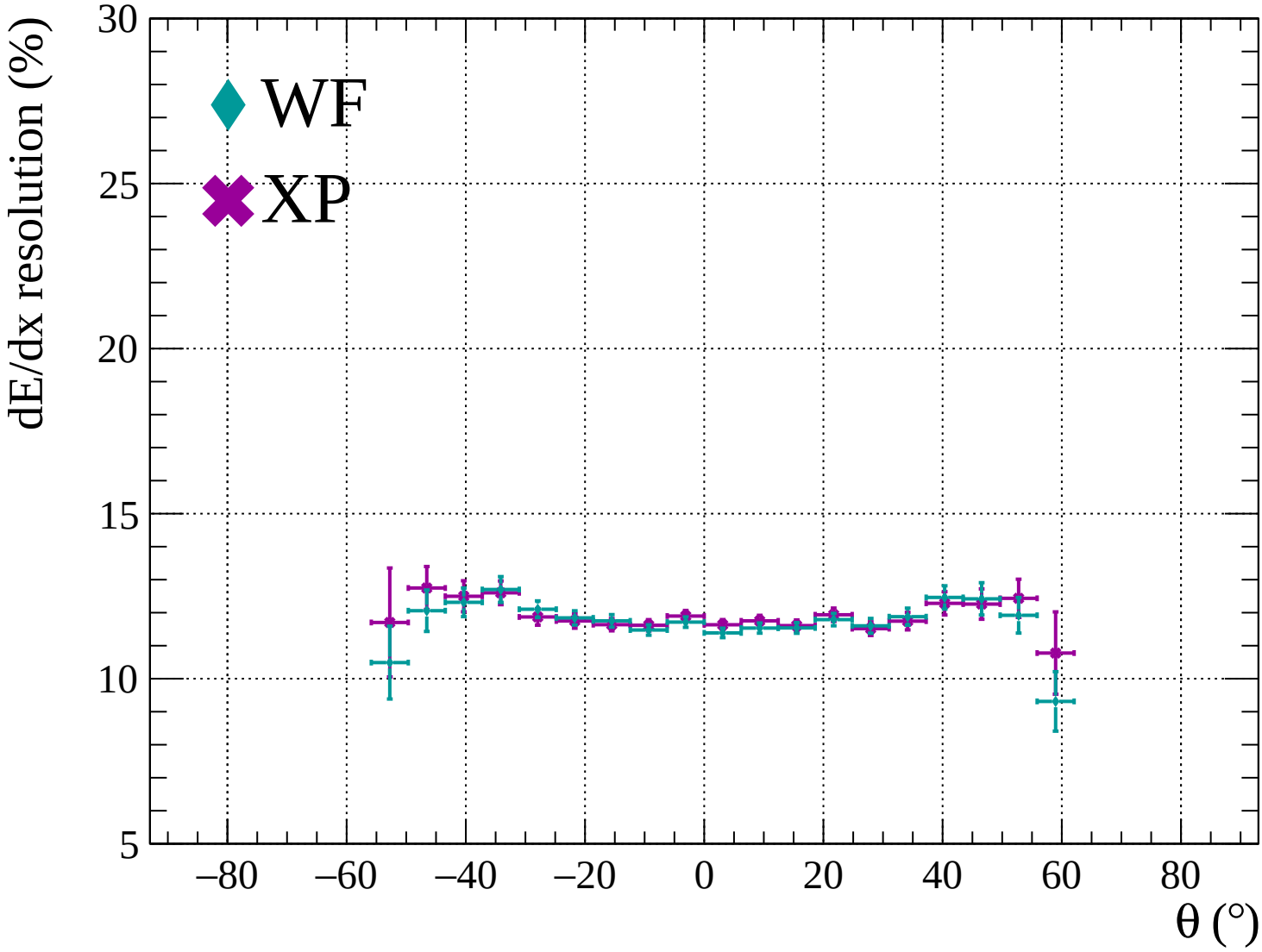


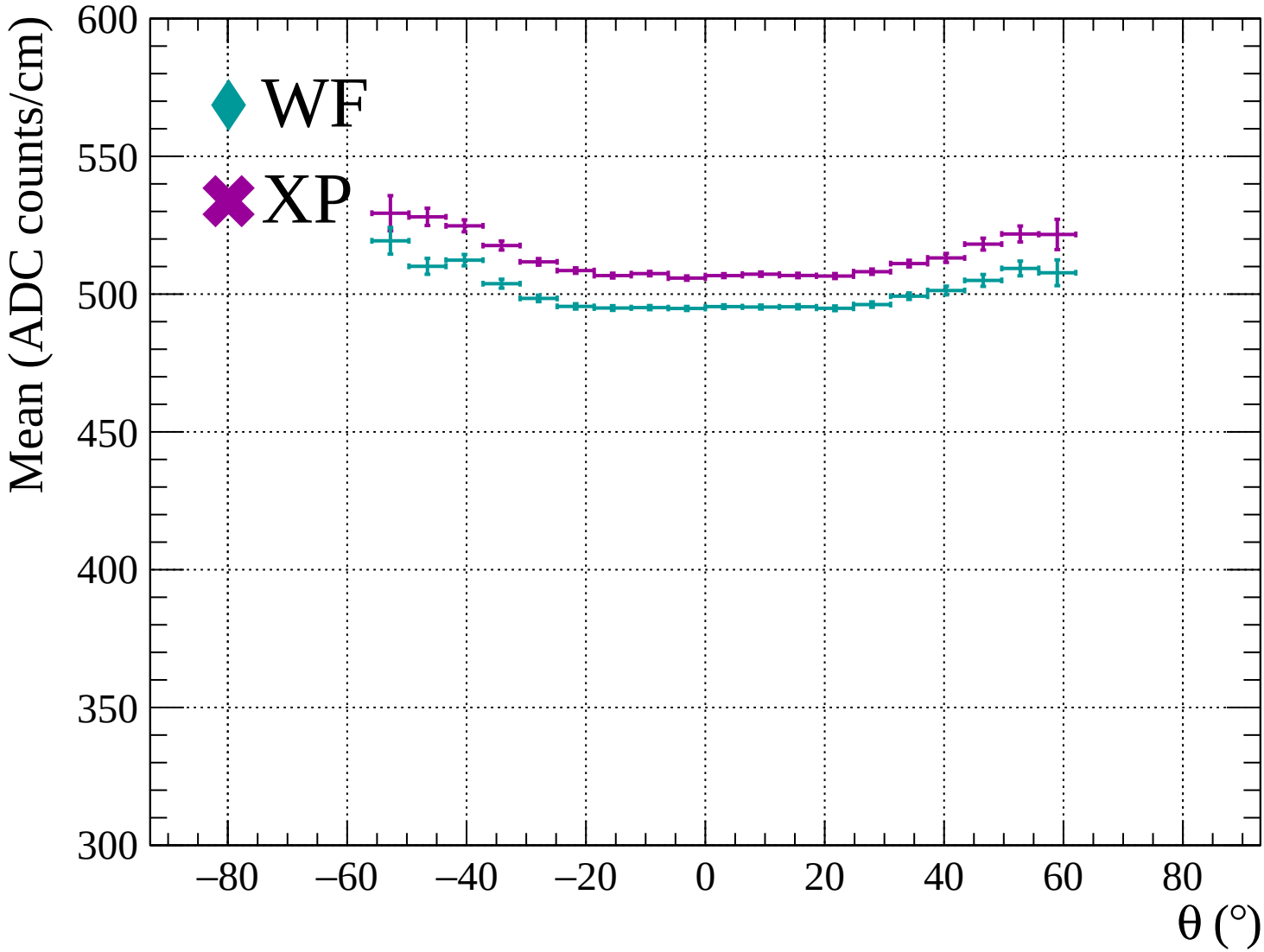


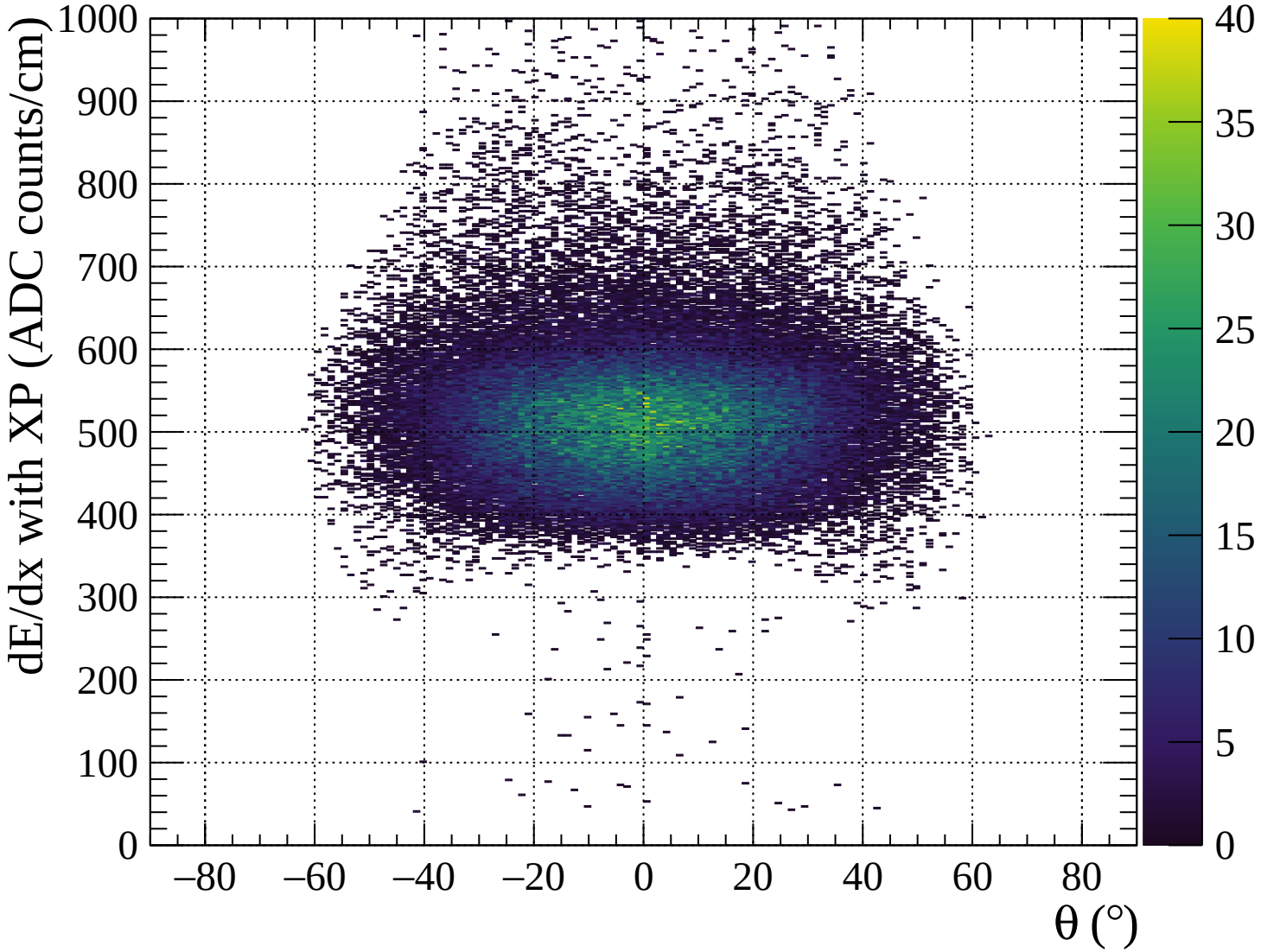


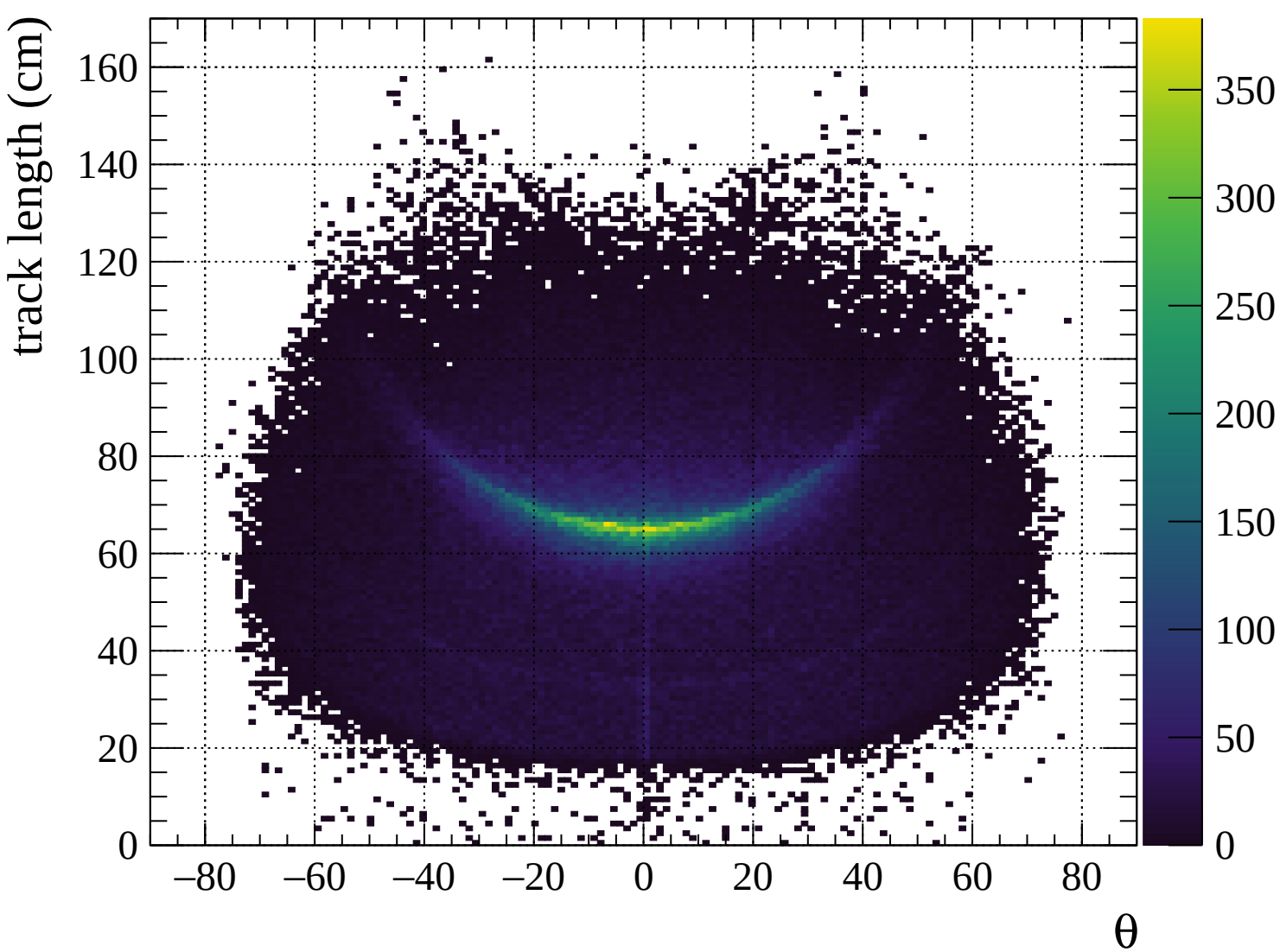


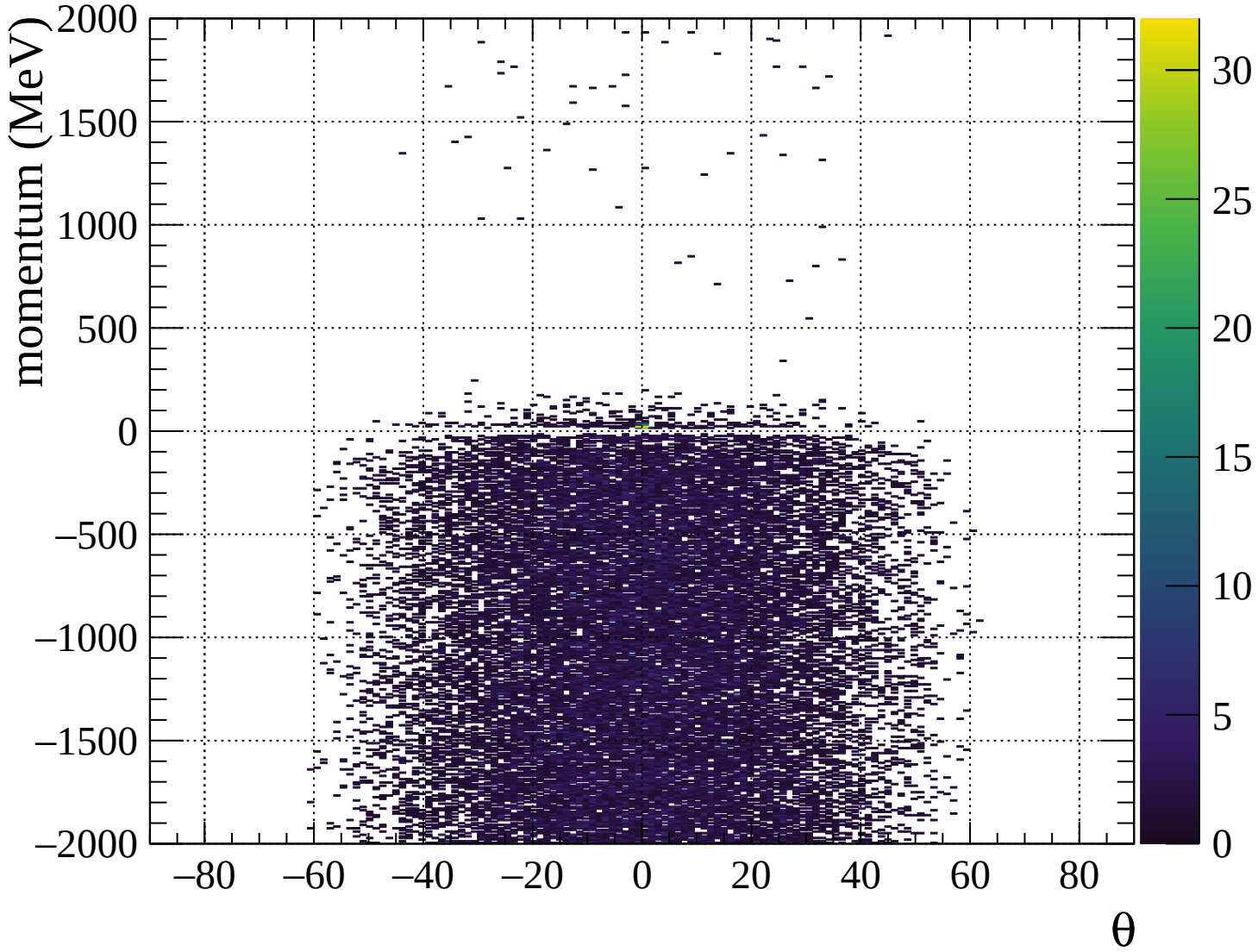


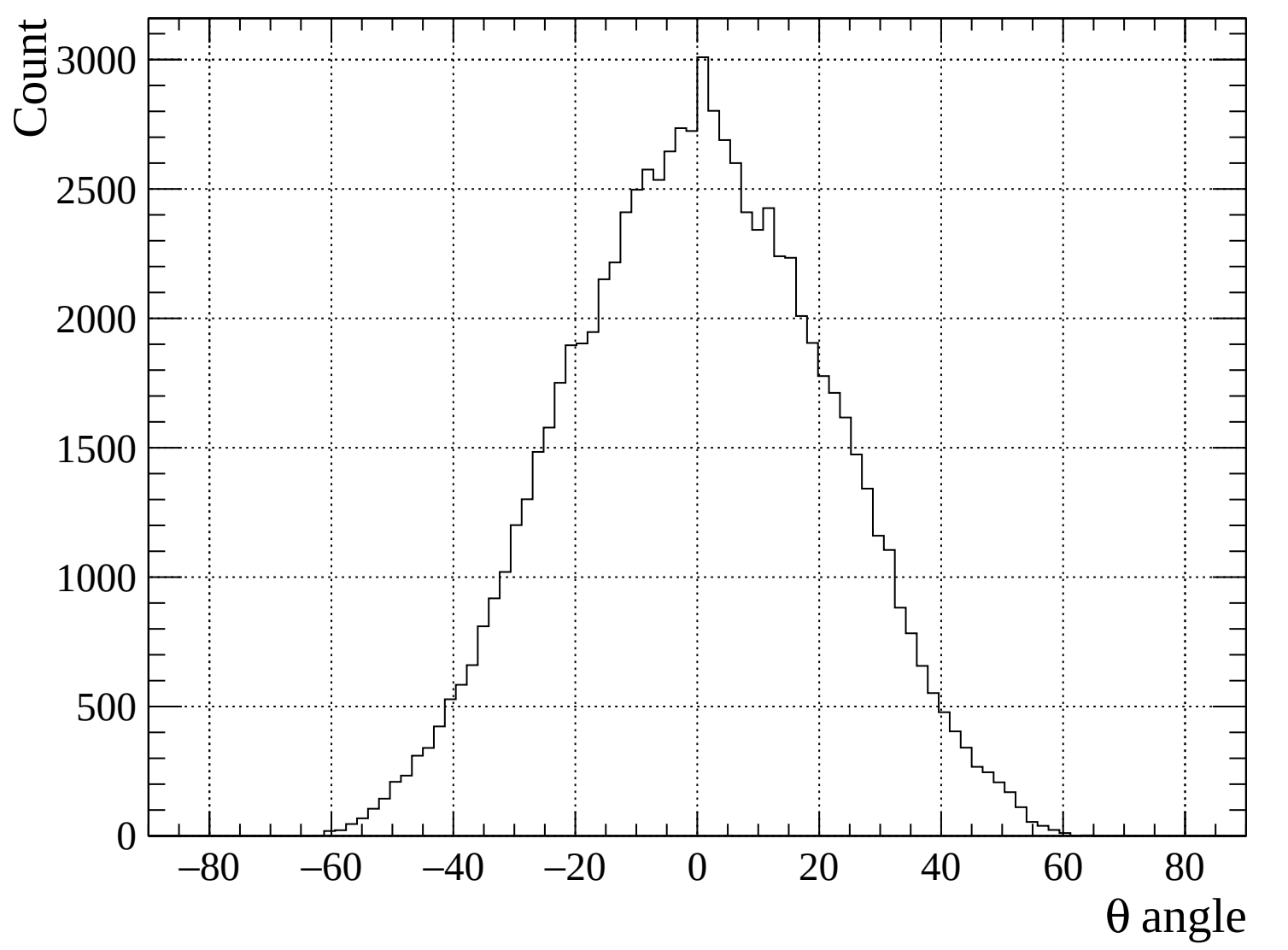


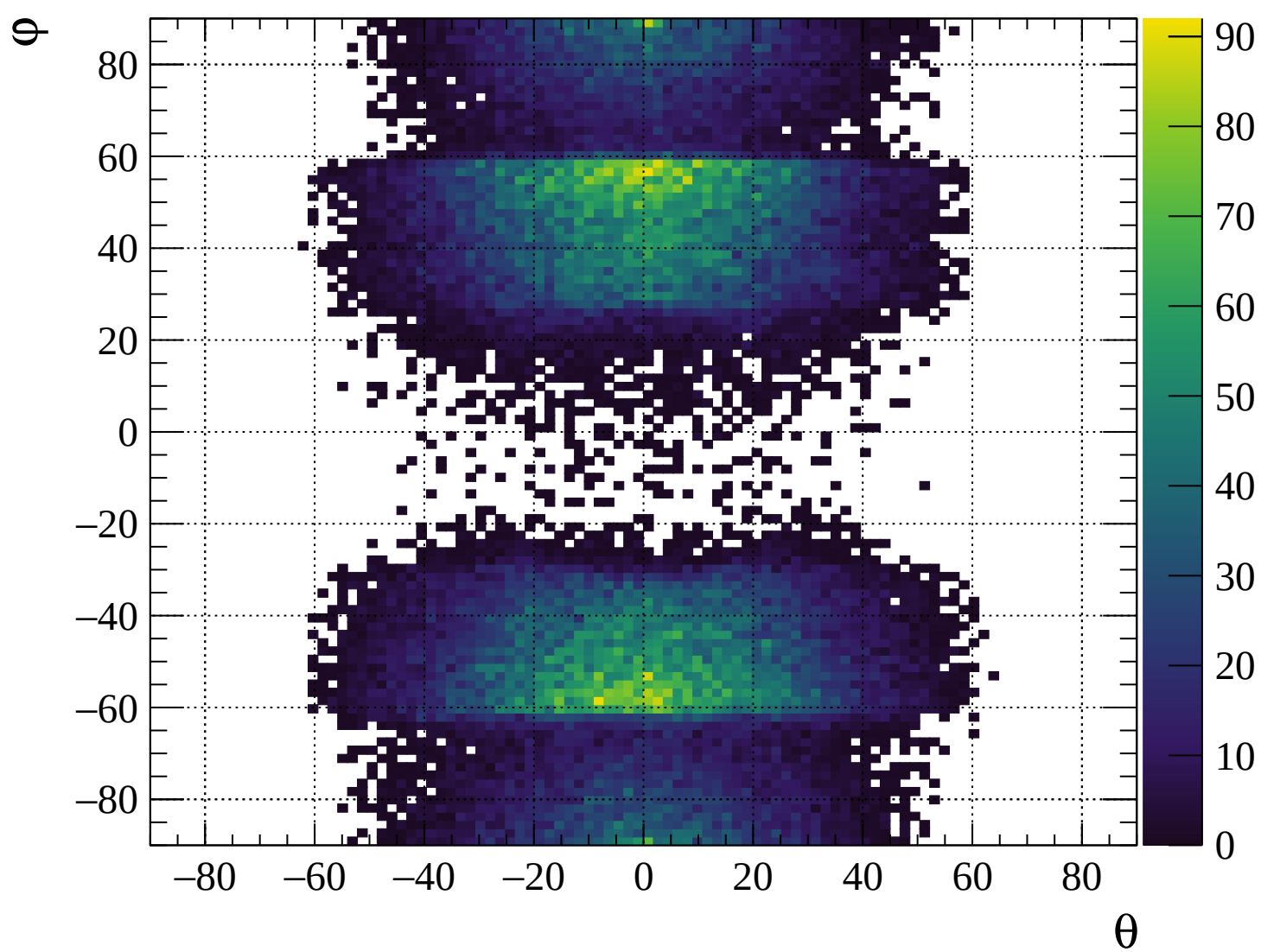


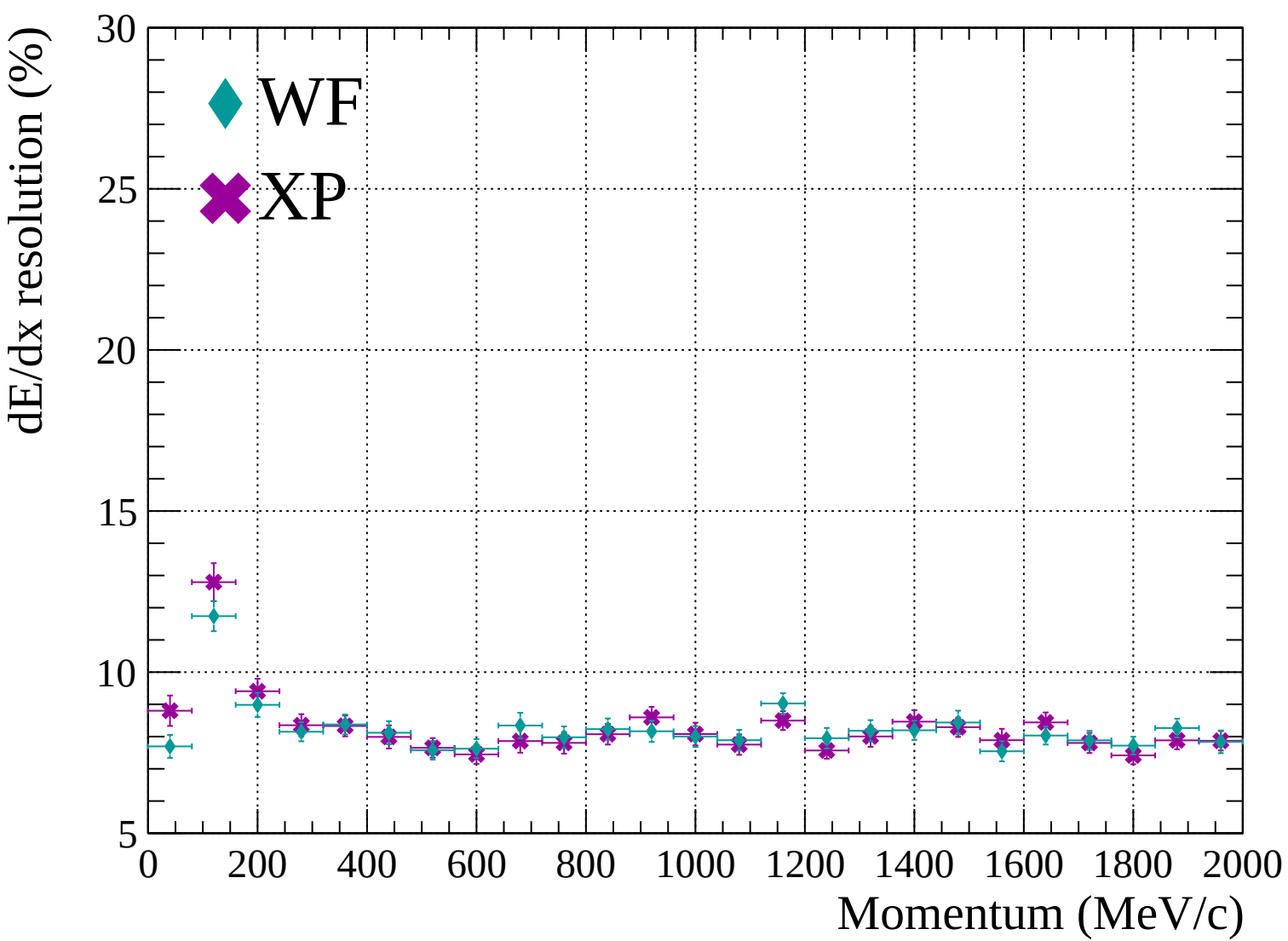


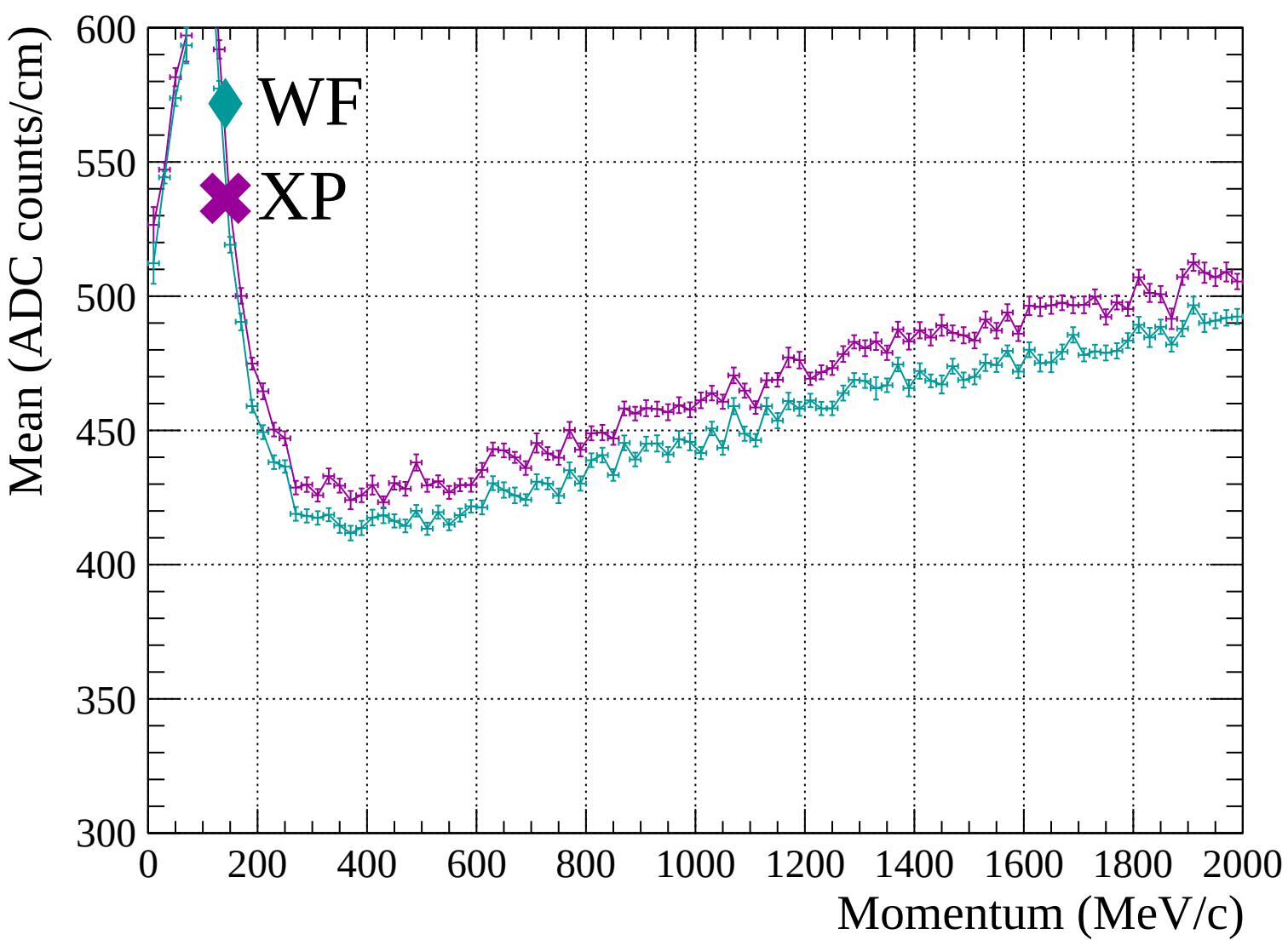


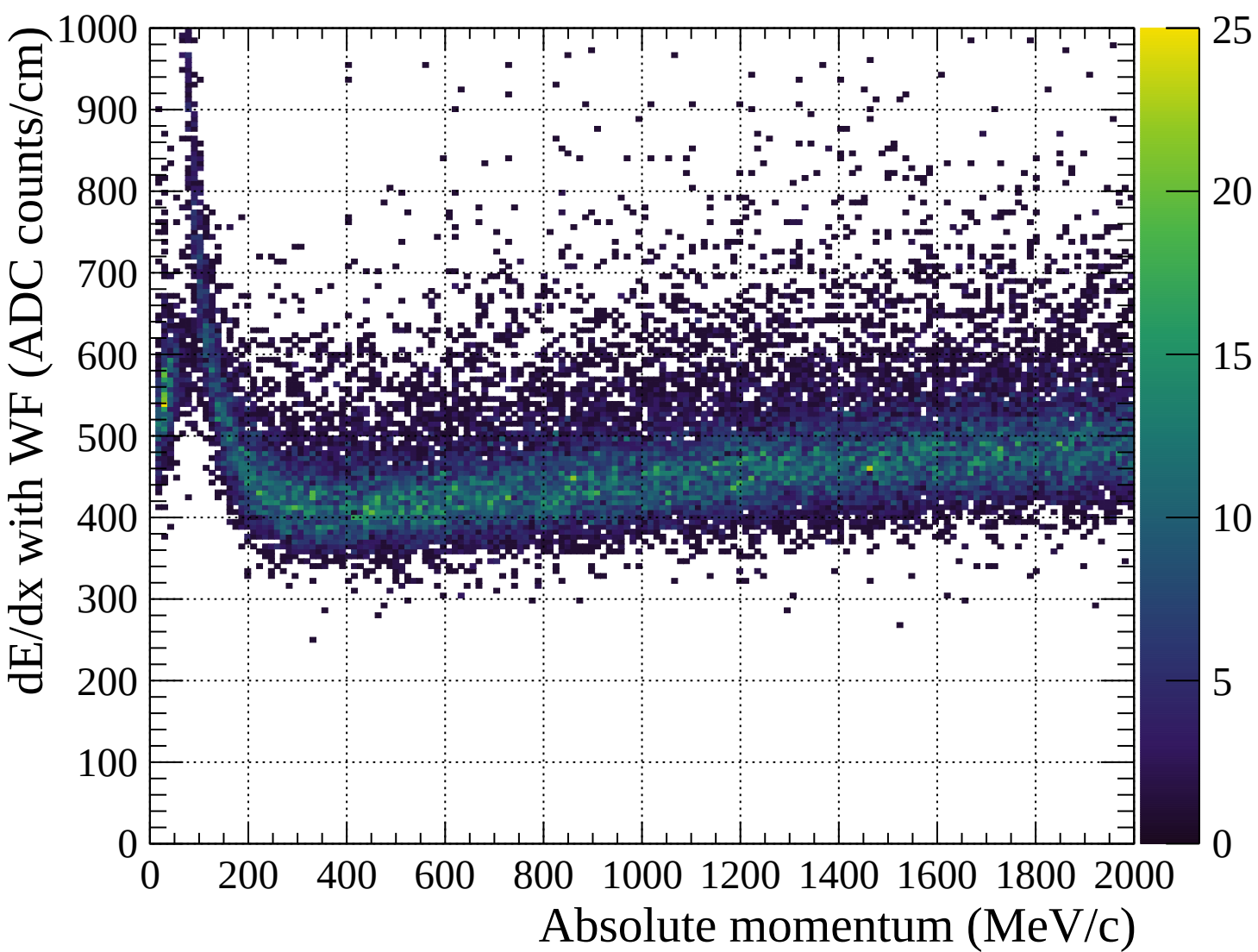


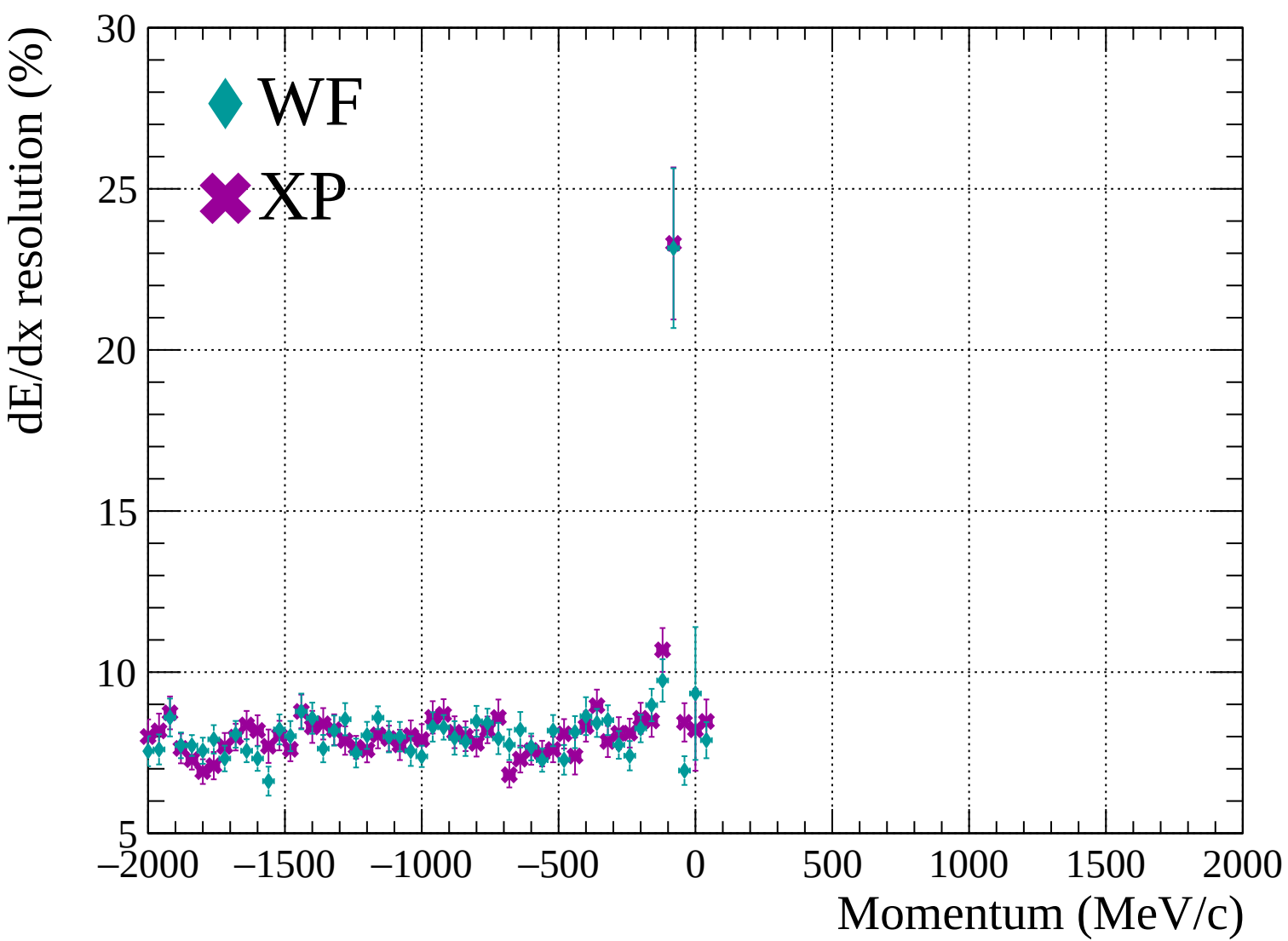


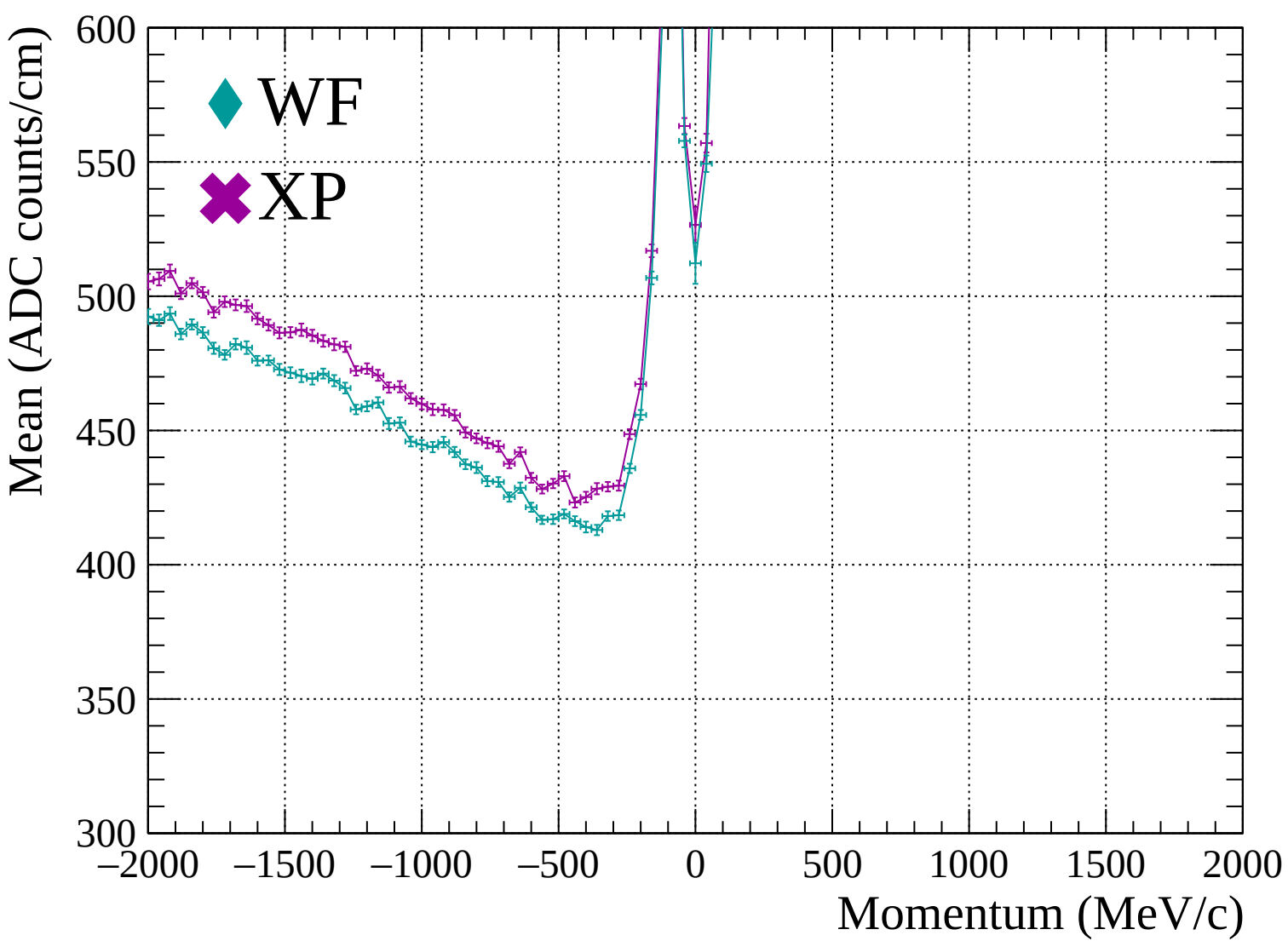


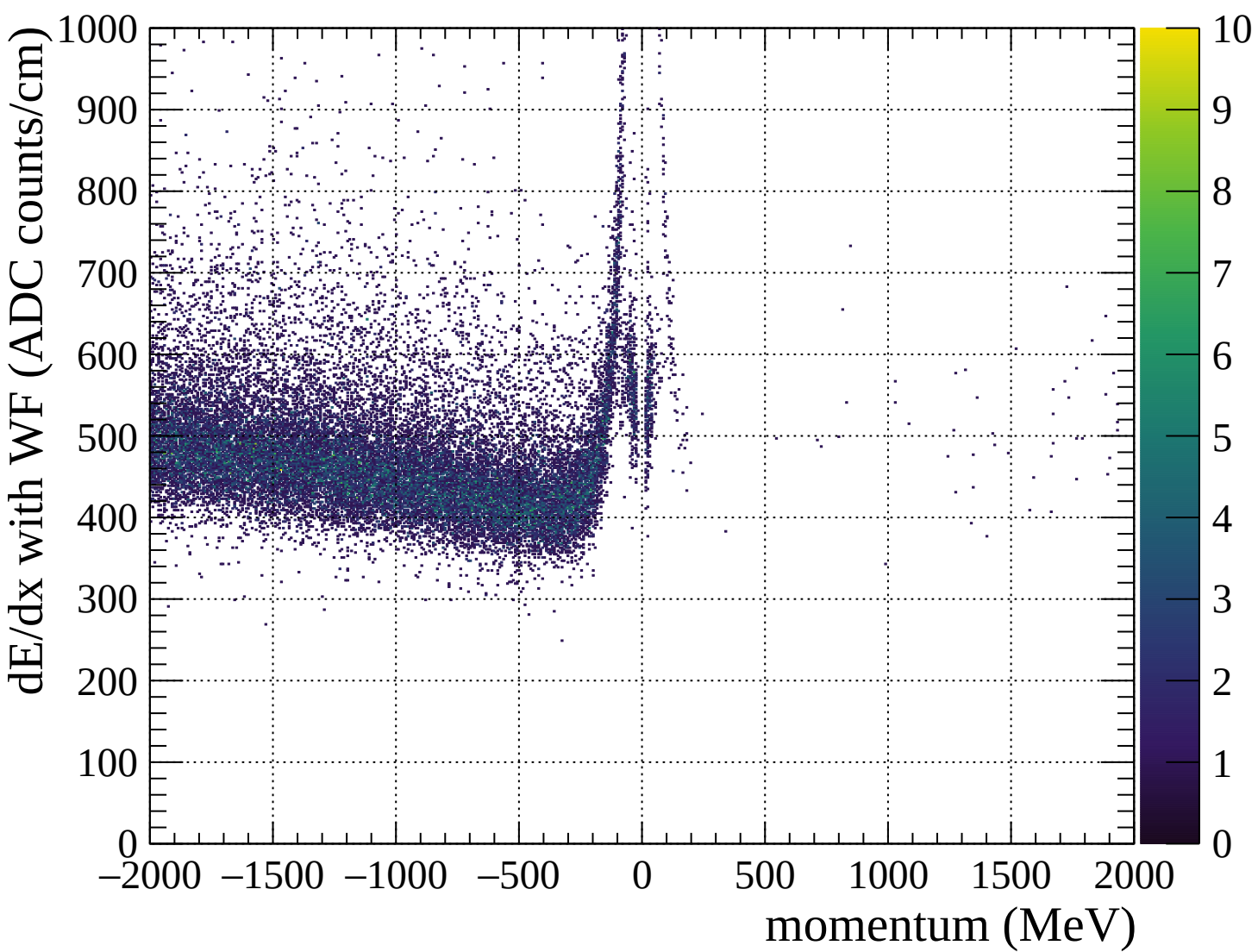


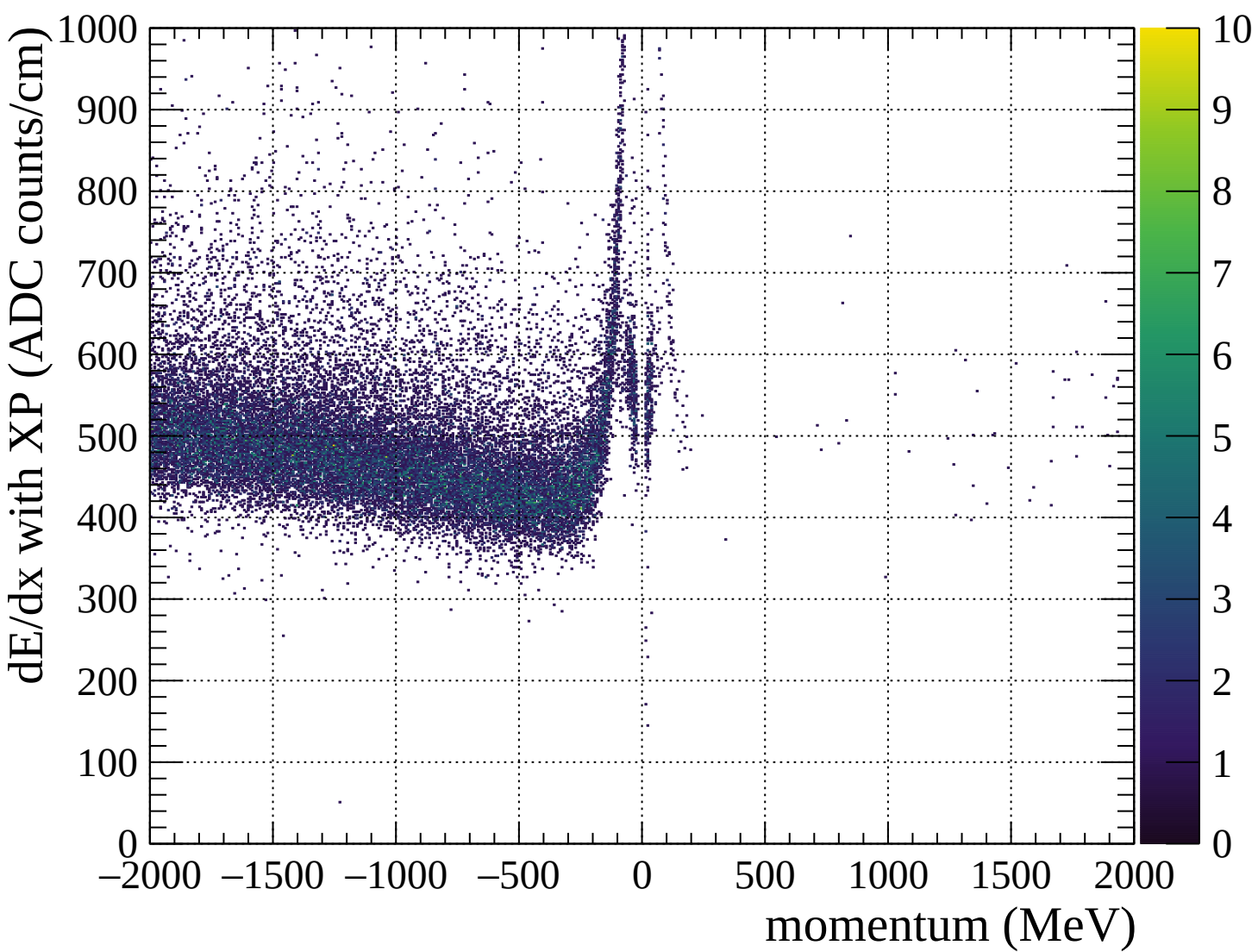


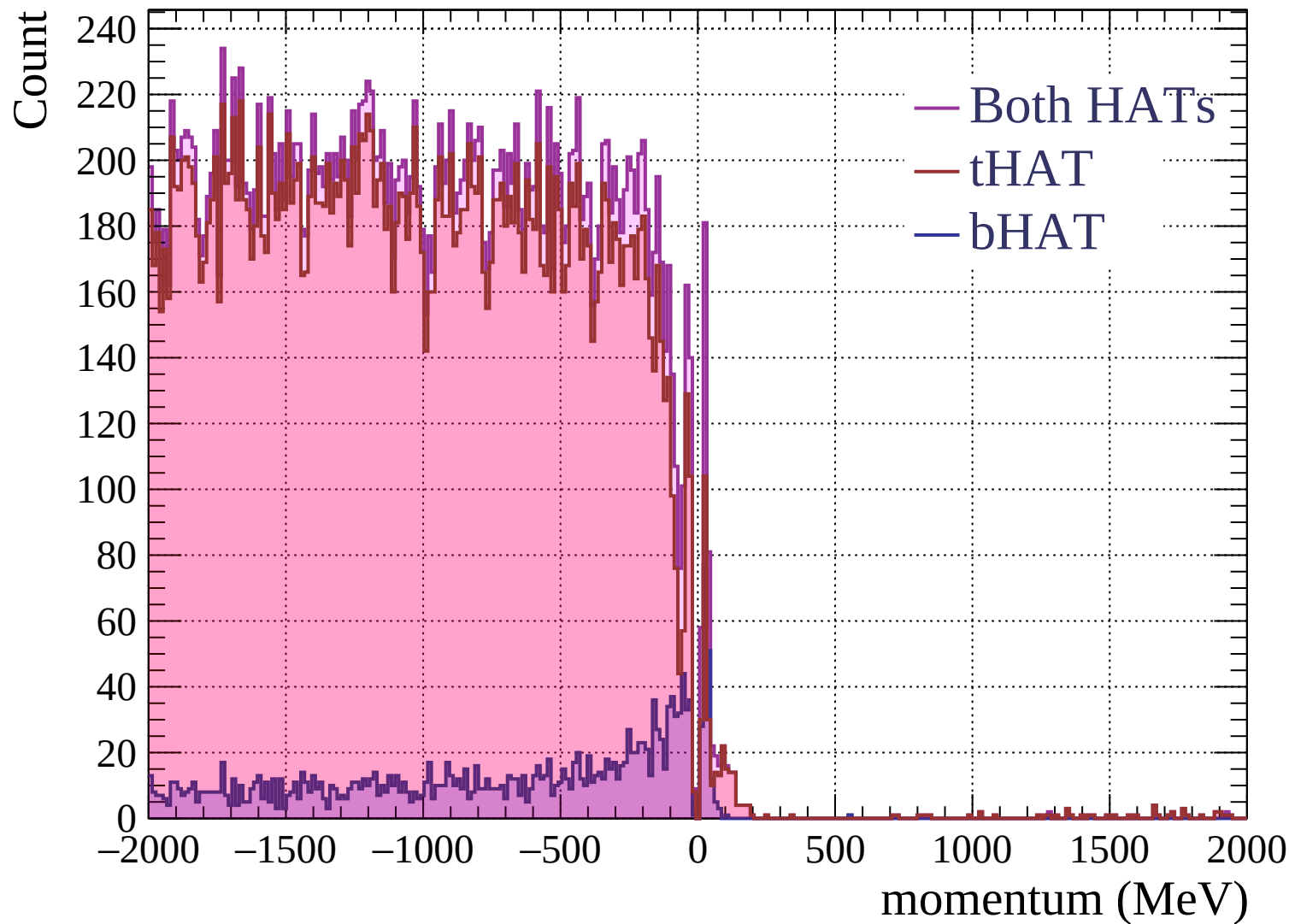


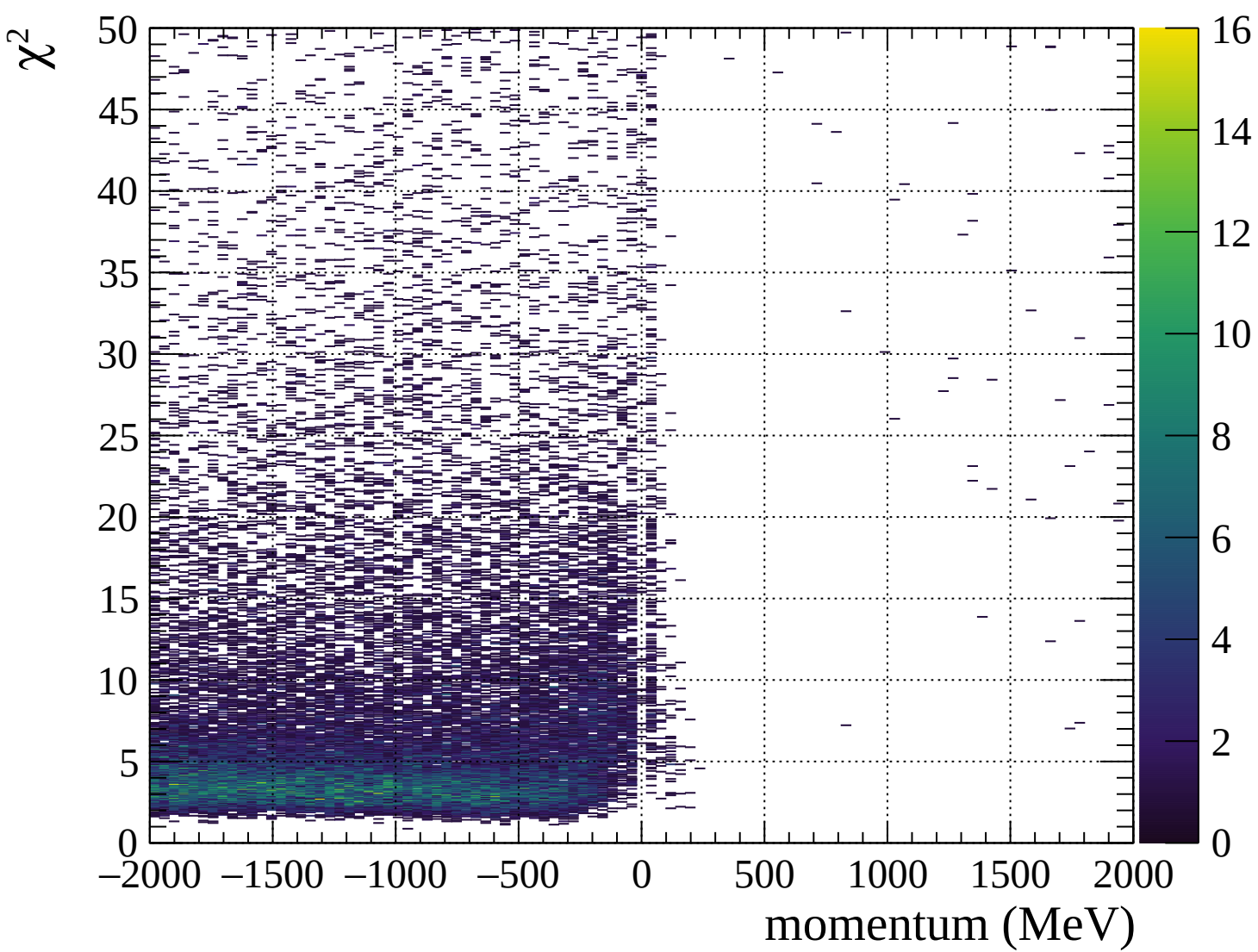




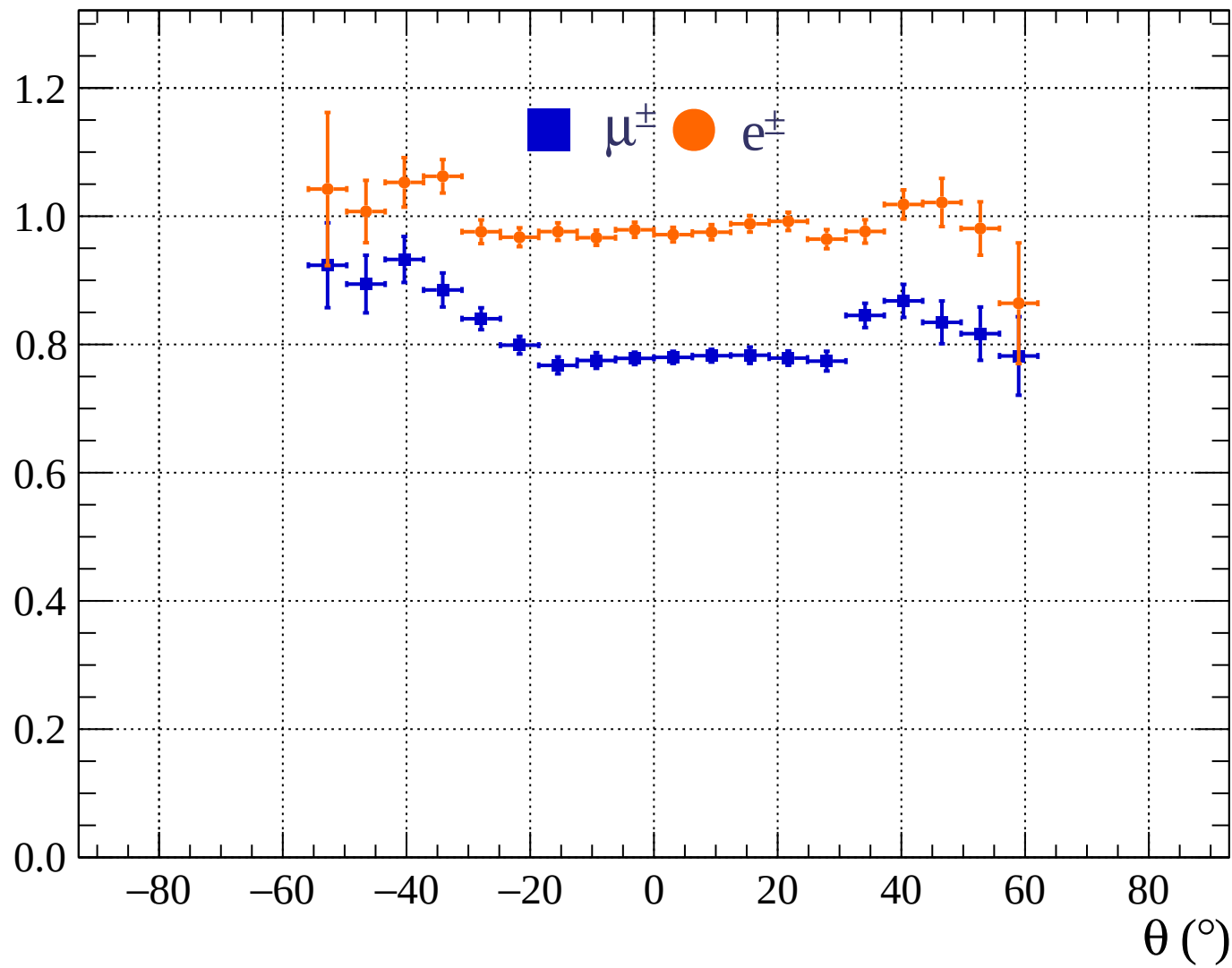


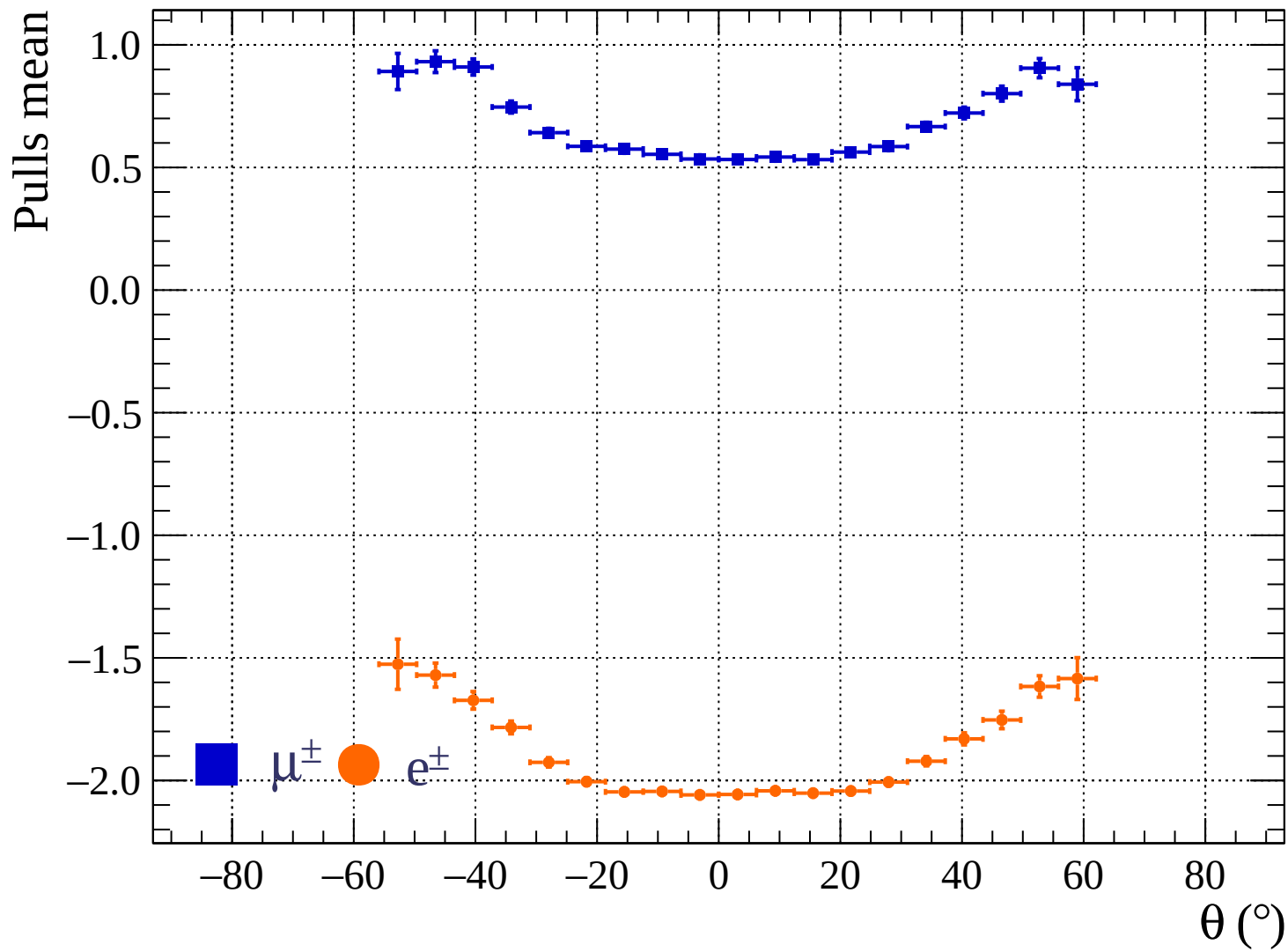




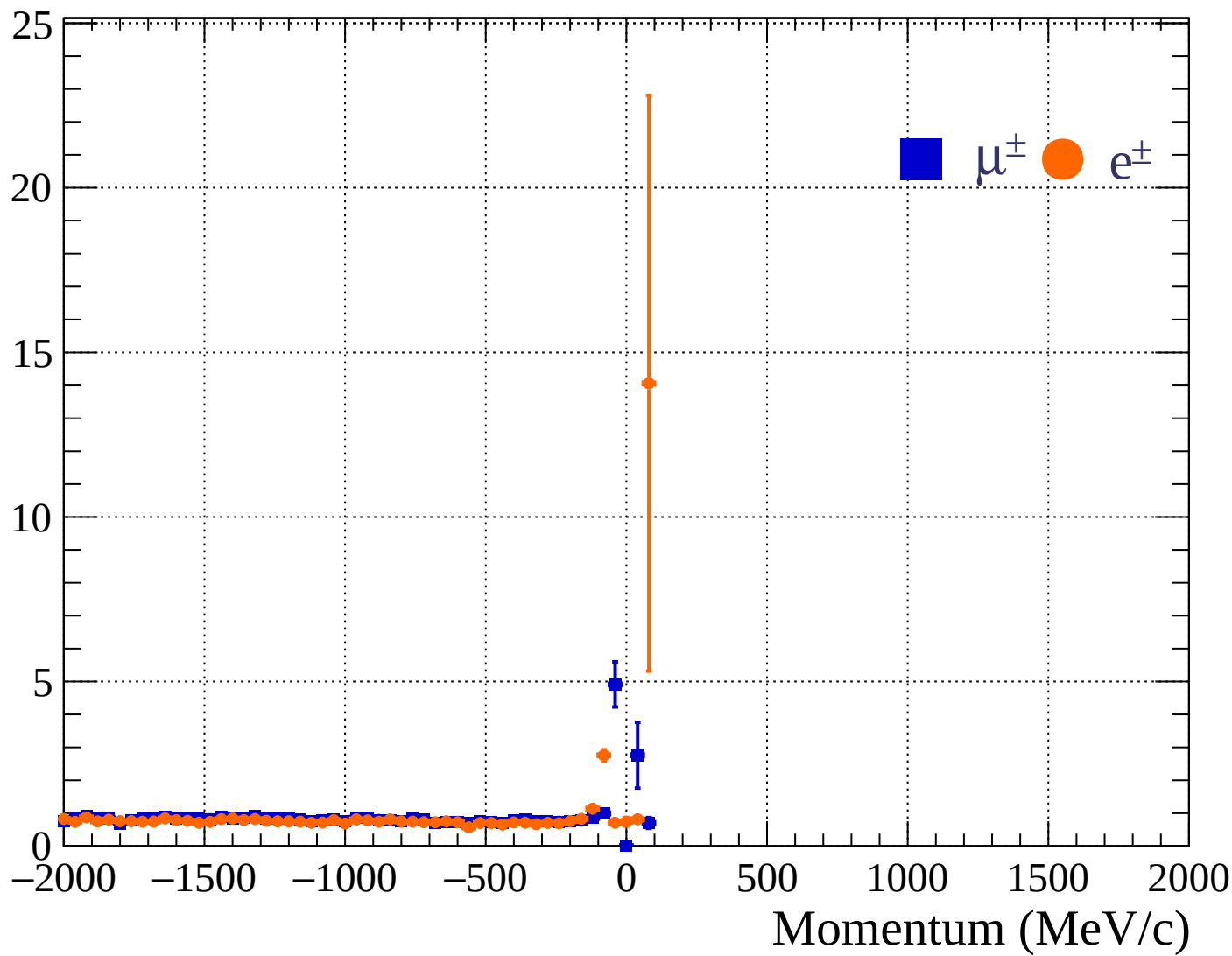


Pulls standard deviation

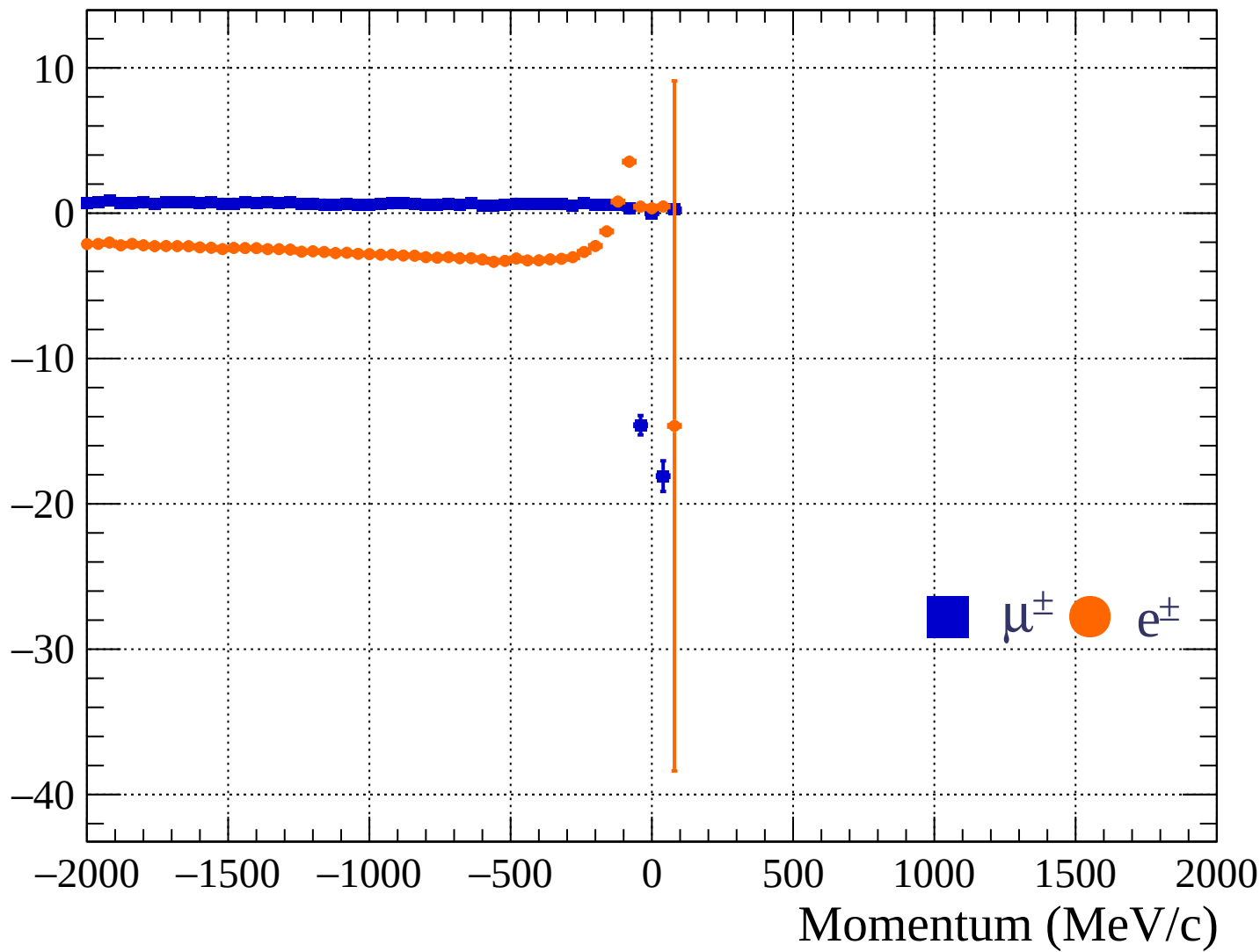


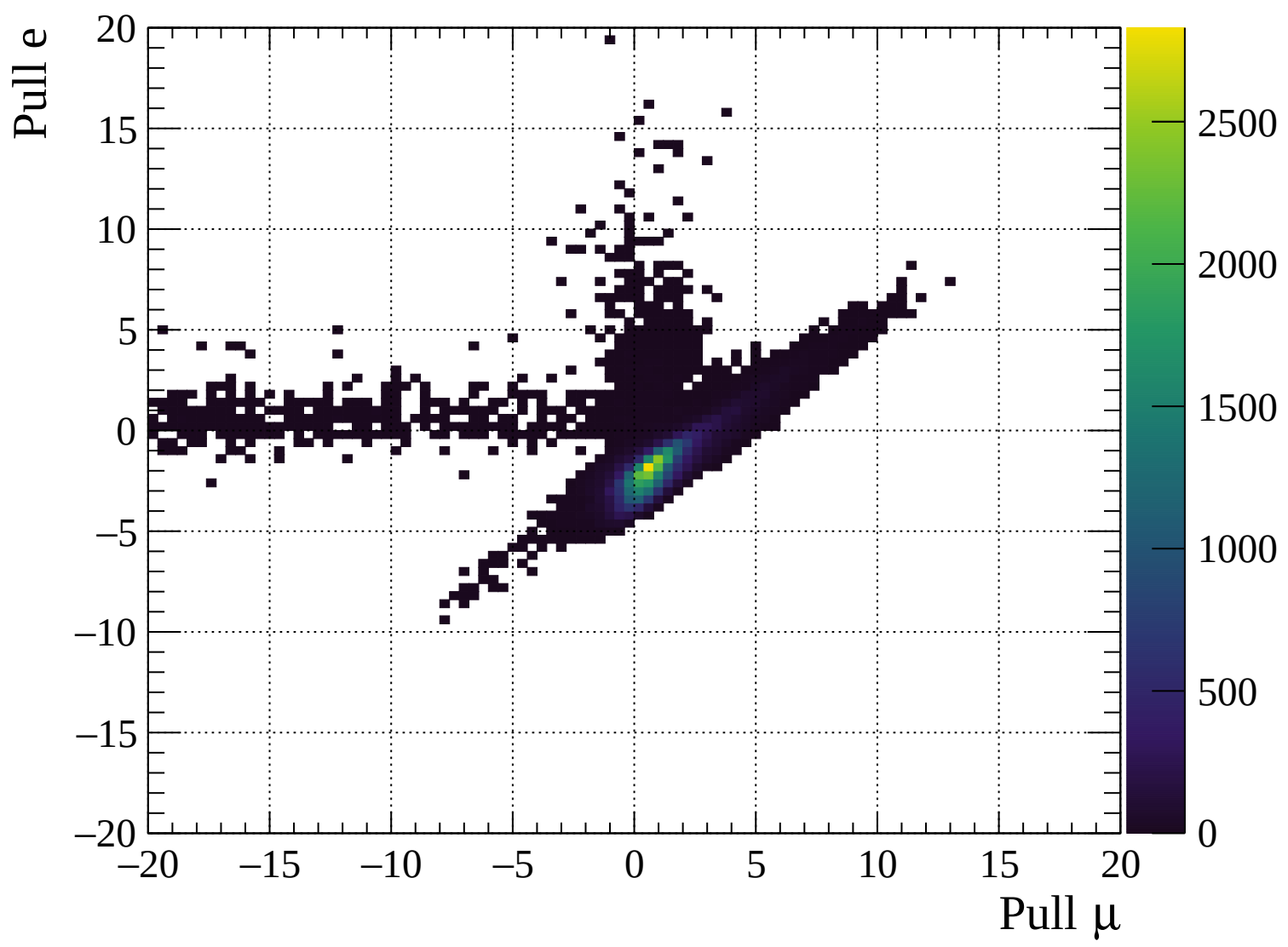


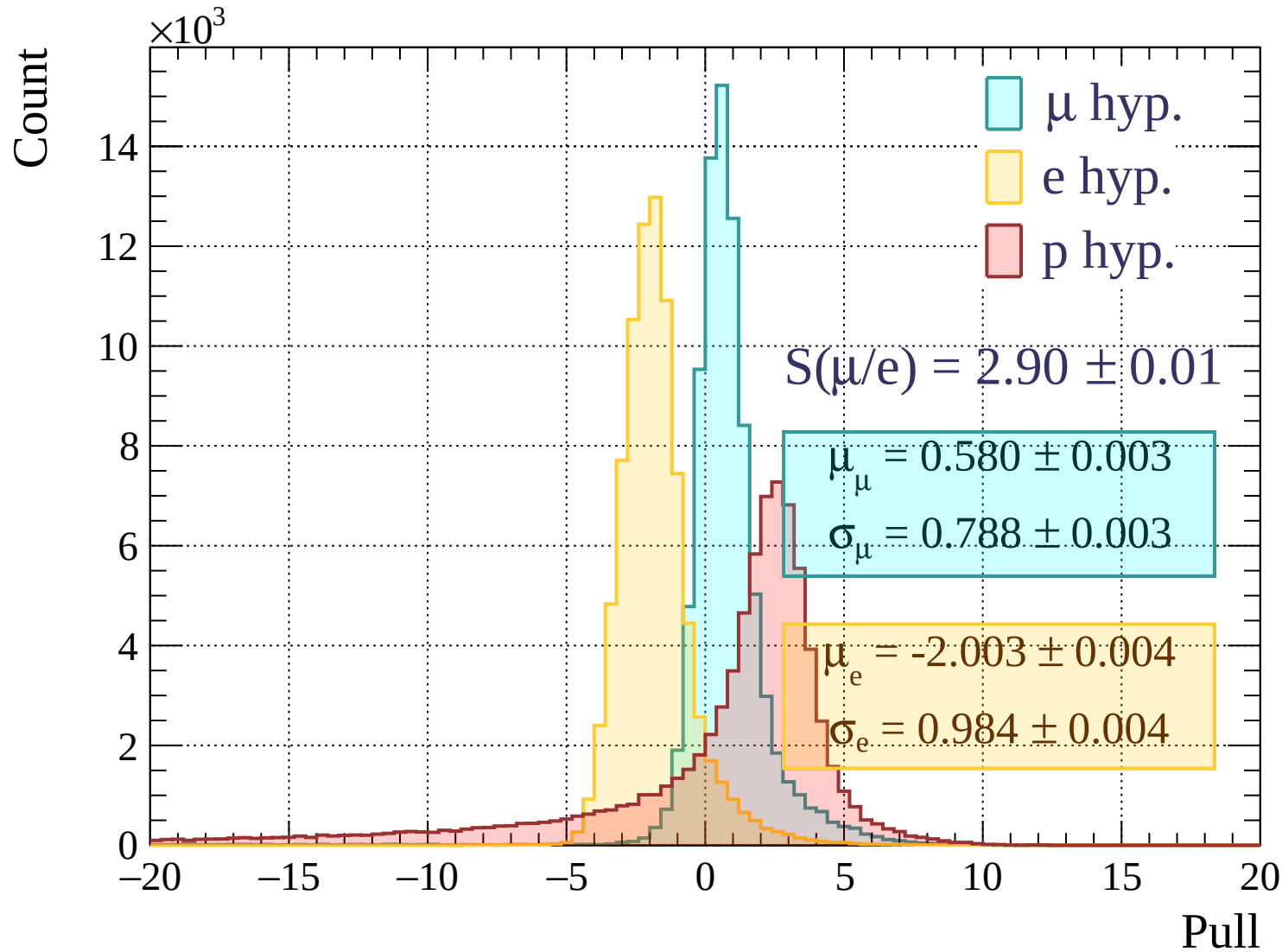
Pulls standard deviation

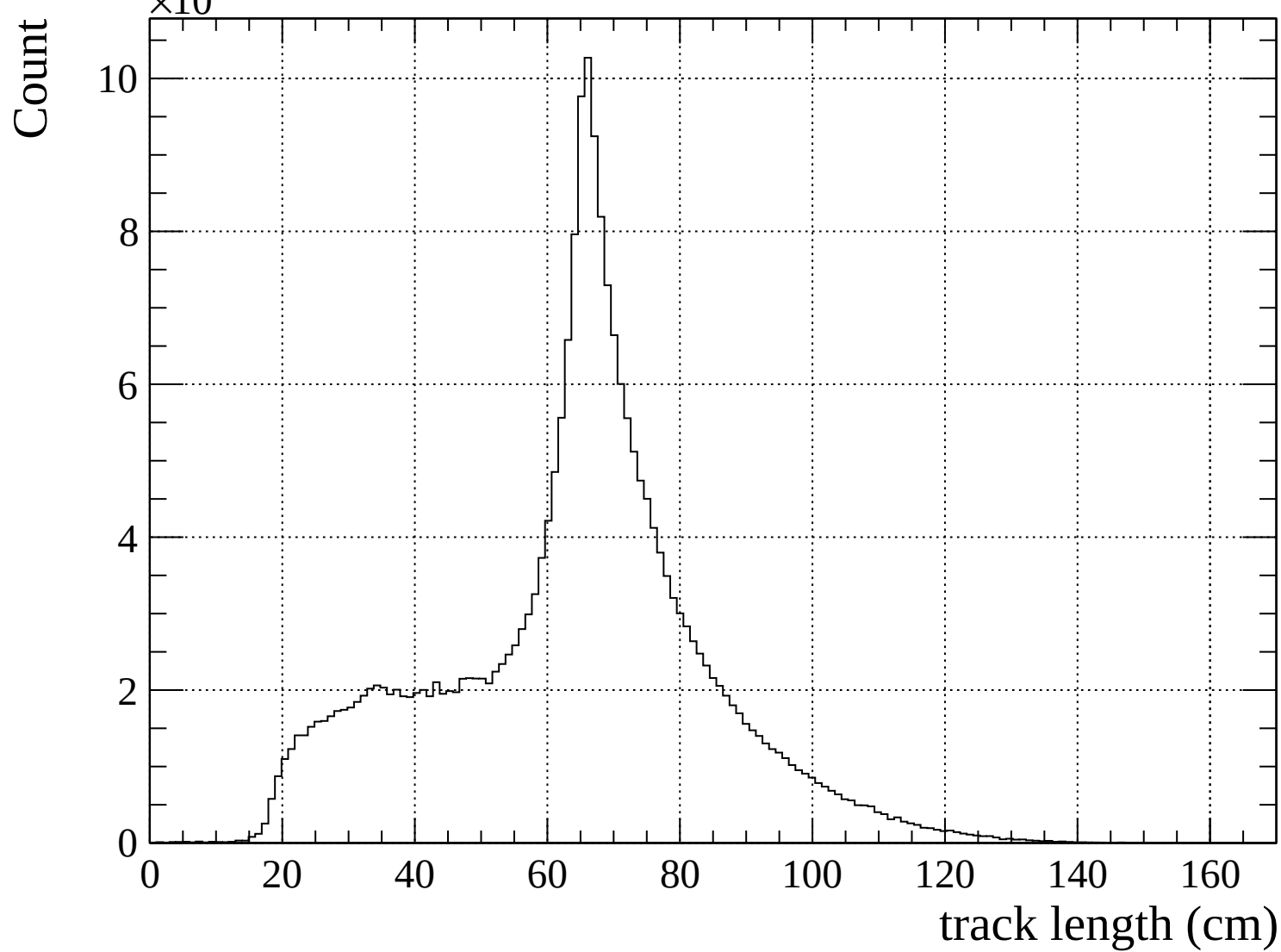


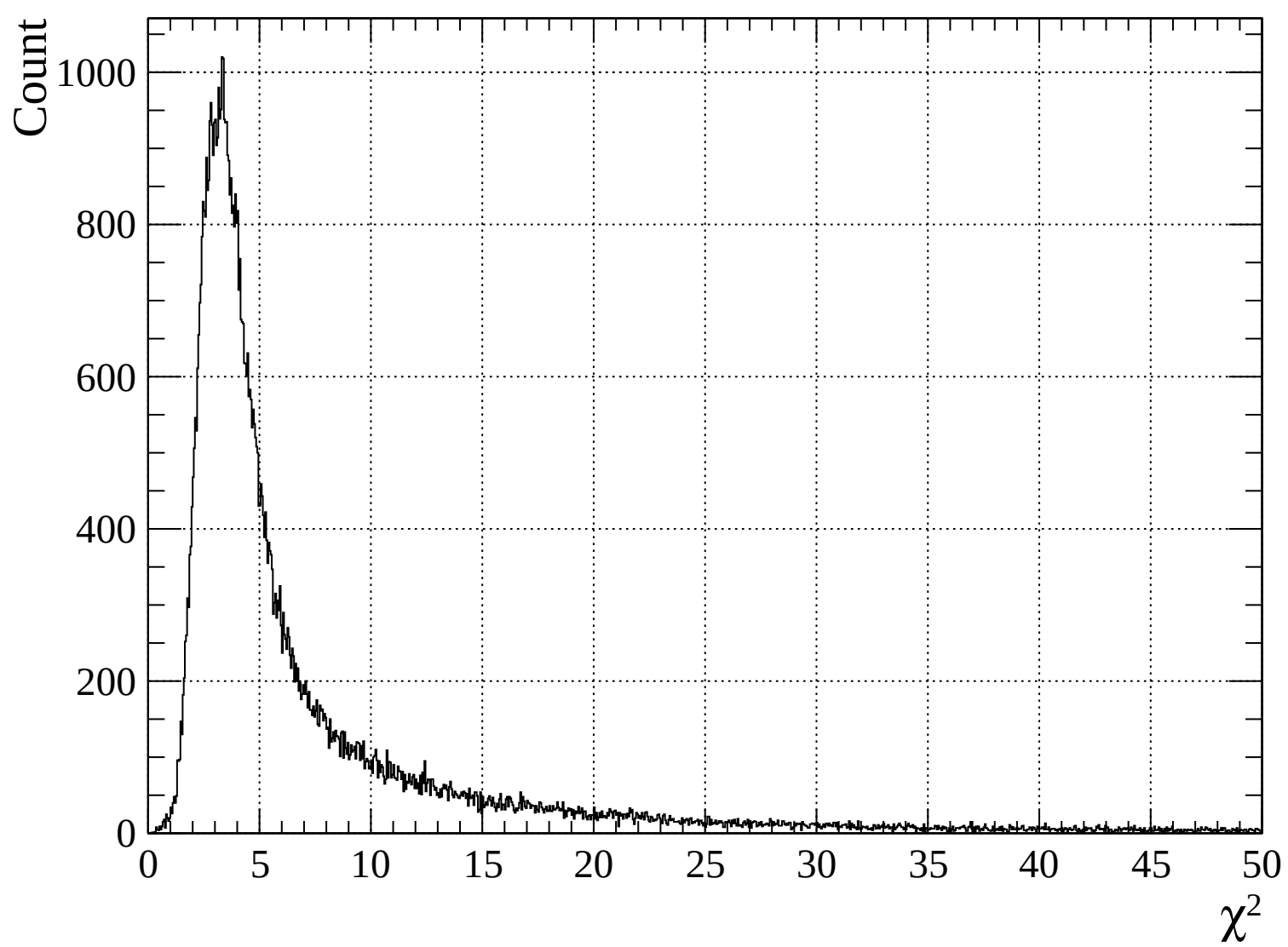
Pulls mean

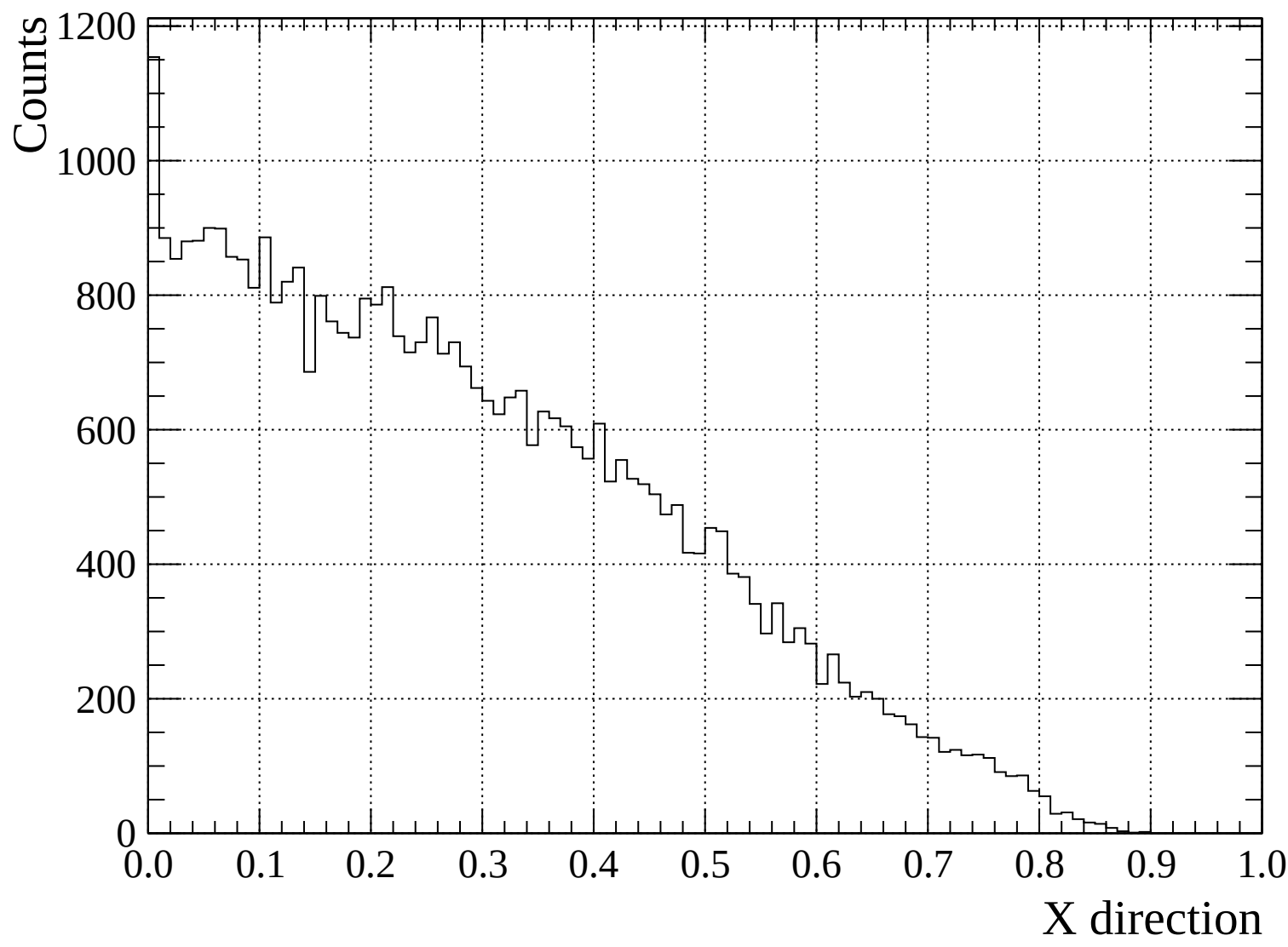


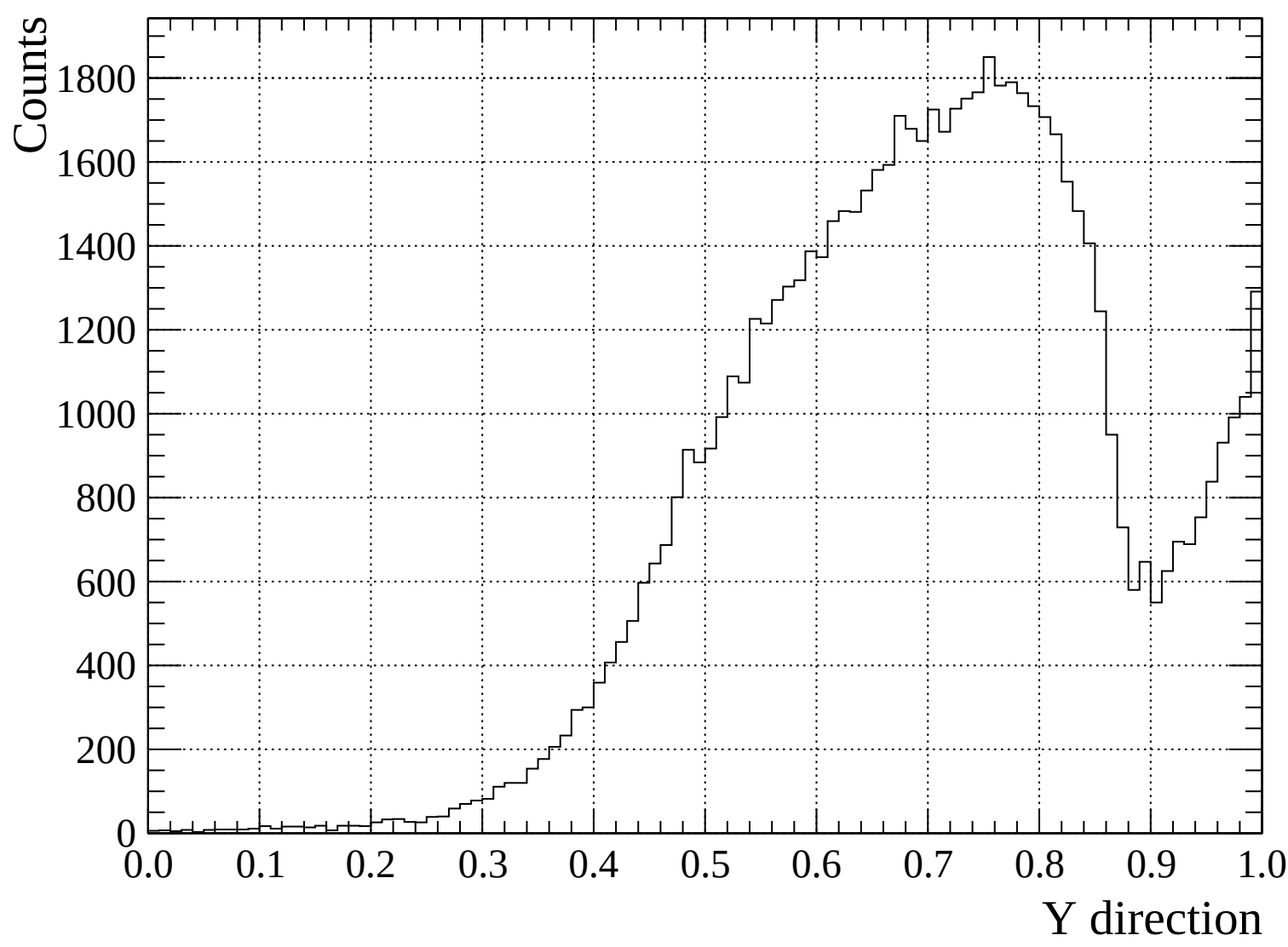


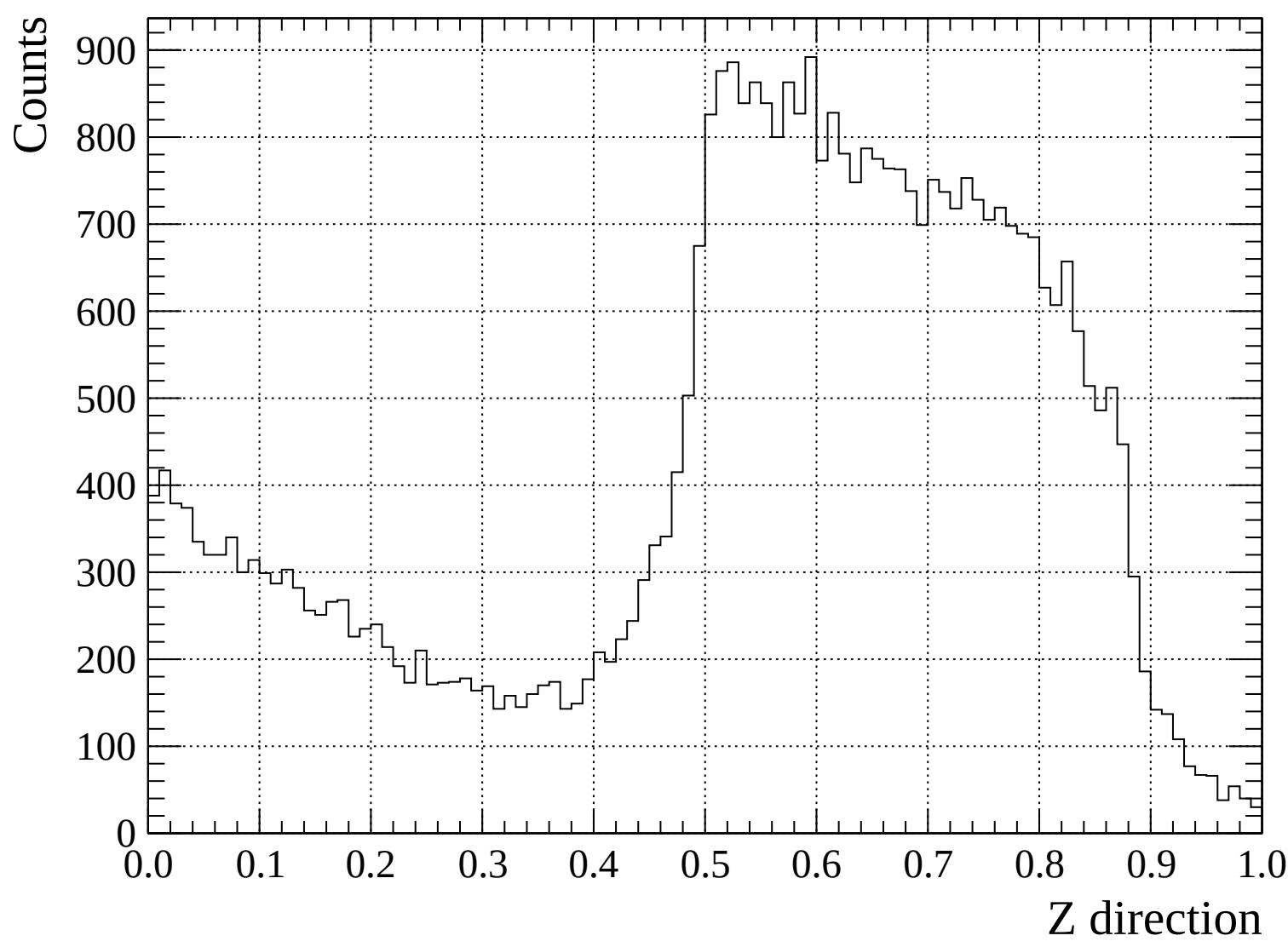


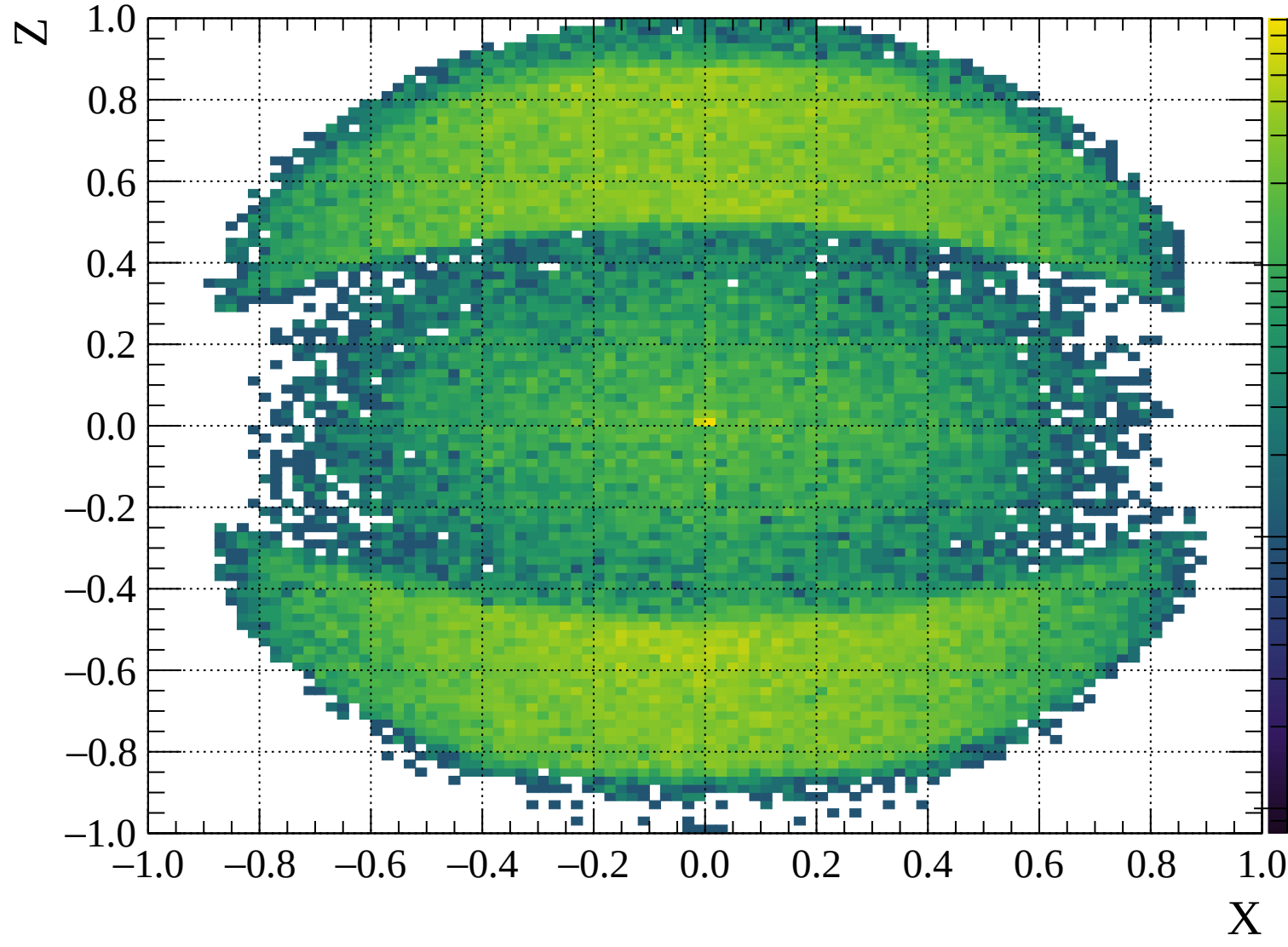


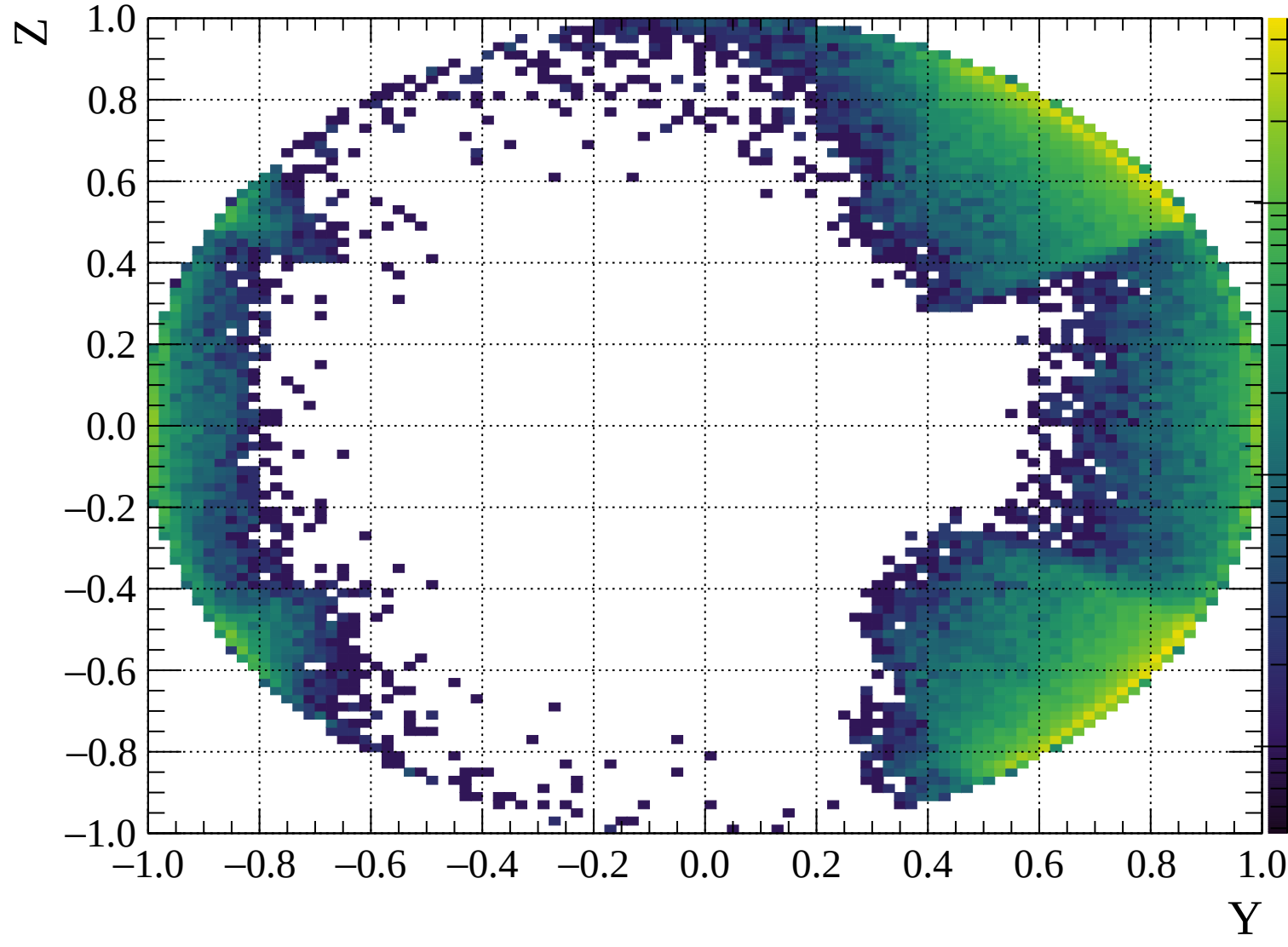


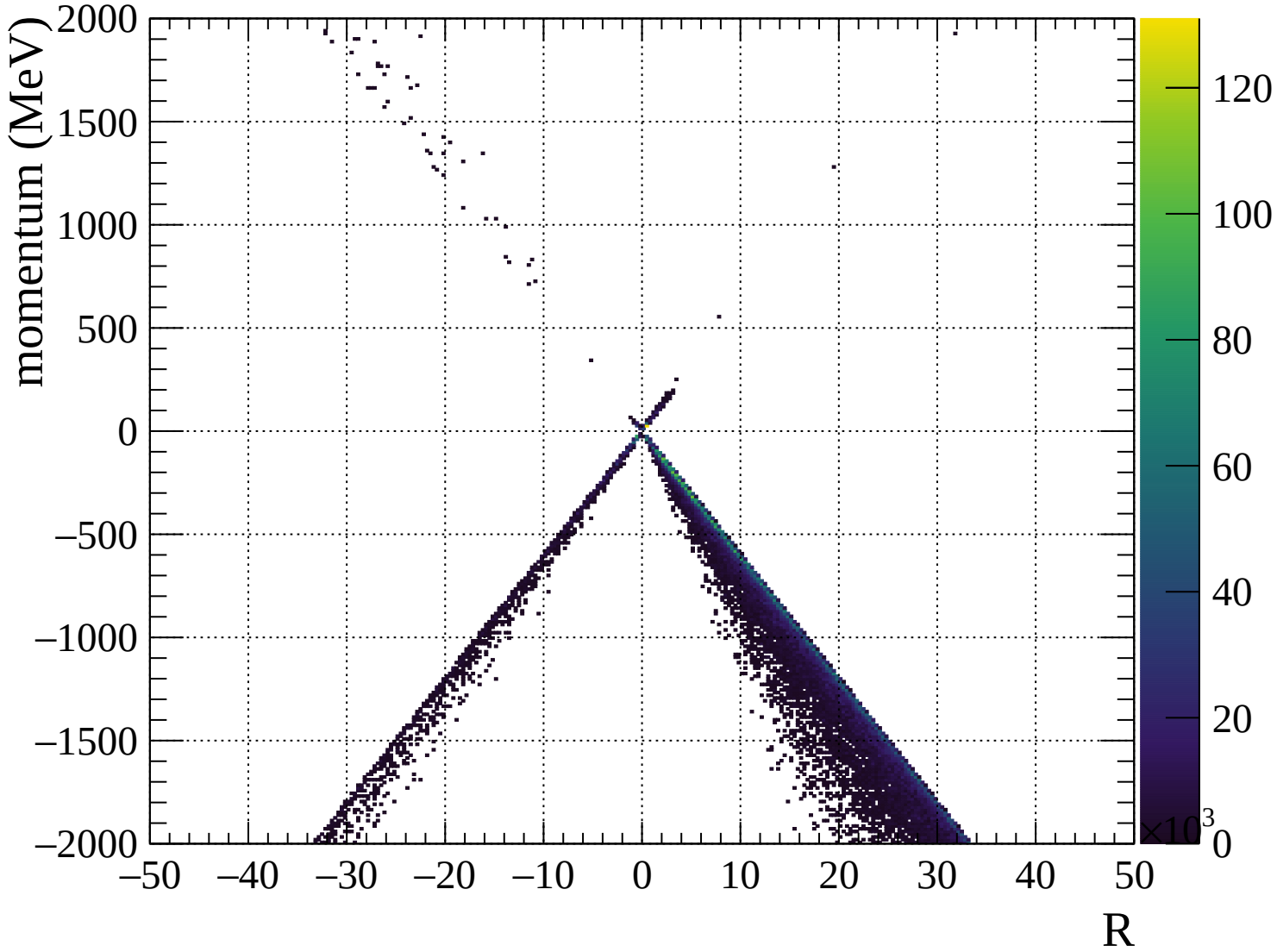


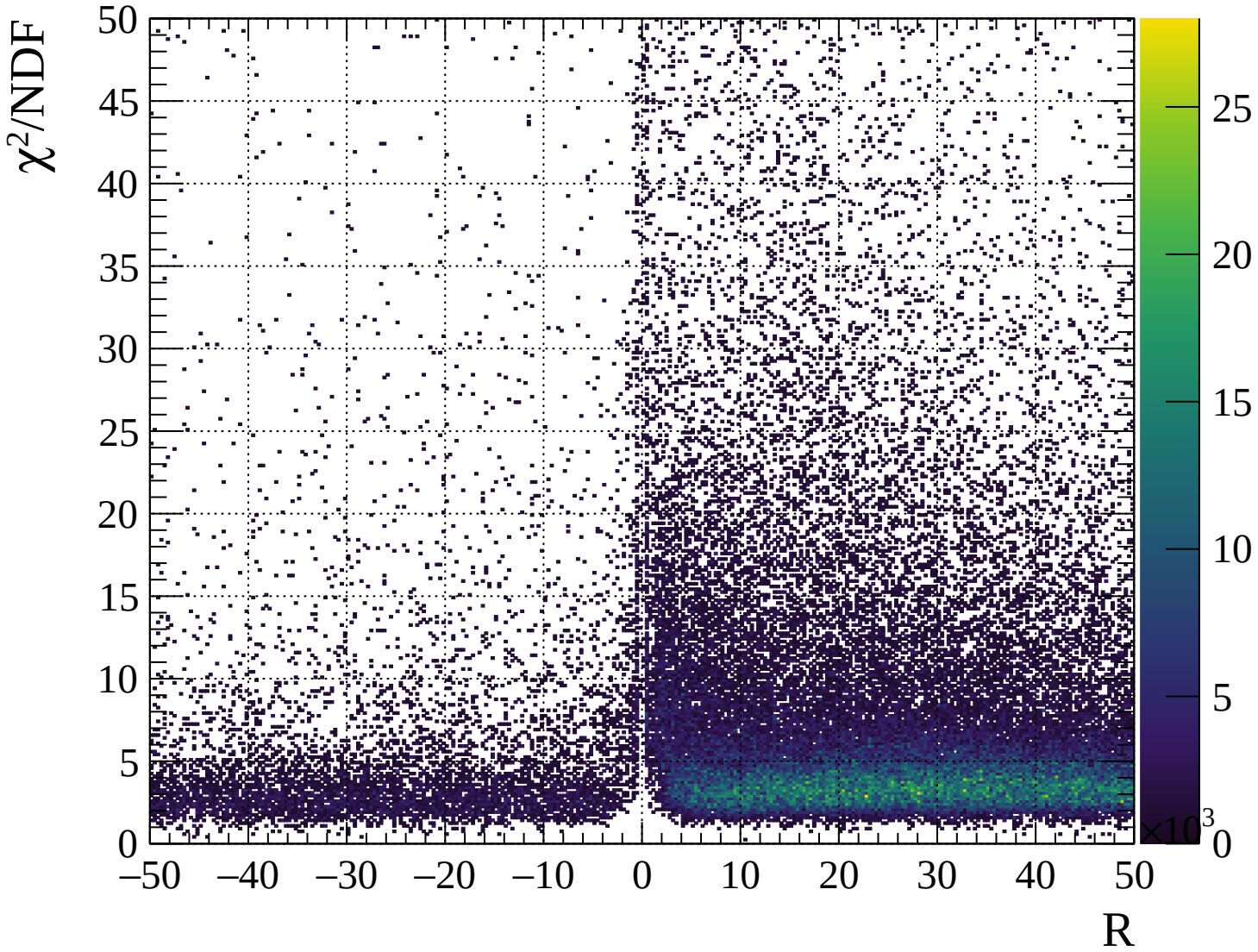






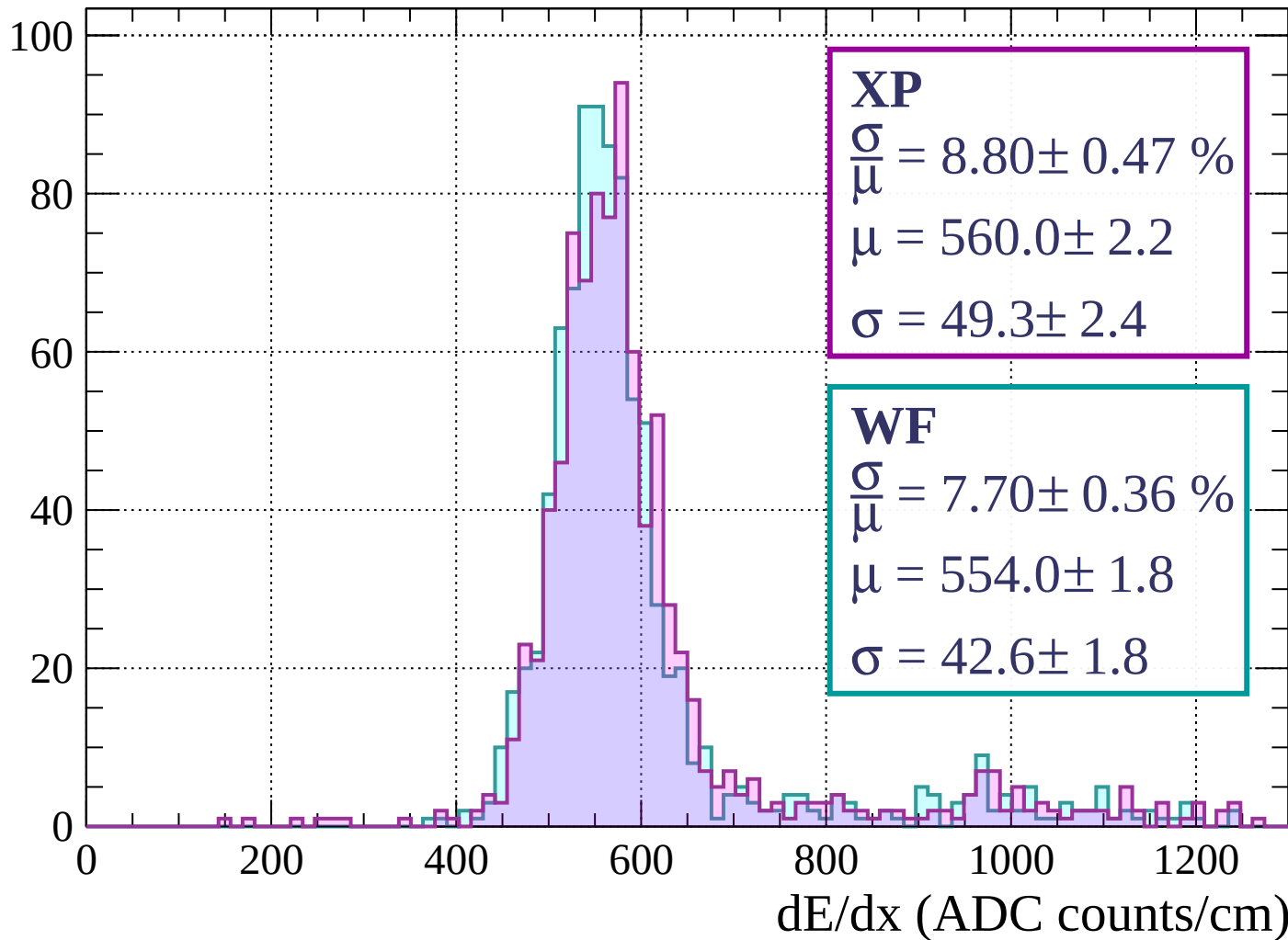






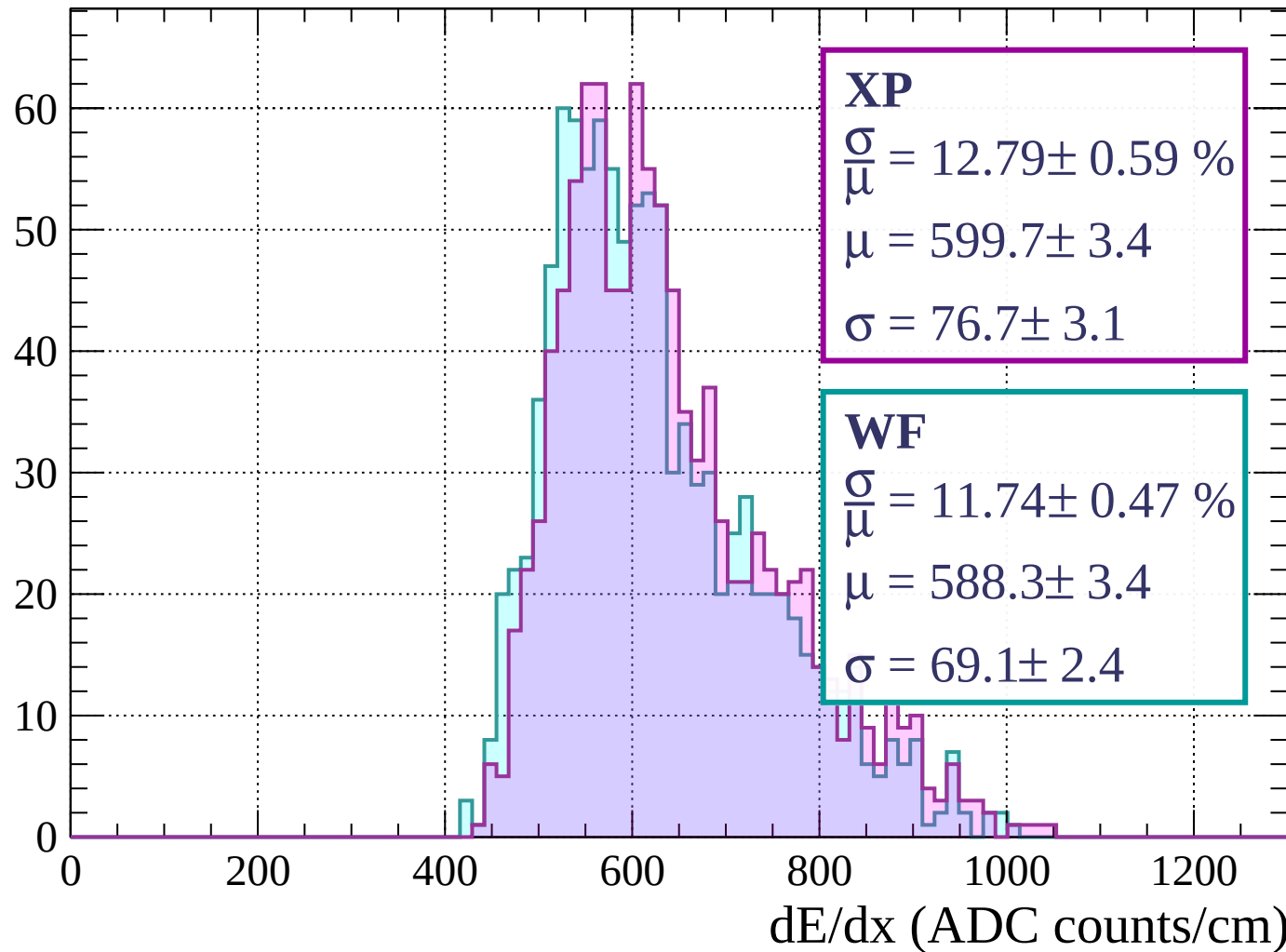
Energy loss | $0 < p < 80$

Count



Energy loss | $80 < p < 160$

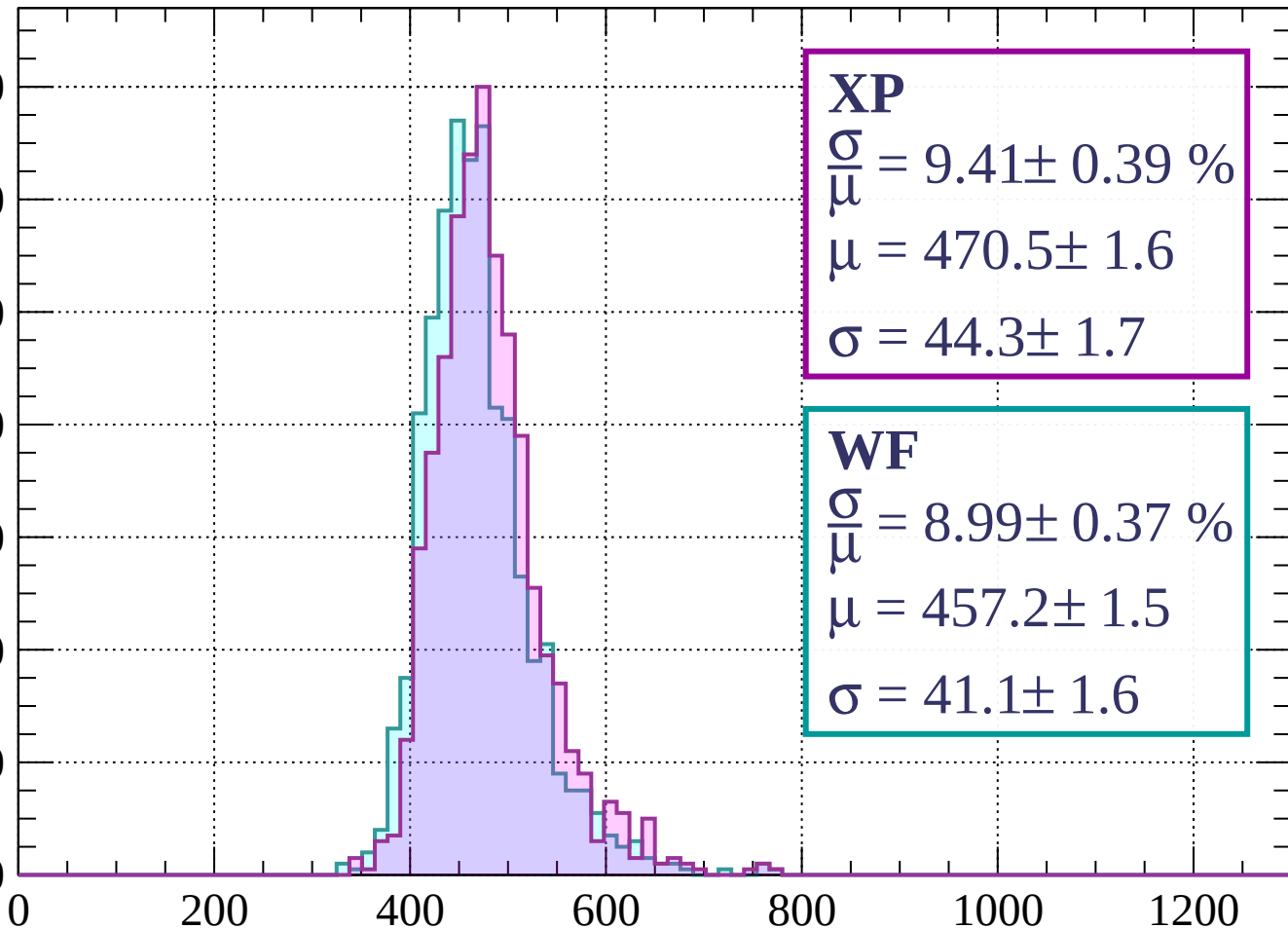
Count



Energy loss | $160 < p < 240$

Count

140
120
100
80
60
40
20
0



XP

$$\frac{\sigma}{\mu} = 9.41 \pm 0.39 \%$$

$$\mu = 470.5 \pm 1.6$$

$$\sigma = 44.3 \pm 1.7$$

WF

$$\frac{\sigma}{\mu} = 8.99 \pm 0.37 \%$$

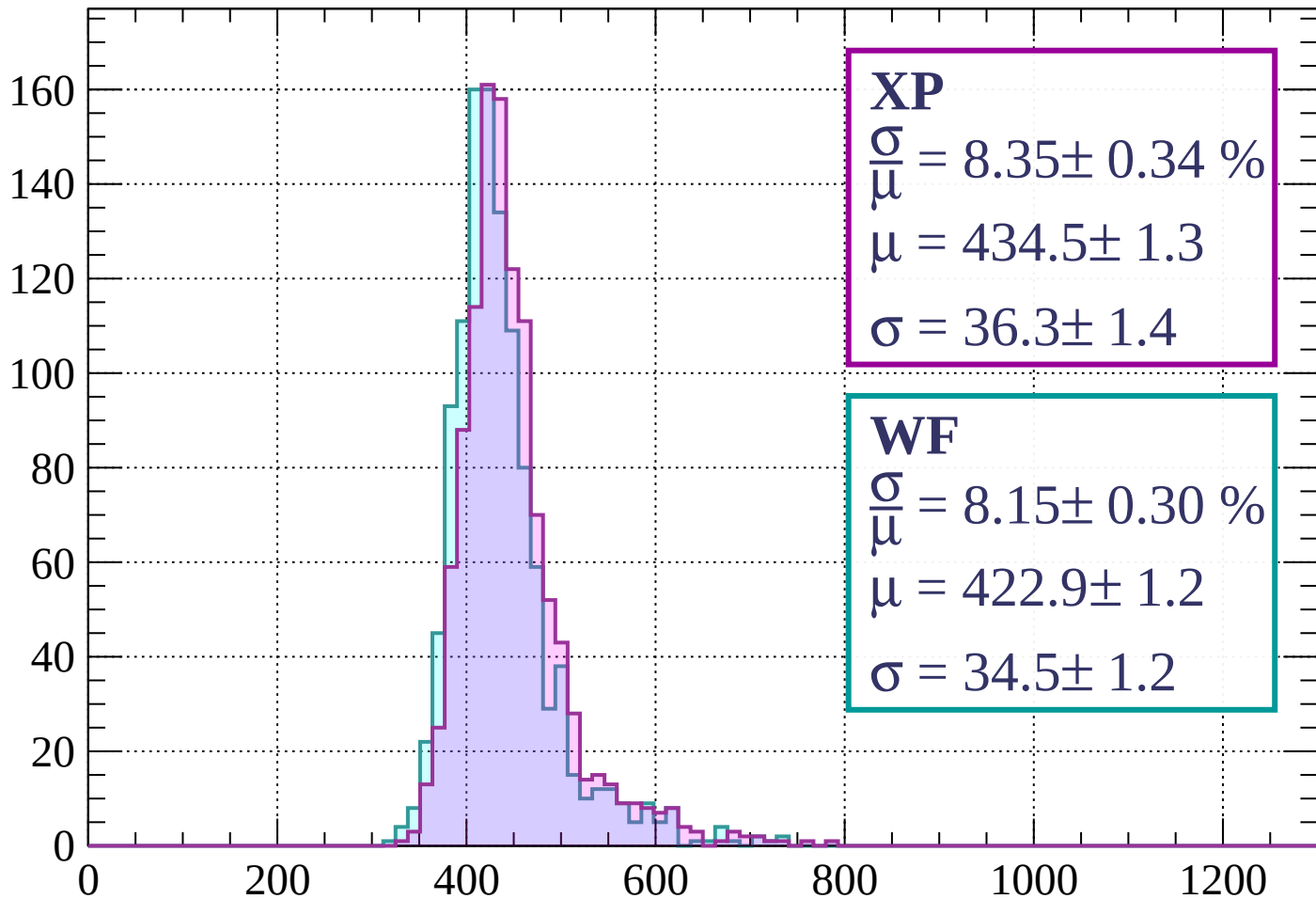
$$\mu = 457.2 \pm 1.5$$

$$\sigma = 41.1 \pm 1.6$$

dE/dx (ADC counts/cm)

Energy loss | $240 < p < 320$

Count



XP

$$\frac{\sigma}{\mu} = 8.35 \pm 0.34 \%$$

$$\mu = 434.5 \pm 1.3$$

$$\sigma = 36.3 \pm 1.4$$

WF

$$\frac{\sigma}{\mu} = 8.15 \pm 0.30 \%$$

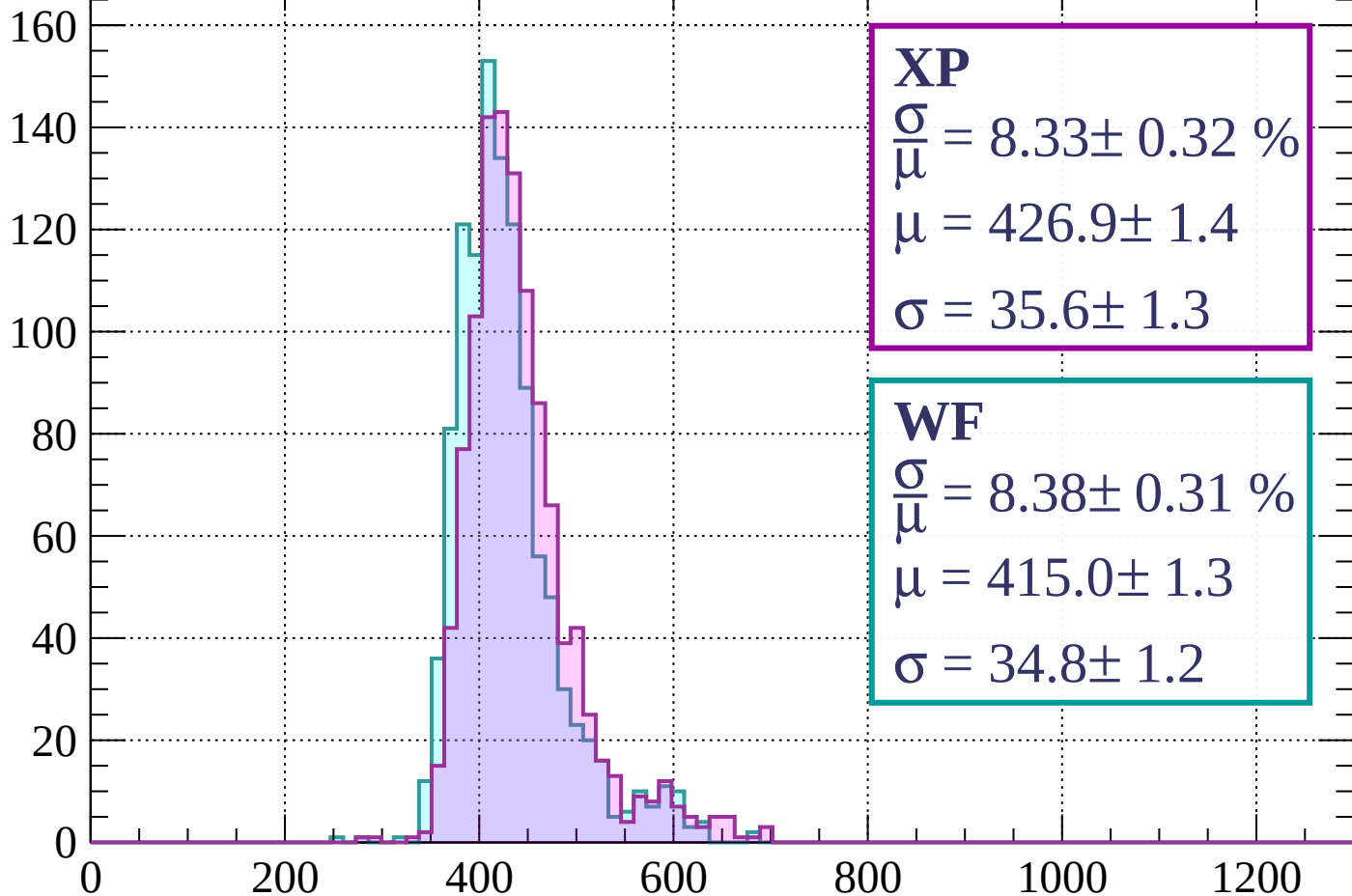
$$\mu = 422.9 \pm 1.2$$

$$\sigma = 34.5 \pm 1.2$$

dE/dx (ADC counts/cm)

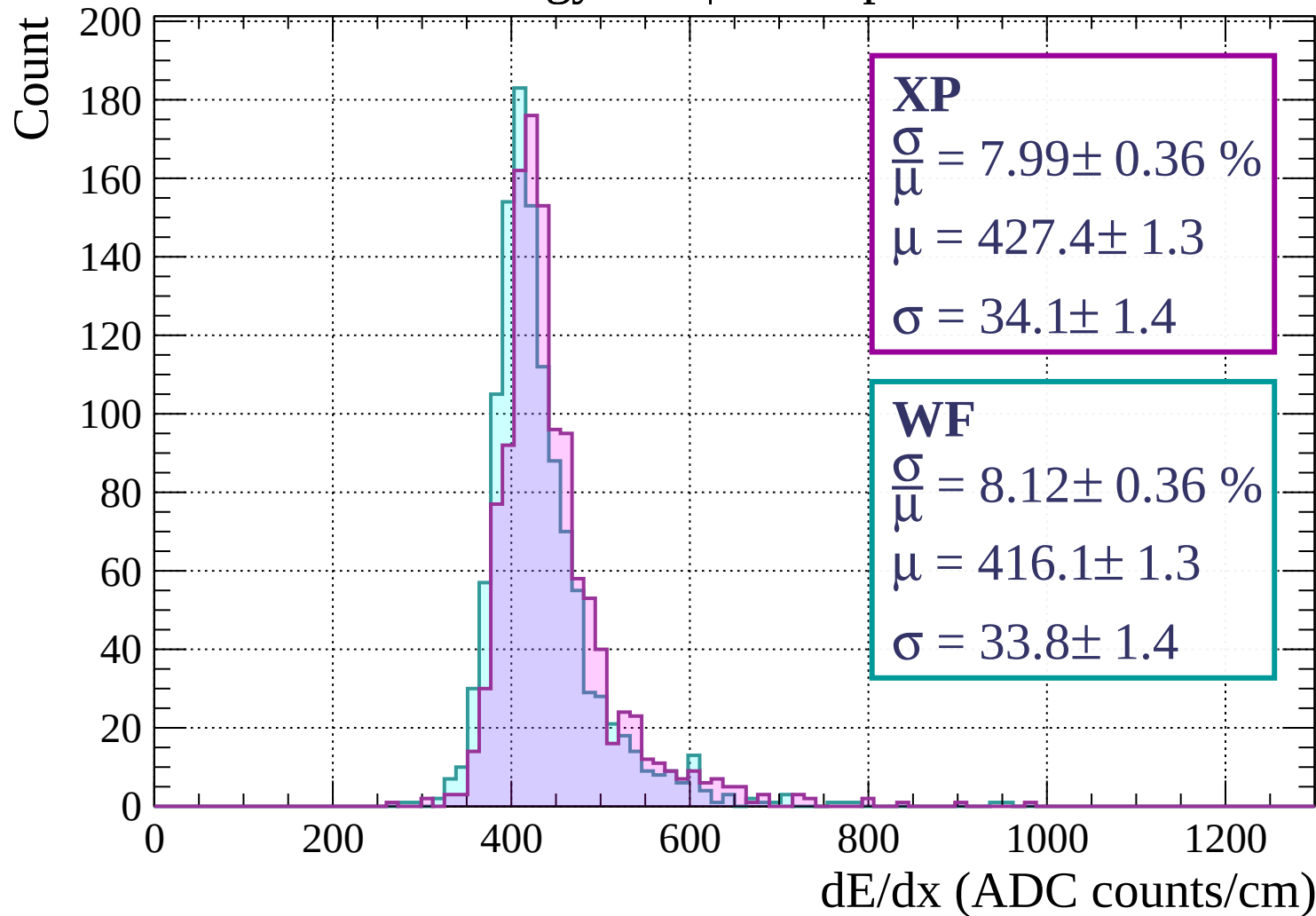
Energy loss | $320 < p < 400$

Count



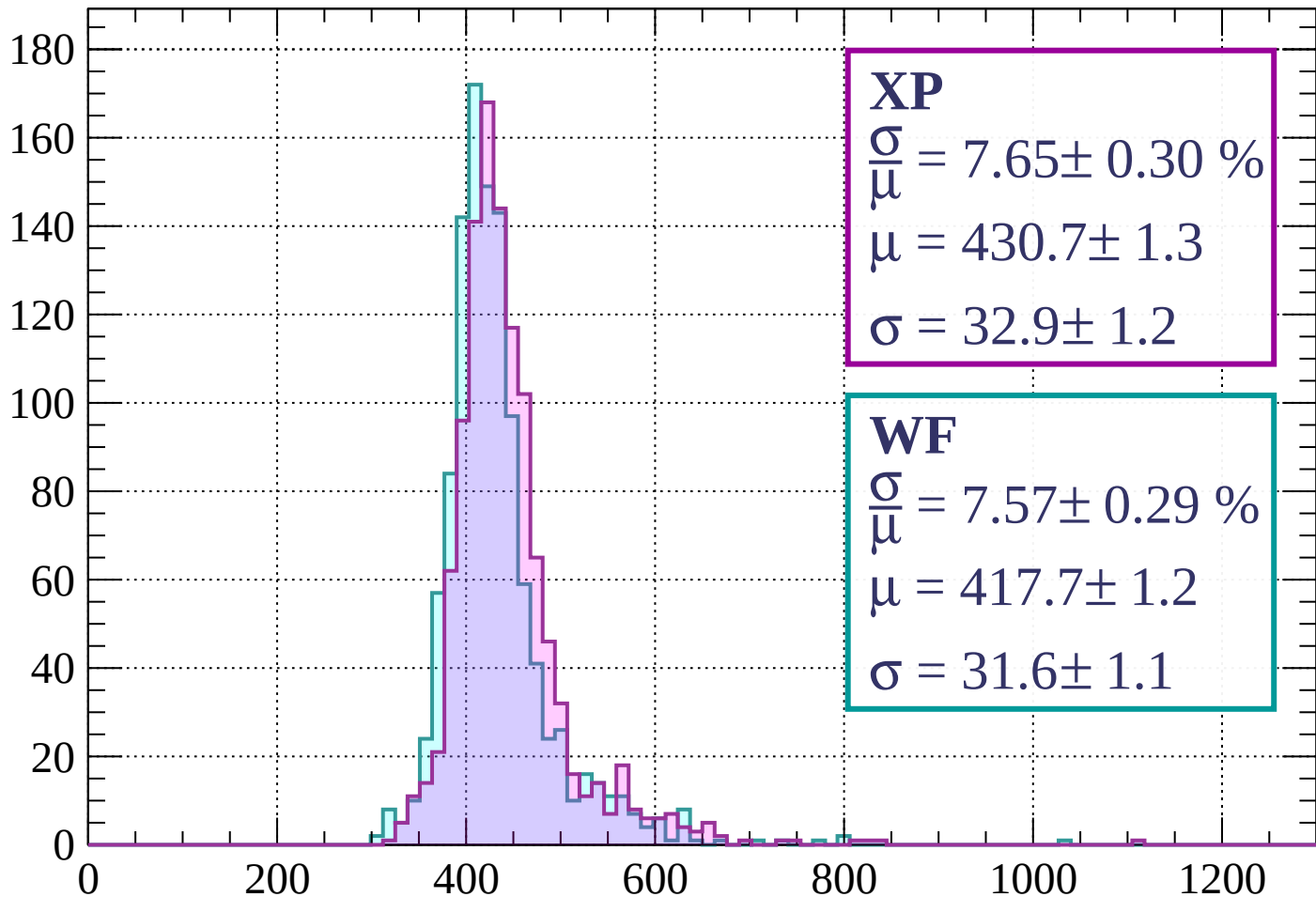
dE/dx (ADC counts/cm)

Energy loss | $400 < p < 480$



Energy loss | $480 < p < 560$

Count



XP

$$\frac{\sigma}{\mu} = 7.65 \pm 0.30 \%$$

$$\mu = 430.7 \pm 1.3$$

$$\sigma = 32.9 \pm 1.2$$

WF

$$\frac{\sigma}{\mu} = 7.57 \pm 0.29 \%$$

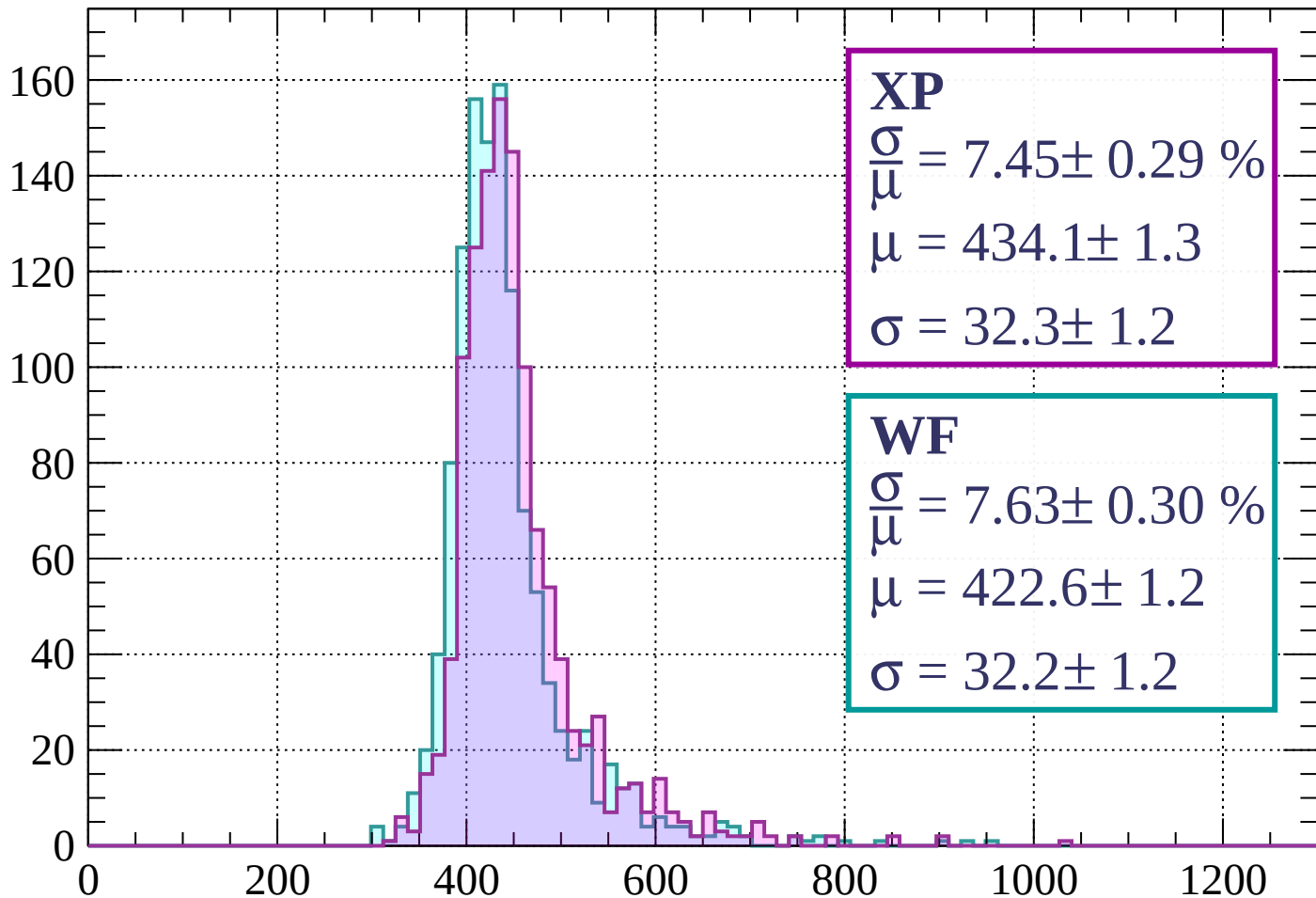
$$\mu = 417.7 \pm 1.2$$

$$\sigma = 31.6 \pm 1.1$$

dE/dx (ADC counts/cm)

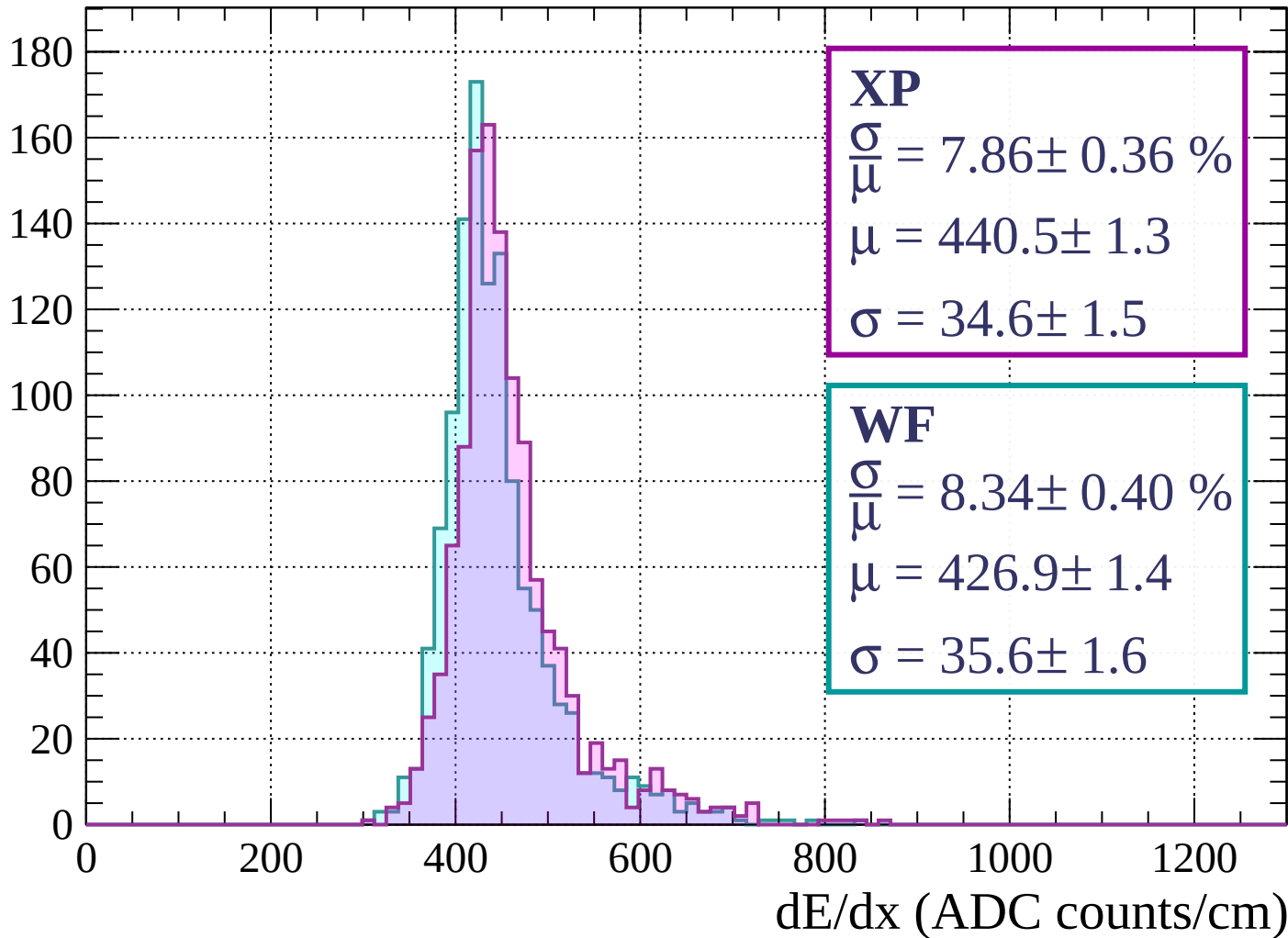
Energy loss | $560 < p < 640$

Count



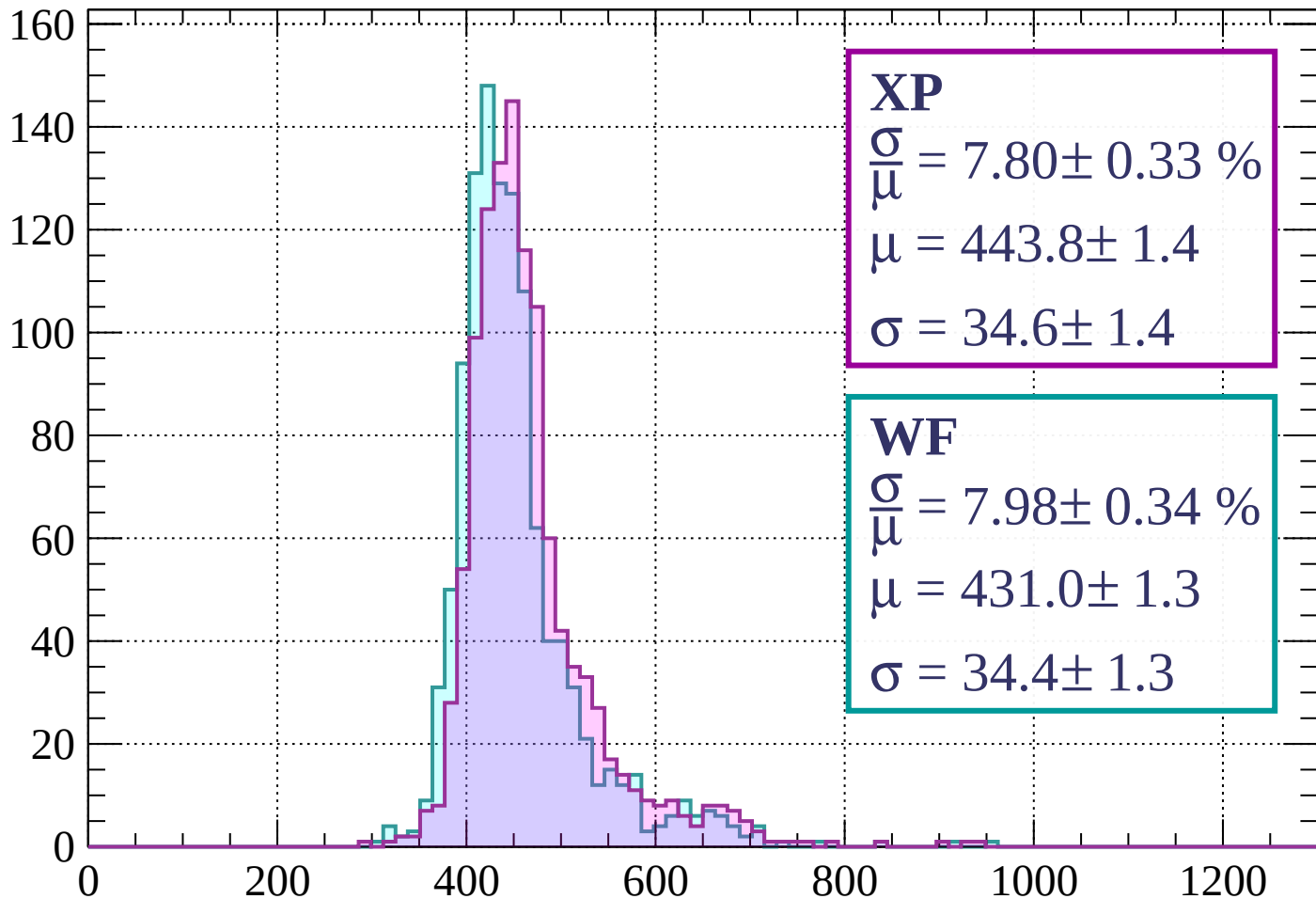
Energy loss | $640 < p < 720$

Count



Energy loss | $720 < p < 800$

Count



XP

$$\frac{\sigma}{\bar{\mu}} = 7.80 \pm 0.33 \%$$

$$\mu = 443.8 \pm 1.4$$

$$\sigma = 34.6 \pm 1.4$$

WF

$$\frac{\sigma}{\bar{\mu}} = 7.98 \pm 0.34 \%$$

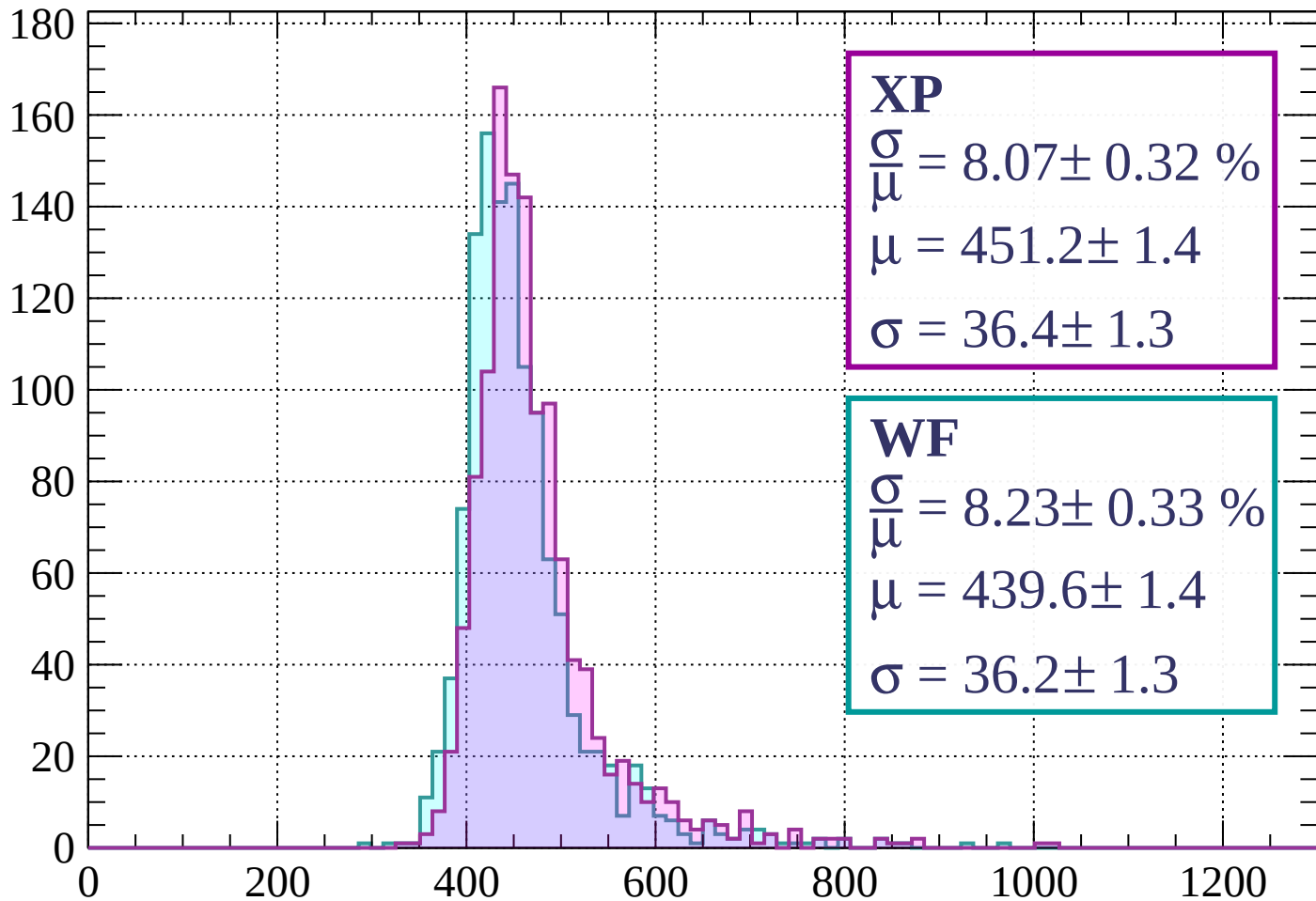
$$\mu = 431.0 \pm 1.3$$

$$\sigma = 34.4 \pm 1.3$$

dE/dx (ADC counts/cm)

Energy loss | $800 < p < 880$

Count



XP

$$\frac{\sigma}{\mu} = 8.07 \pm 0.32 \%$$

$$\mu = 451.2 \pm 1.4$$

$$\sigma = 36.4 \pm 1.3$$

WF

$$\frac{\sigma}{\mu} = 8.23 \pm 0.33 \%$$

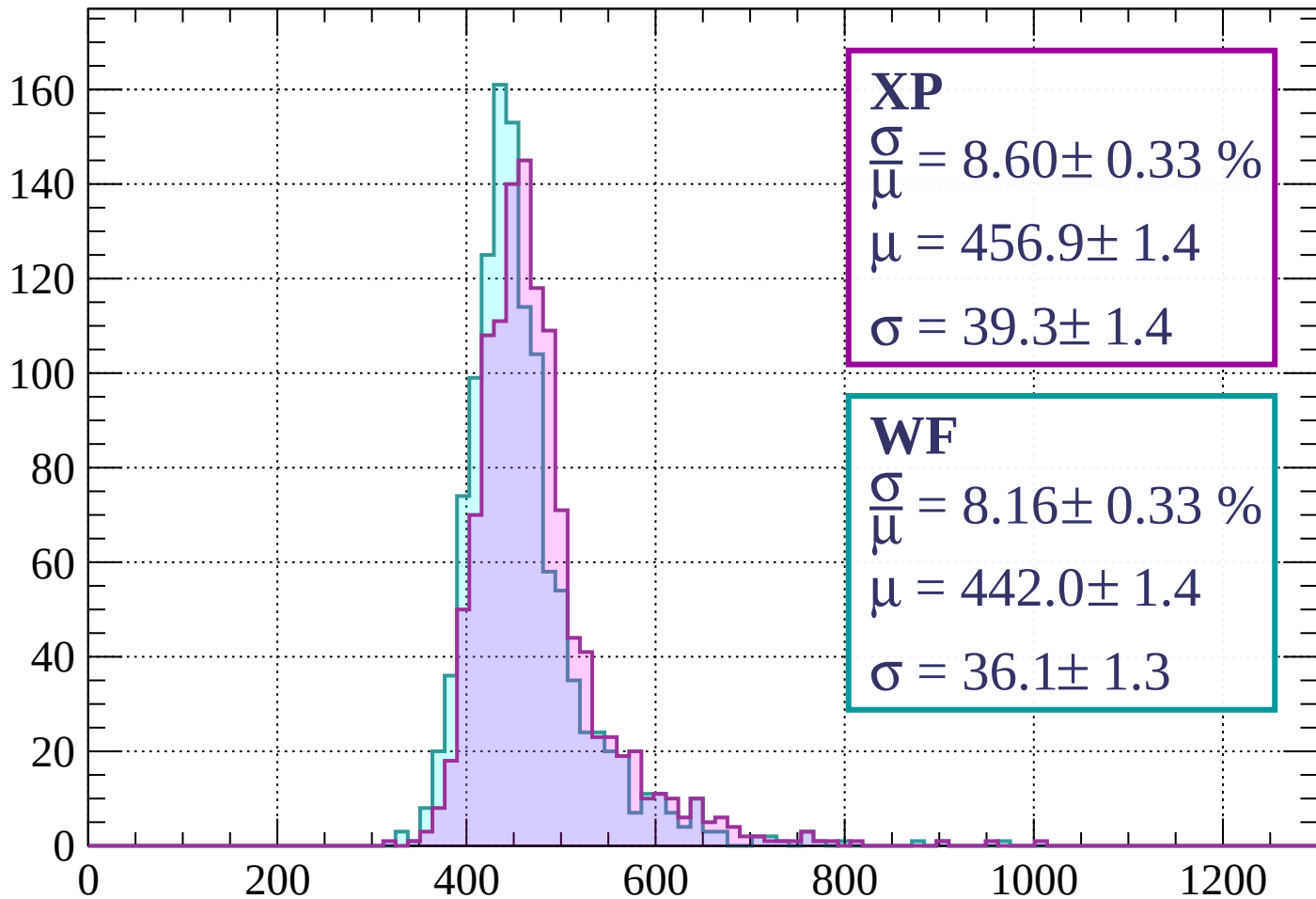
$$\mu = 439.6 \pm 1.4$$

$$\sigma = 36.2 \pm 1.3$$

dE/dx (ADC counts/cm)

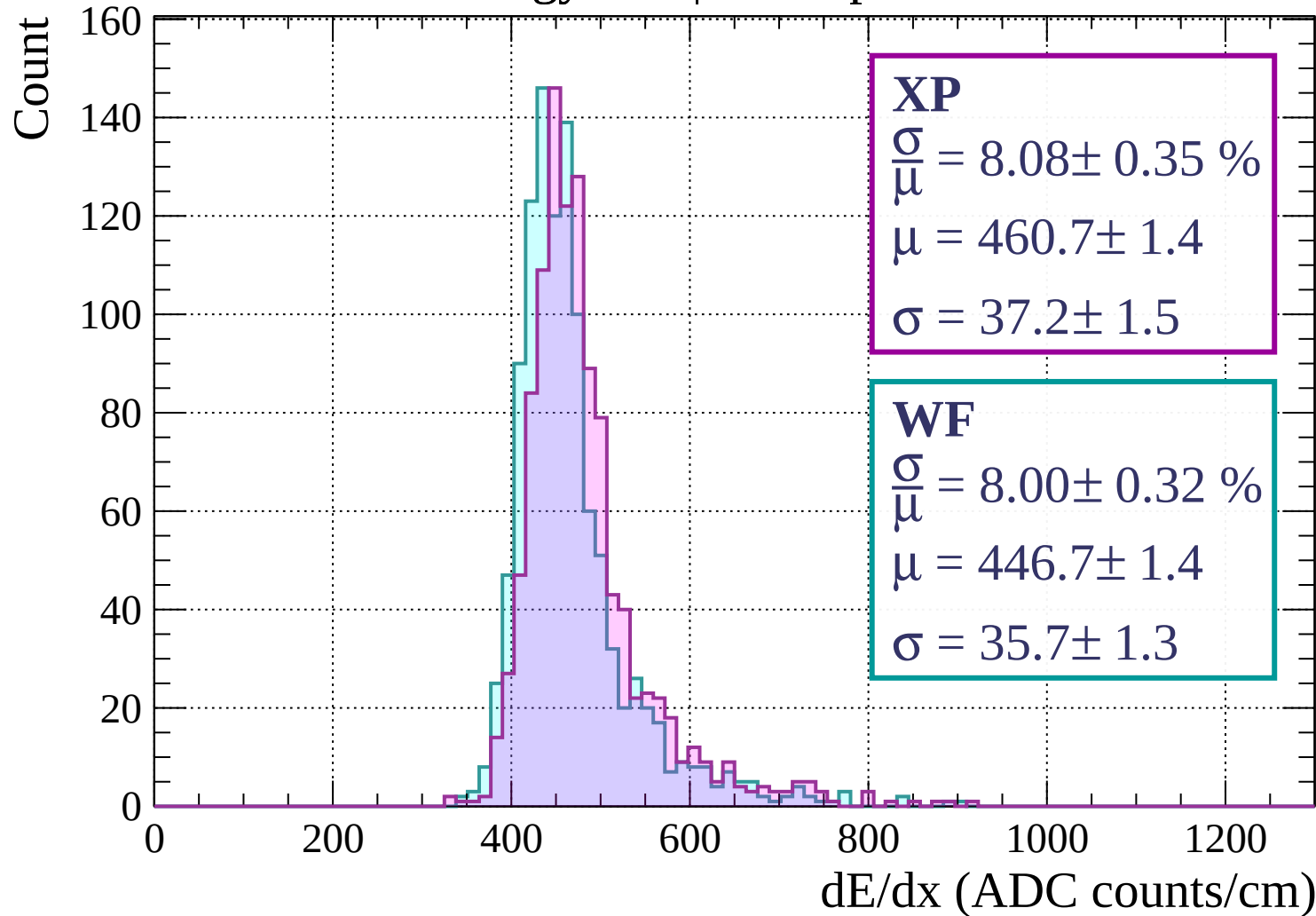
Energy loss | $880 < p < 960$

Count



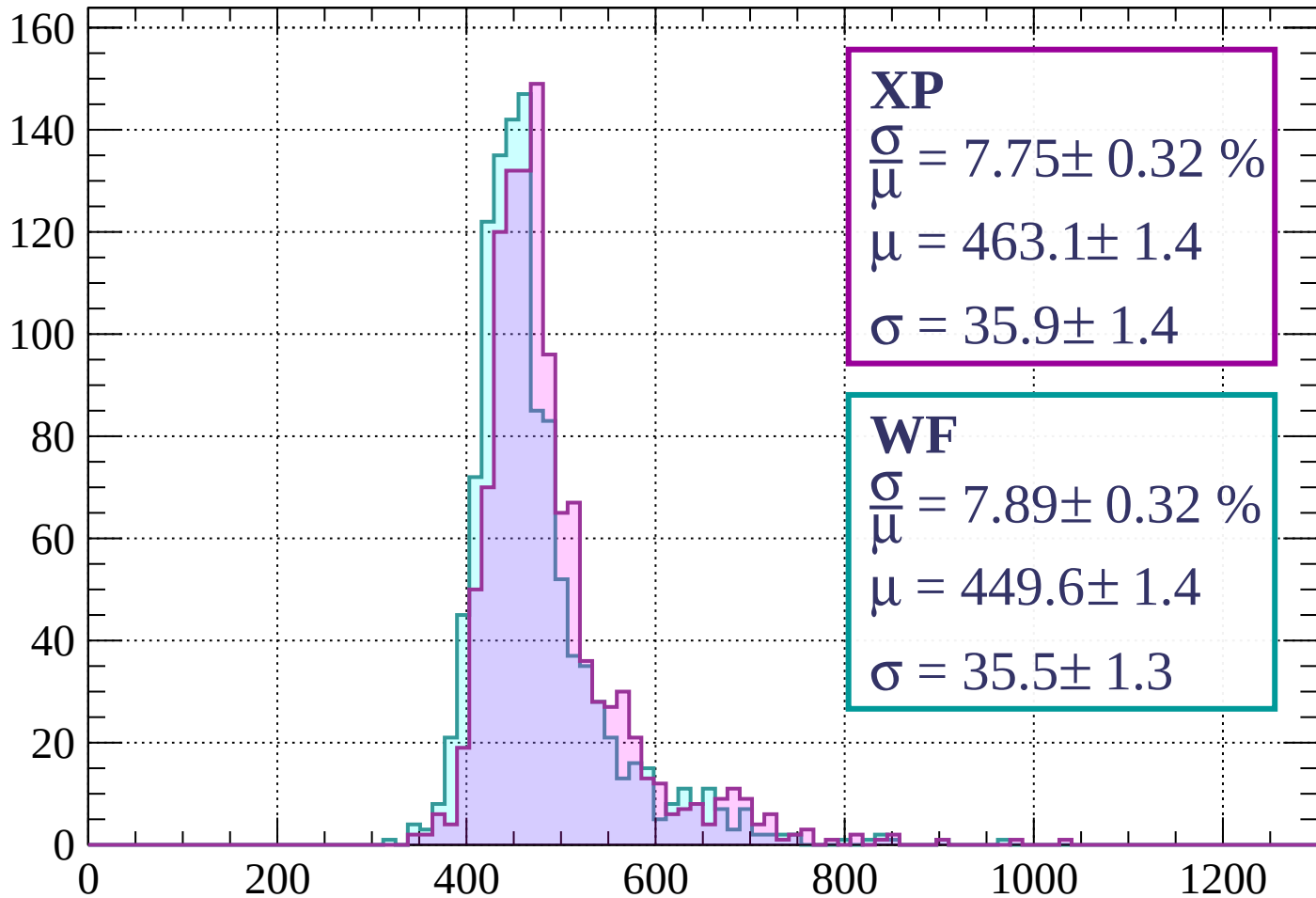
dE/dx (ADC counts/cm)

Energy loss | $960 < p < 1040$



Energy loss | $1040 < p < 1120$

Count



XP

$$\frac{\sigma}{\mu} = 7.75 \pm 0.32 \%$$

$$\mu = 463.1 \pm 1.4$$

$$\sigma = 35.9 \pm 1.4$$

WF

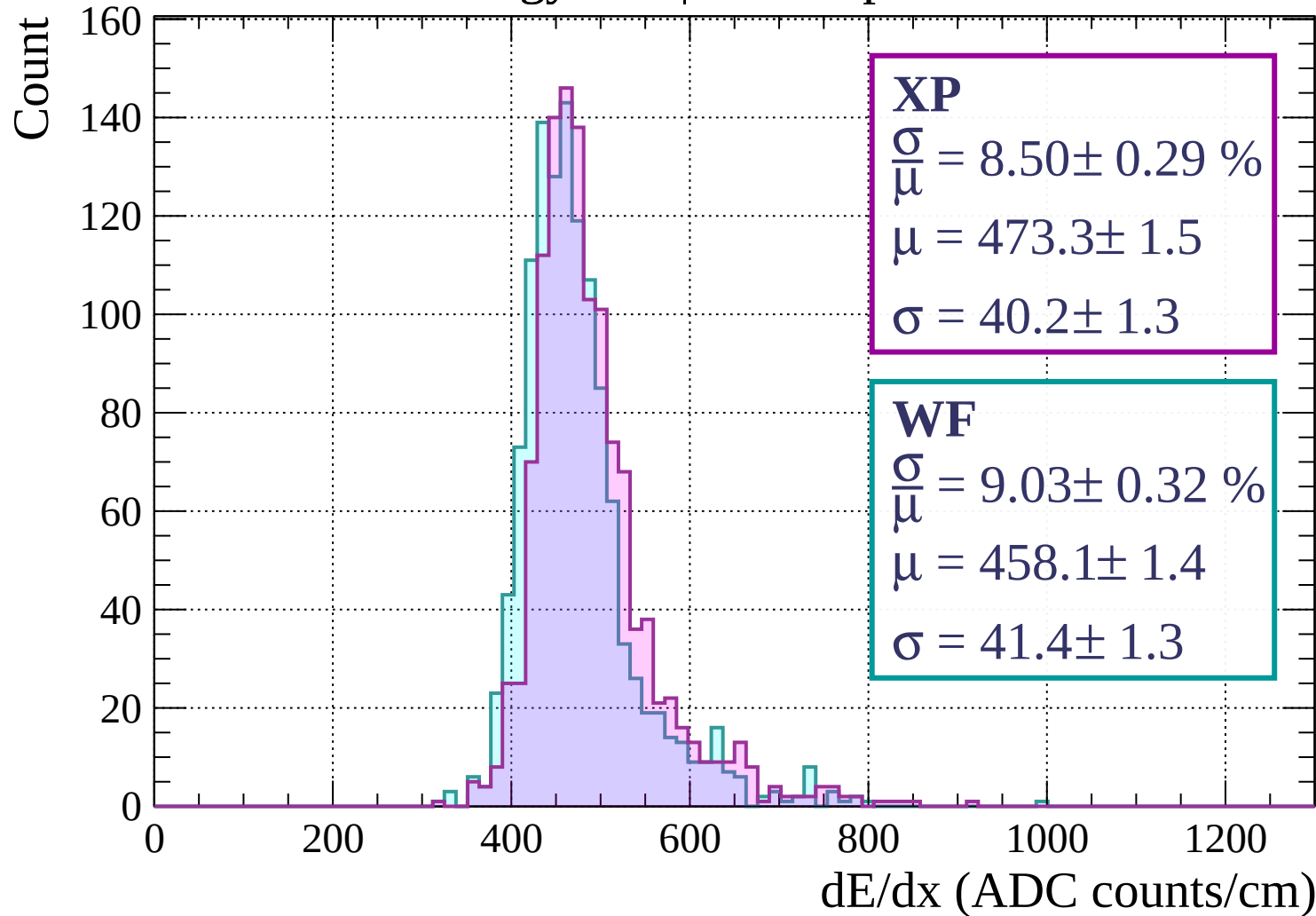
$$\frac{\sigma}{\mu} = 7.89 \pm 0.32 \%$$

$$\mu = 449.6 \pm 1.4$$

$$\sigma = 35.5 \pm 1.3$$

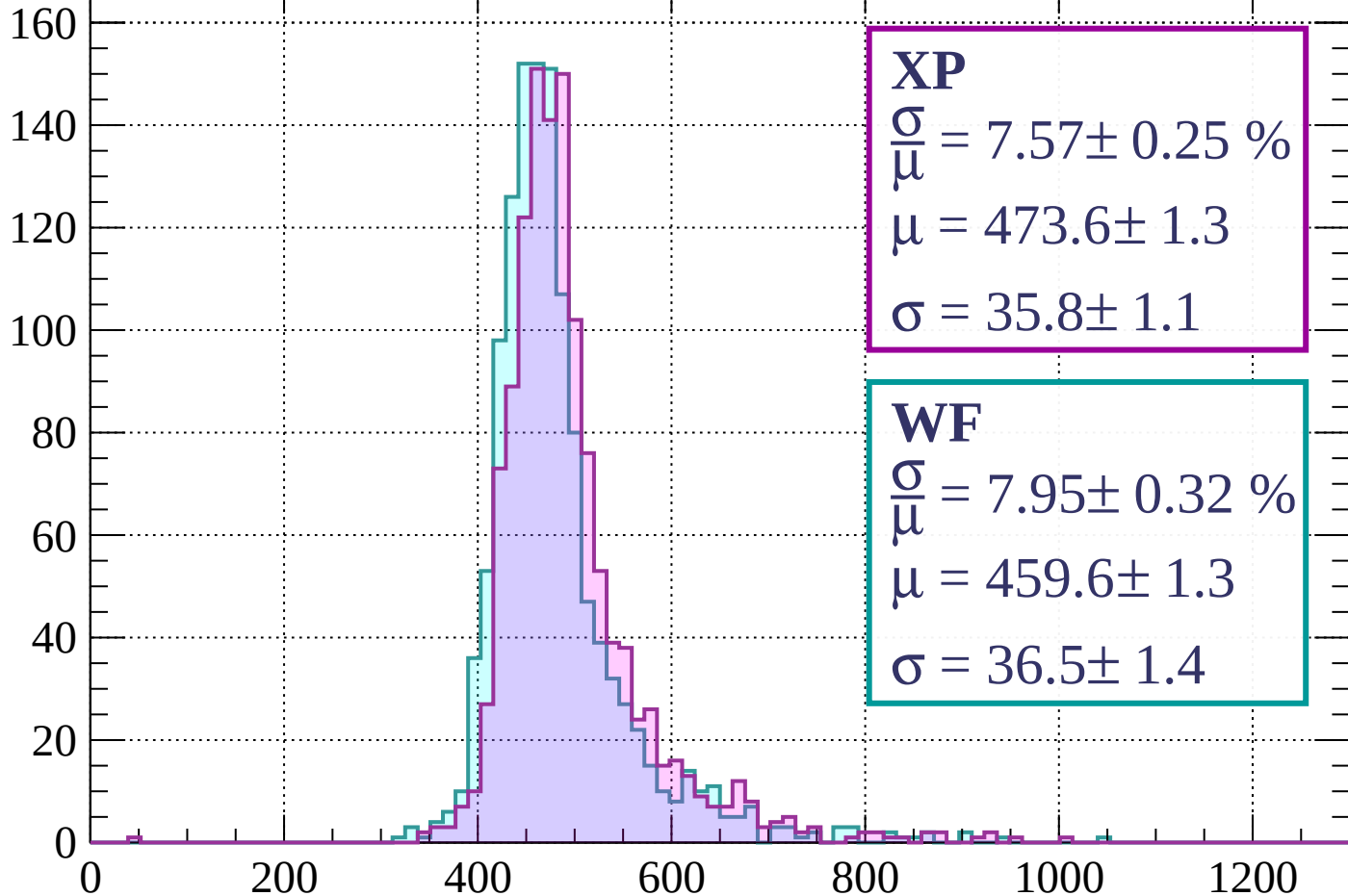
dE/dx (ADC counts/cm)

Energy loss | $1120 < p < 1200$



Energy loss | $1200 < p < 1280$

Count



XP

$$\frac{\sigma}{\mu} = 7.57 \pm 0.25 \%$$

$$\mu = 473.6 \pm 1.3$$

$$\sigma = 35.8 \pm 1.1$$

WF

$$\frac{\sigma}{\mu} = 7.95 \pm 0.32 \%$$

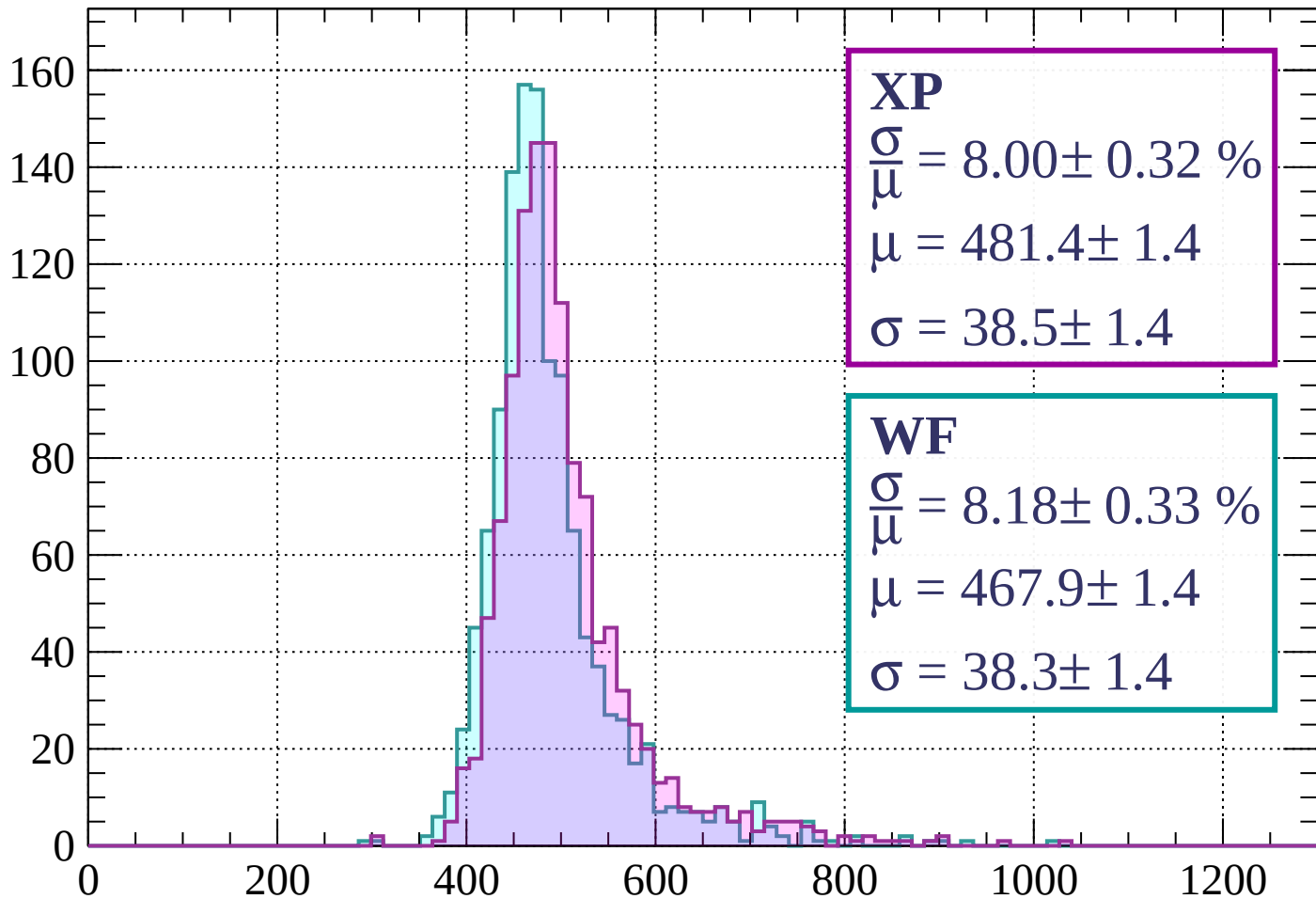
$$\mu = 459.6 \pm 1.3$$

$$\sigma = 36.5 \pm 1.4$$

dE/dx (ADC counts/cm)

Energy loss | $1280 < p < 1360$

Count



XP

$$\frac{\sigma}{\mu} = 8.00 \pm 0.32 \%$$

$$\mu = 481.4 \pm 1.4$$

$$\sigma = 38.5 \pm 1.4$$

WF

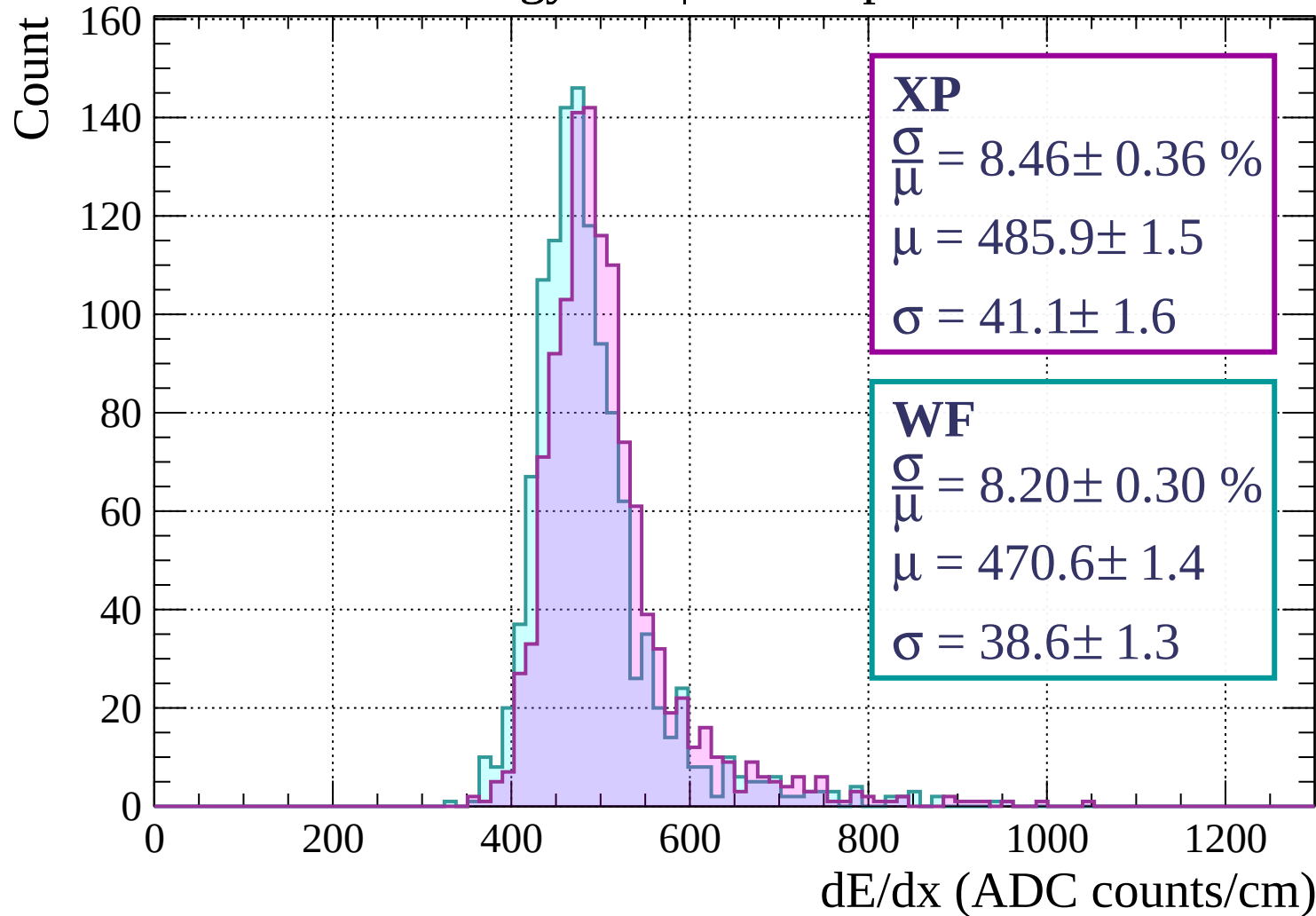
$$\frac{\sigma}{\mu} = 8.18 \pm 0.33 \%$$

$$\mu = 467.9 \pm 1.4$$

$$\sigma = 38.3 \pm 1.4$$

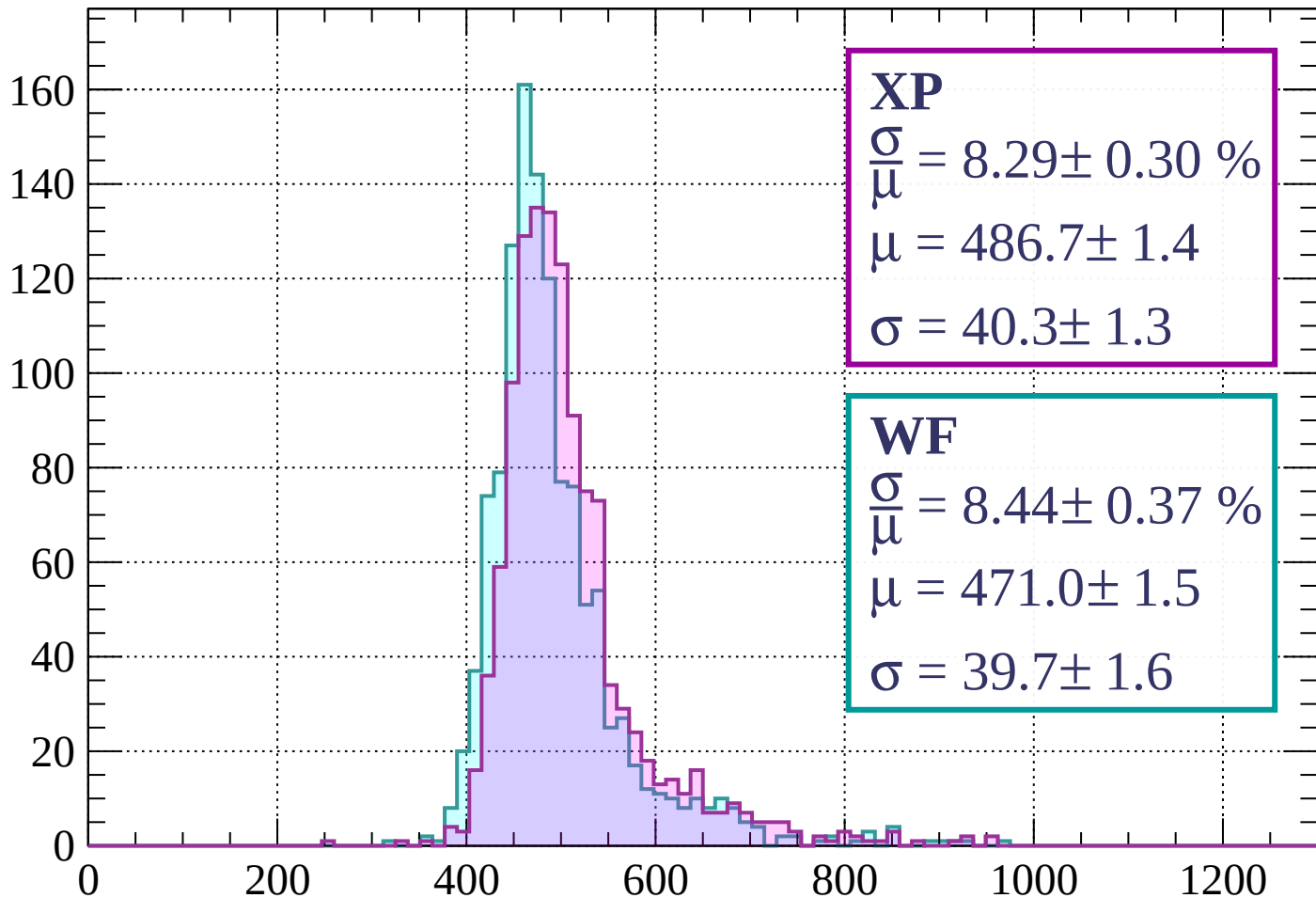
dE/dx (ADC counts/cm)

Energy loss | $1360 < p < 1440$



Energy loss | $1440 < p < 1520$

Count



XP

$$\frac{\sigma}{\mu} = 8.29 \pm 0.30 \%$$

$$\mu = 486.7 \pm 1.4$$

$$\sigma = 40.3 \pm 1.3$$

WF

$$\frac{\sigma}{\mu} = 8.44 \pm 0.37 \%$$

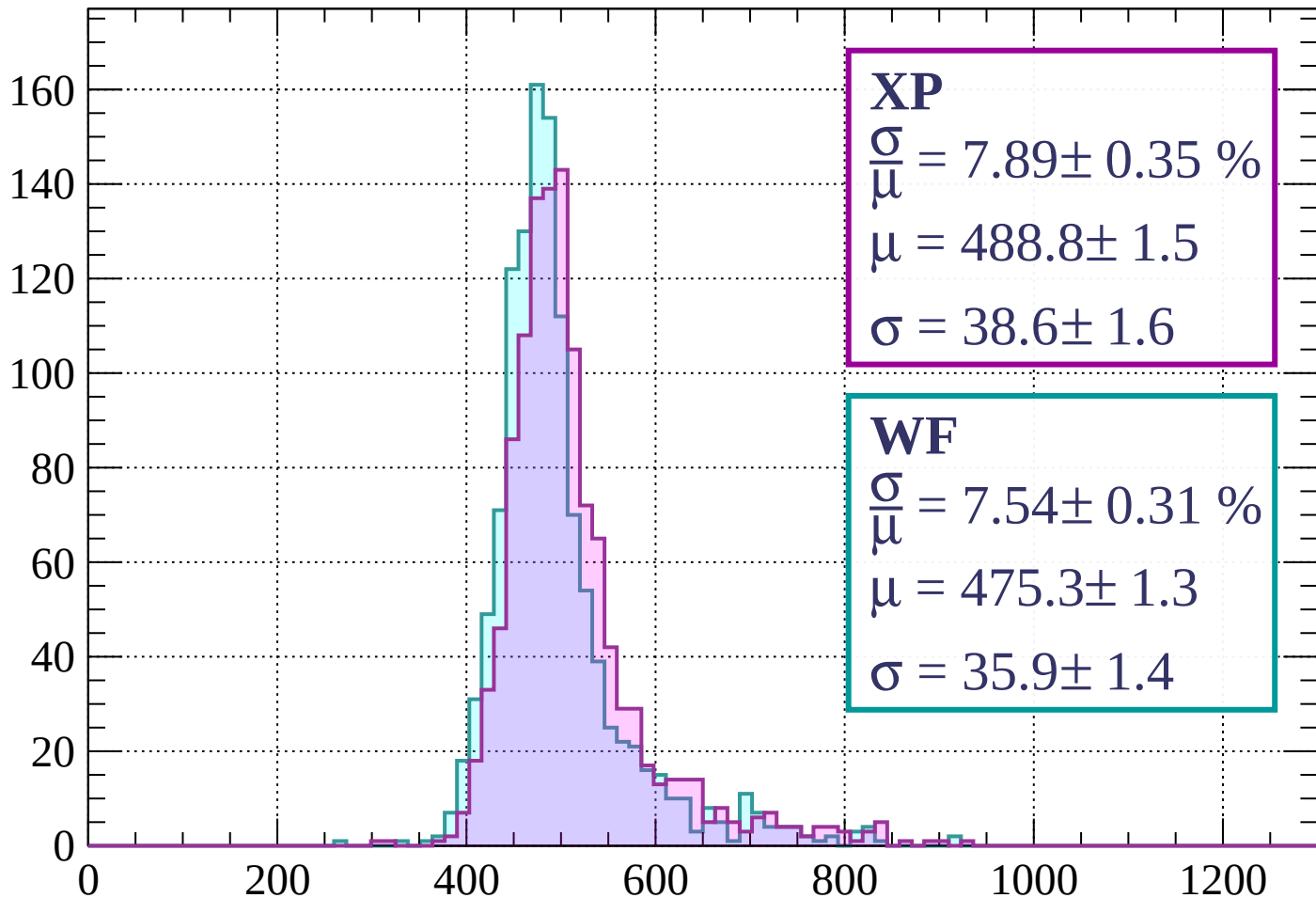
$$\mu = 471.0 \pm 1.5$$

$$\sigma = 39.7 \pm 1.6$$

dE/dx (ADC counts/cm)

Energy loss | $1520 < p < 1600$

Count



XP

$$\frac{\sigma}{\mu} = 7.89 \pm 0.35 \%$$

$$\mu = 488.8 \pm 1.5$$

$$\sigma = 38.6 \pm 1.6$$

WF

$$\frac{\sigma}{\mu} = 7.54 \pm 0.31 \%$$

$$\mu = 475.3 \pm 1.3$$

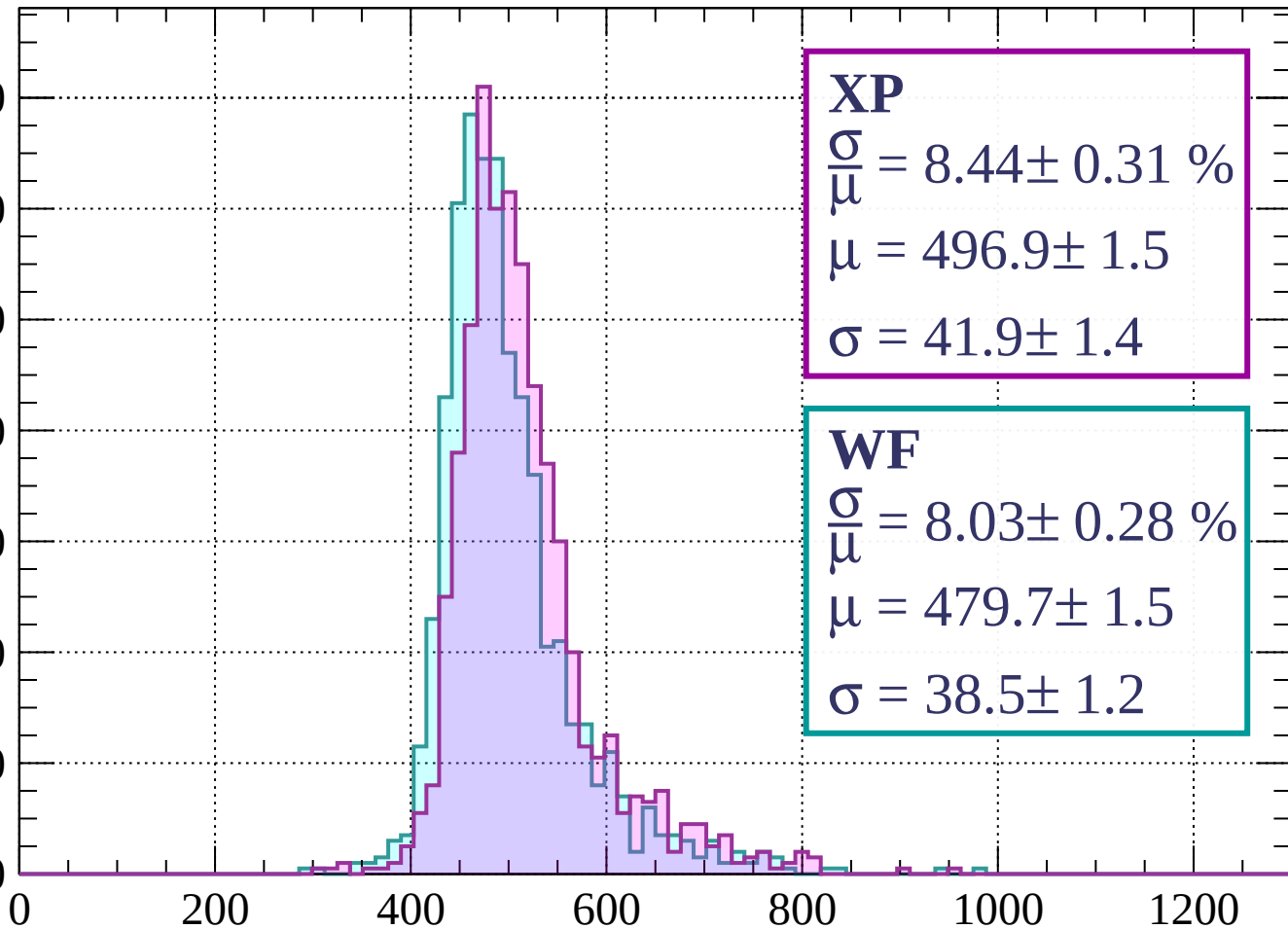
$$\sigma = 35.9 \pm 1.4$$

dE/dx (ADC counts/cm)

Energy loss | $1600 < p < 1680$

Count

140
120
100
80
60
40
20
0



XP

$$\frac{\sigma}{\mu} = 8.44 \pm 0.31 \%$$

$$\mu = 496.9 \pm 1.5$$

$$\sigma = 41.9 \pm 1.4$$

WF

$$\frac{\sigma}{\mu} = 8.03 \pm 0.28 \%$$

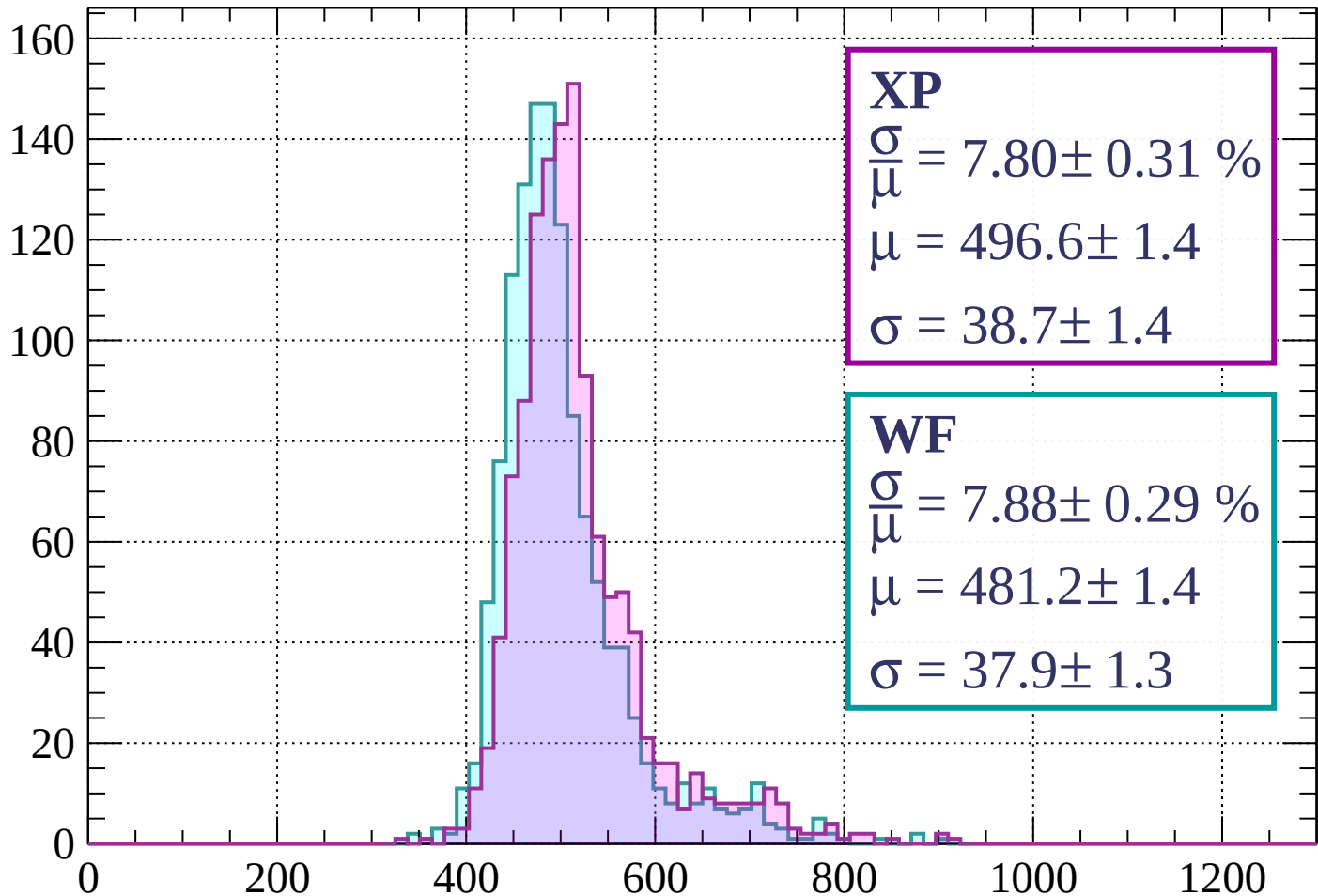
$$\mu = 479.7 \pm 1.5$$

$$\sigma = 38.5 \pm 1.2$$

dE/dx (ADC counts/cm)

Energy loss | $1680 < p < 1760$

Count



XP

$$\frac{\sigma}{\mu} = 7.80 \pm 0.31 \%$$

$$\mu = 496.6 \pm 1.4$$

$$\sigma = 38.7 \pm 1.4$$

WF

$$\frac{\sigma}{\mu} = 7.88 \pm 0.29 \%$$

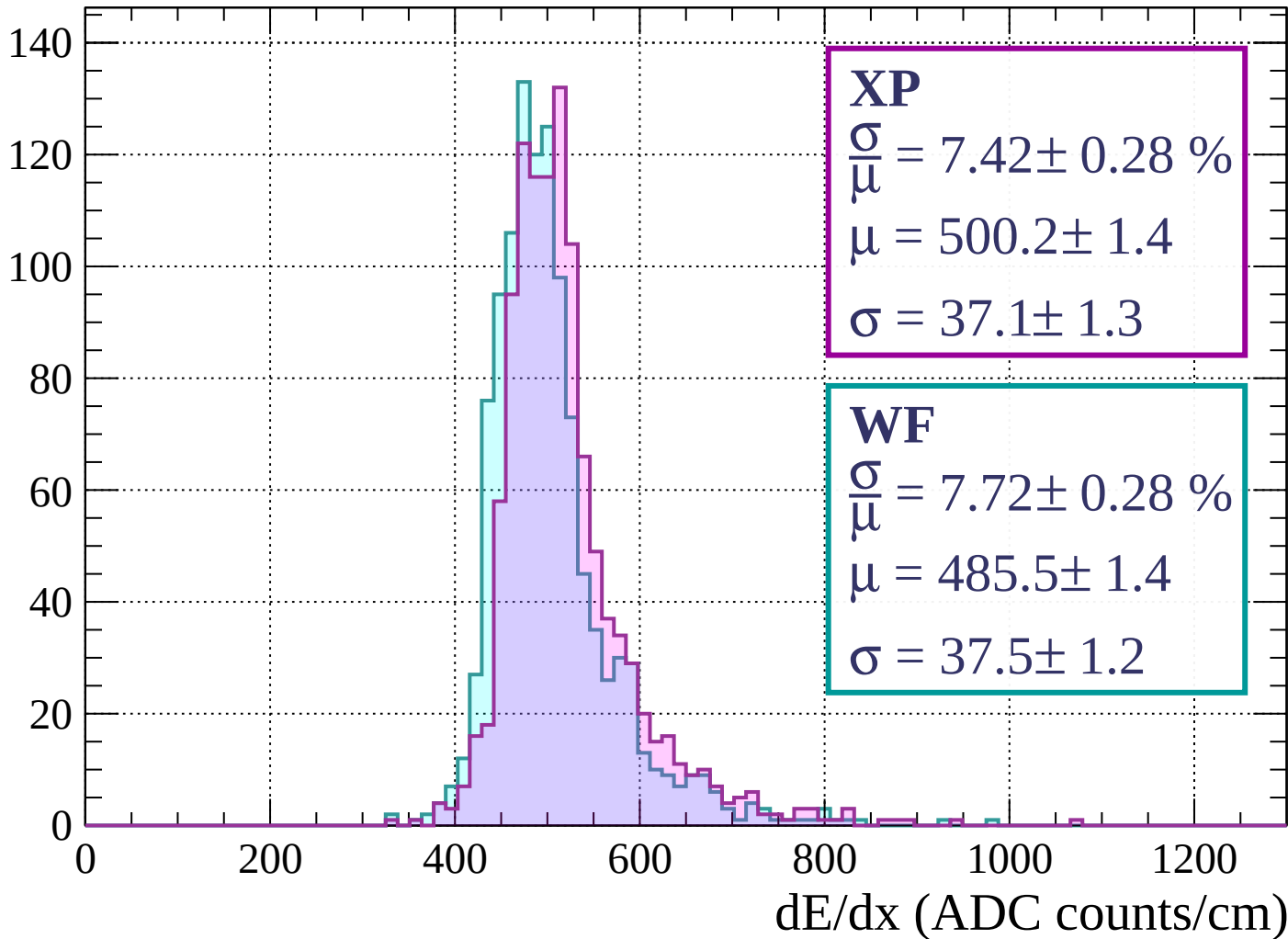
$$\mu = 481.2 \pm 1.4$$

$$\sigma = 37.9 \pm 1.3$$

dE/dx (ADC counts/cm)

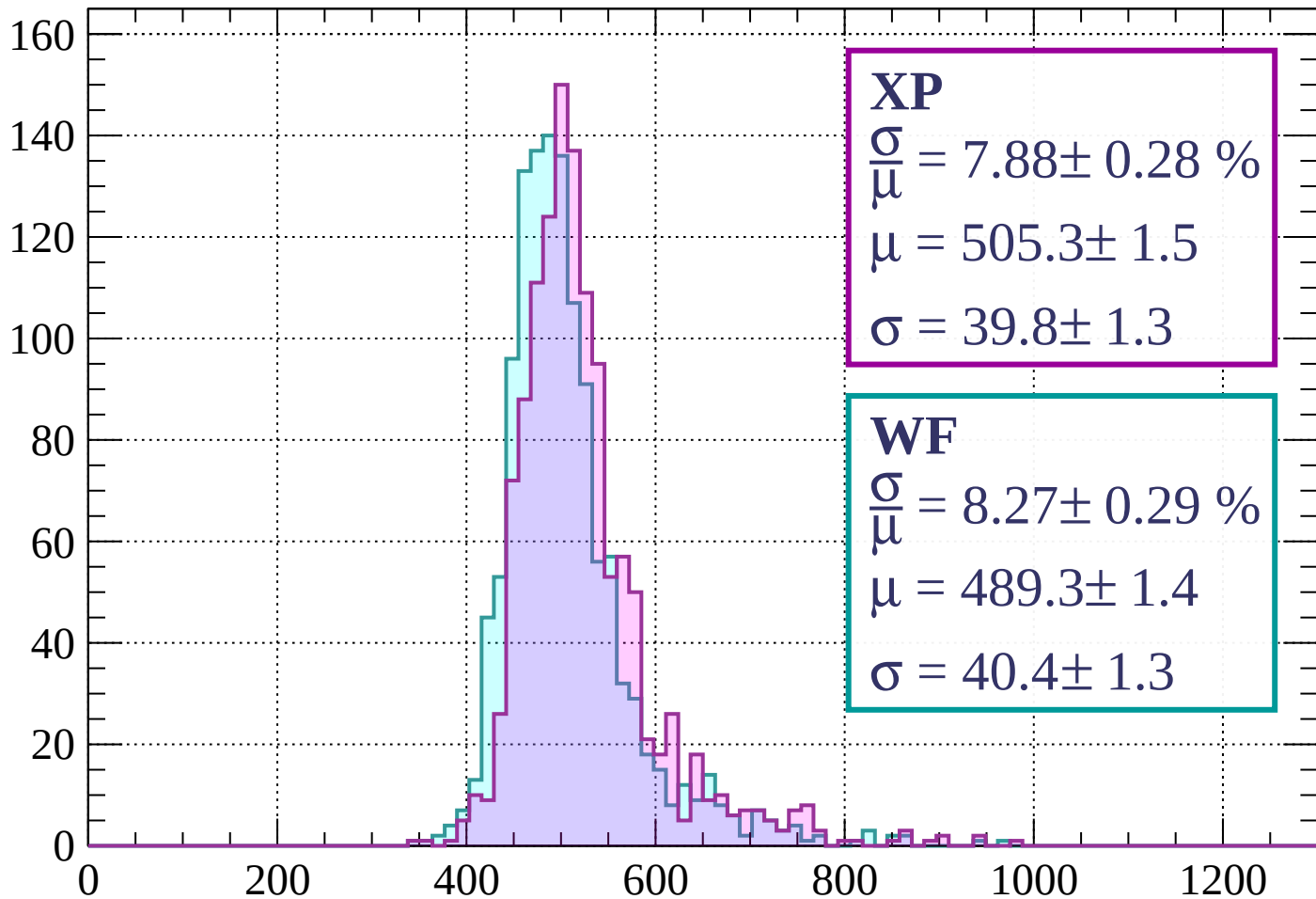
Energy loss | $1760 < p < 1840$

Count



Energy loss | $1840 < p < 1920$

Count



XP

$$\frac{\sigma}{\mu} = 7.88 \pm 0.28 \%$$

$$\mu = 505.3 \pm 1.5$$

$$\sigma = 39.8 \pm 1.3$$

WF

$$\frac{\sigma}{\mu} = 8.27 \pm 0.29 \%$$

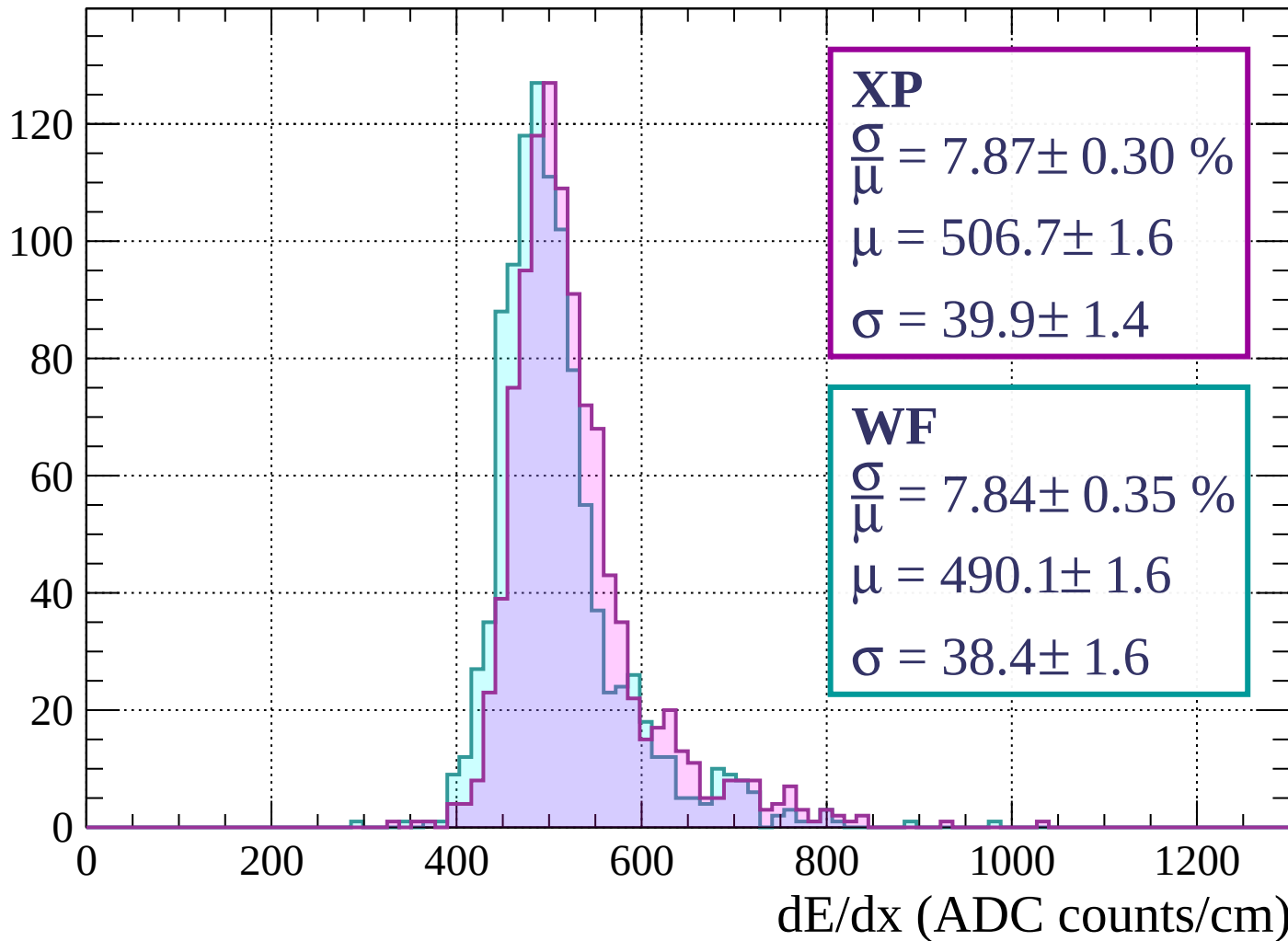
$$\mu = 489.3 \pm 1.4$$

$$\sigma = 40.4 \pm 1.3$$

dE/dx (ADC counts/cm)

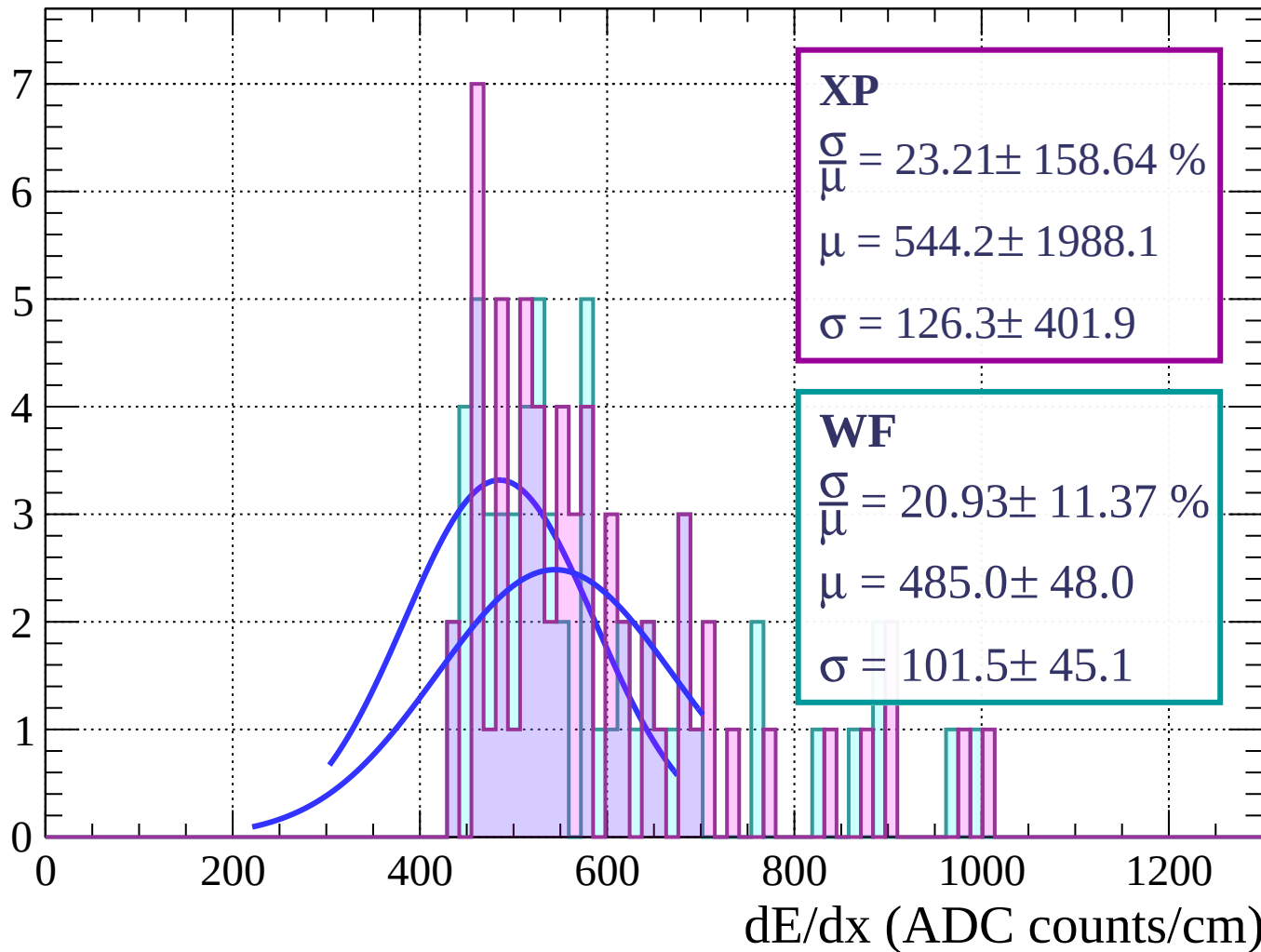
Energy loss | $1920 < p < 2000$

Count



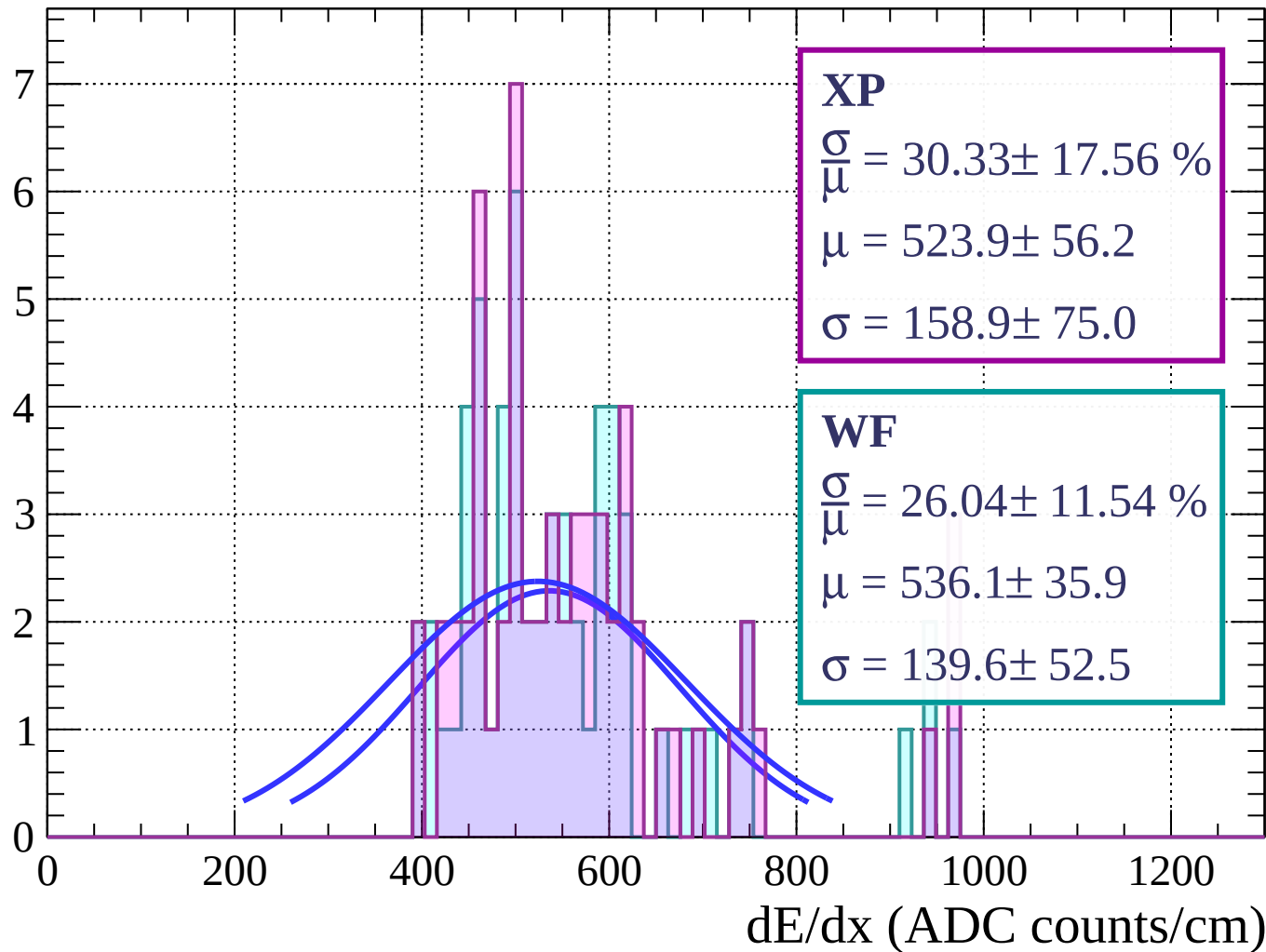
Energy loss | $0 < \phi < 3$

Count



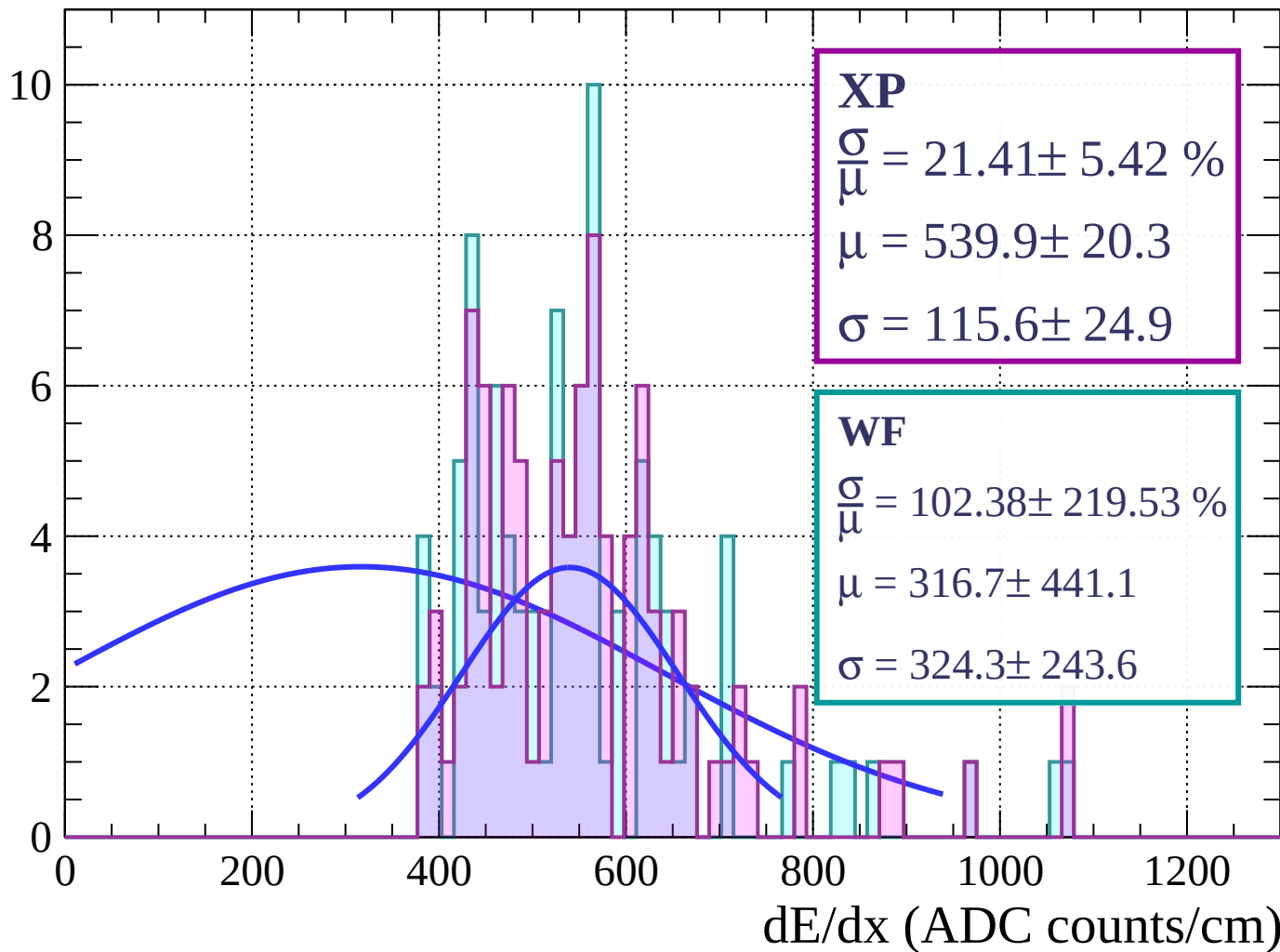
Energy loss | $3 < \phi < 6$

Count



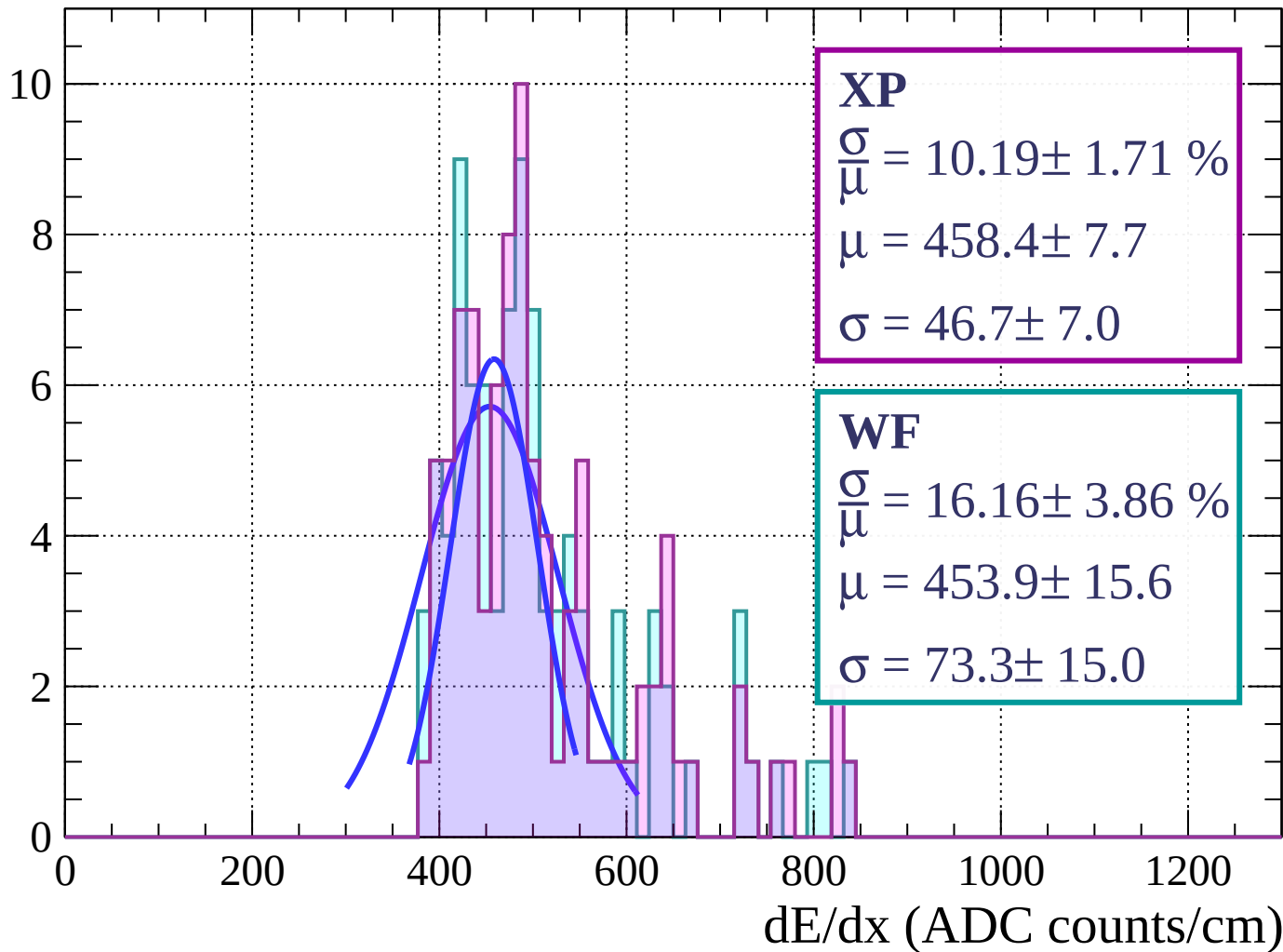
Energy loss | $6 < \phi < 9$

Count



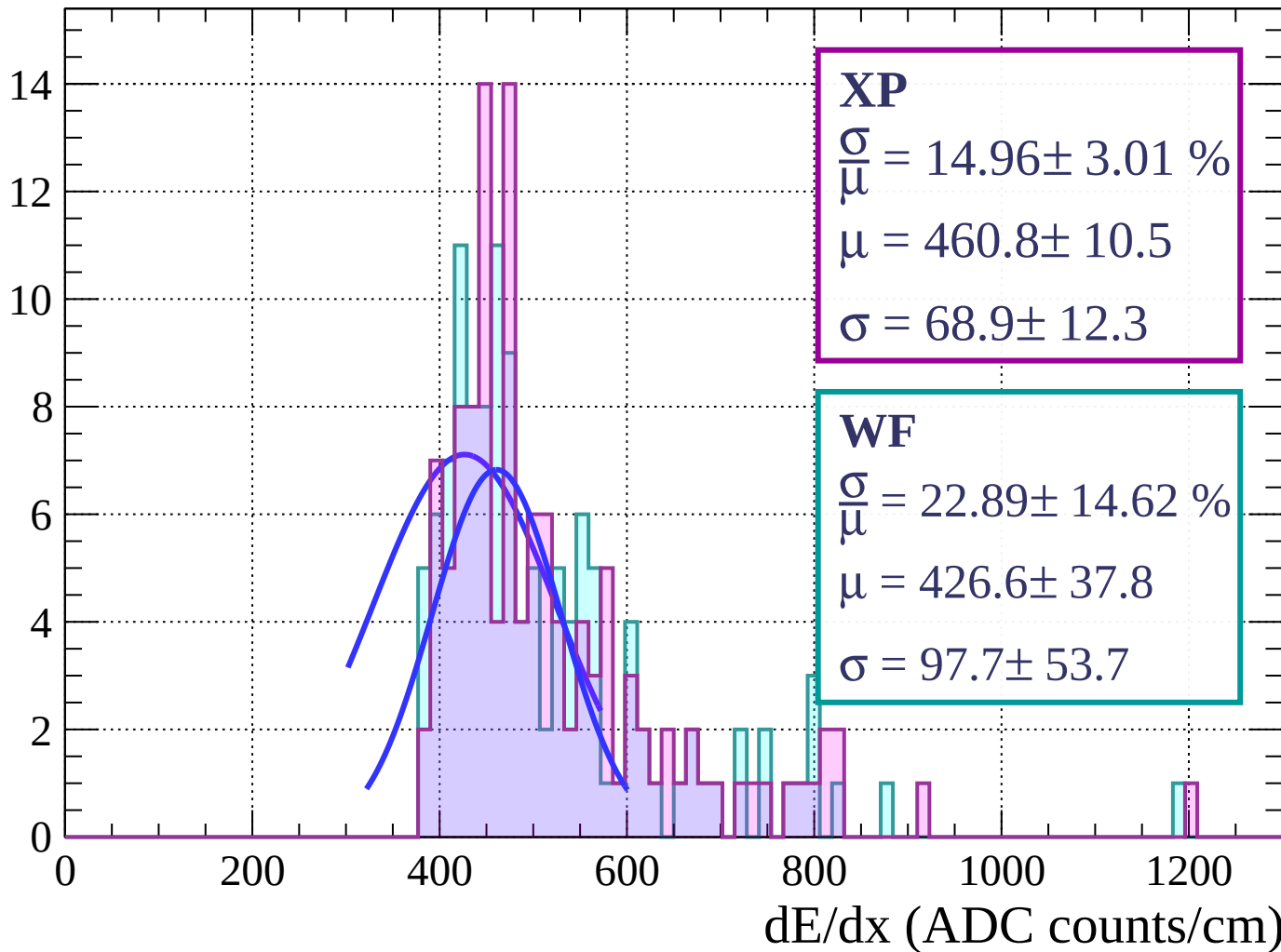
Energy loss | $9 < \phi < 12$

Count



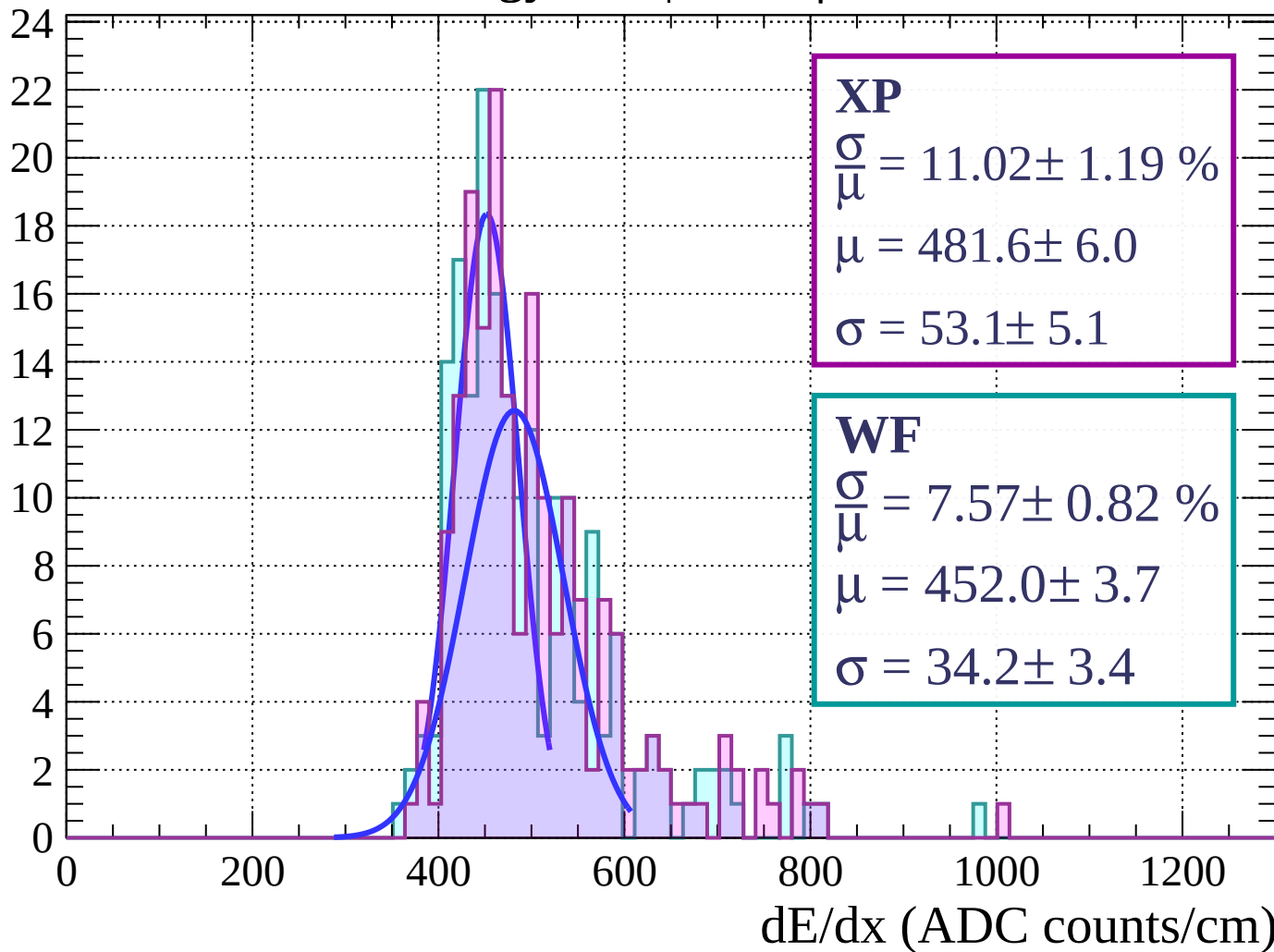
Energy loss | $12 < \phi < 15$

Count



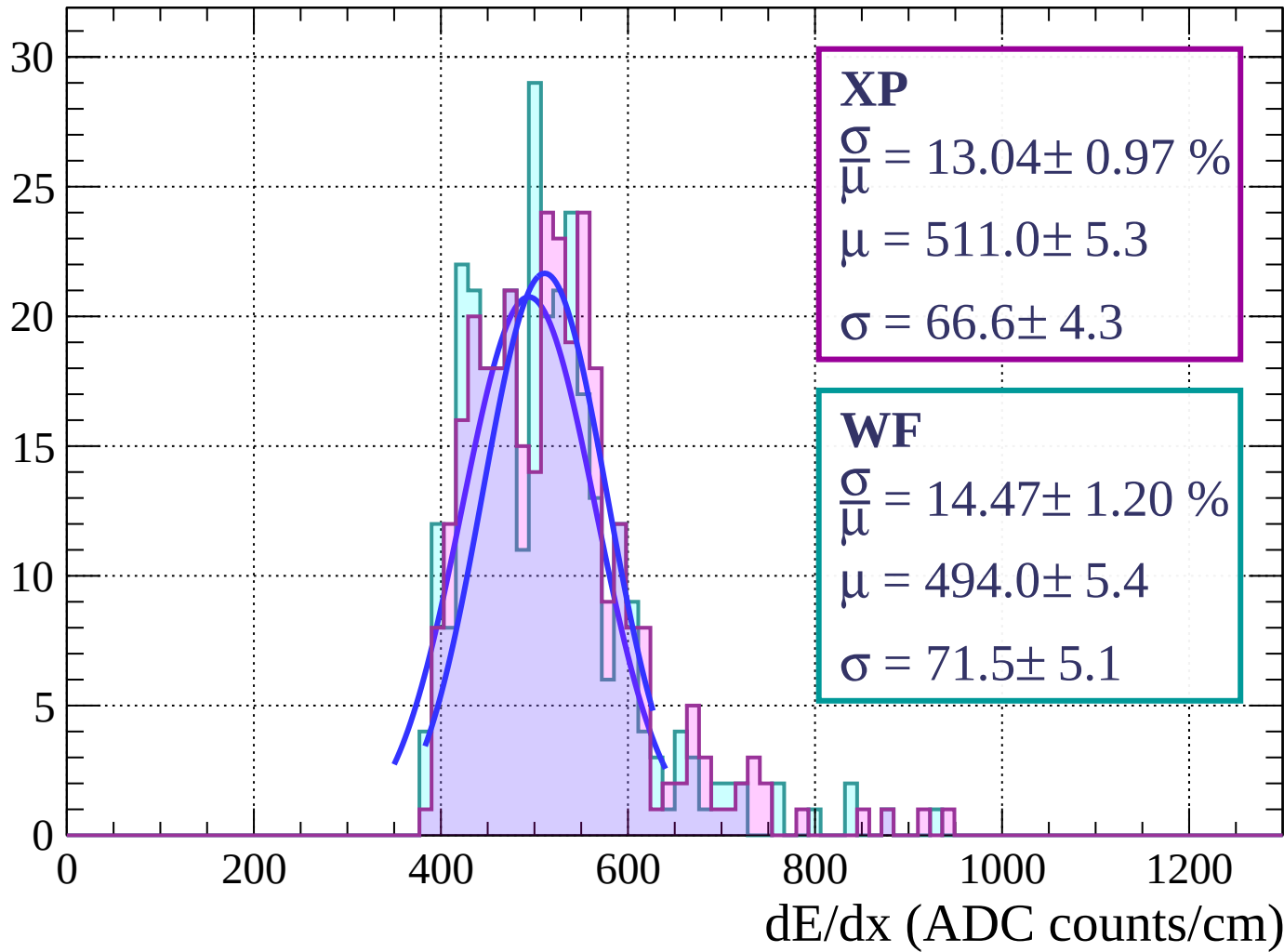
Energy loss | $15 < \phi < 18$

Count



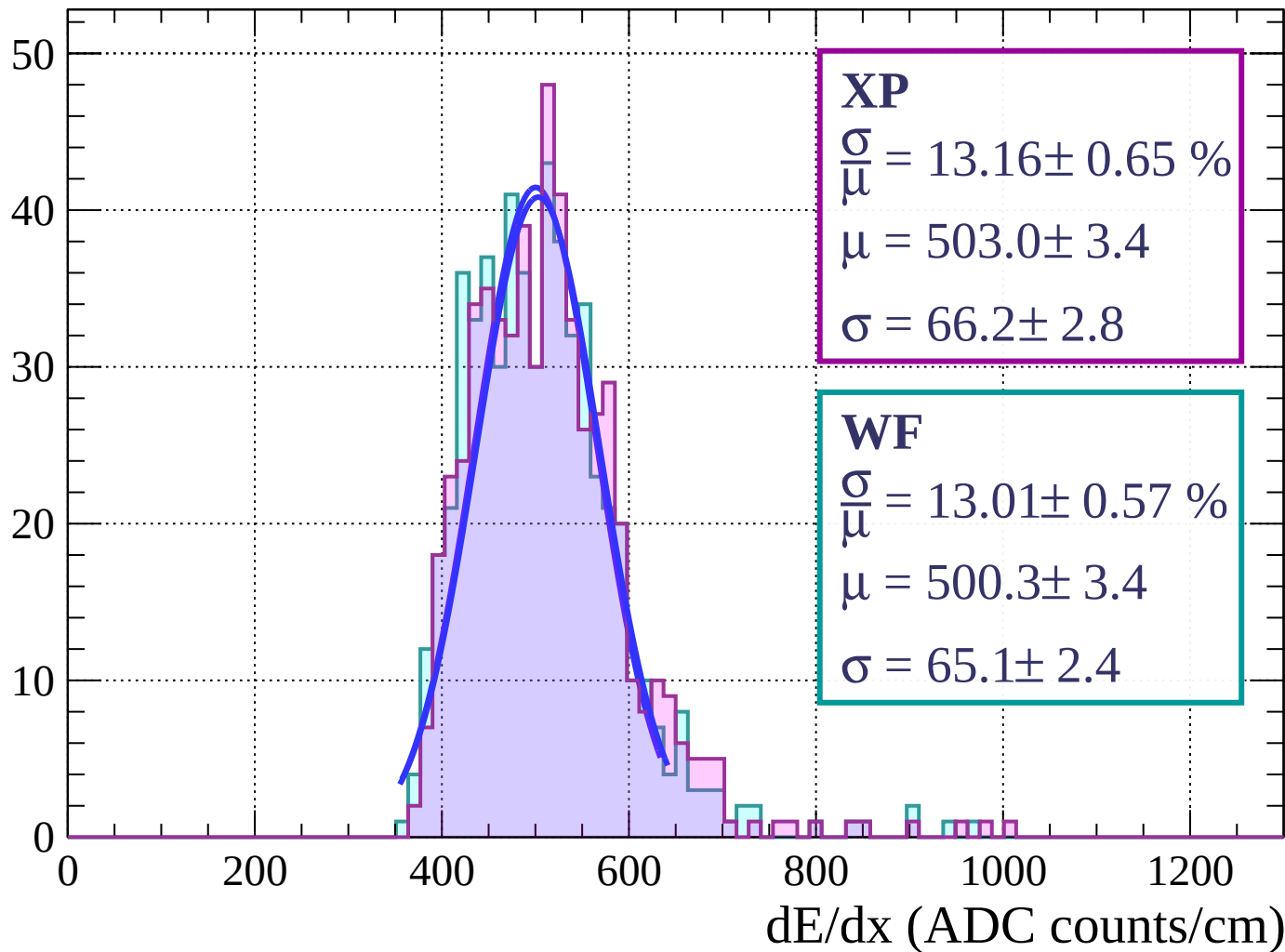
Energy loss | $18 < \phi < 21$

Count



Energy loss | $21 < \phi < 24$

Count



Energy loss | $24 < \phi < 27$

Count

XP

$$\frac{\sigma}{\mu} = 12.68 \pm 0.53 \%$$

$$\mu = 507.4 \pm 2.5$$

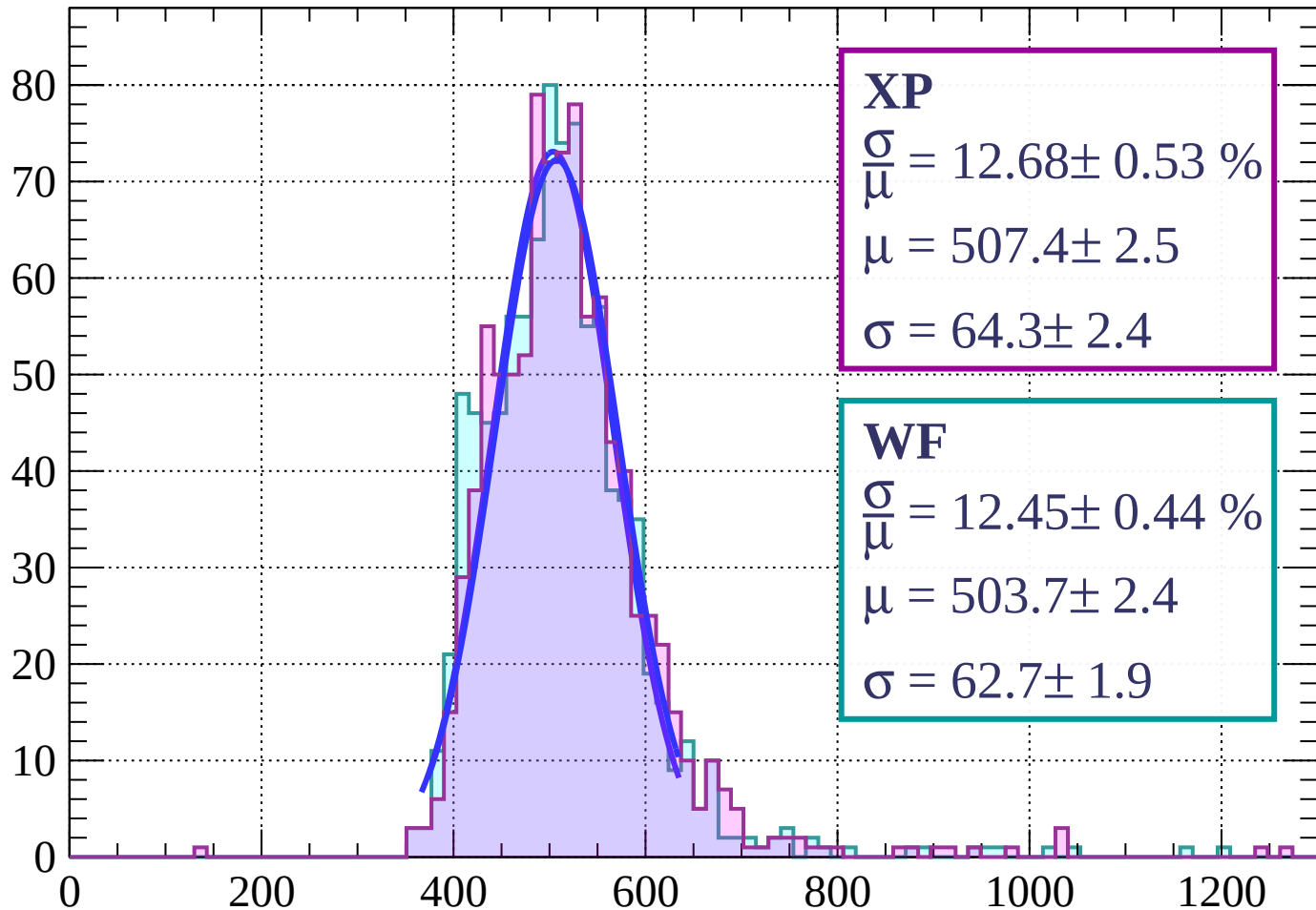
$$\sigma = 64.3 \pm 2.4$$

WF

$$\frac{\sigma}{\mu} = 12.45 \pm 0.44 \%$$

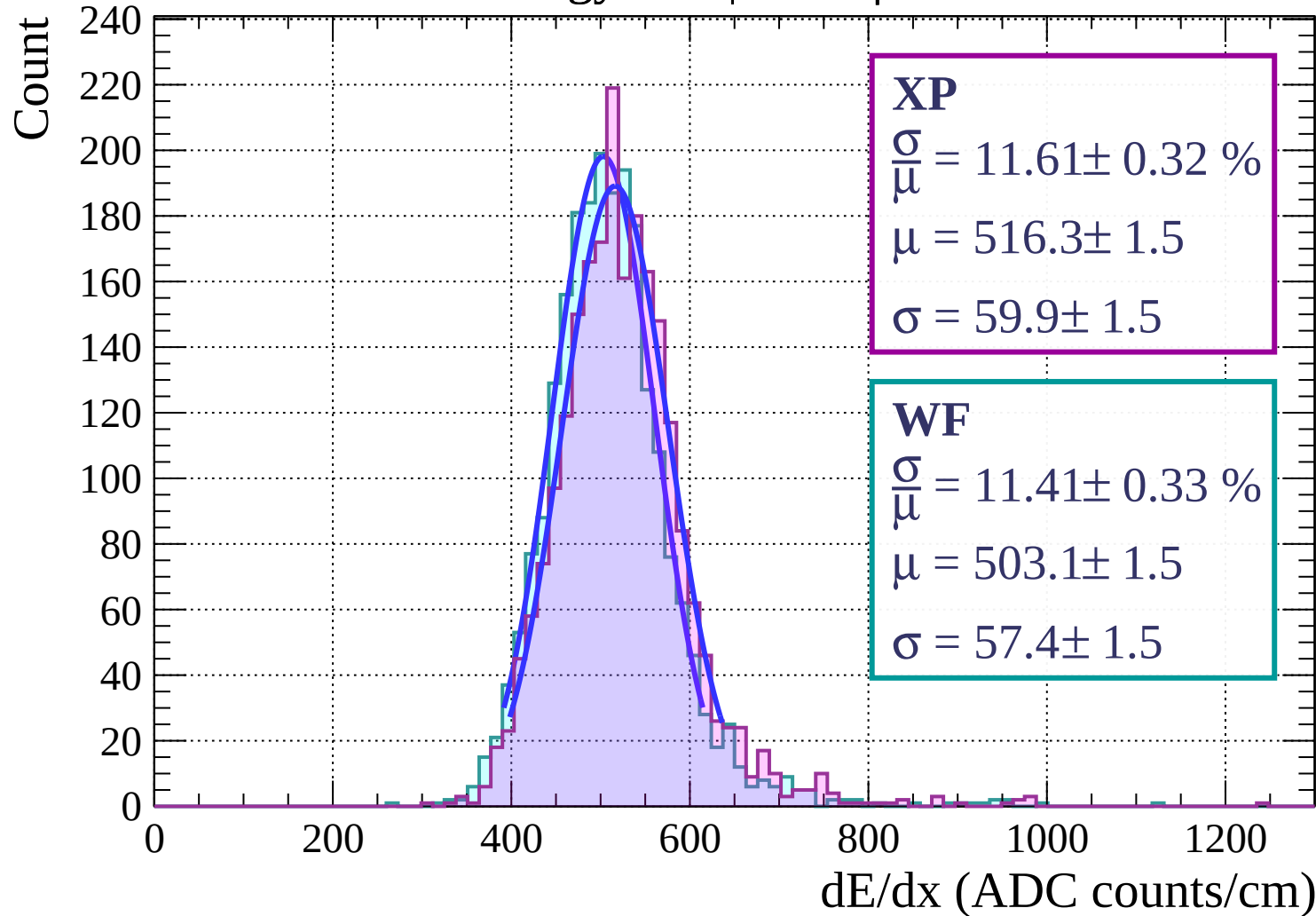
$$\mu = 503.7 \pm 2.4$$

$$\sigma = 62.7 \pm 1.9$$

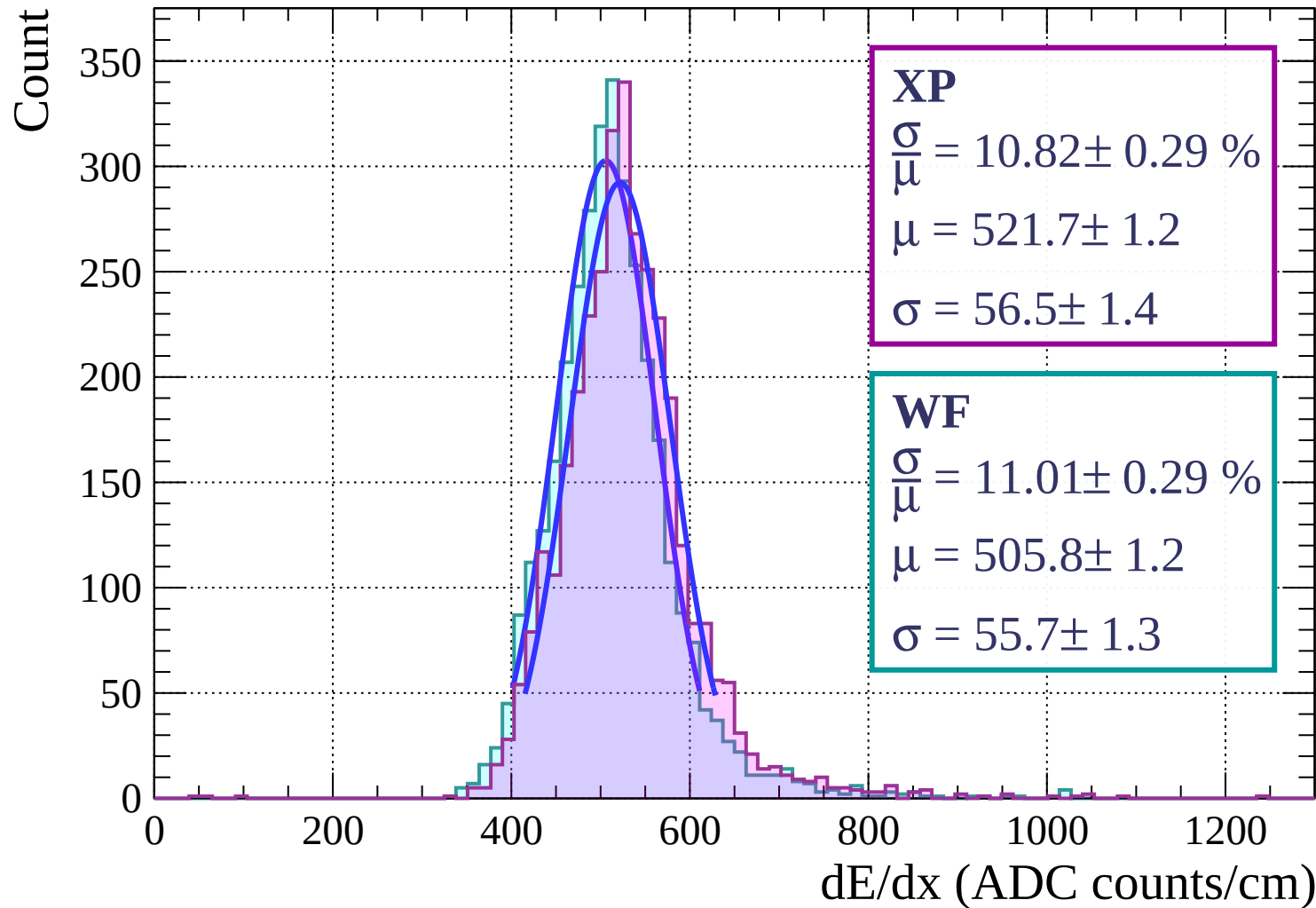


dE/dx (ADC counts/cm)

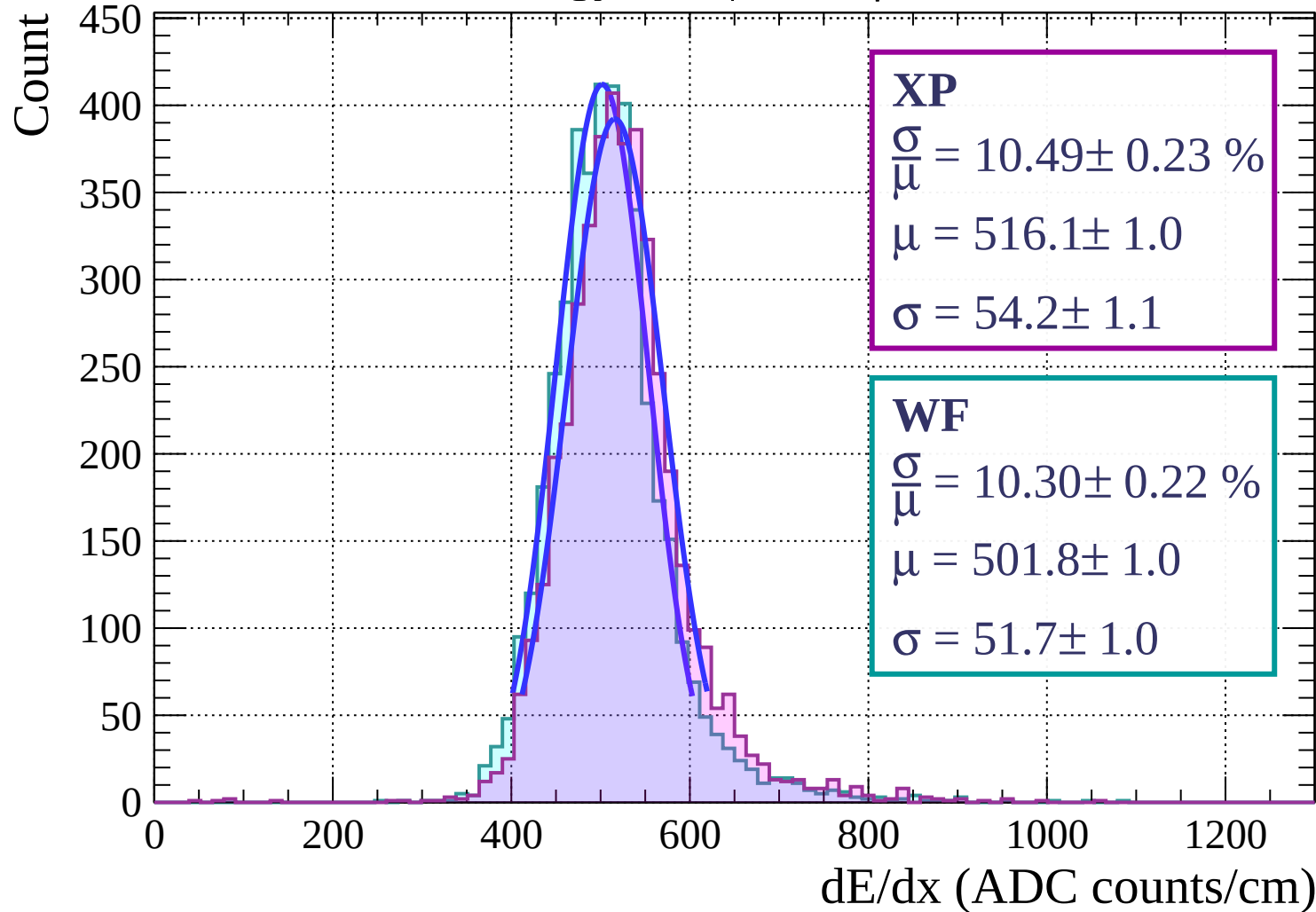
Energy loss | $27 < \phi < 30$



Energy loss | $30 < \phi < 33$



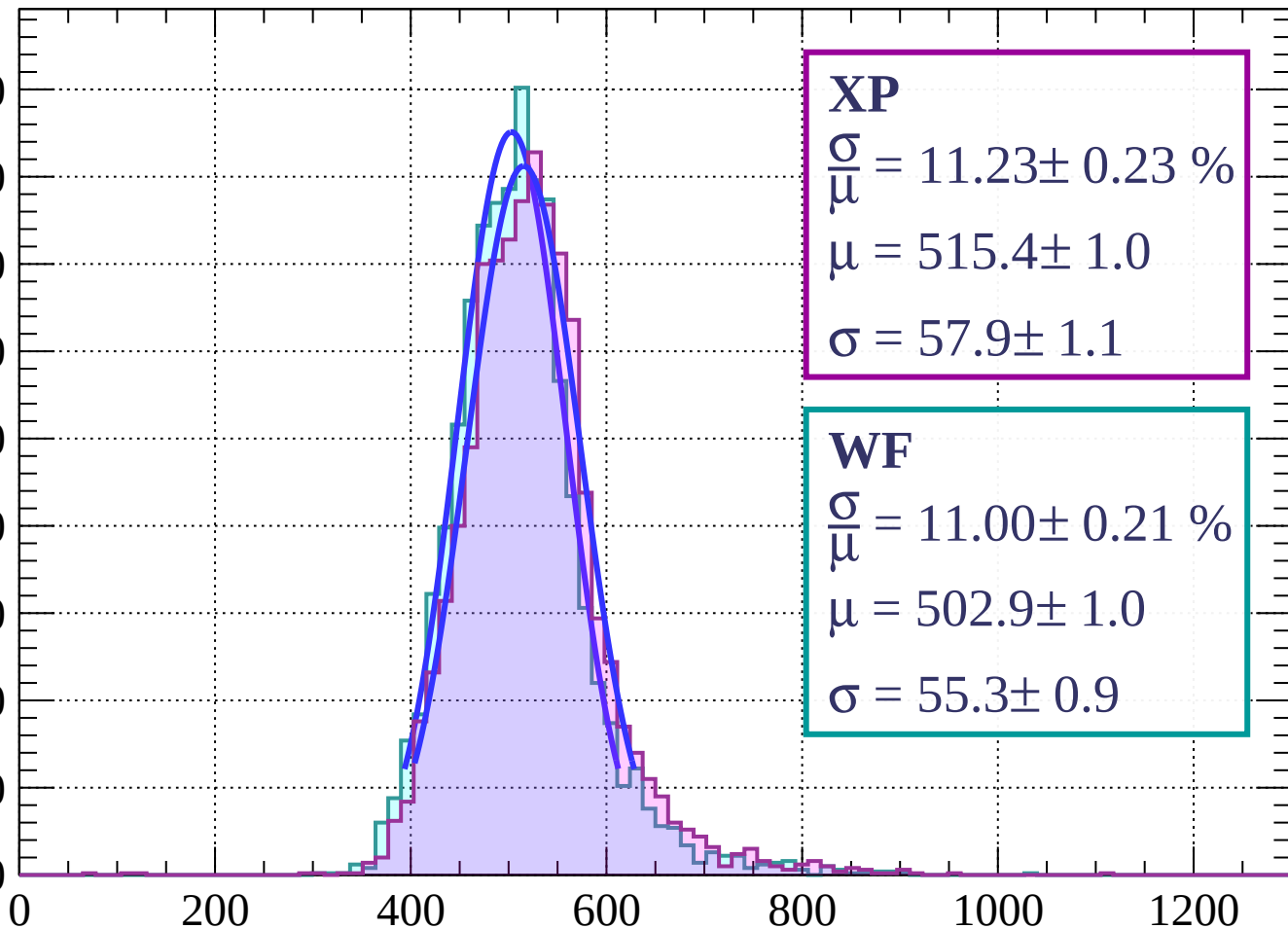
Energy loss | $33 < \phi < 36$



Energy loss | $36 < \phi < 39$

Count

450
400
350
300
250
200
150
100
50
0



XP

$$\frac{\sigma}{\mu} = 11.23 \pm 0.23 \%$$

$$\mu = 515.4 \pm 1.0$$

$$\sigma = 57.9 \pm 1.1$$

WF

$$\frac{\sigma}{\mu} = 11.00 \pm 0.21 \%$$

$$\mu = 502.9 \pm 1.0$$

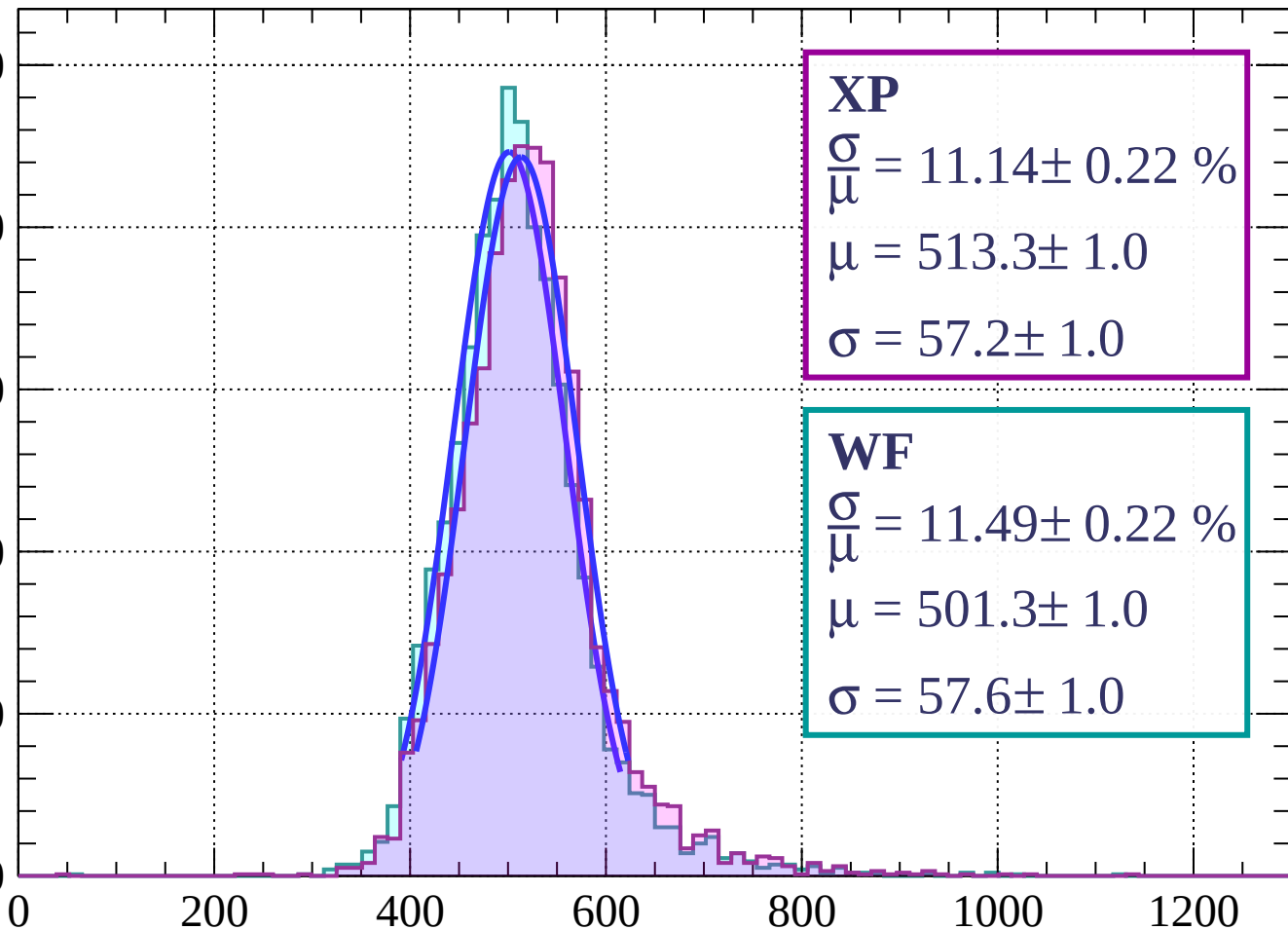
$$\sigma = 55.3 \pm 0.9$$

dE/dx (ADC counts/cm)

Energy loss | $39 < \phi < 42$

Count

500
400
300
200
100
0



XP

$$\frac{\sigma}{\mu} = 11.14 \pm 0.22 \%$$

$$\mu = 513.3 \pm 1.0$$

$$\sigma = 57.2 \pm 1.0$$

WF

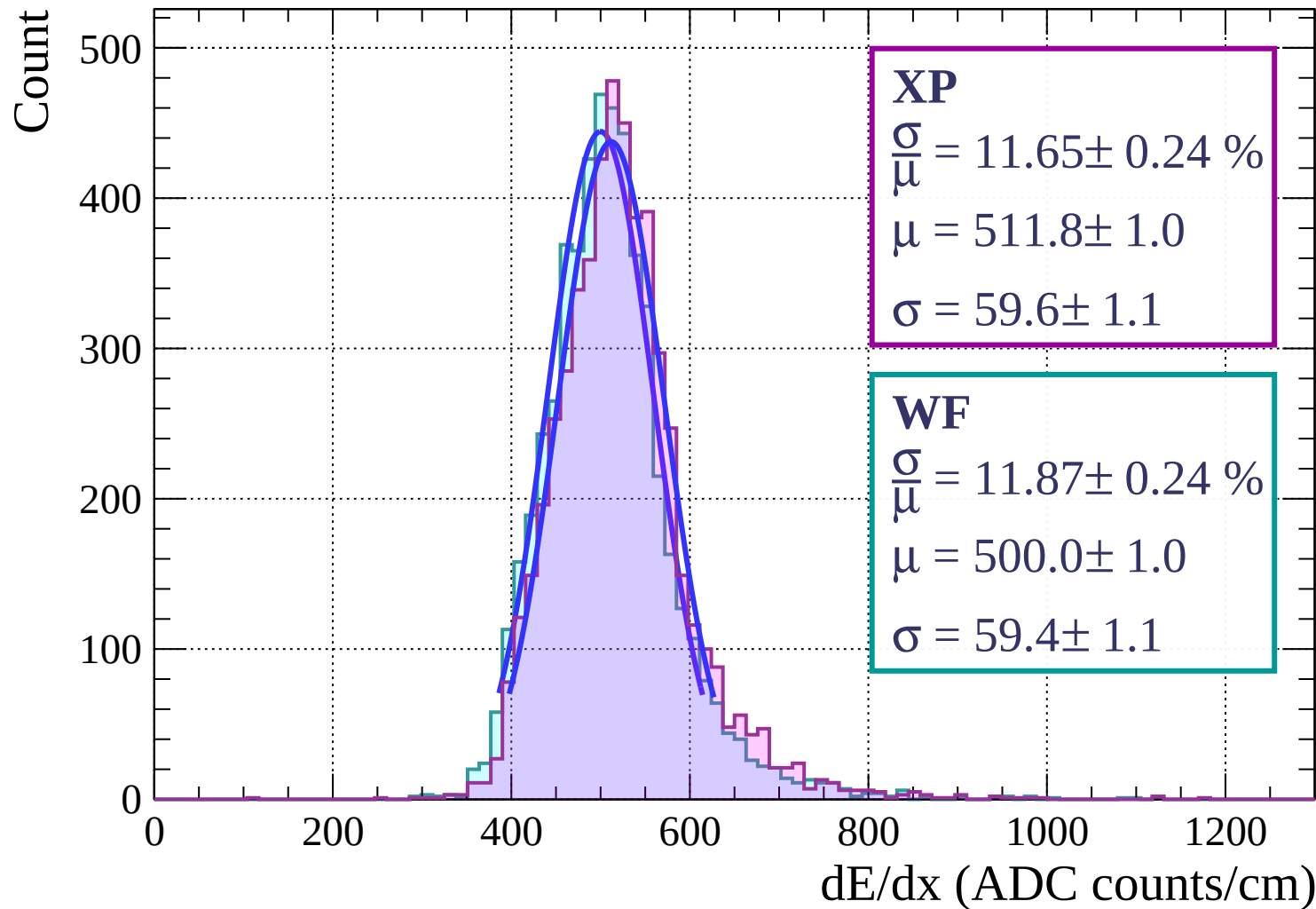
$$\frac{\sigma}{\mu} = 11.49 \pm 0.22 \%$$

$$\mu = 501.3 \pm 1.0$$

$$\sigma = 57.6 \pm 1.0$$

dE/dx (ADC counts/cm)

Energy loss | $42 < \phi < 45$



Energy loss | $45 < \phi < 48$

Count

500

400

300

200

100

0

0

200

400

600

800

1000

1200

dE/dx (ADC counts/cm)

XP

$$\frac{\sigma}{\mu} = 11.63 \pm 0.20 \%$$

$$\mu = 512.0 \pm 0.9$$

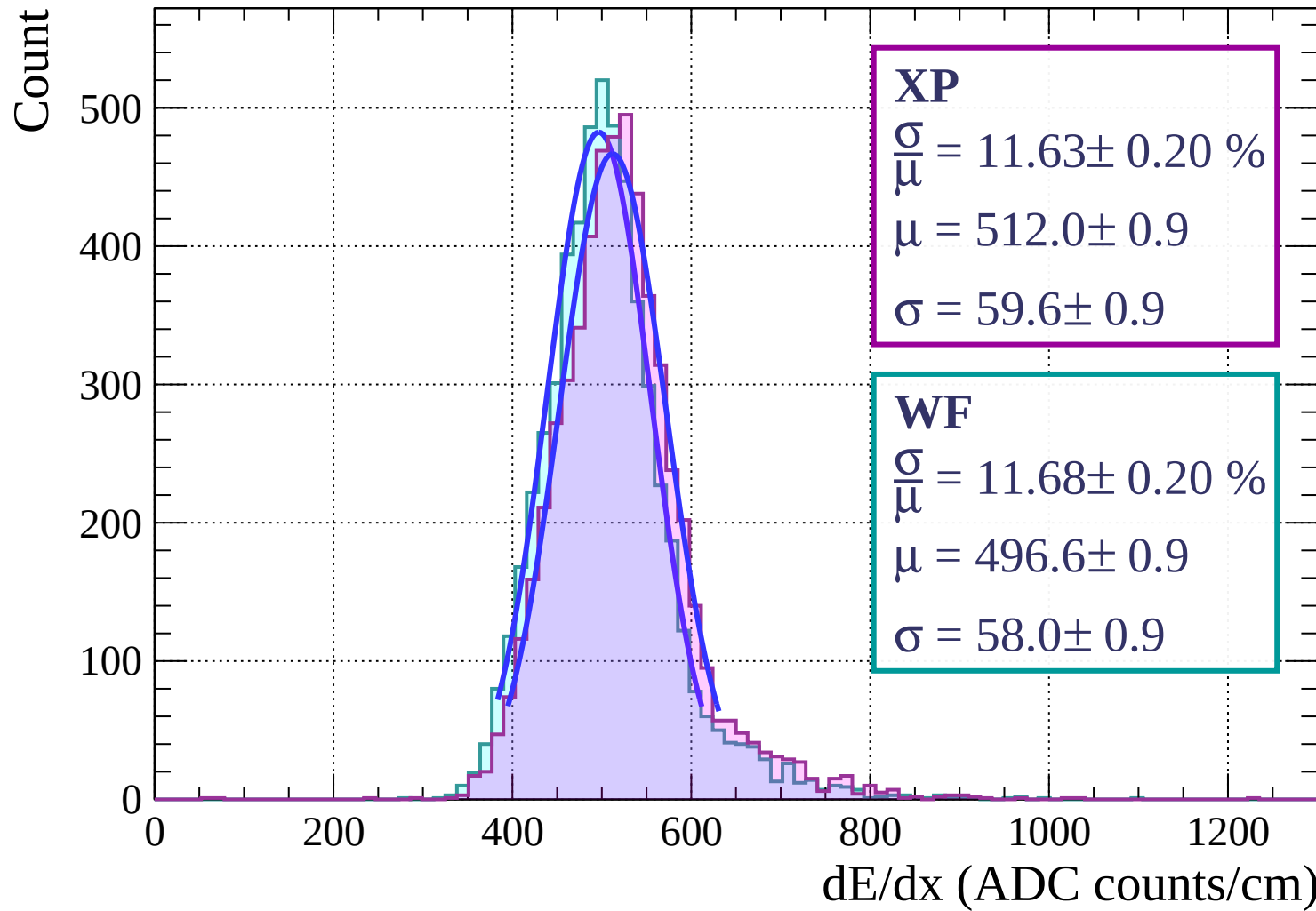
$$\sigma = 59.6 \pm 0.9$$

WF

$$\frac{\sigma}{\mu} = 11.68 \pm 0.20 \%$$

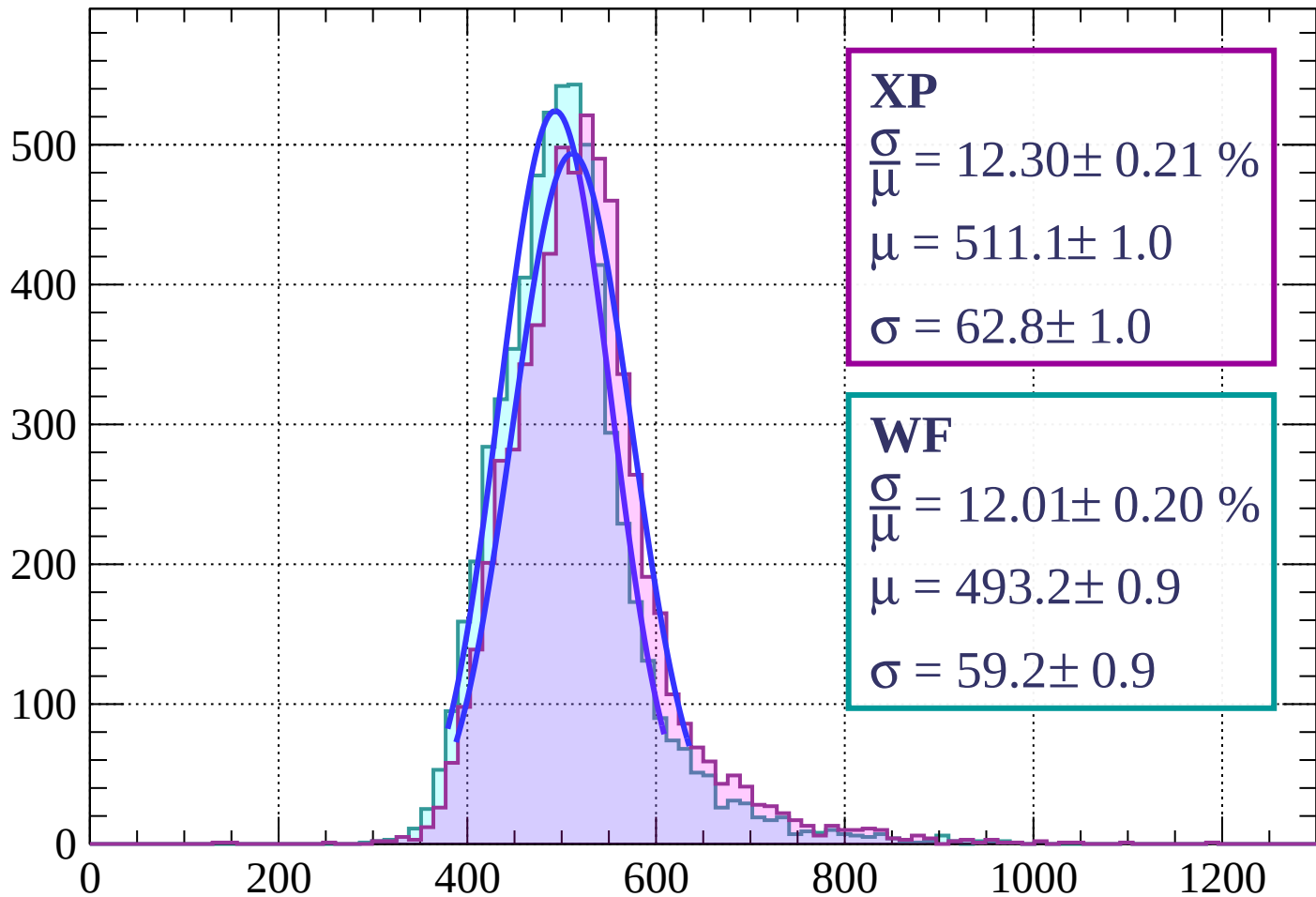
$$\mu = 496.6 \pm 0.9$$

$$\sigma = 58.0 \pm 0.9$$



Energy loss | $48 < \phi < 51$

Count



XP

$$\frac{\sigma}{\mu} = 12.30 \pm 0.21 \%$$

$$\mu = 511.1 \pm 1.0$$

$$\sigma = 62.8 \pm 1.0$$

WF

$$\frac{\sigma}{\mu} = 12.01 \pm 0.20 \%$$

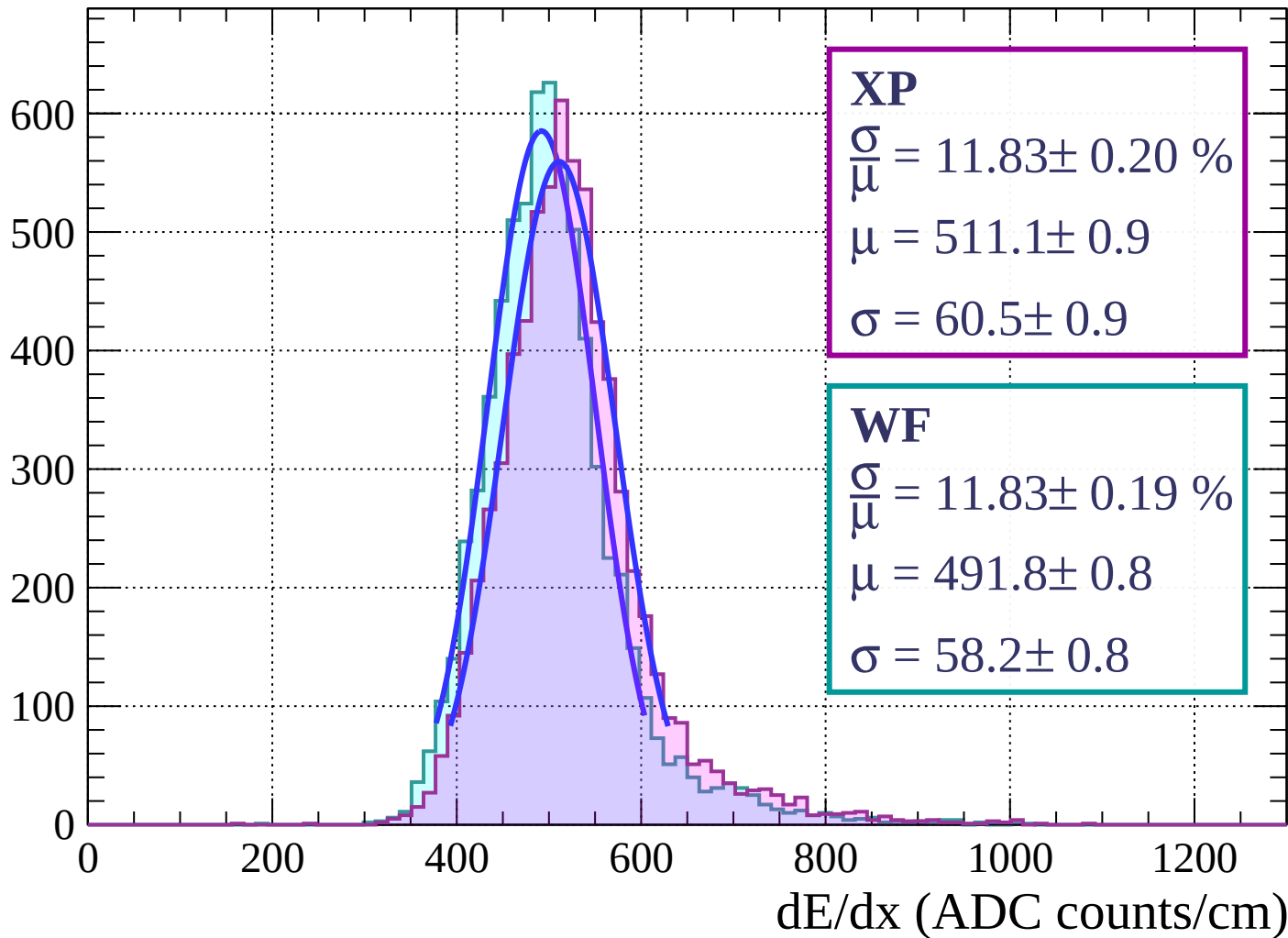
$$\mu = 493.2 \pm 0.9$$

$$\sigma = 59.2 \pm 0.9$$

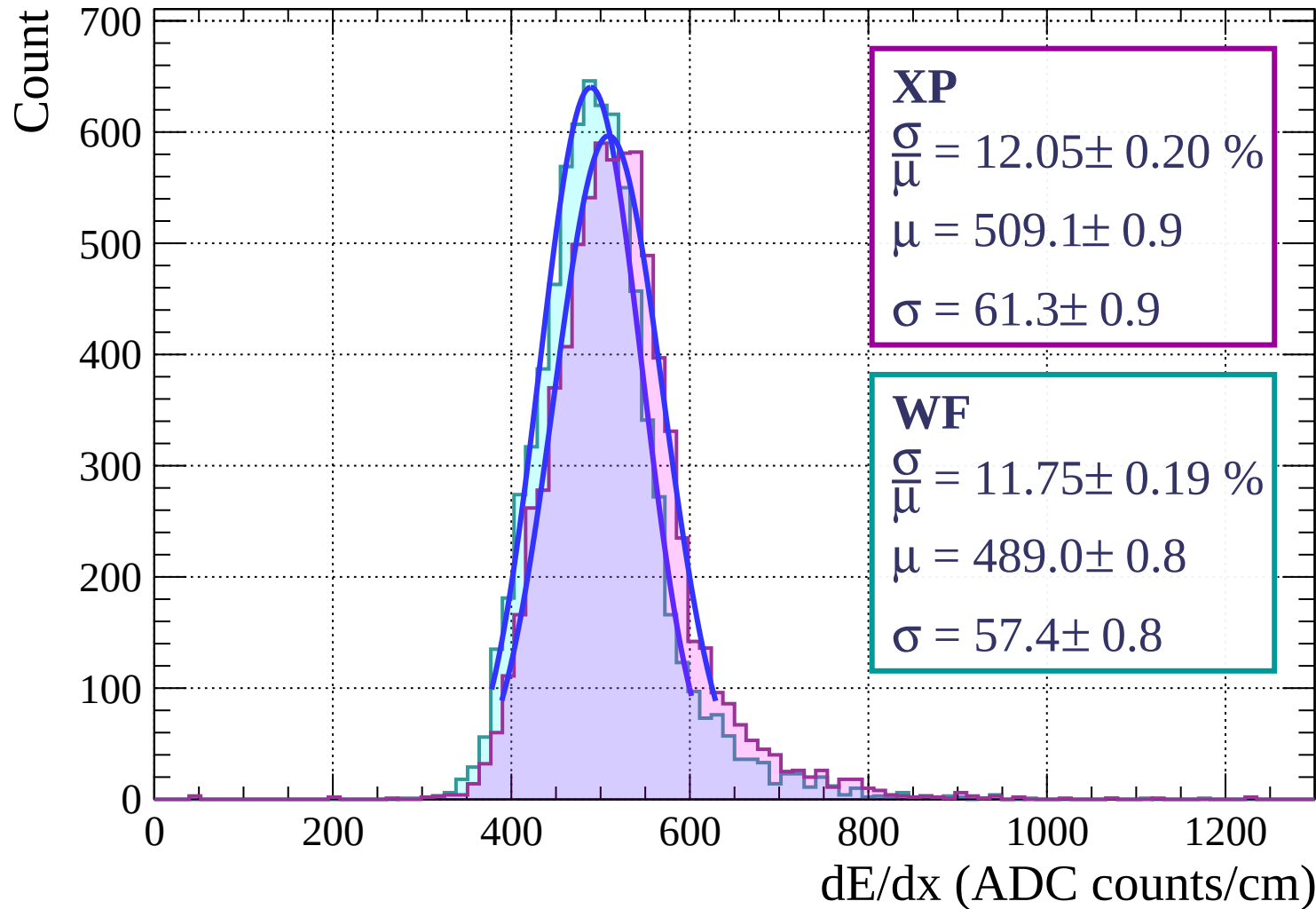
dE/dx (ADC counts/cm)

Energy loss | $51 < \phi < 54$

Count

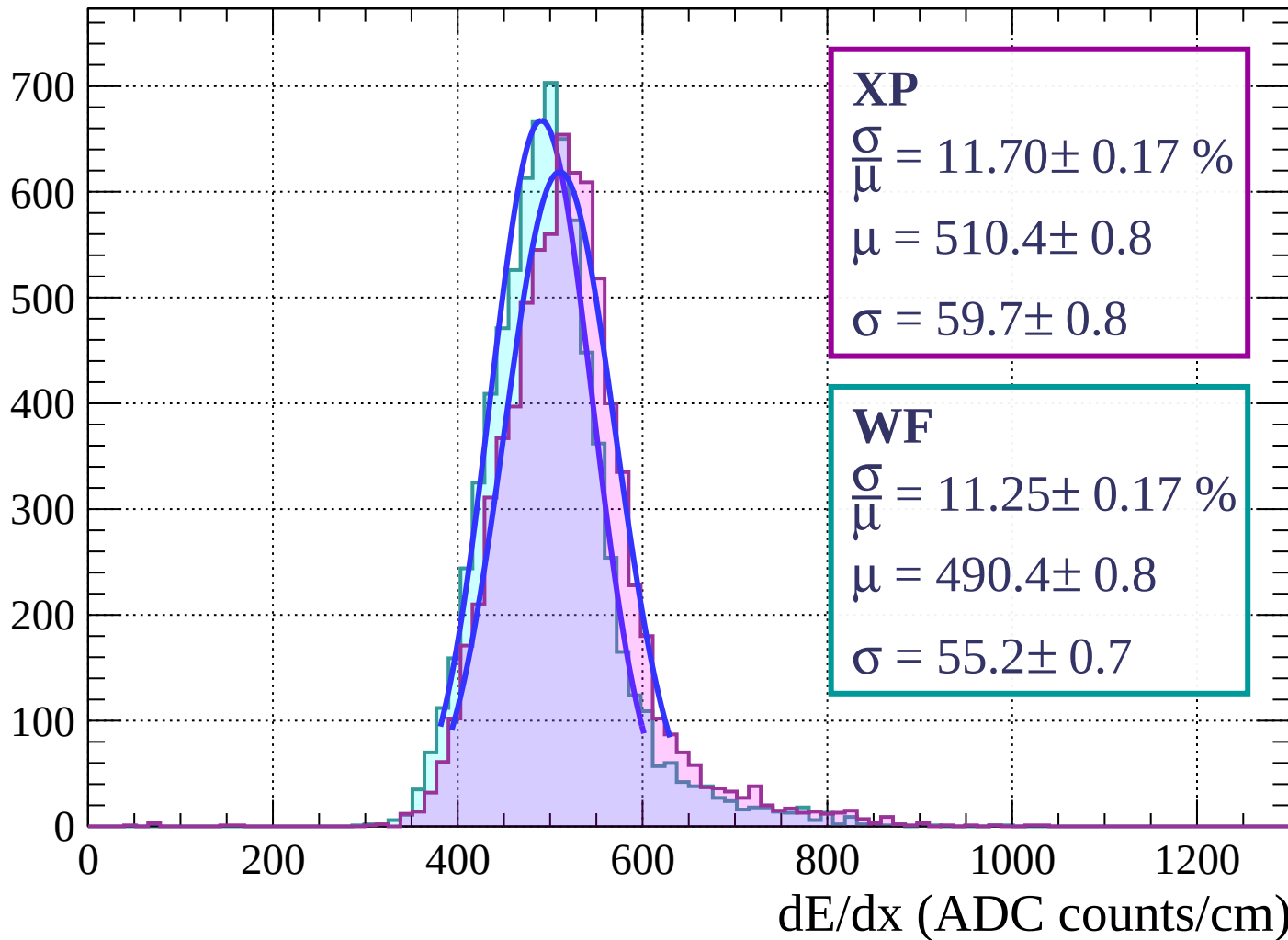


Energy loss | $54 < \phi < 57$



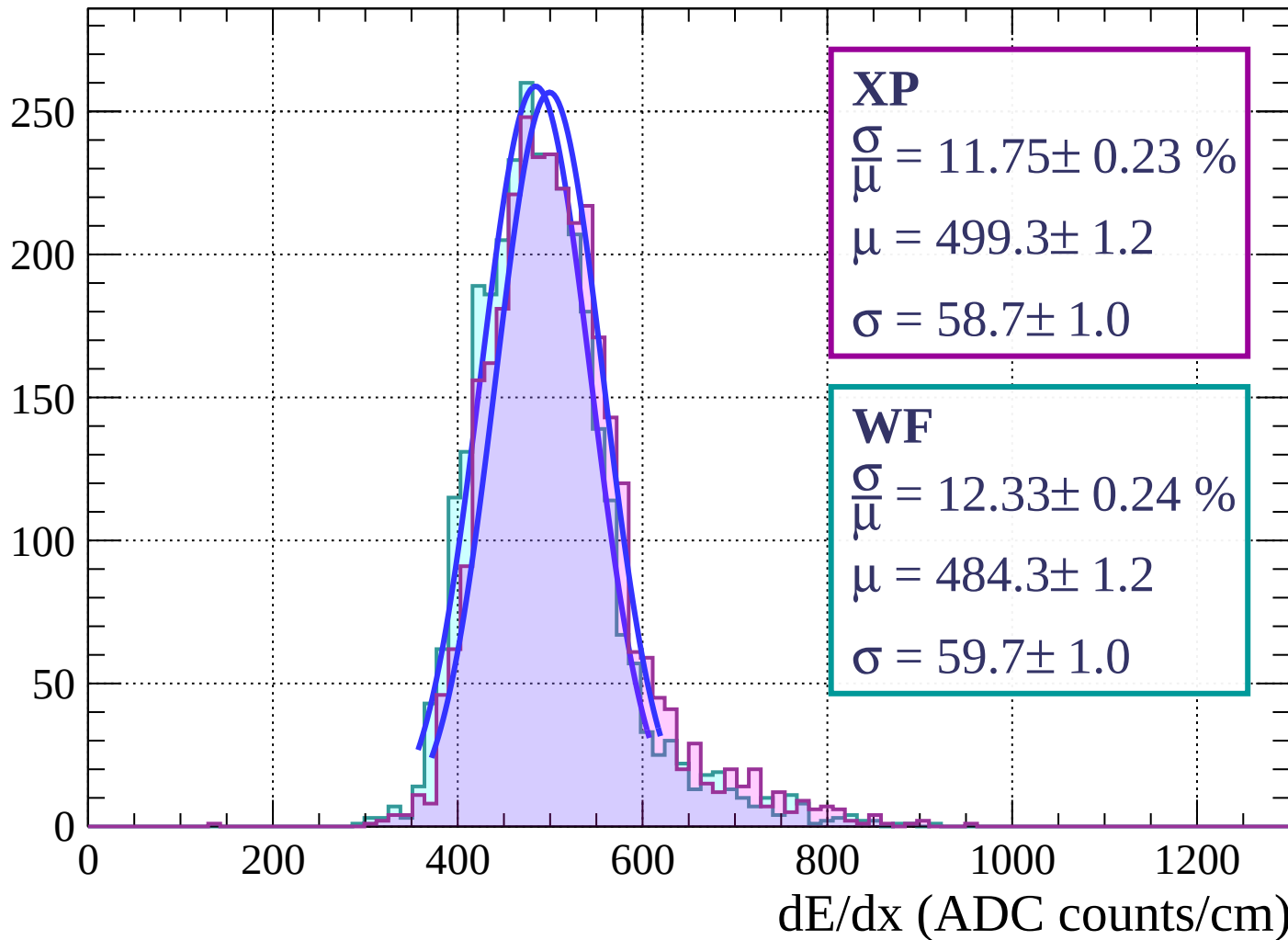
Energy loss | $57 < \phi < 60$

Count



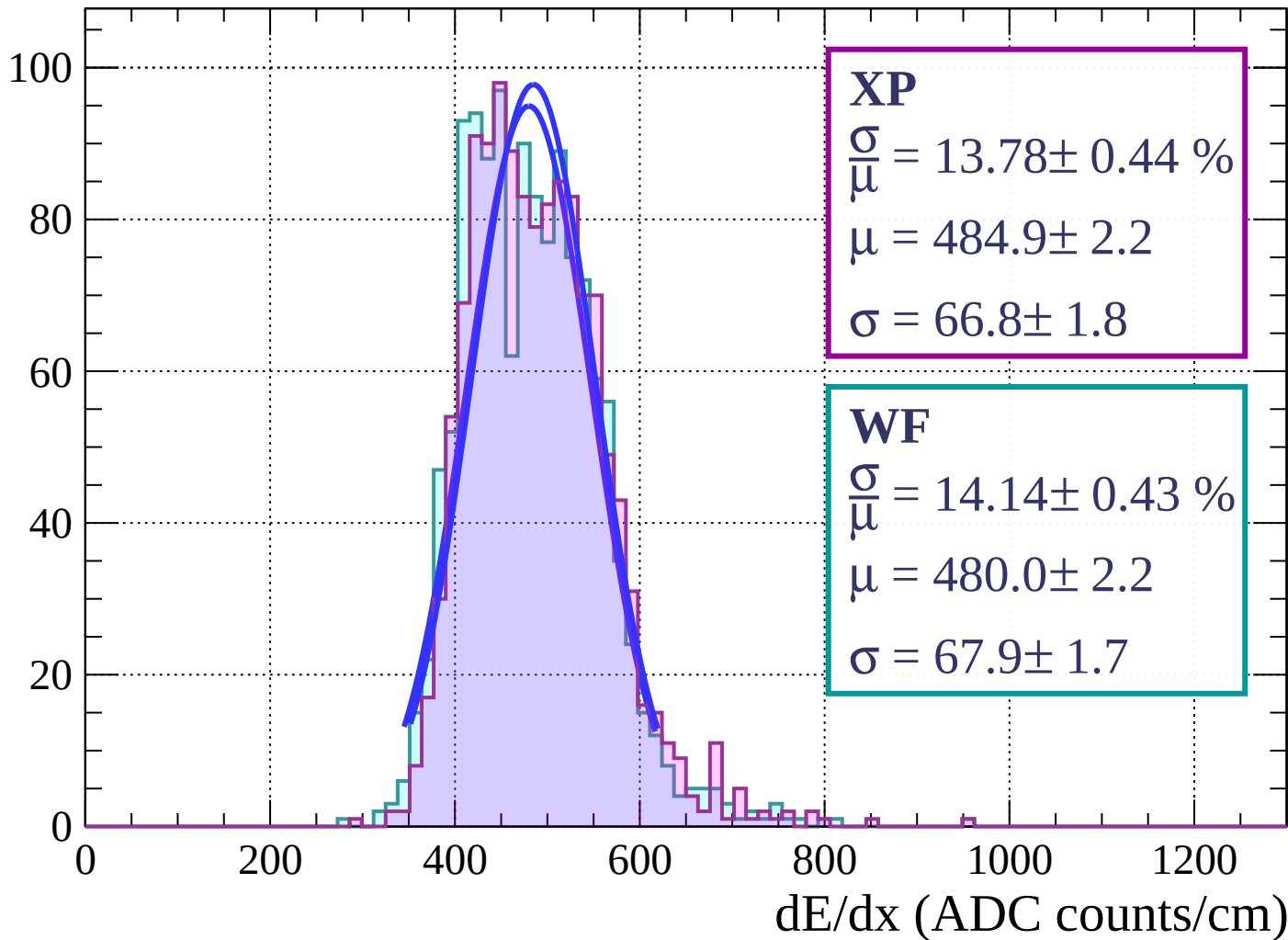
Energy loss | $60 < \phi < 63$

Count

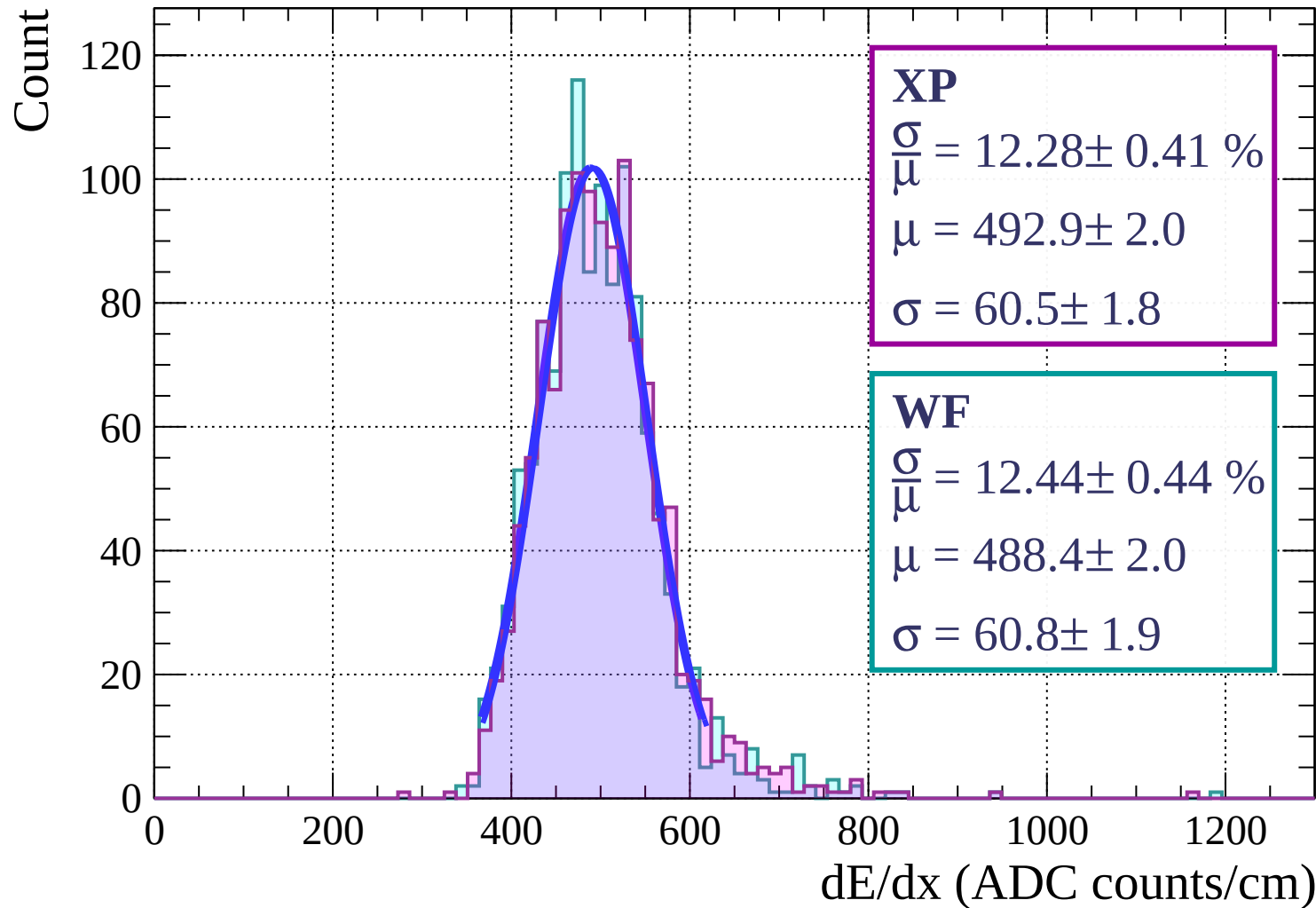


Energy loss | $63 < \phi < 66$

Count

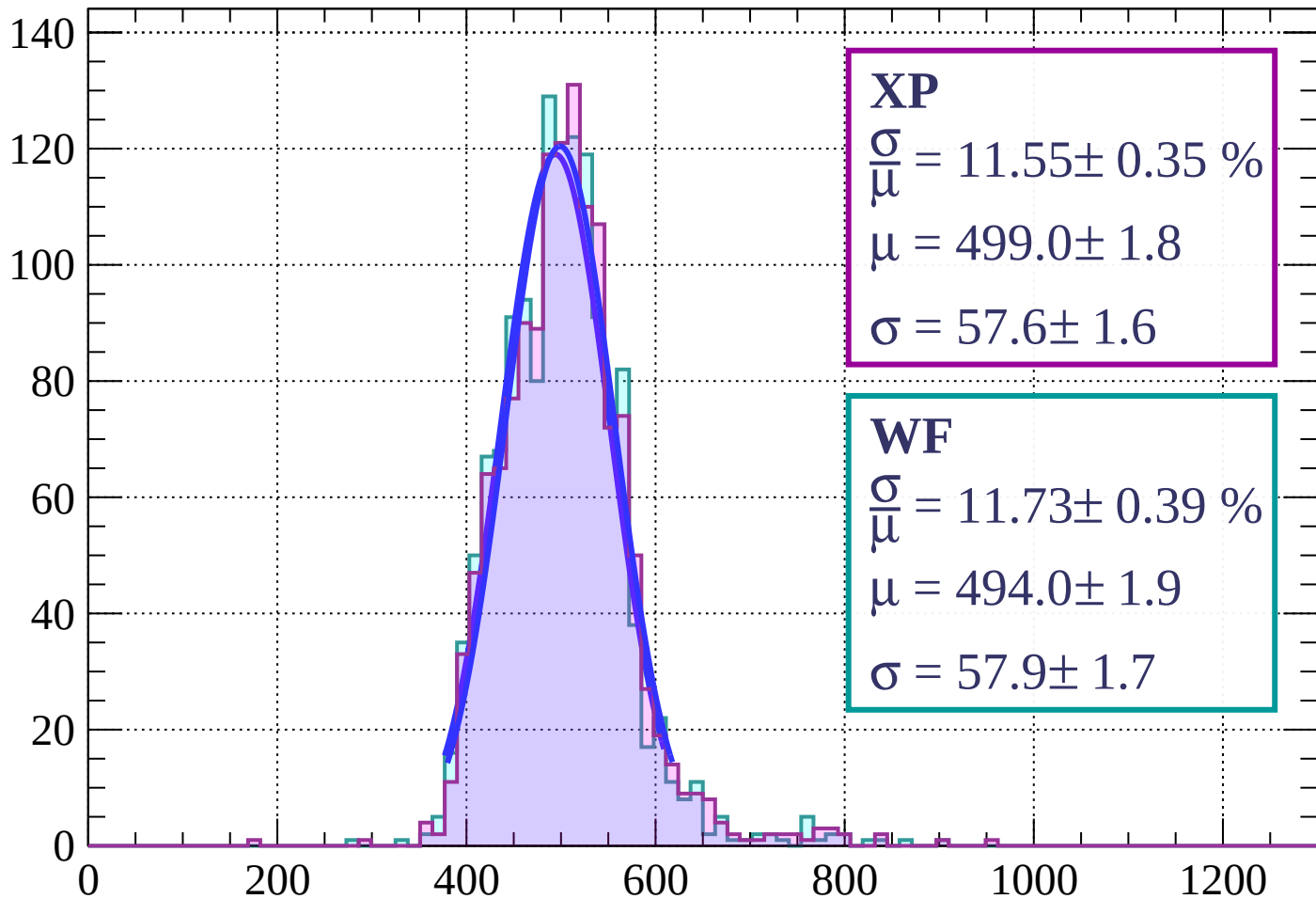


Energy loss | $66 < \phi < 69$



Energy loss | $69 < \phi < 72$

Count



XP

$$\frac{\sigma}{\mu} = 11.55 \pm 0.35 \%$$

$$\mu = 499.0 \pm 1.8$$

$$\sigma = 57.6 \pm 1.6$$

WF

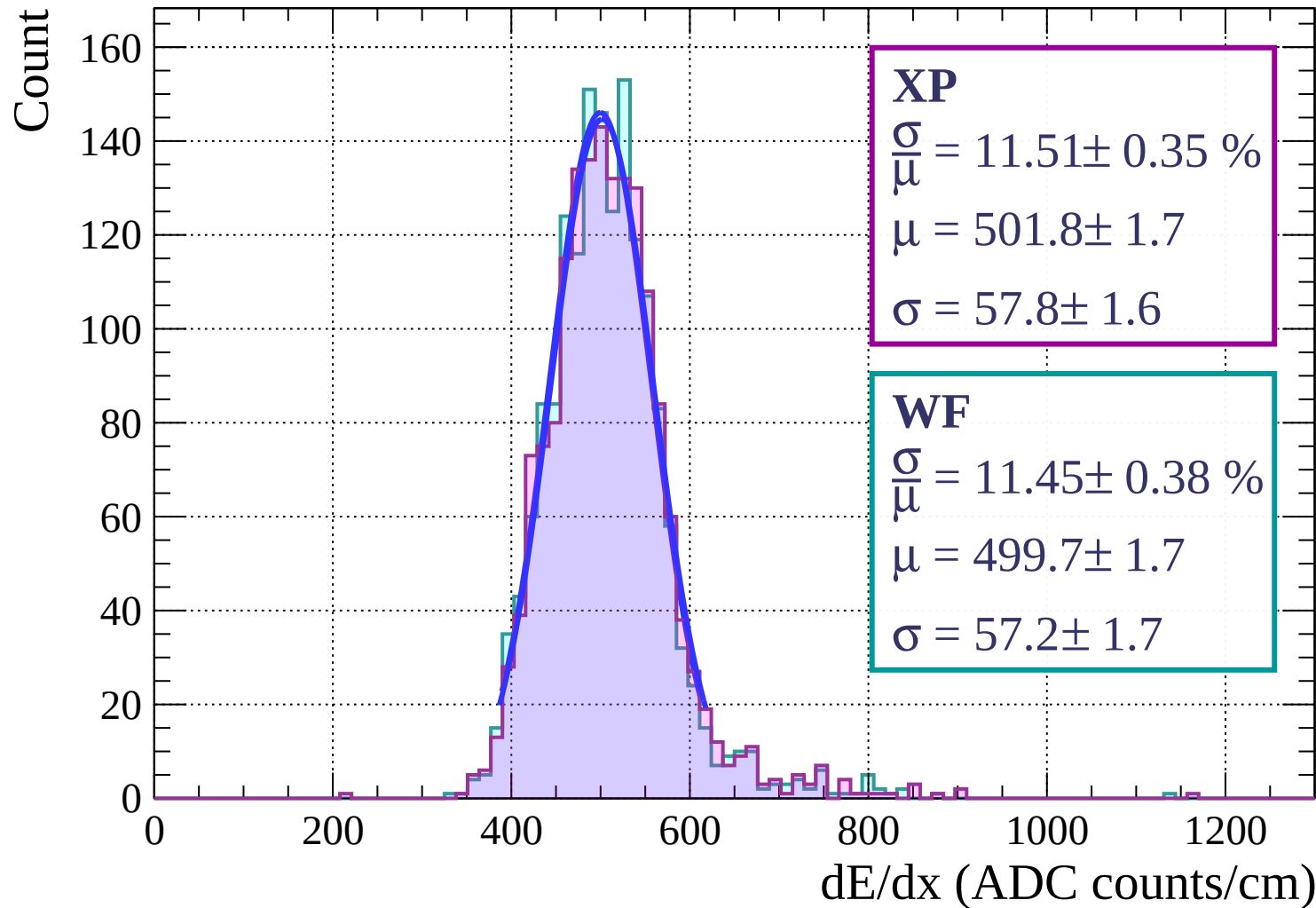
$$\frac{\sigma}{\mu} = 11.73 \pm 0.39 \%$$

$$\mu = 494.0 \pm 1.9$$

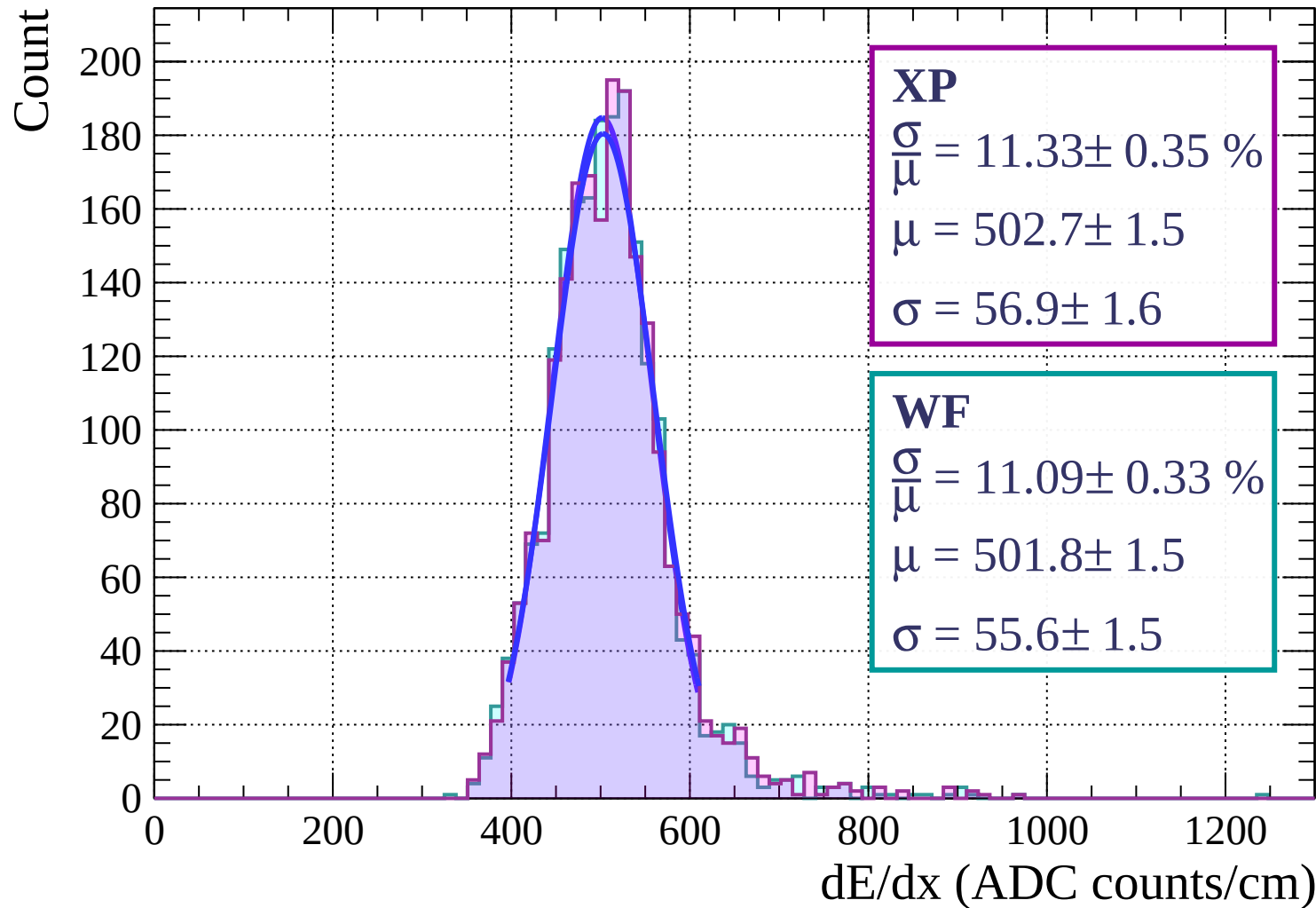
$$\sigma = 57.9 \pm 1.7$$

dE/dx (ADC counts/cm)

Energy loss | $72 < \phi < 75$

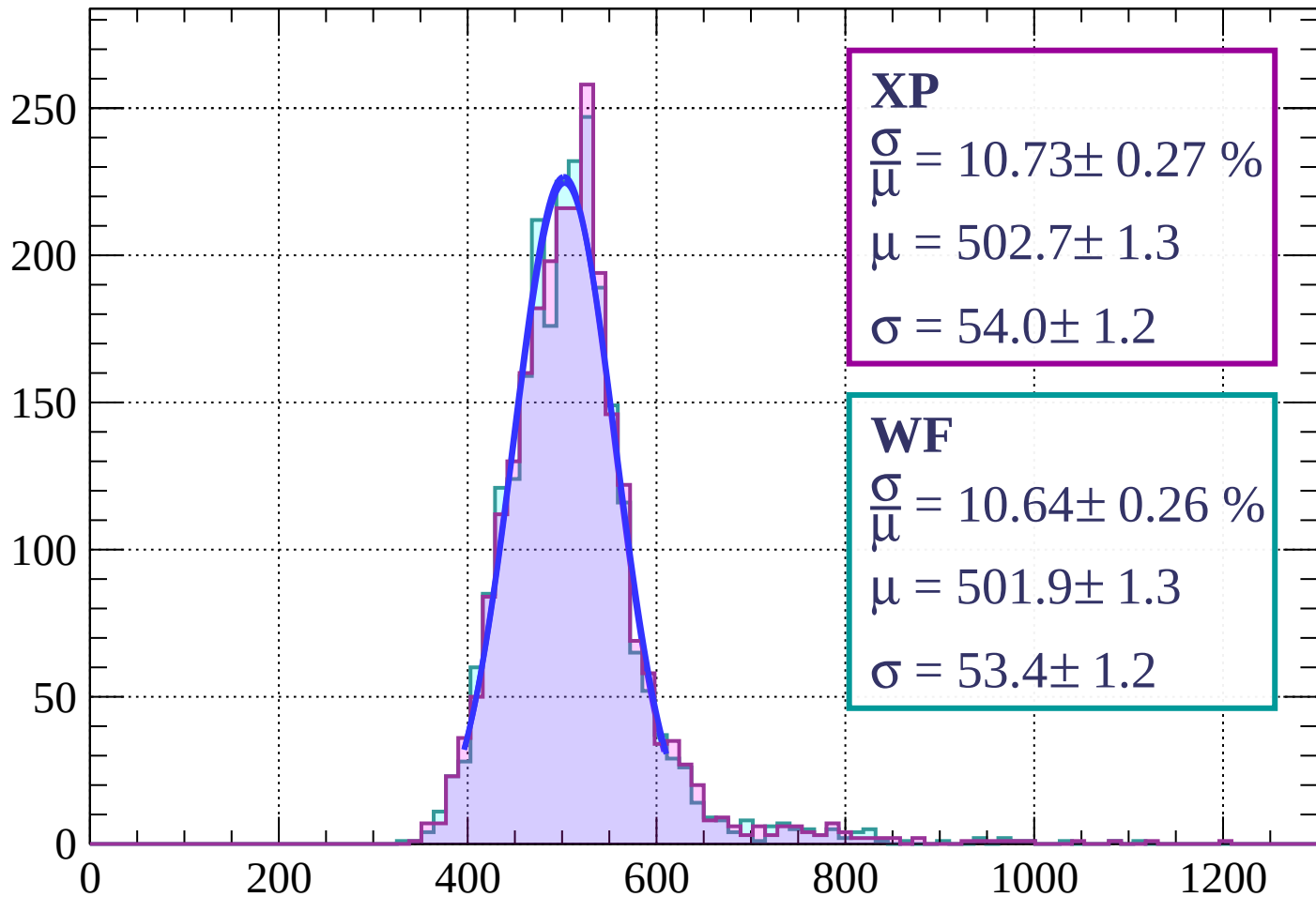


Energy loss | $75 < \phi < 78$



Energy loss | $78 < \phi < 81$

Count



XP

$$\frac{\sigma}{\mu} = 10.73 \pm 0.27 \%$$

$$\mu = 502.7 \pm 1.3$$

$$\sigma = 54.0 \pm 1.2$$

WF

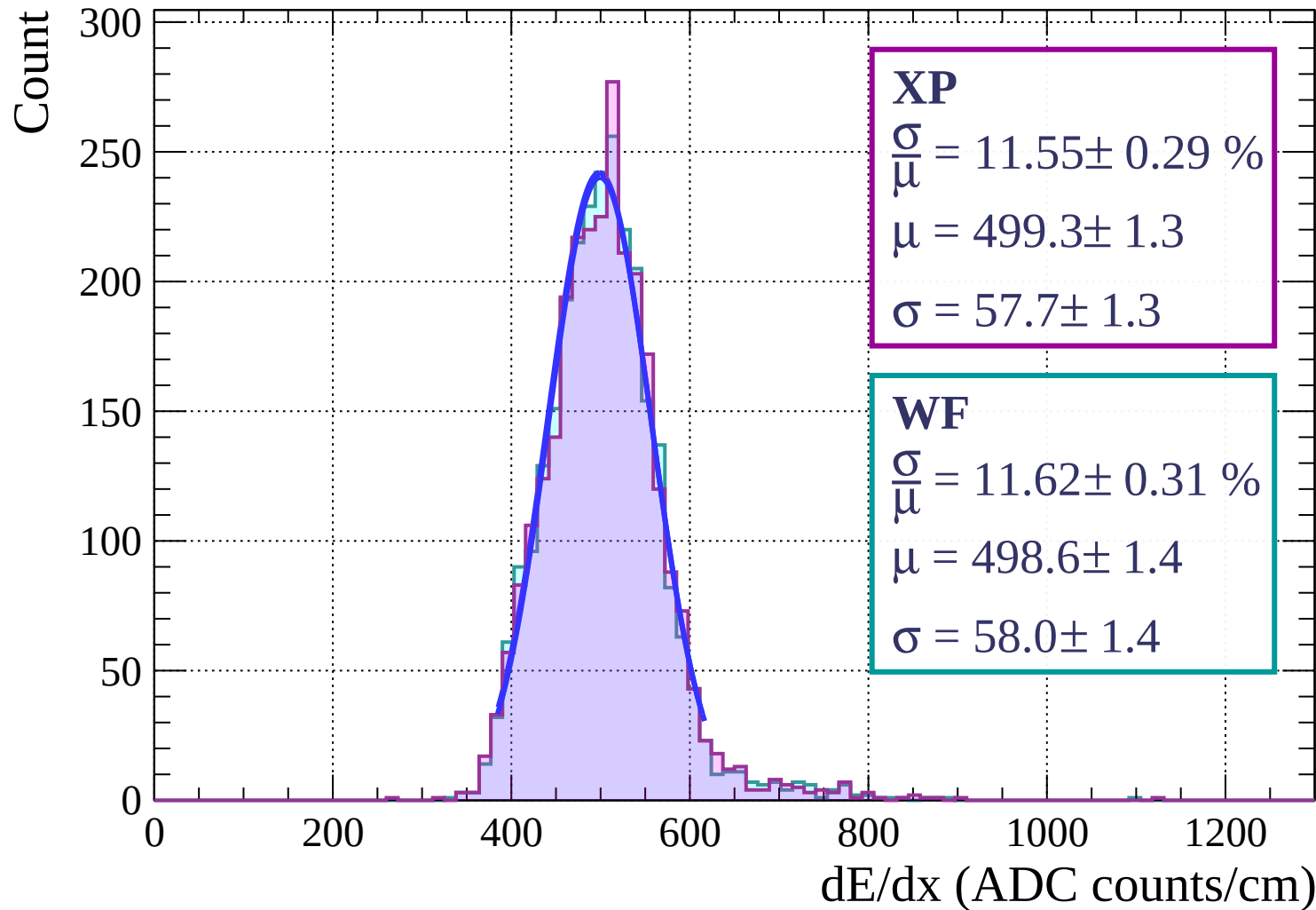
$$\frac{\sigma}{\mu} = 10.64 \pm 0.26 \%$$

$$\mu = 501.9 \pm 1.3$$

$$\sigma = 53.4 \pm 1.2$$

dE/dx (ADC counts/cm)

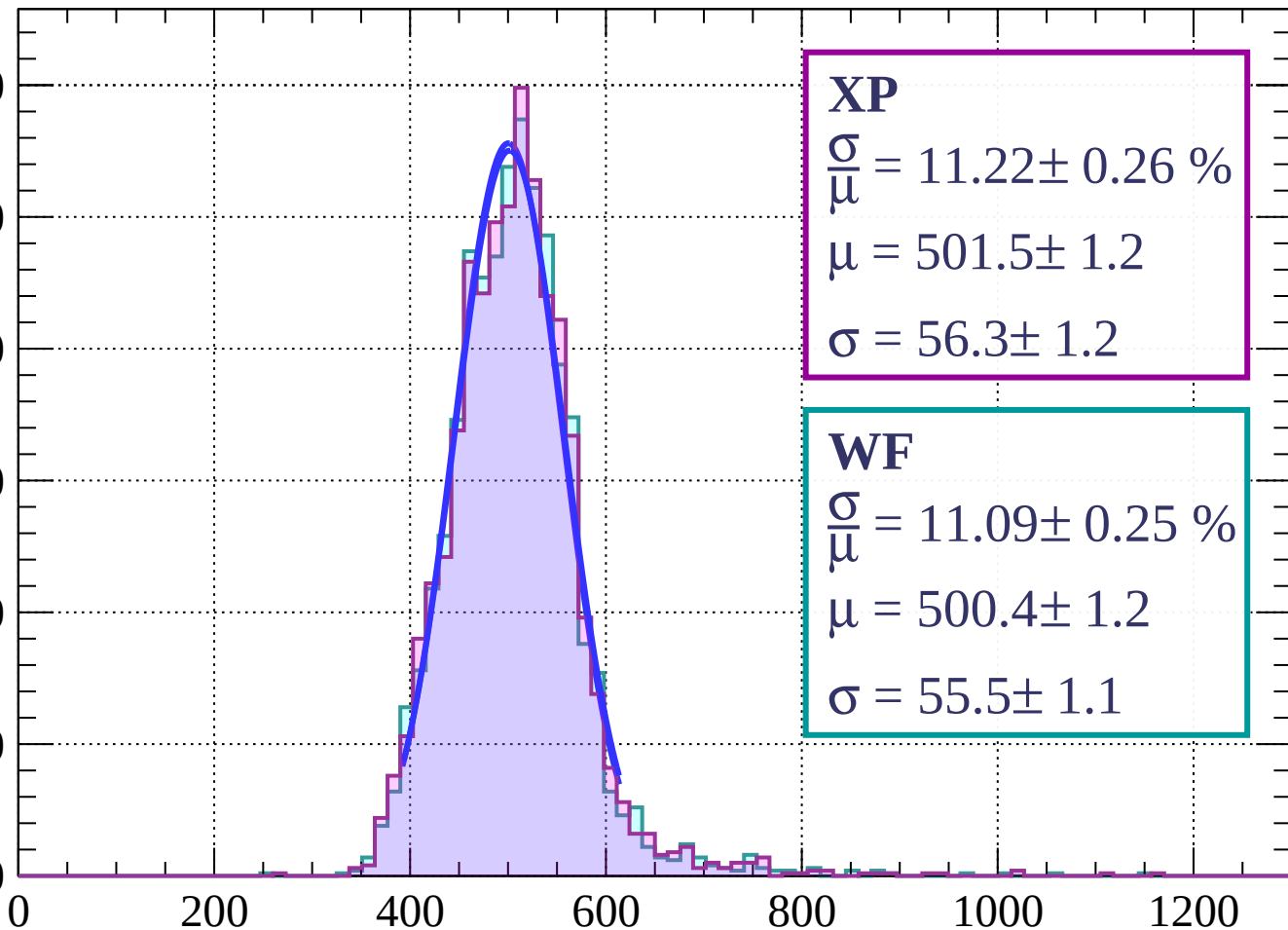
Energy loss | $81 < \phi < 84$



Energy loss | $84 < \phi < 87$

Count

300
250
200
150
100
50
0



XP

$$\frac{\sigma}{\mu} = 11.22 \pm 0.26 \%$$

$$\mu = 501.5 \pm 1.2$$

$$\sigma = 56.3 \pm 1.2$$

WF

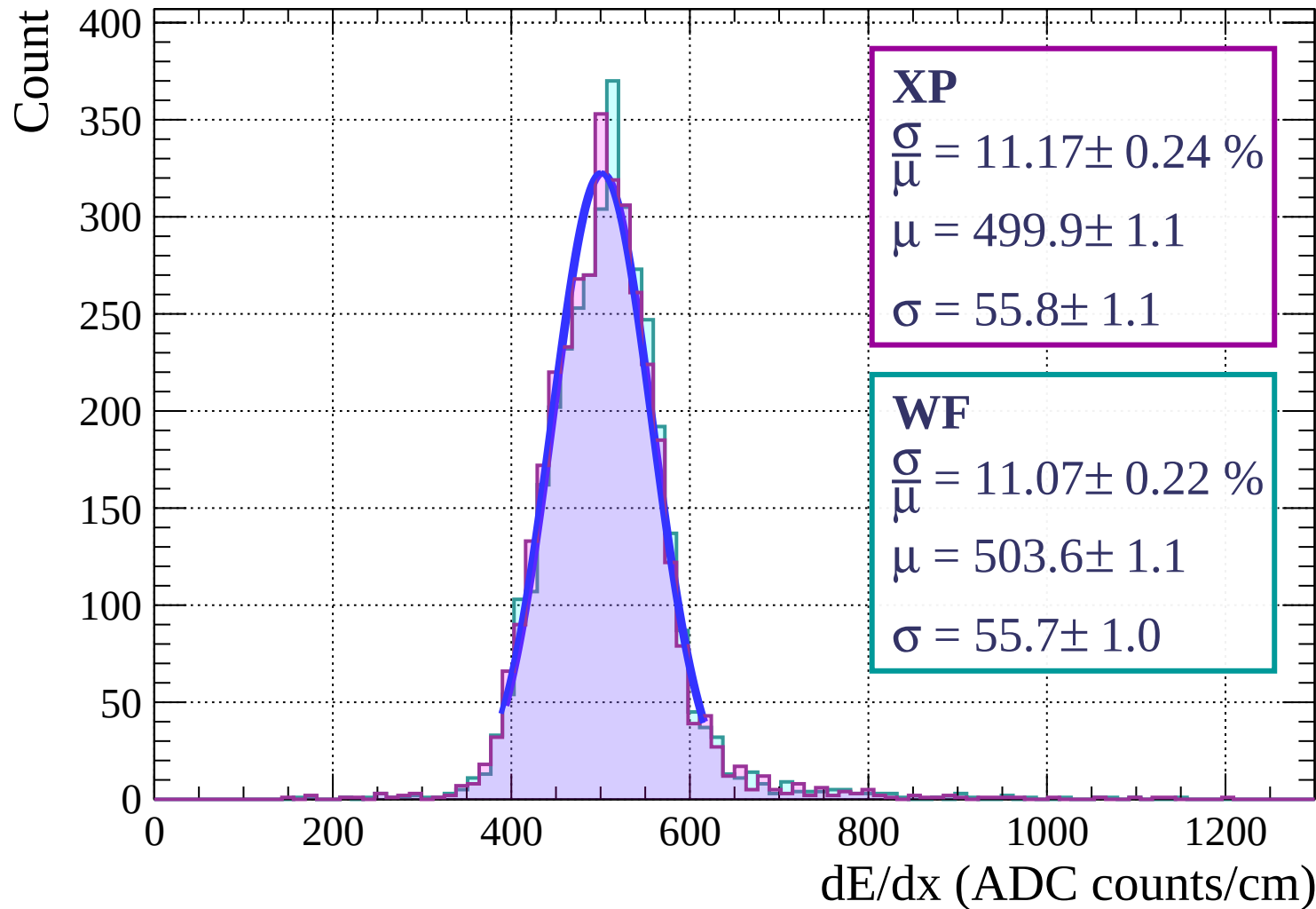
$$\frac{\sigma}{\mu} = 11.09 \pm 0.25 \%$$

$$\mu = 500.4 \pm 1.2$$

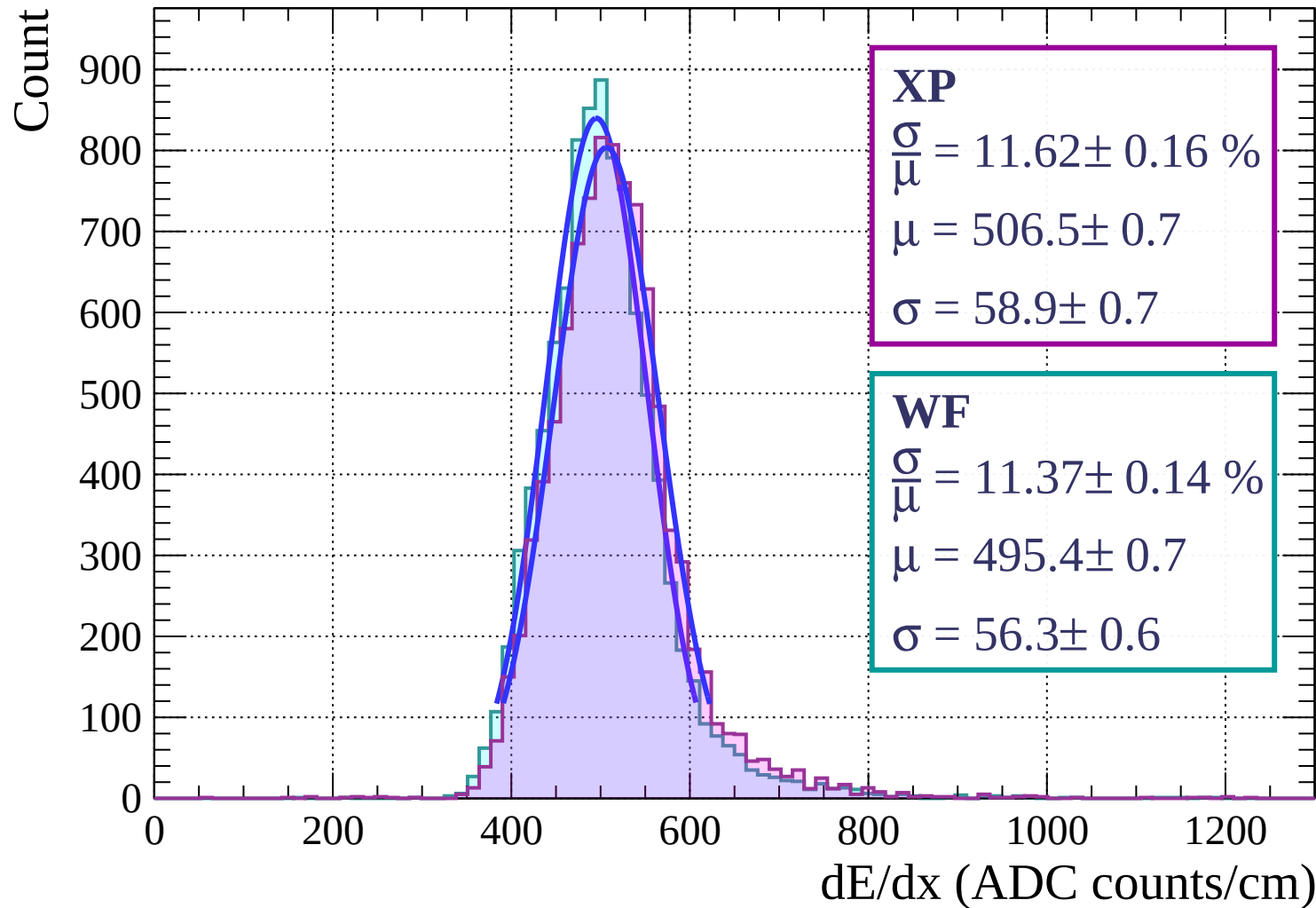
$$\sigma = 55.5 \pm 1.1$$

dE/dx (ADC counts/cm)

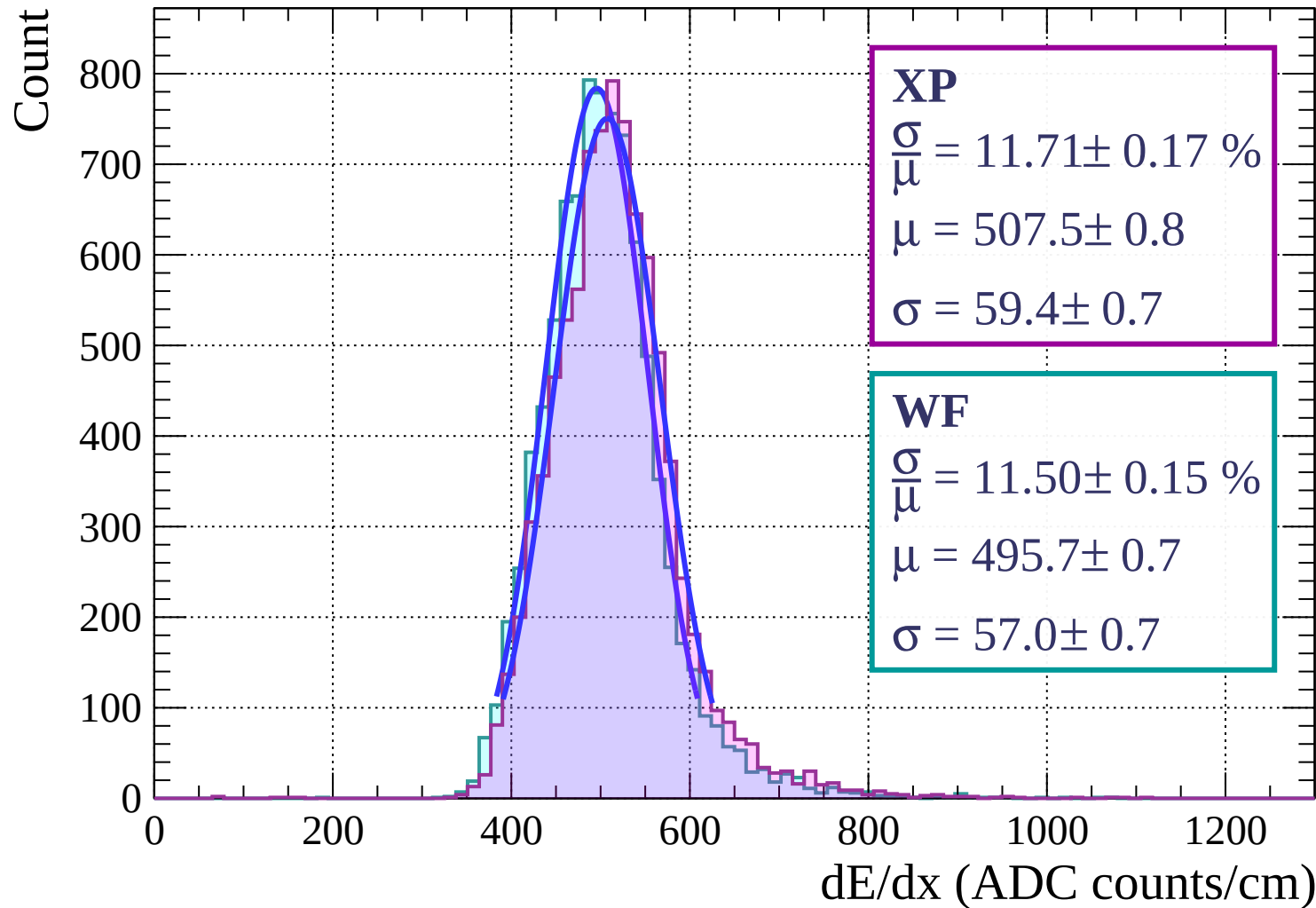
Energy loss | $87 < \phi < 90$



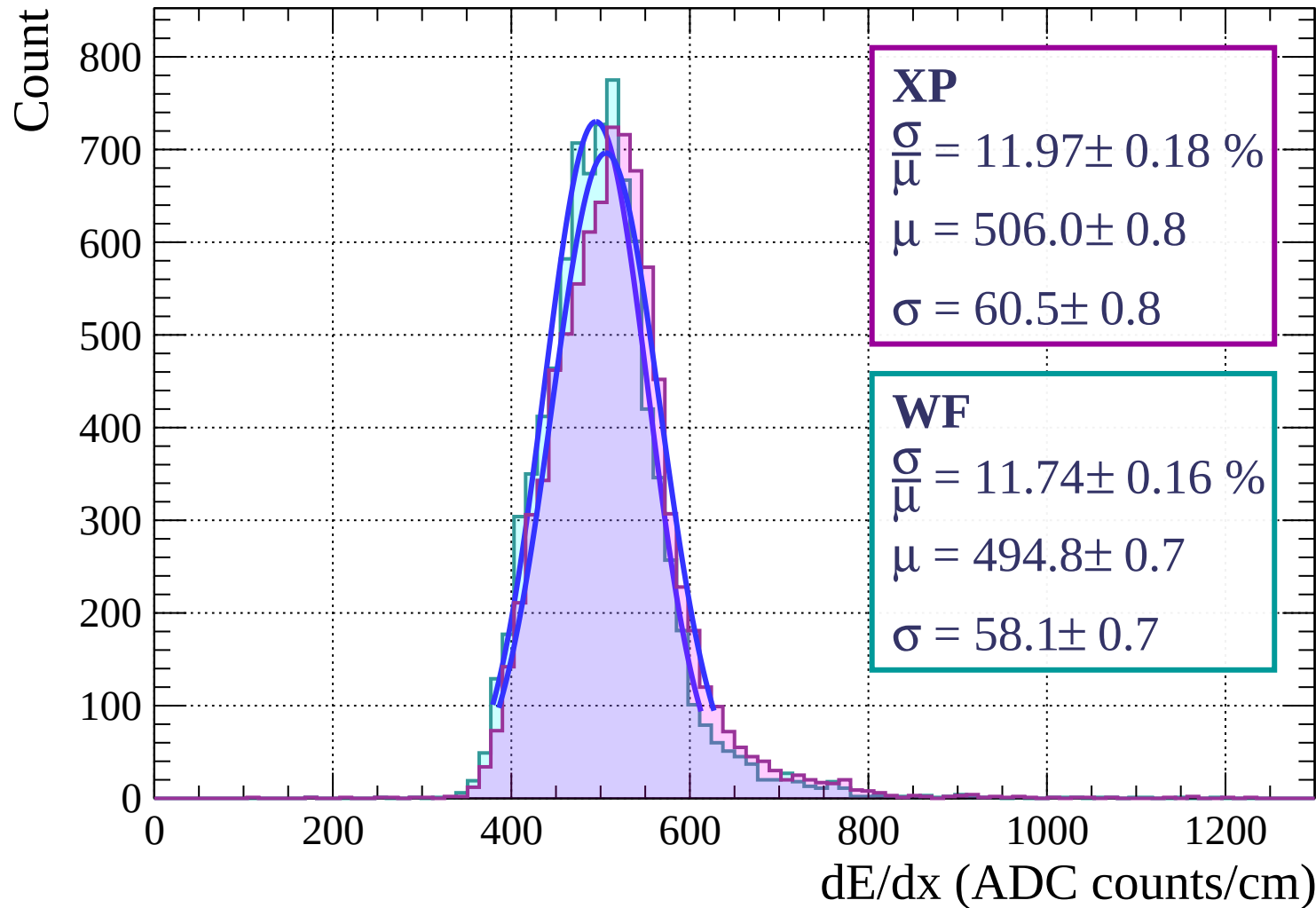
Energy loss | $0 < \theta < 3$



Energy loss | $3 < \theta < 6$

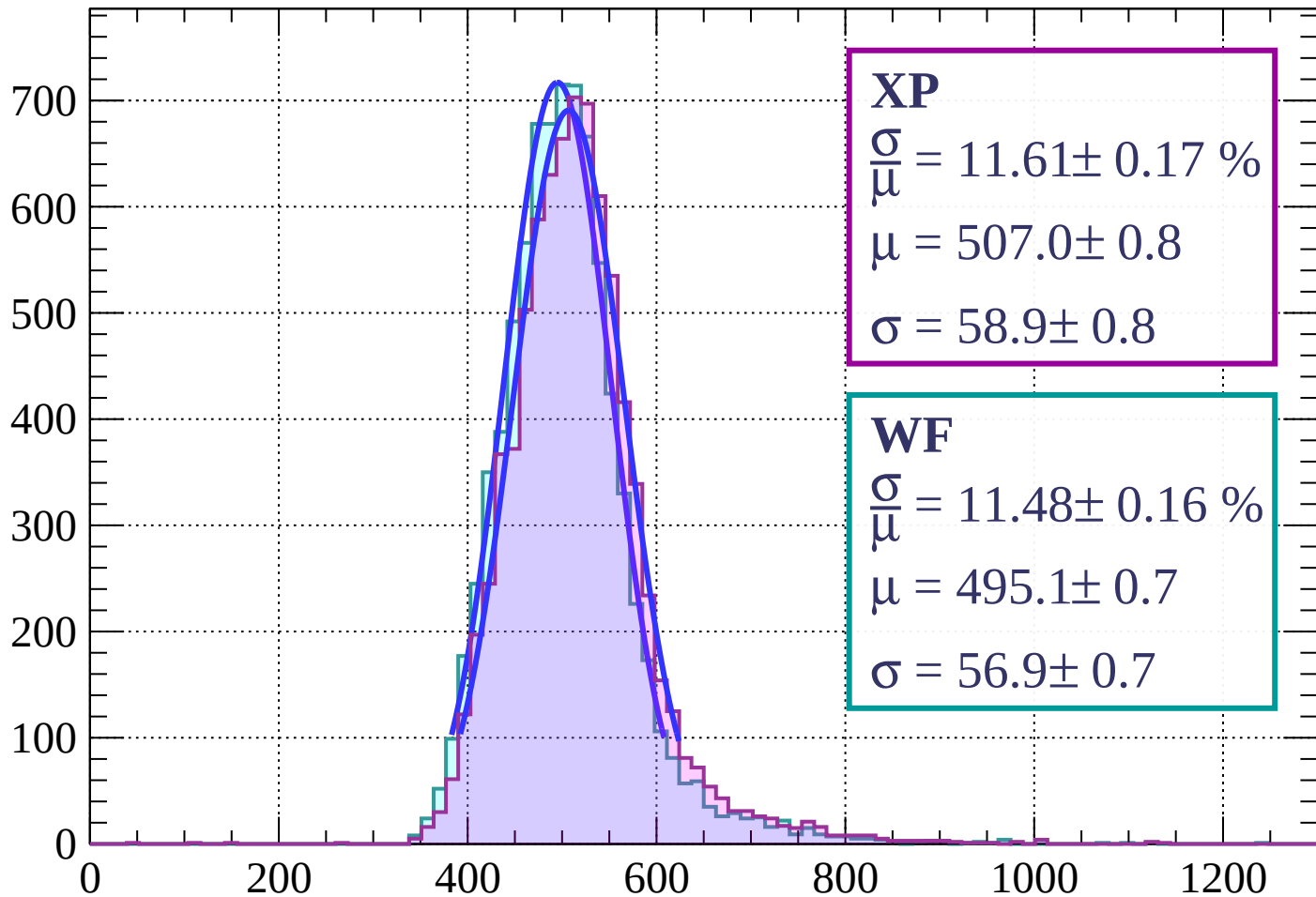


Energy loss | $6 < \theta < 9$



Energy loss | $9 < \theta < 12$

Count



XP

$$\frac{\sigma}{\mu} = 11.61 \pm 0.17 \%$$

$$\mu = 507.0 \pm 0.8$$

$$\sigma = 58.9 \pm 0.8$$

WF

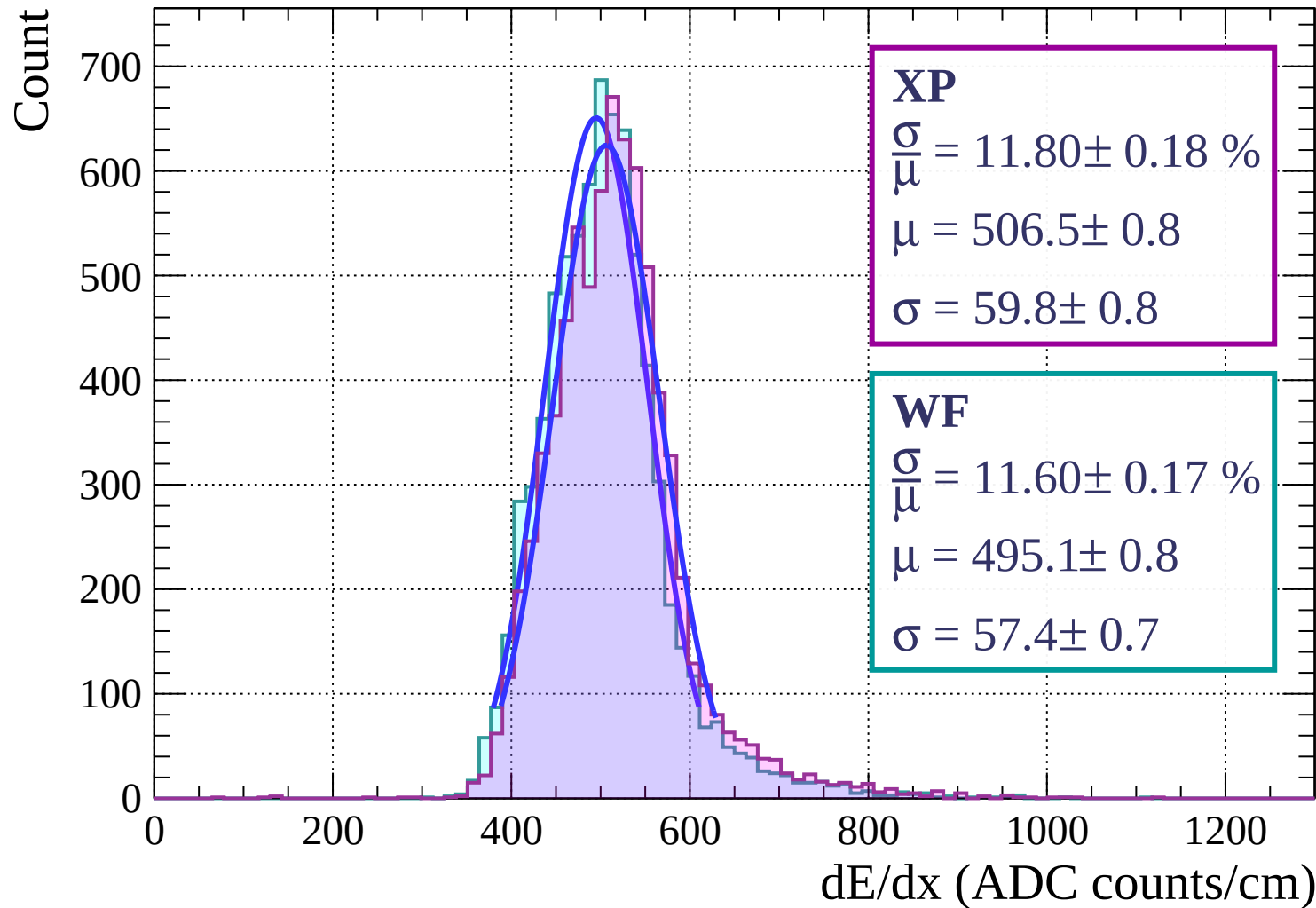
$$\frac{\sigma}{\mu} = 11.48 \pm 0.16 \%$$

$$\mu = 495.1 \pm 0.7$$

$$\sigma = 56.9 \pm 0.7$$

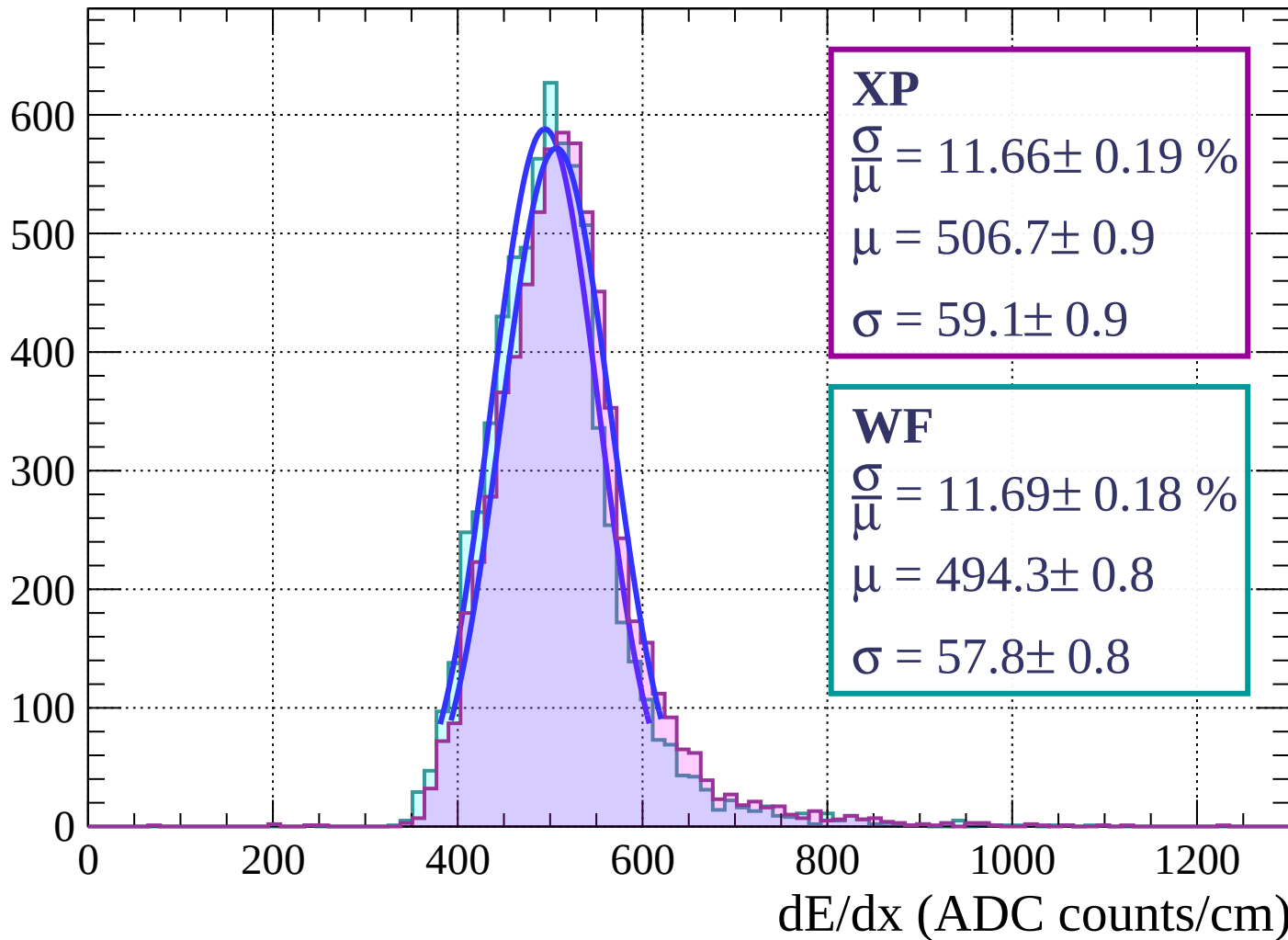
dE/dx (ADC counts/cm)

Energy loss | $12 < \theta < 15$



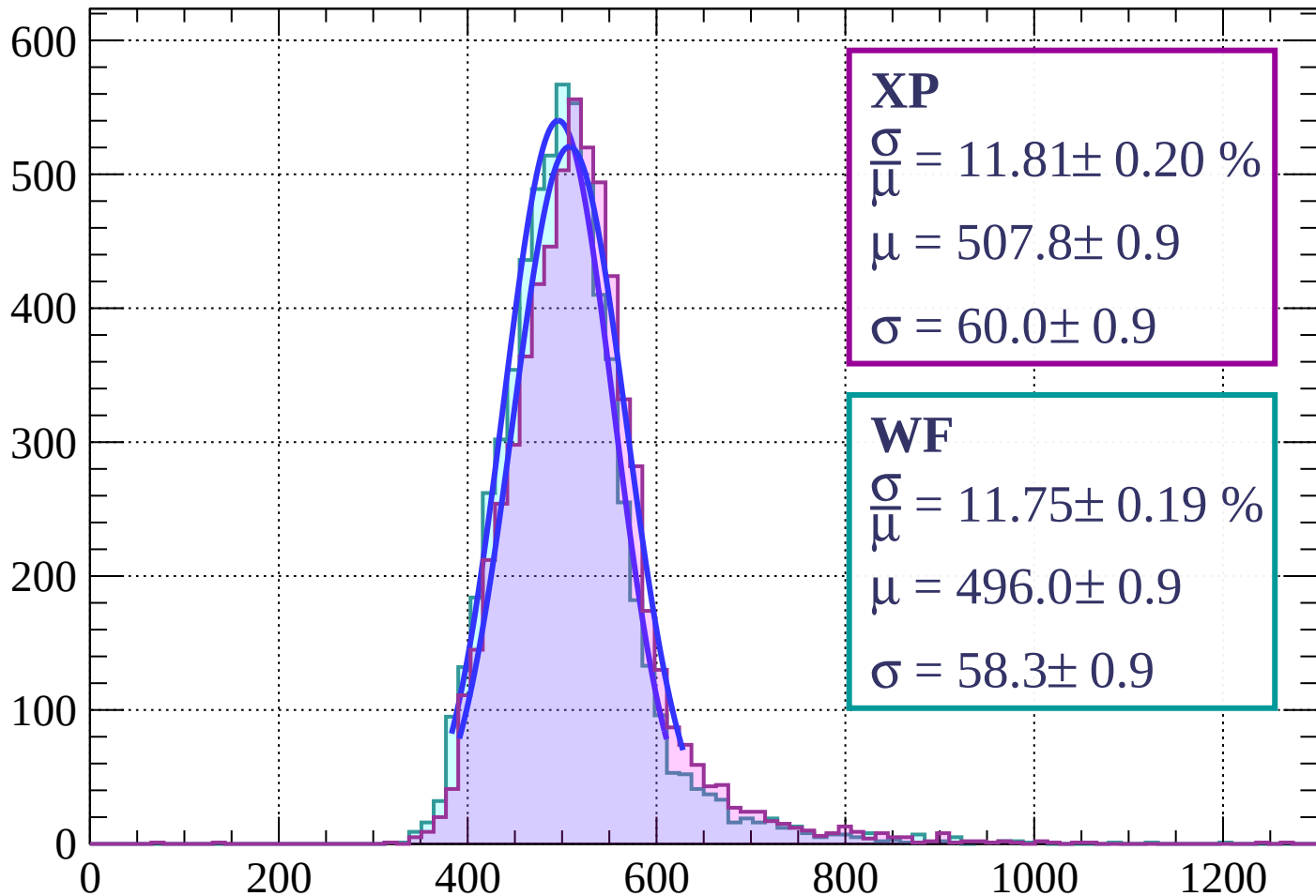
Energy loss | $15 < \theta < 18$

Count



Energy loss | $18 < \theta < 21$

Count



XP

$$\frac{\sigma}{\mu} = 11.81 \pm 0.20 \%$$

$$\mu = 507.8 \pm 0.9$$

$$\sigma = 60.0 \pm 0.9$$

WF

$$\frac{\sigma}{\mu} = 11.75 \pm 0.19 \%$$

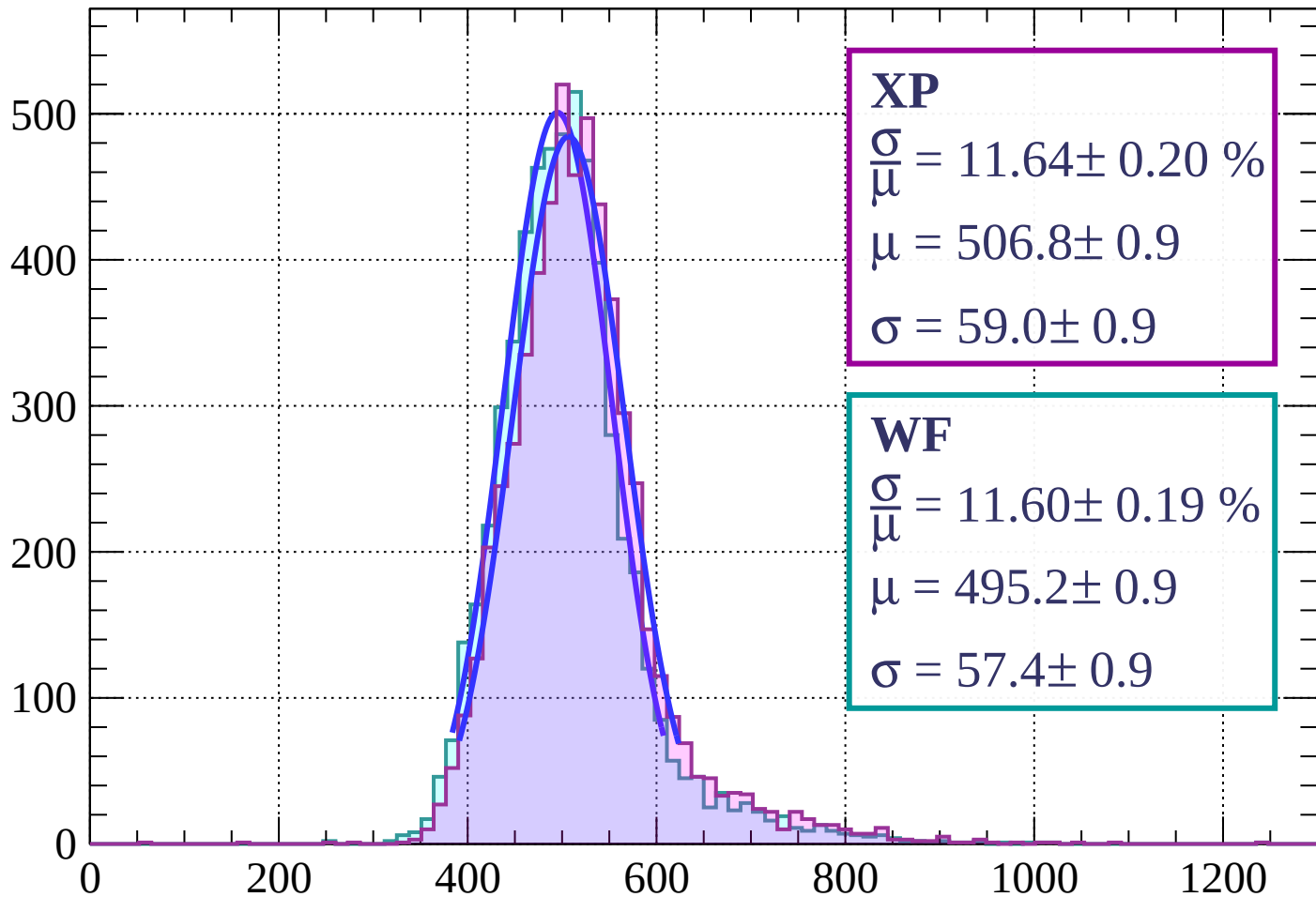
$$\mu = 496.0 \pm 0.9$$

$$\sigma = 58.3 \pm 0.9$$

dE/dx (ADC counts/cm)

Energy loss | $21 < \theta < 24$

Count



XP

$$\frac{\sigma}{\mu} = 11.64 \pm 0.20 \%$$

$$\mu = 506.8 \pm 0.9$$

$$\sigma = 59.0 \pm 0.9$$

WF

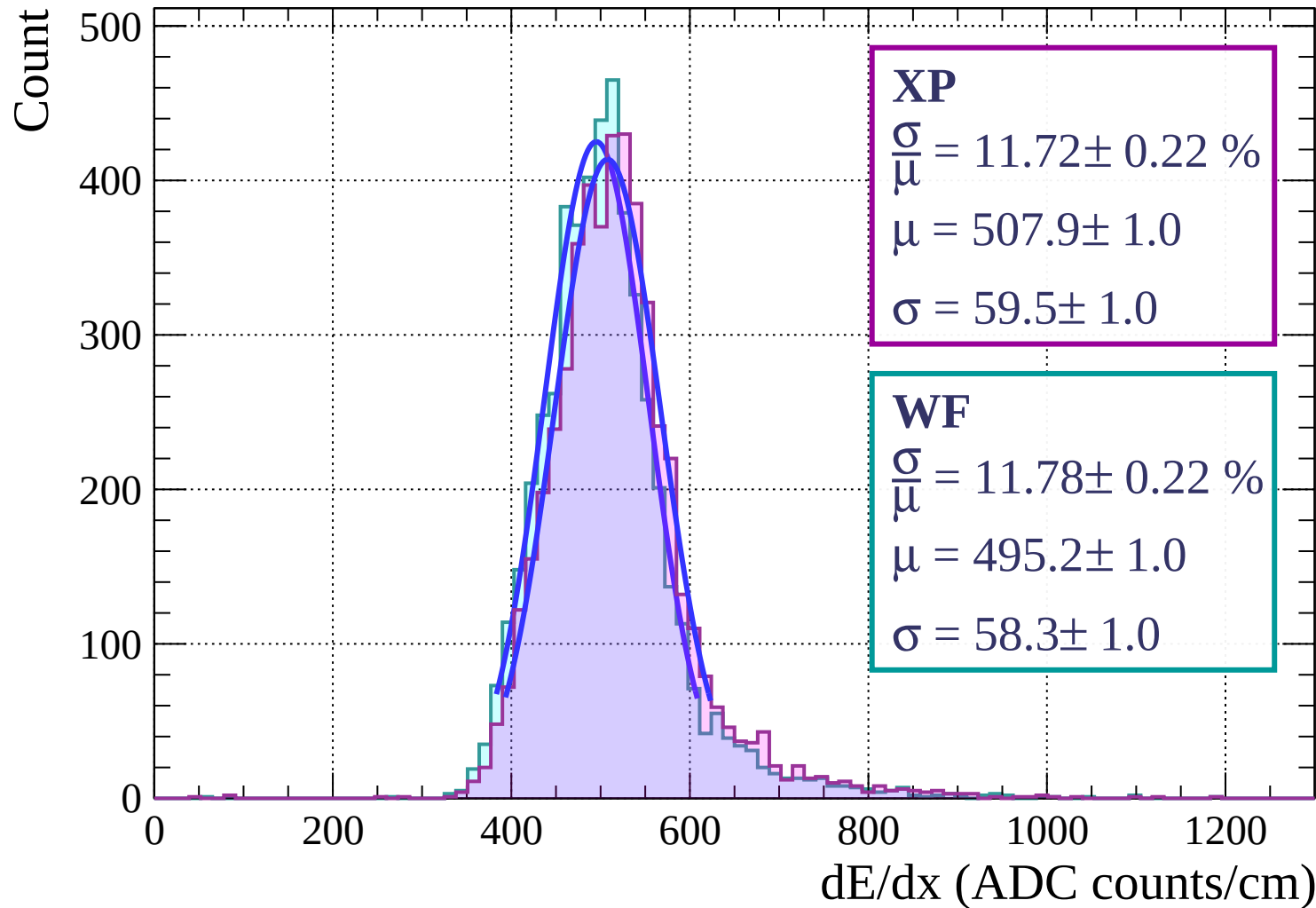
$$\frac{\sigma}{\mu} = 11.60 \pm 0.19 \%$$

$$\mu = 495.2 \pm 0.9$$

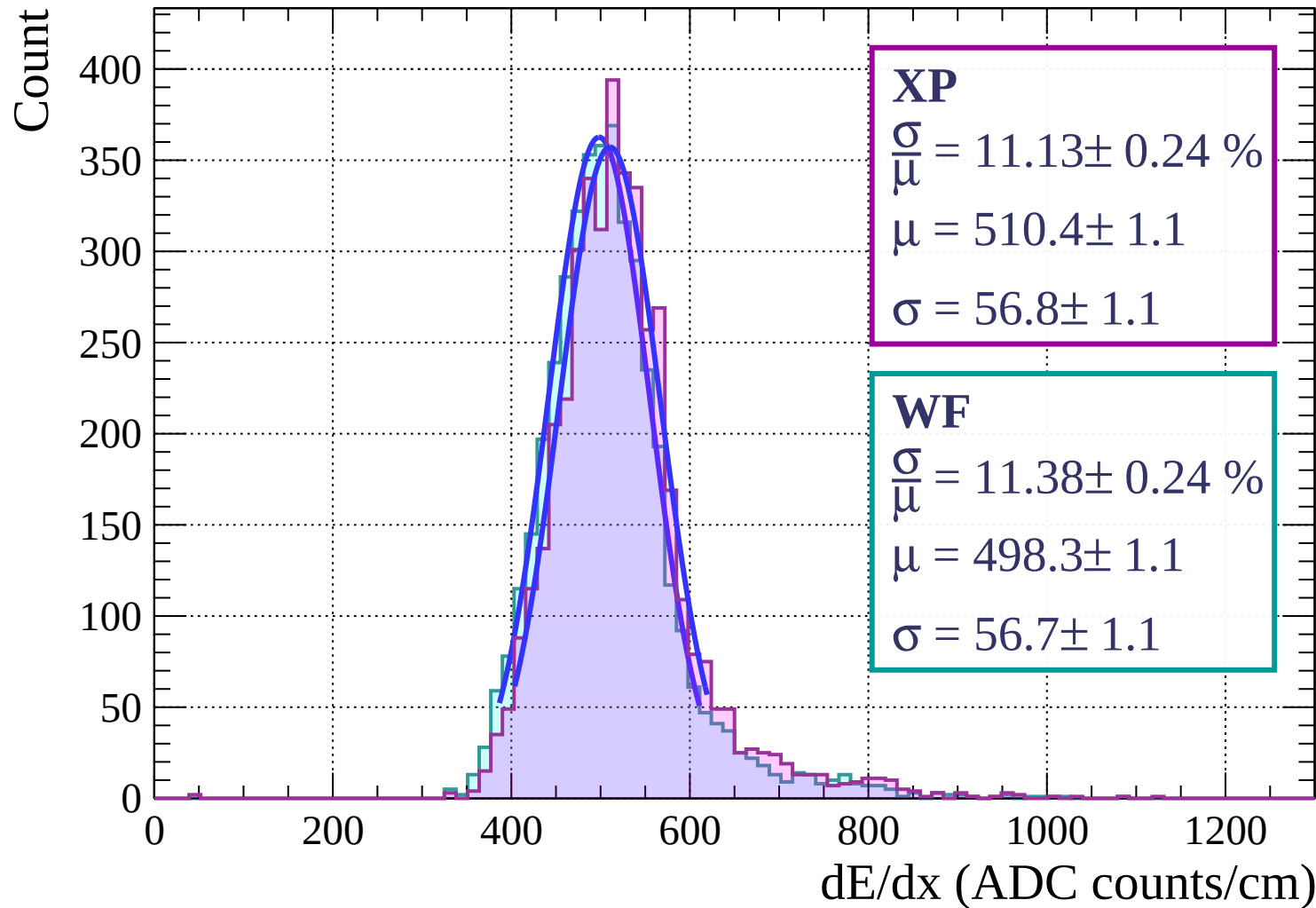
$$\sigma = 57.4 \pm 0.9$$

dE/dx (ADC counts/cm)

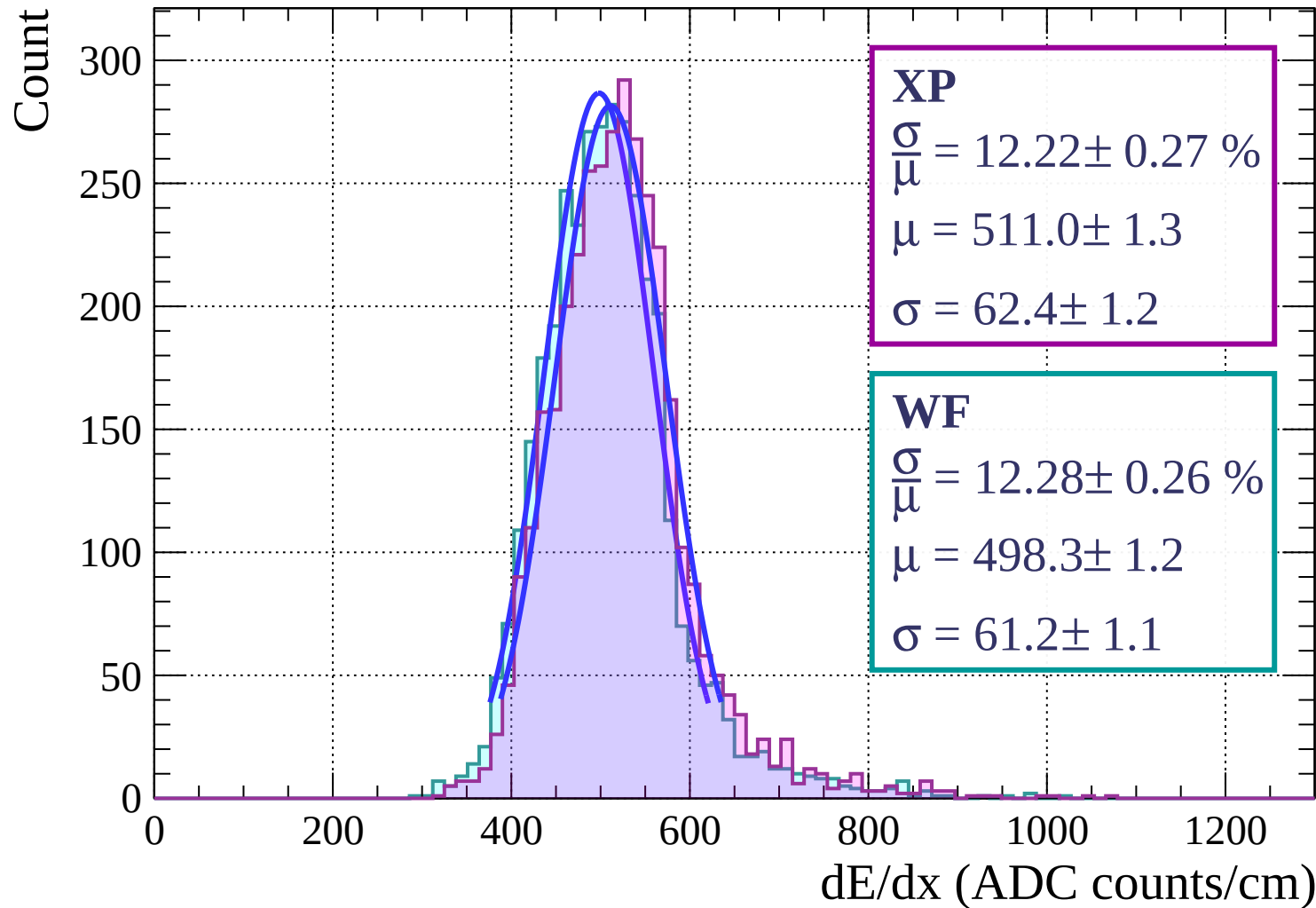
Energy loss | $24 < \theta < 27$



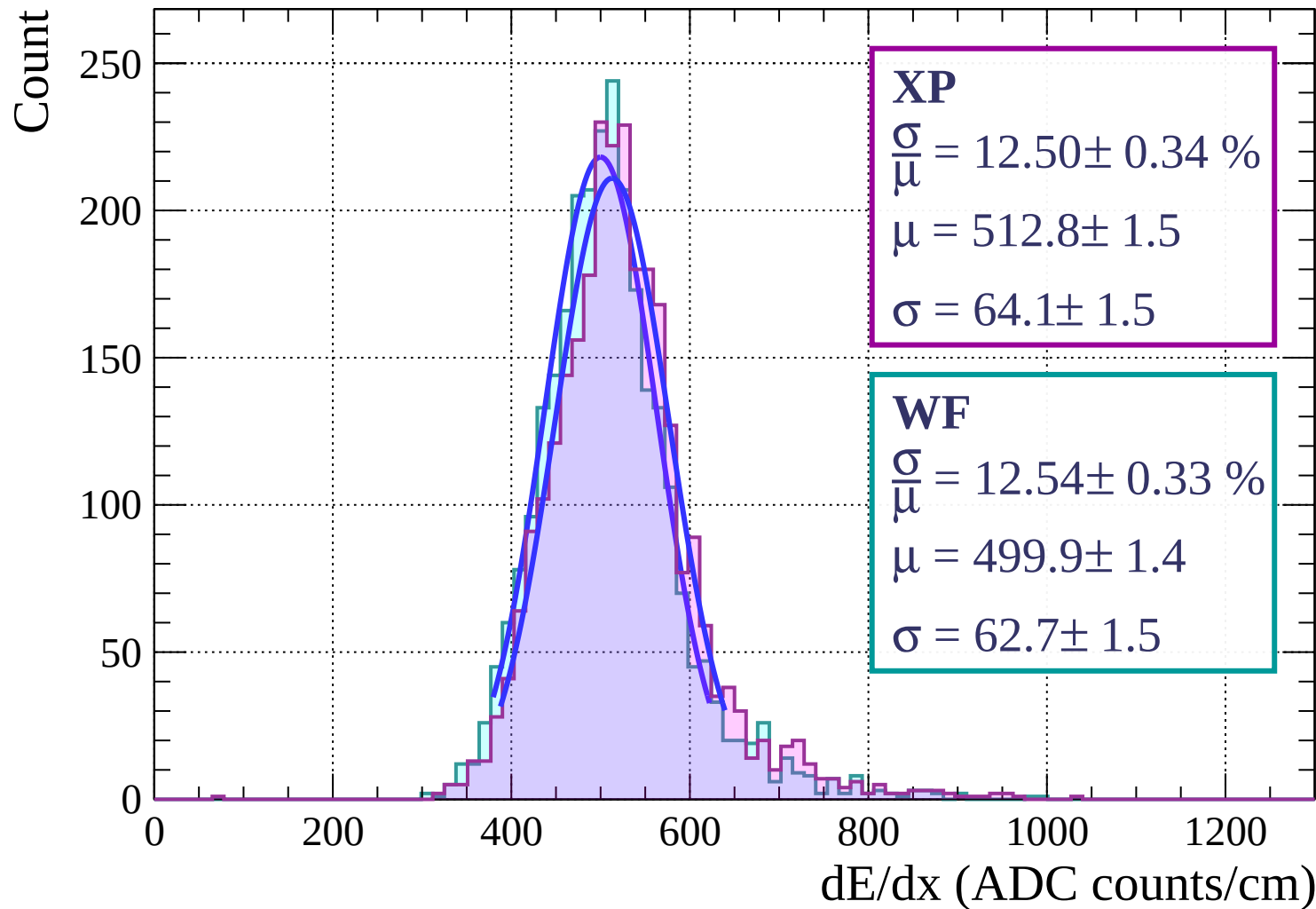
Energy loss | $27 < \theta < 30$



Energy loss | $30 < \theta < 33$

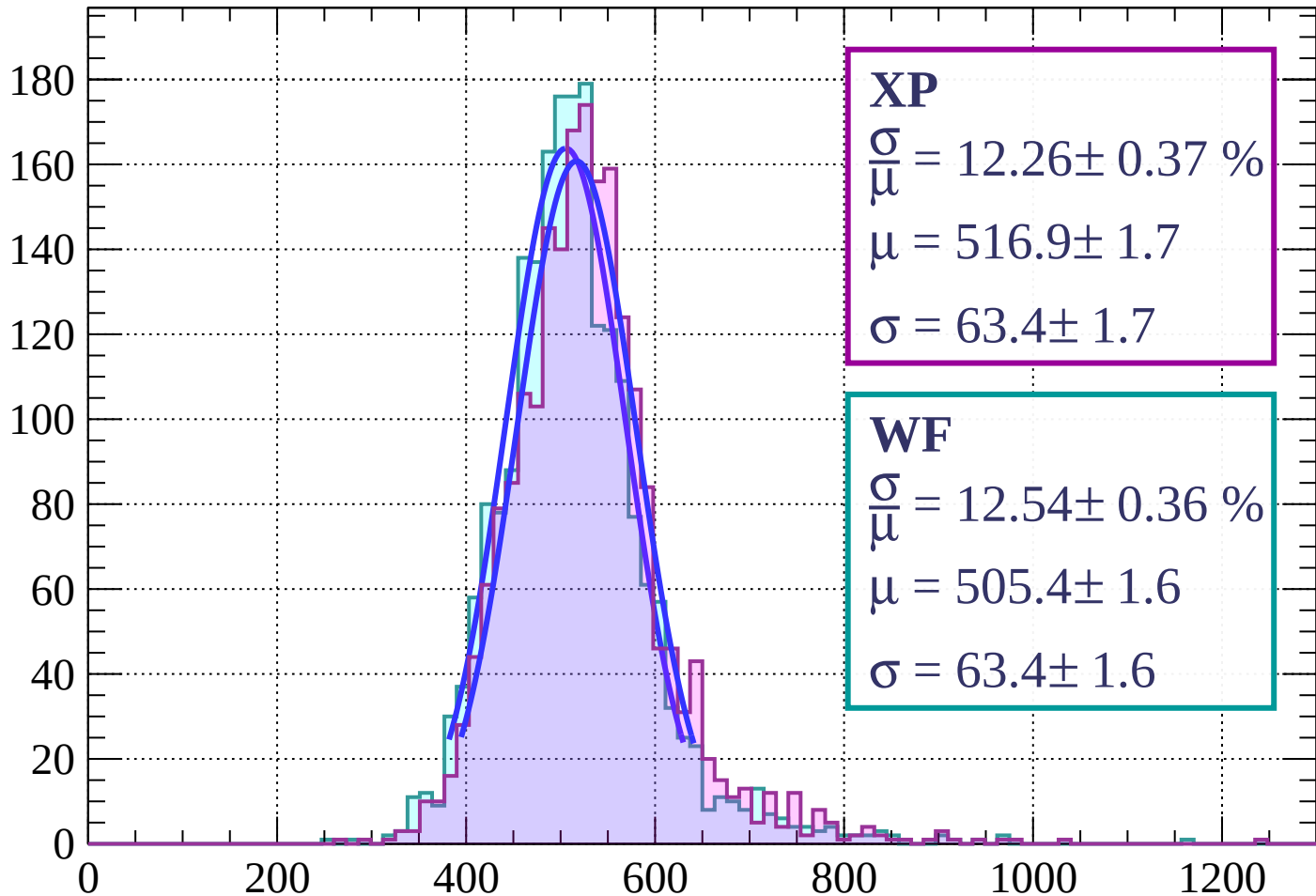


Energy loss | $33 < \theta < 36$



Energy loss | $36 < \theta < 39$

Count



XP

$$\frac{\sigma}{\mu} = 12.26 \pm 0.37 \%$$

$$\mu = 516.9 \pm 1.7$$

$$\sigma = 63.4 \pm 1.7$$

WF

$$\frac{\sigma}{\mu} = 12.54 \pm 0.36 \%$$

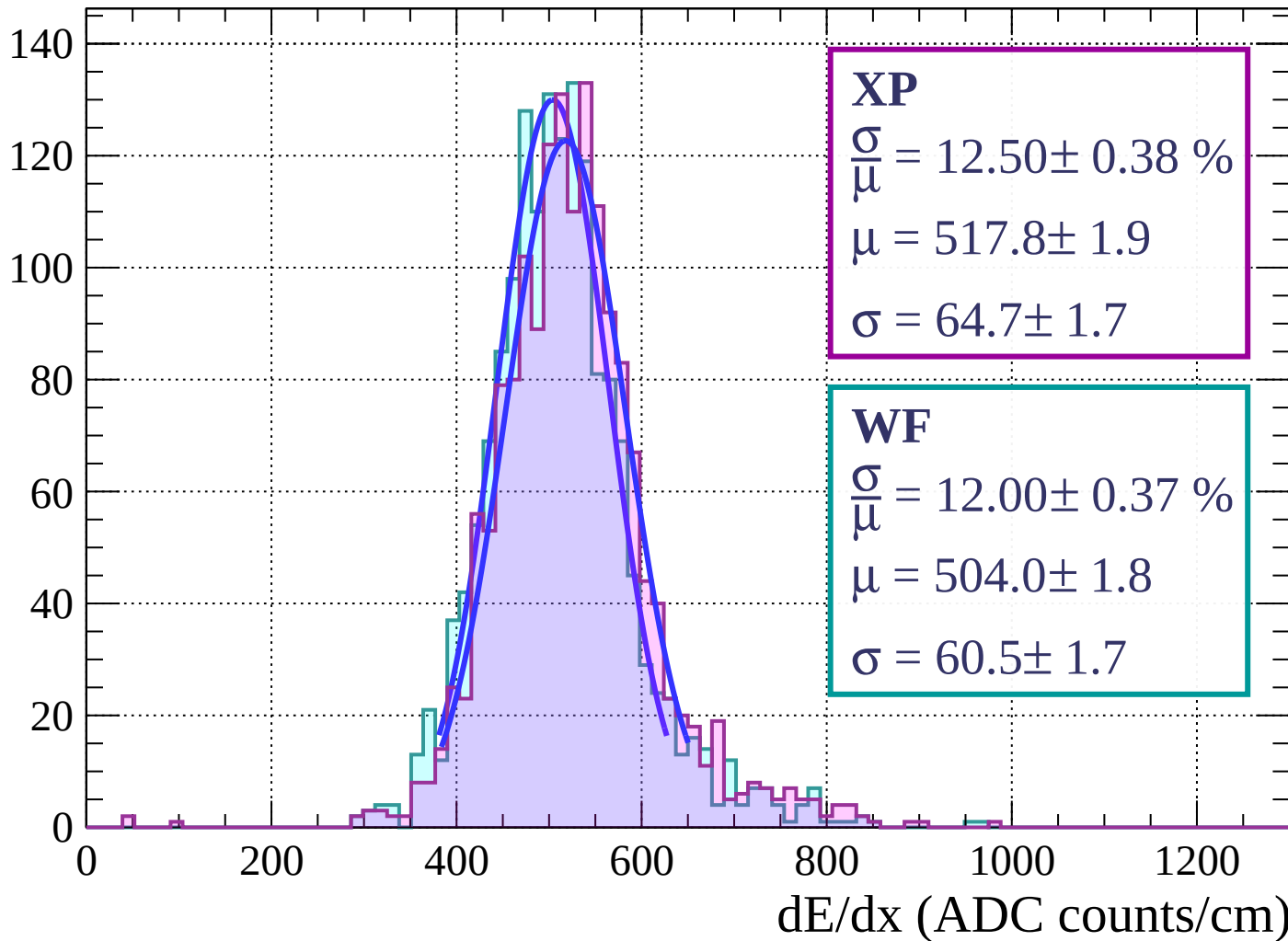
$$\mu = 505.4 \pm 1.6$$

$$\sigma = 63.4 \pm 1.6$$

dE/dx (ADC counts/cm)

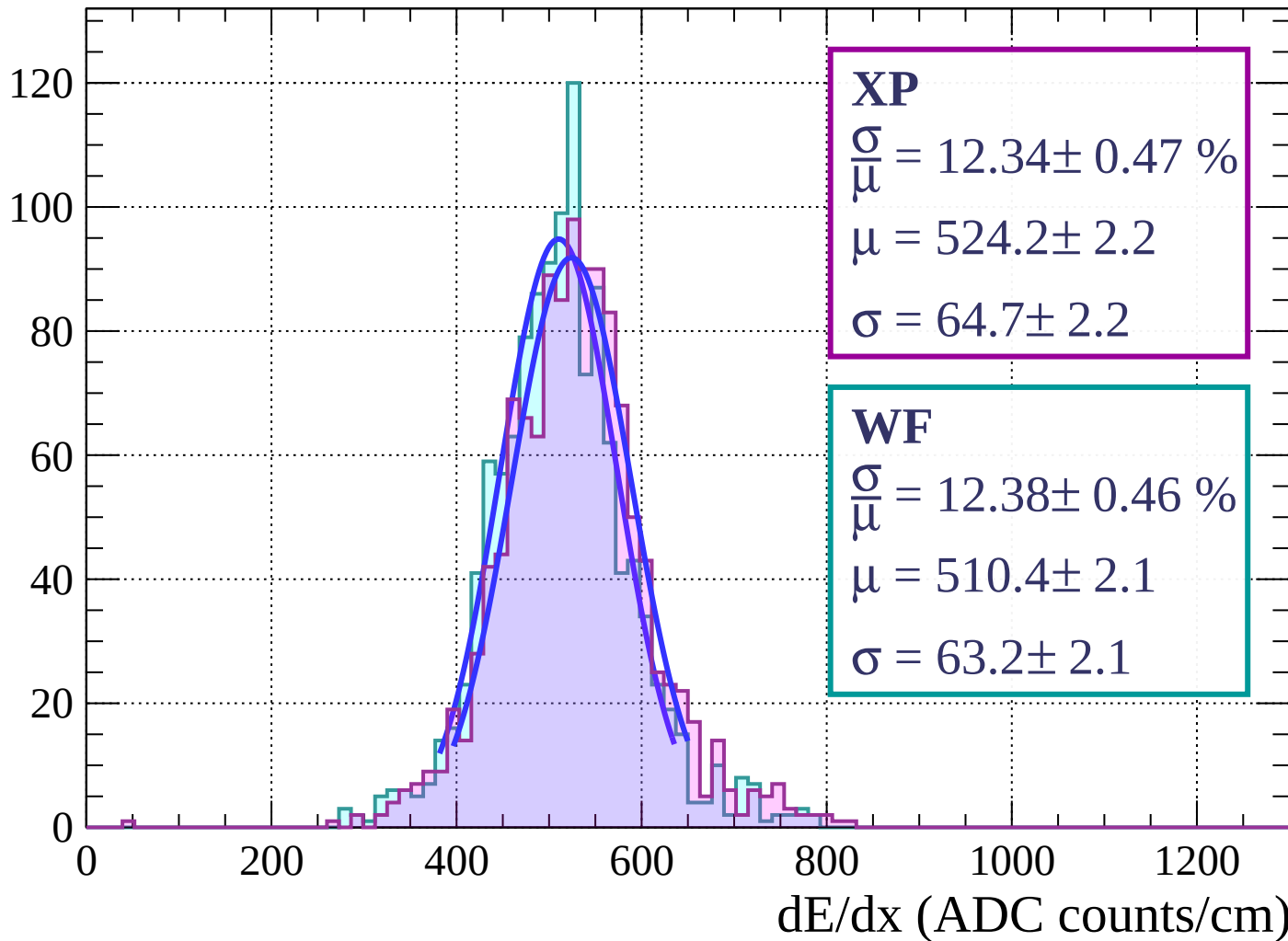
Energy loss | $39 < \theta < 42$

Count



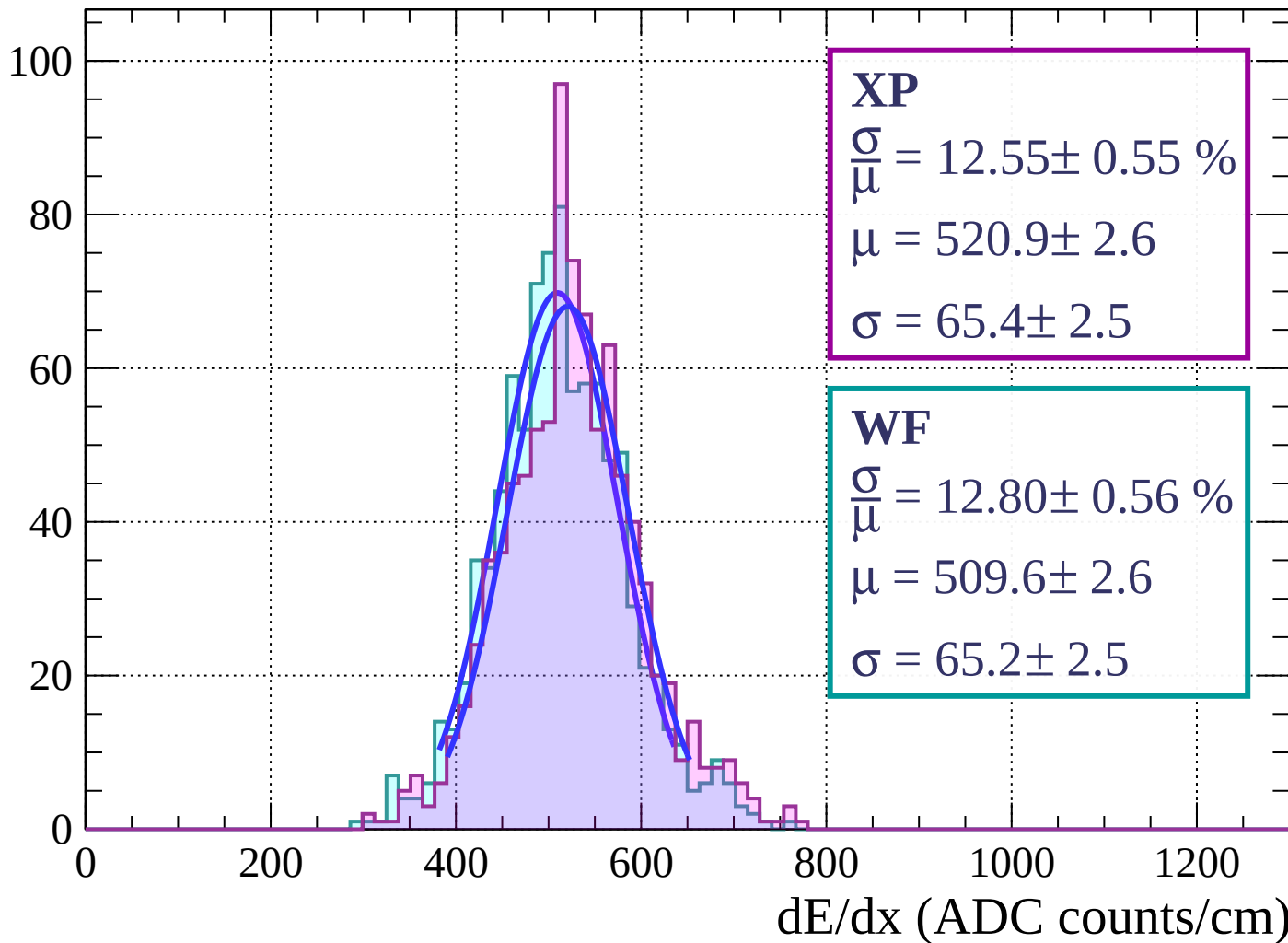
Energy loss | $42 < \theta < 45$

Count



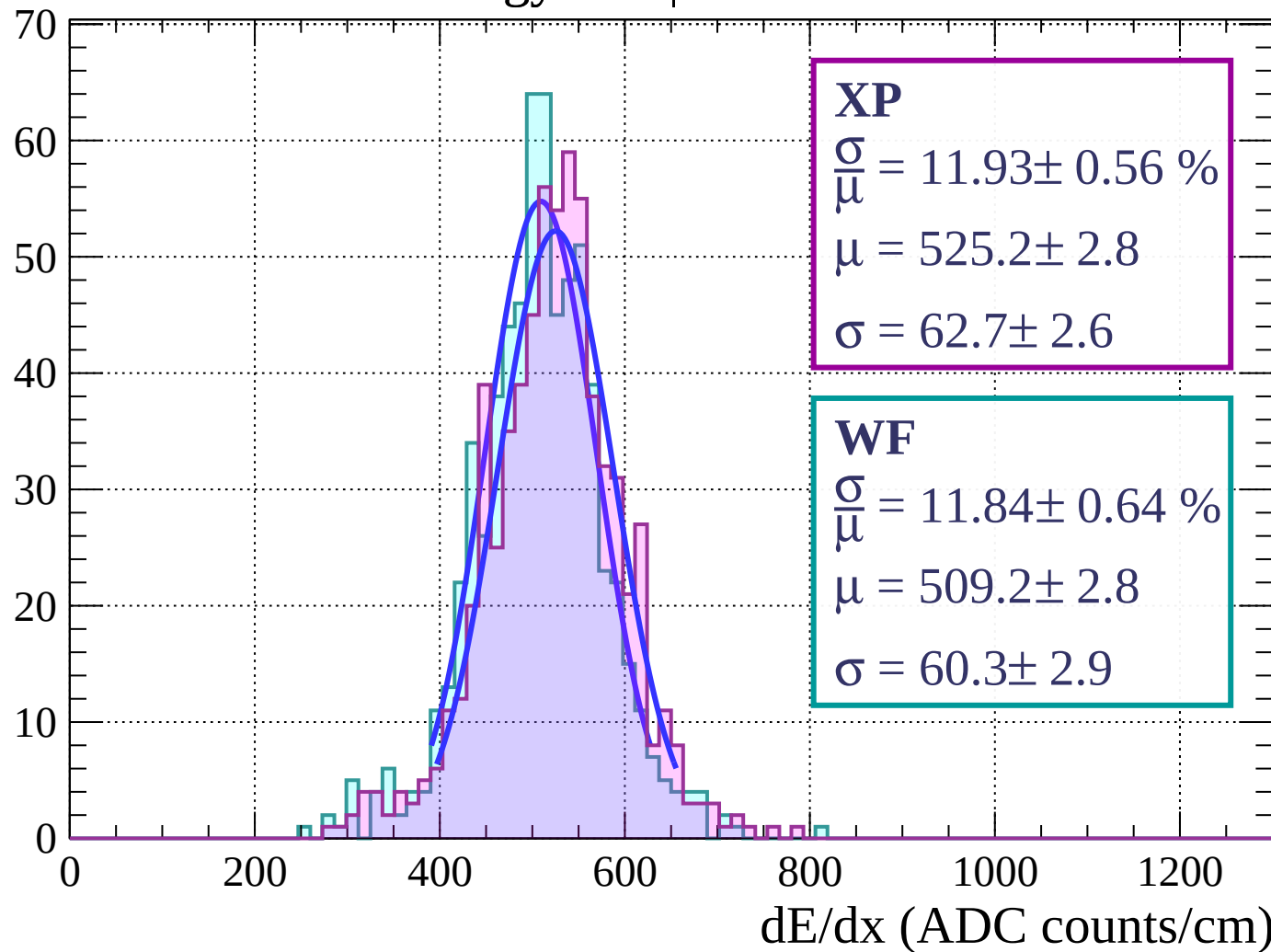
Energy loss | $45 < \theta < 48$

Count



Energy loss | $48 < \theta < 51$

Count



Energy loss | $51 < \theta < 54$

Count

XP

$$\frac{\sigma}{\mu} = 12.66 \pm 0.83 \%$$

$$\mu = 524.4 \pm 4.0$$

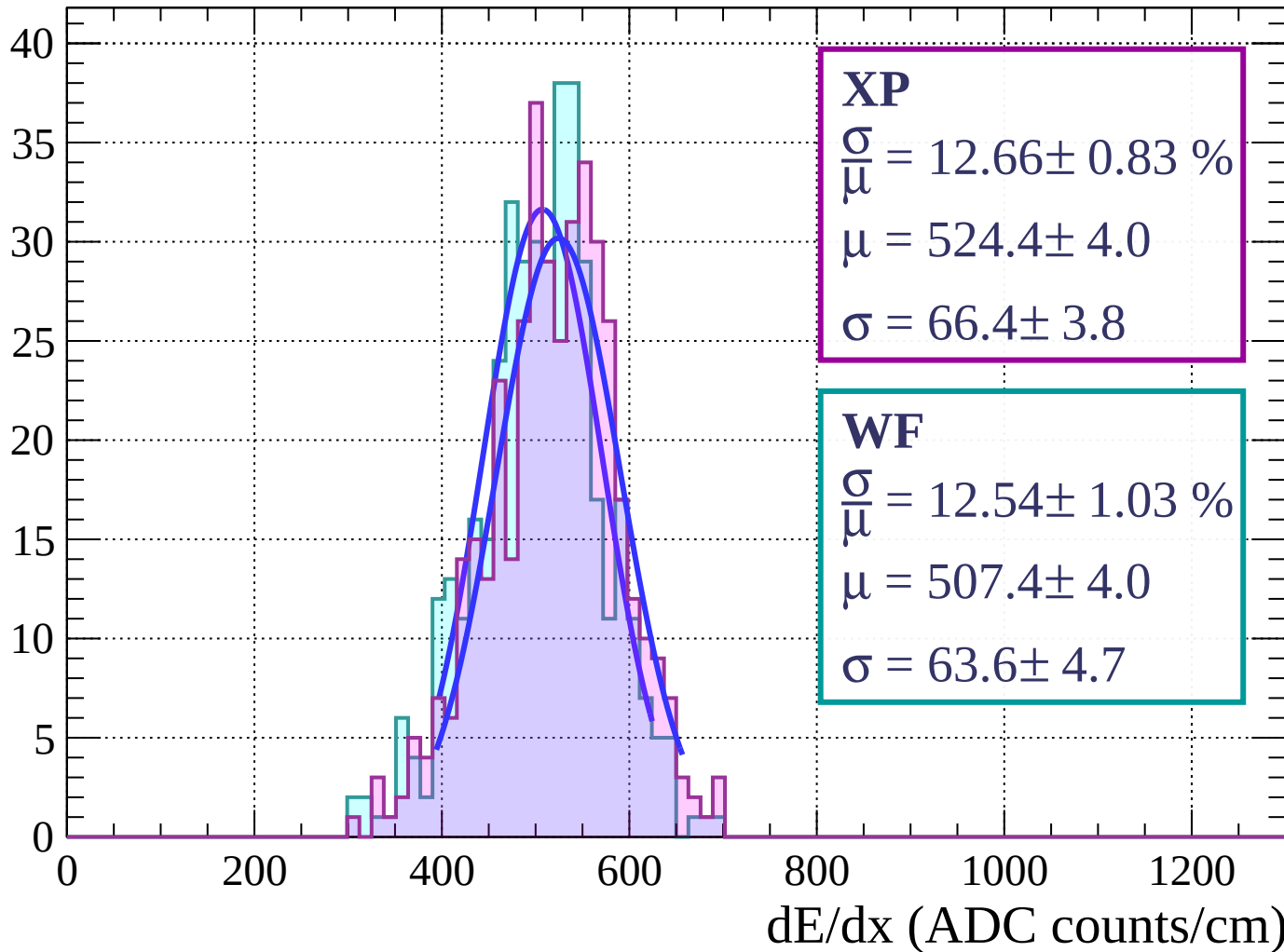
$$\sigma = 66.4 \pm 3.8$$

WF

$$\frac{\sigma}{\mu} = 12.54 \pm 1.03 \%$$

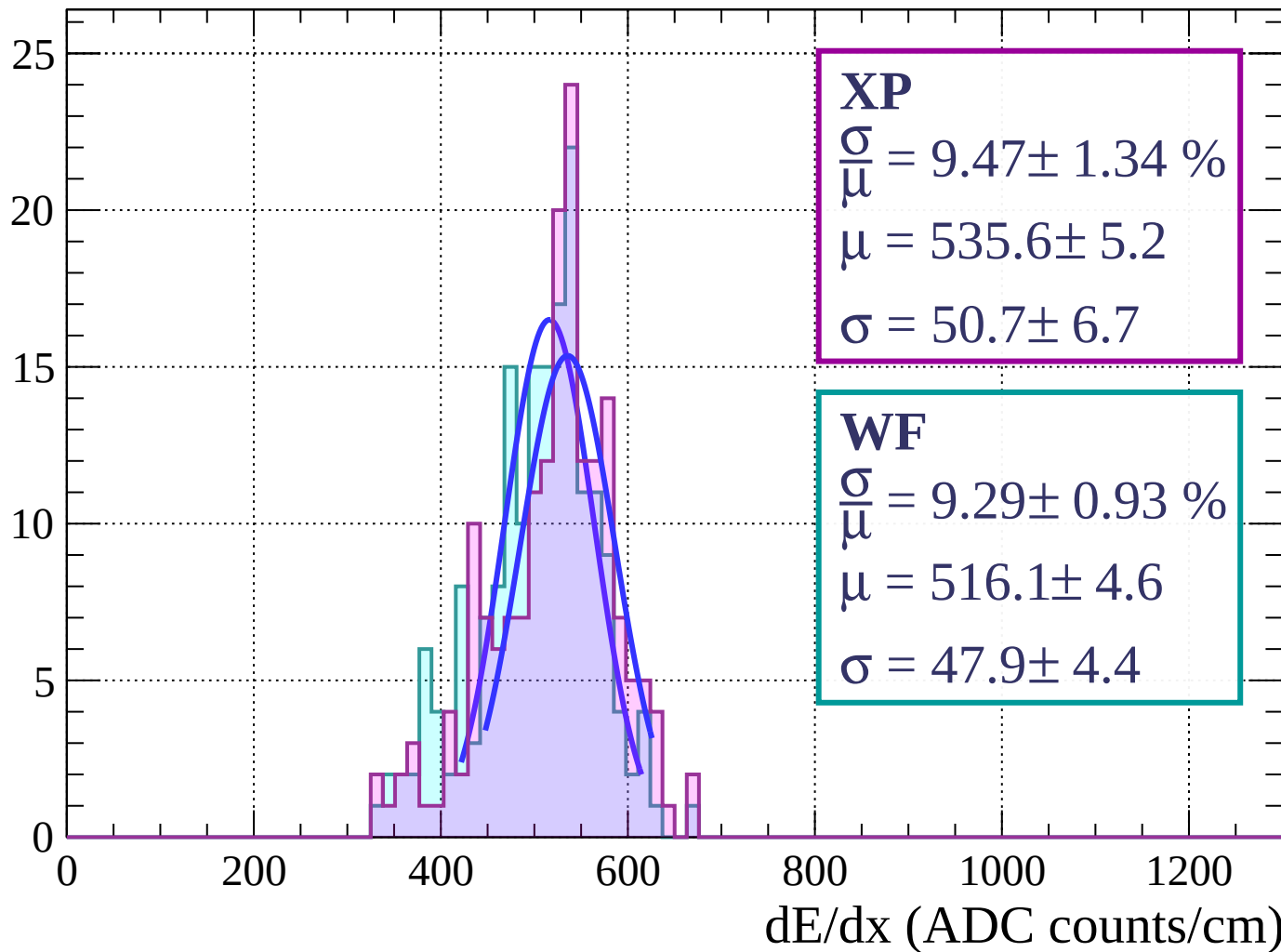
$$\mu = 507.4 \pm 4.0$$

$$\sigma = 63.6 \pm 4.7$$



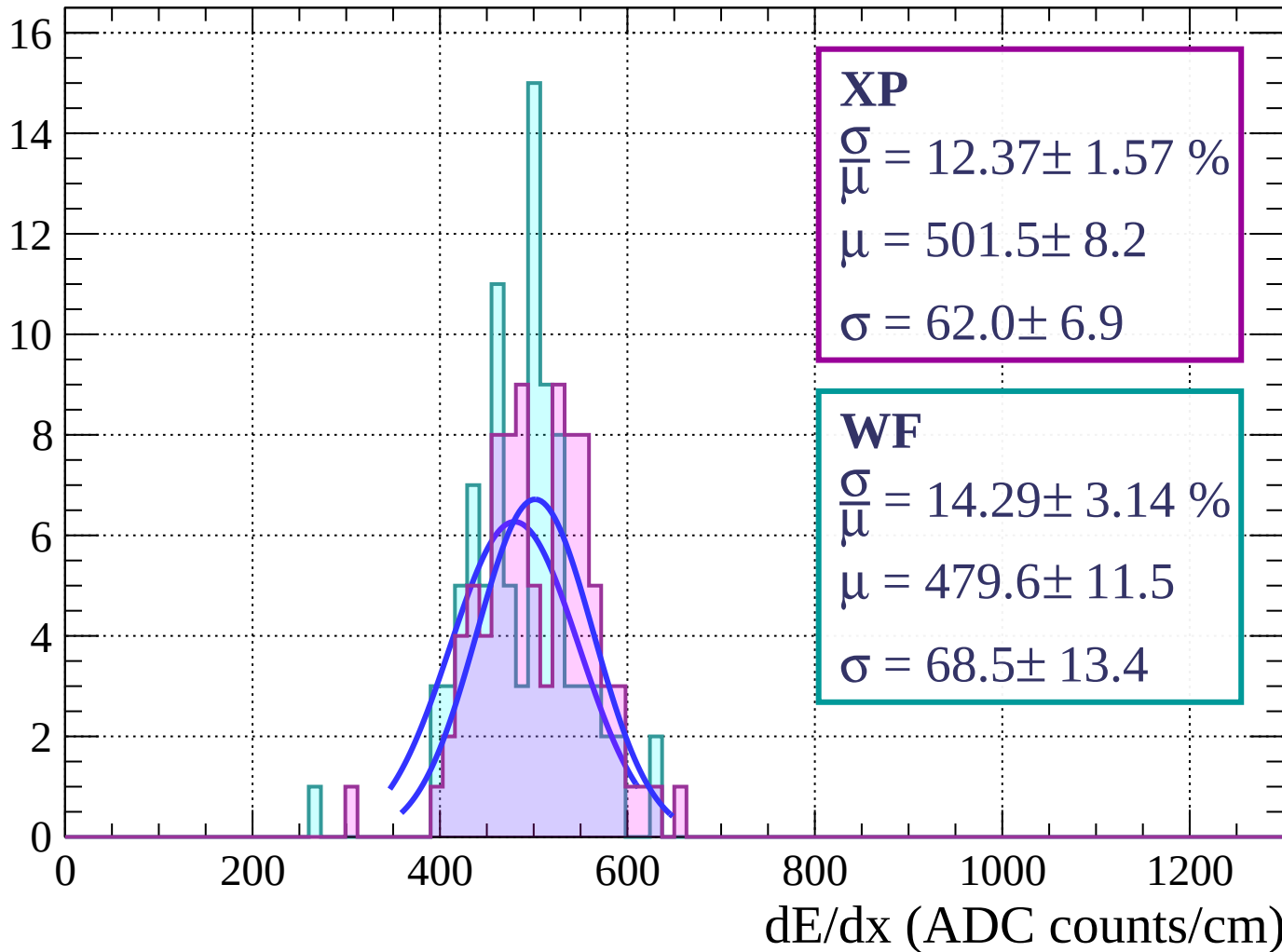
Energy loss | $54 < \theta < 57$

Count



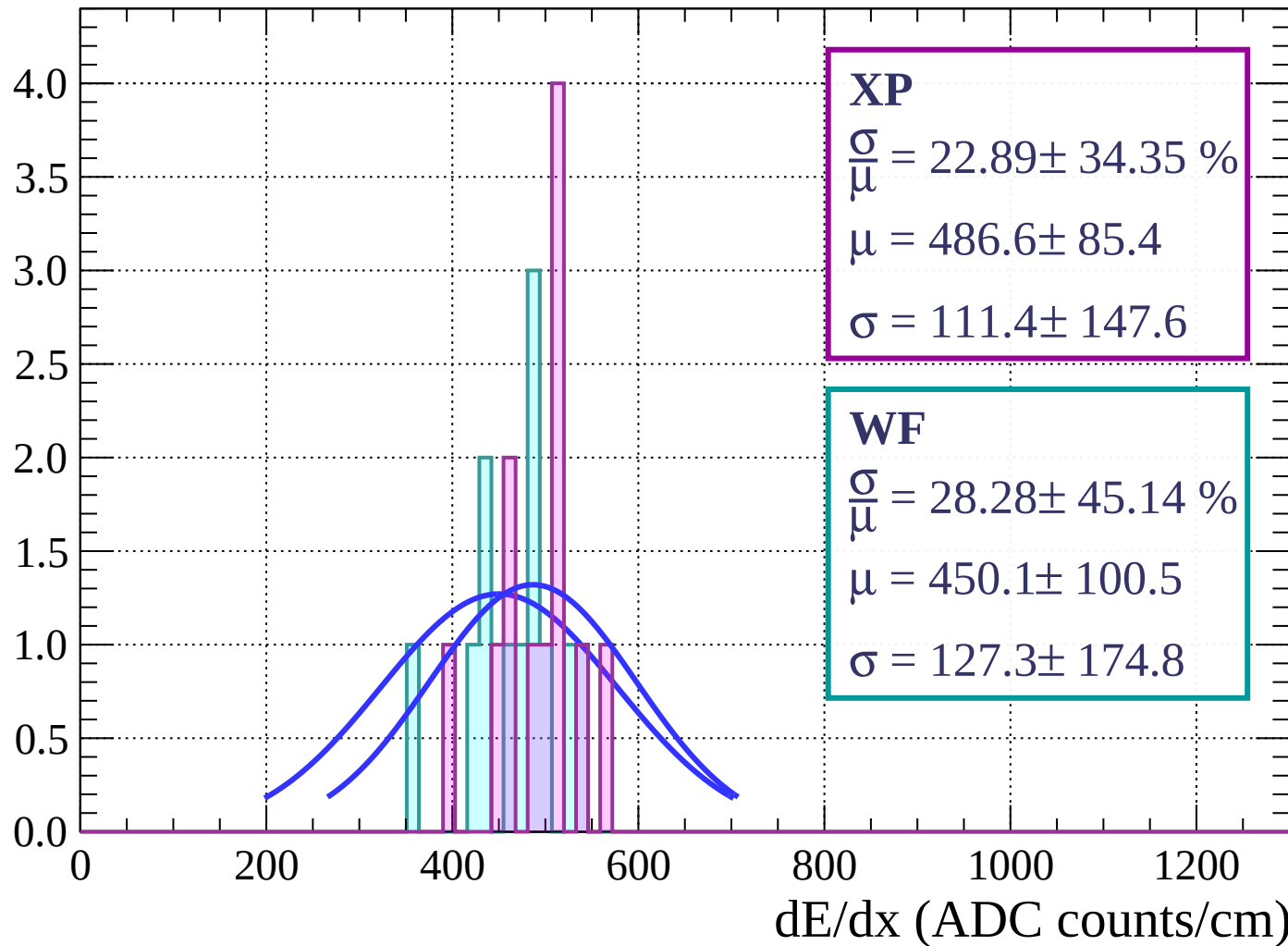
Energy loss | $57 < \theta < 60$

Count



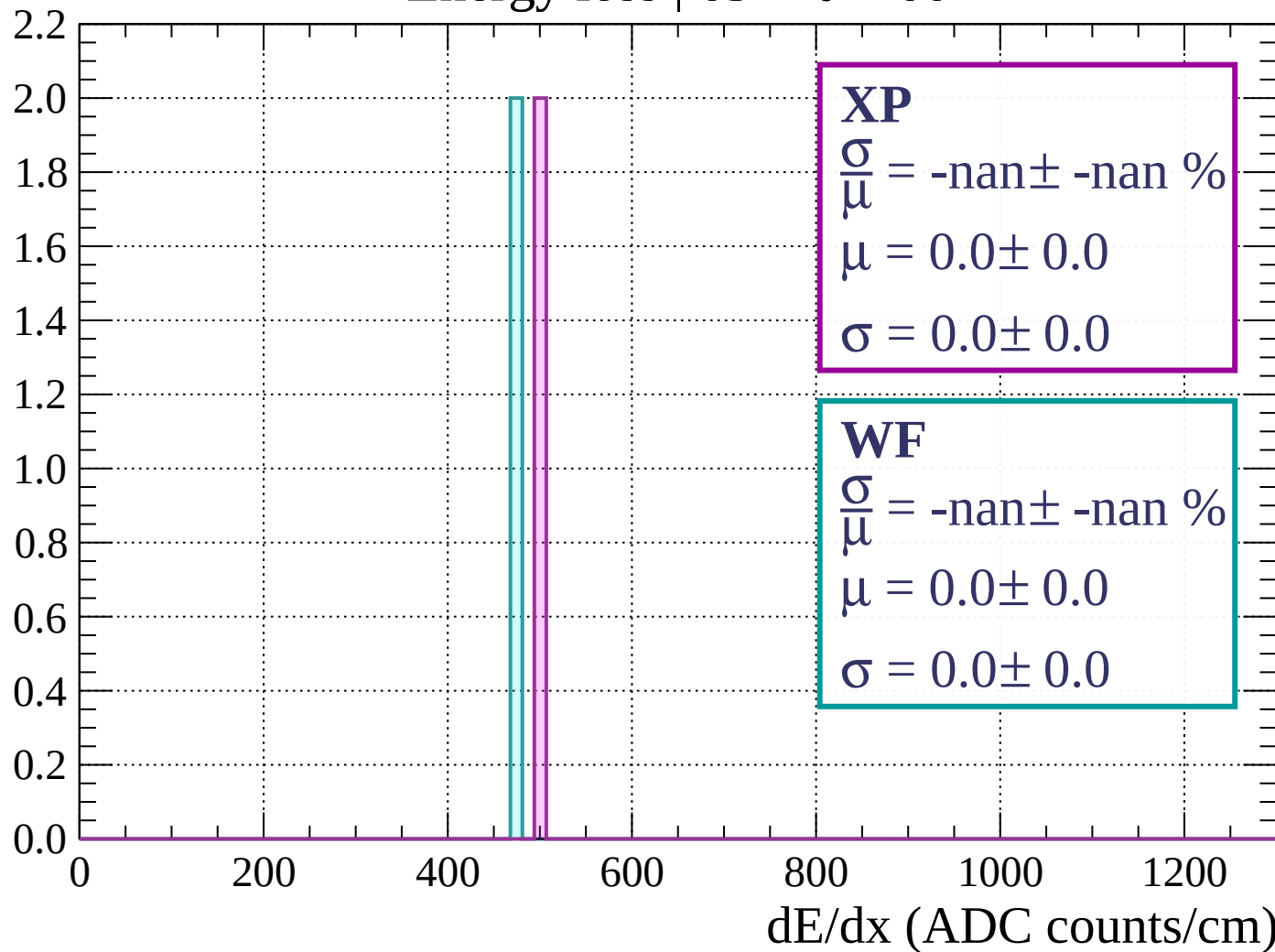
Energy loss | $60 < \theta < 63$

Count



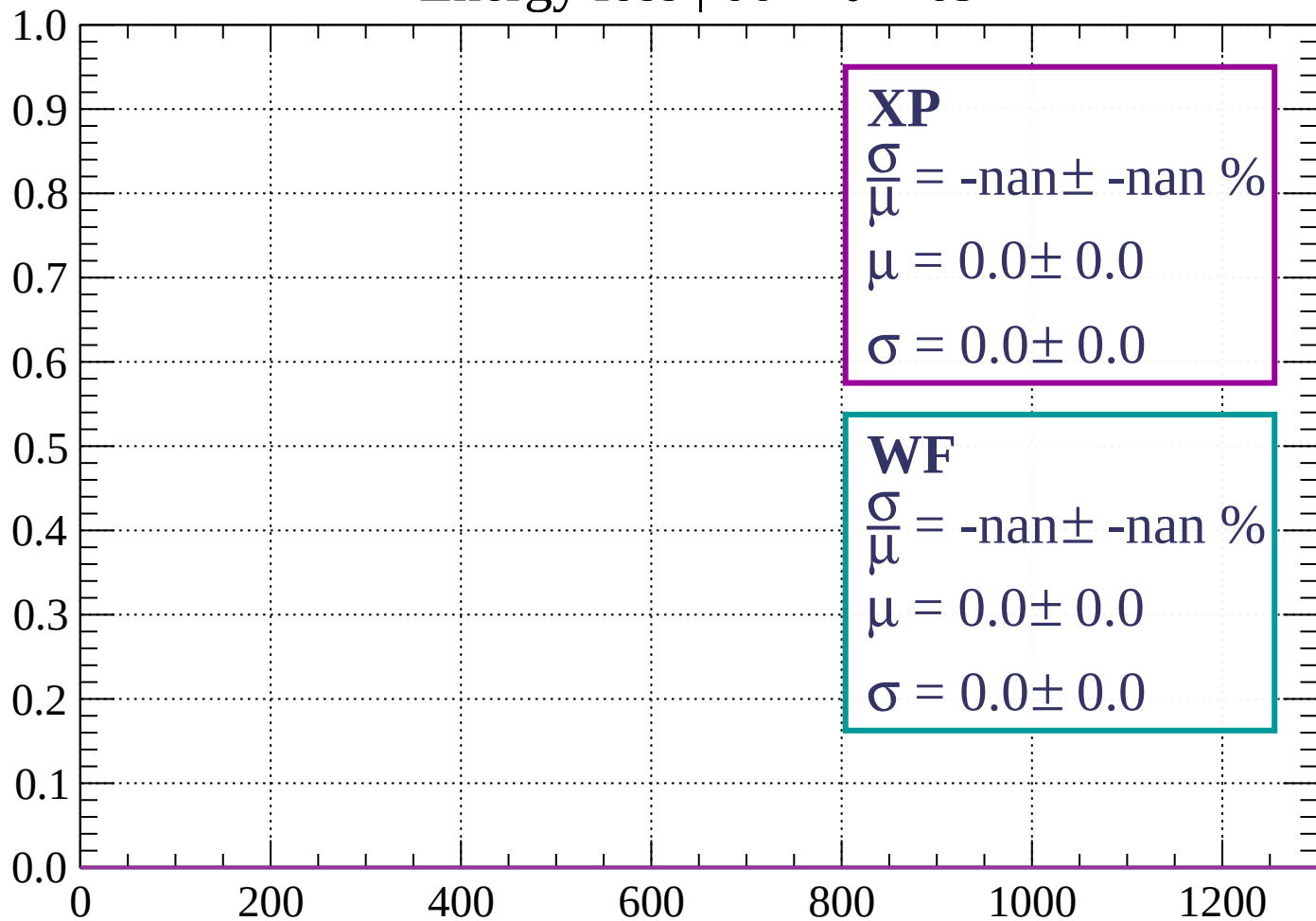
Energy loss | $63 < \theta < 66$

Count



Energy loss | $66 < \theta < 69$

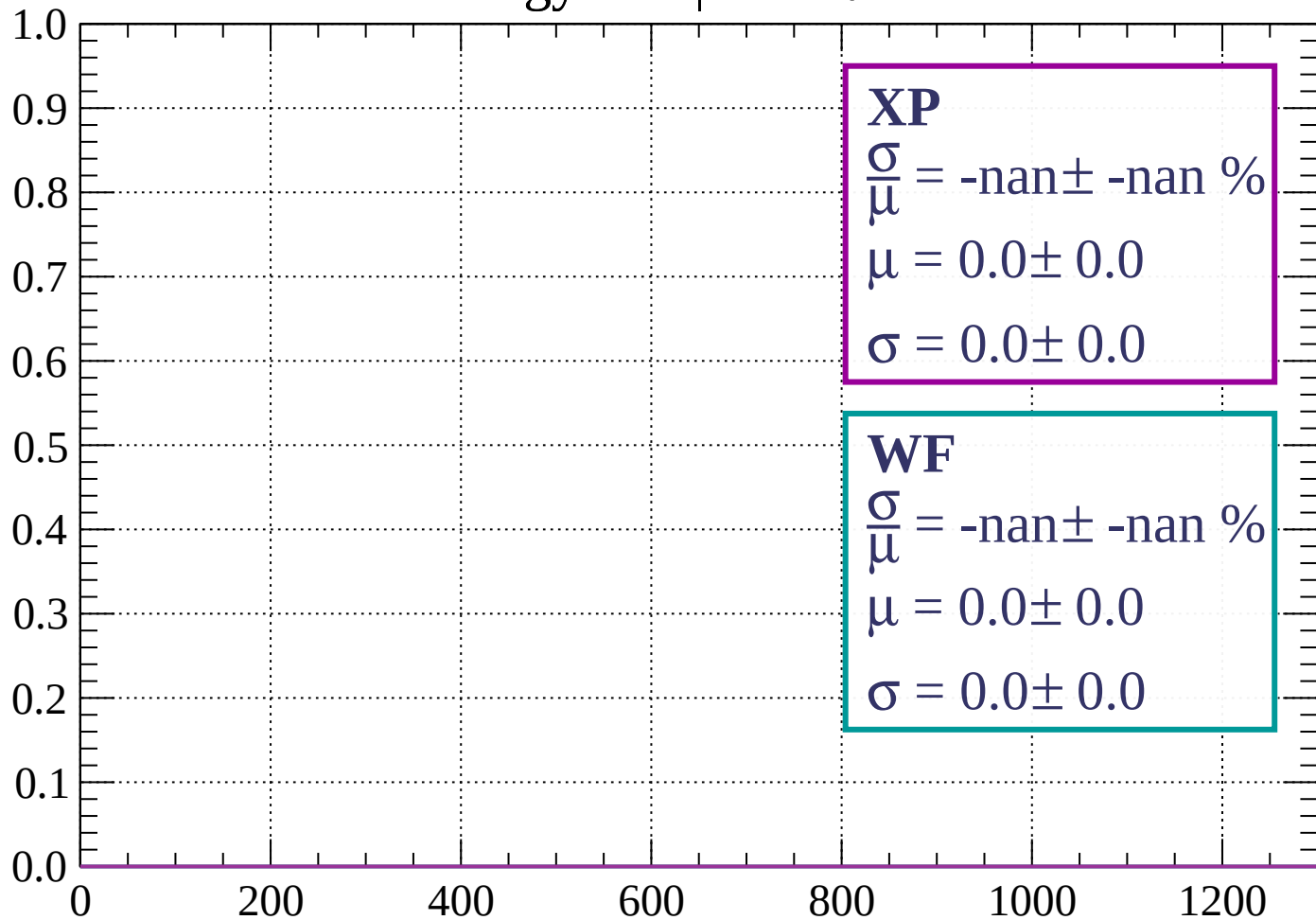
Count



dE/dx (ADC counts/cm)

Energy loss | $69 < \theta < 72$

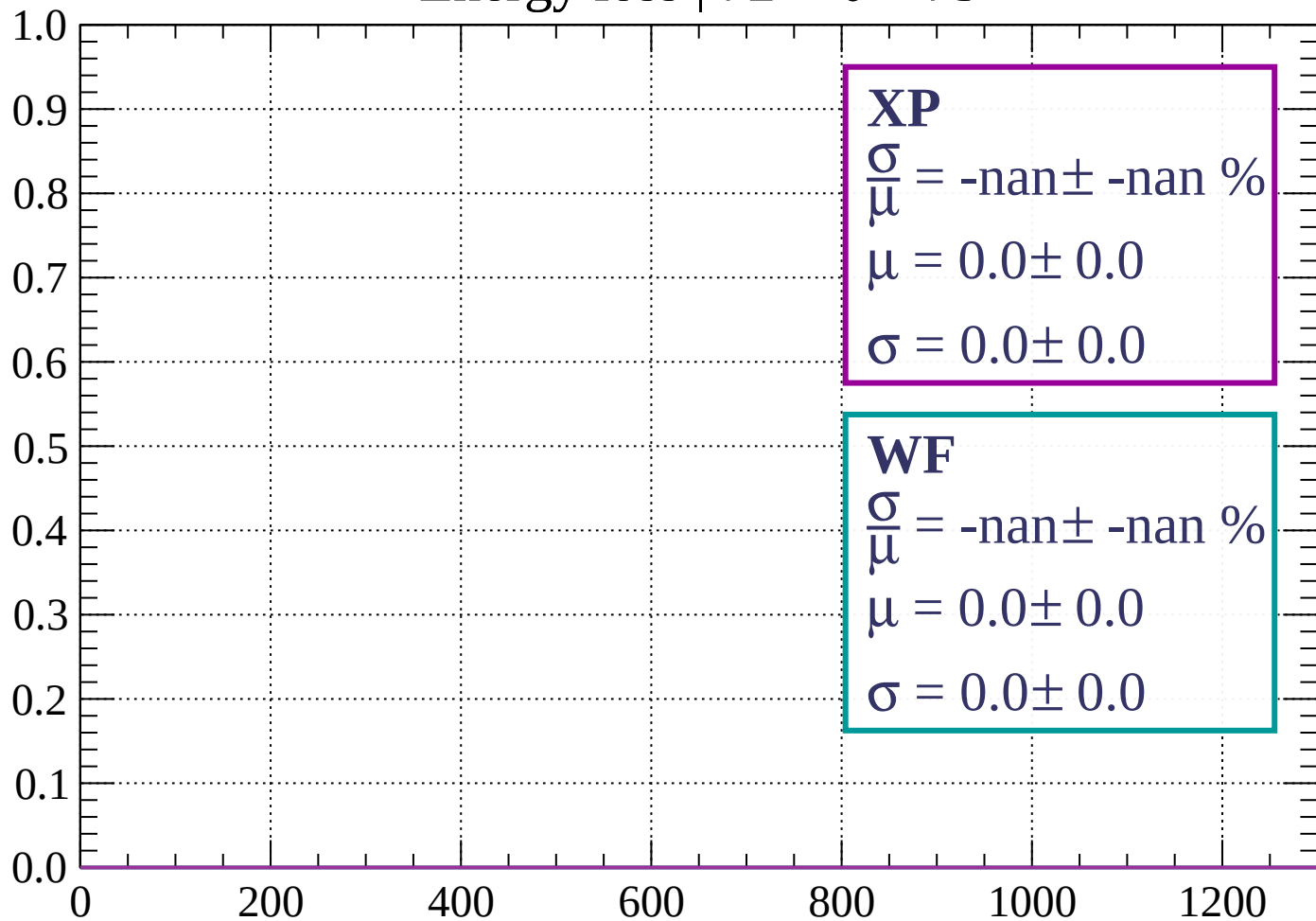
Count



dE/dx (ADC counts/cm)

Energy loss | $72 < \theta < 75$

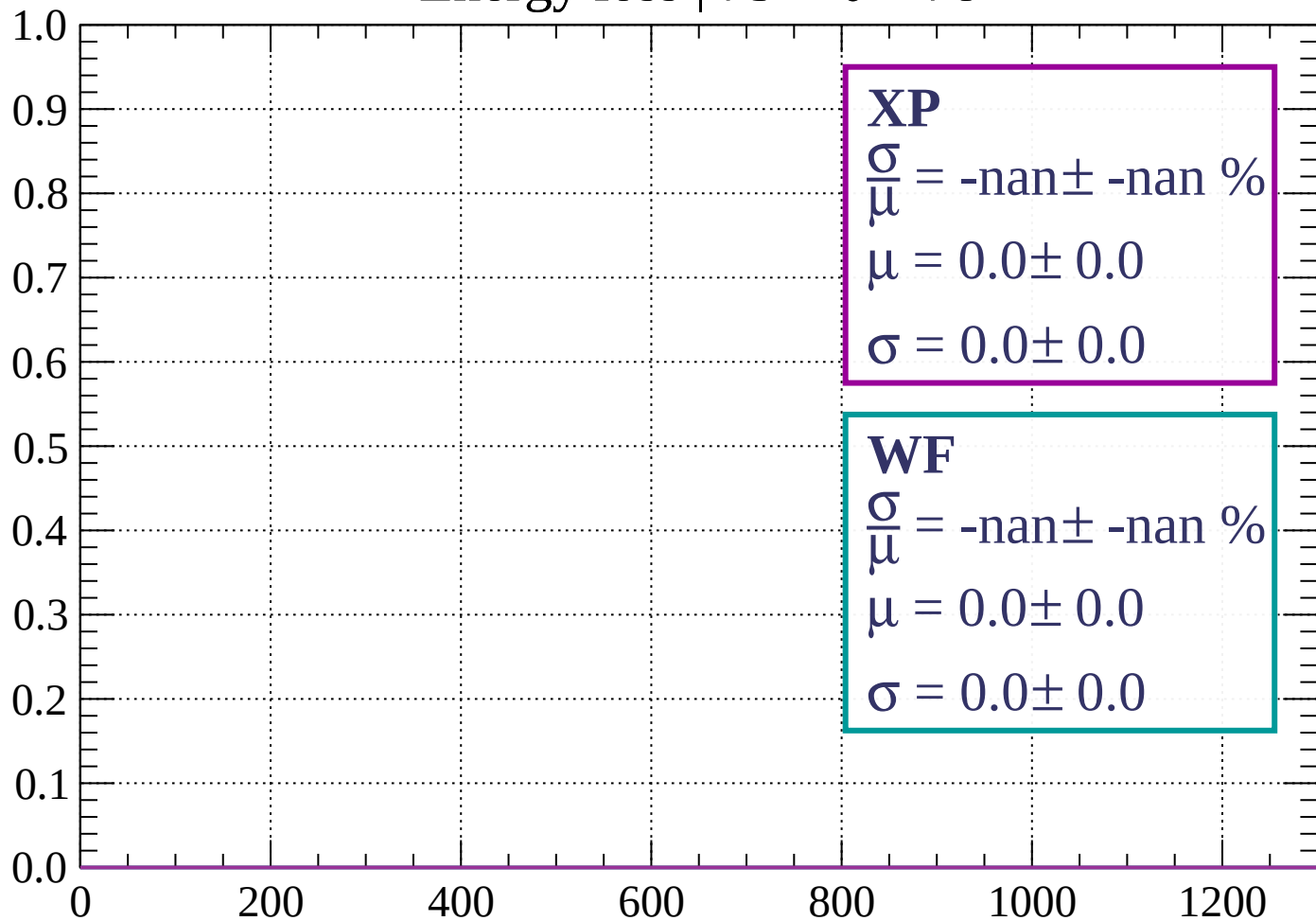
Count



dE/dx (ADC counts/cm)

Energy loss | $75 < \theta < 78$

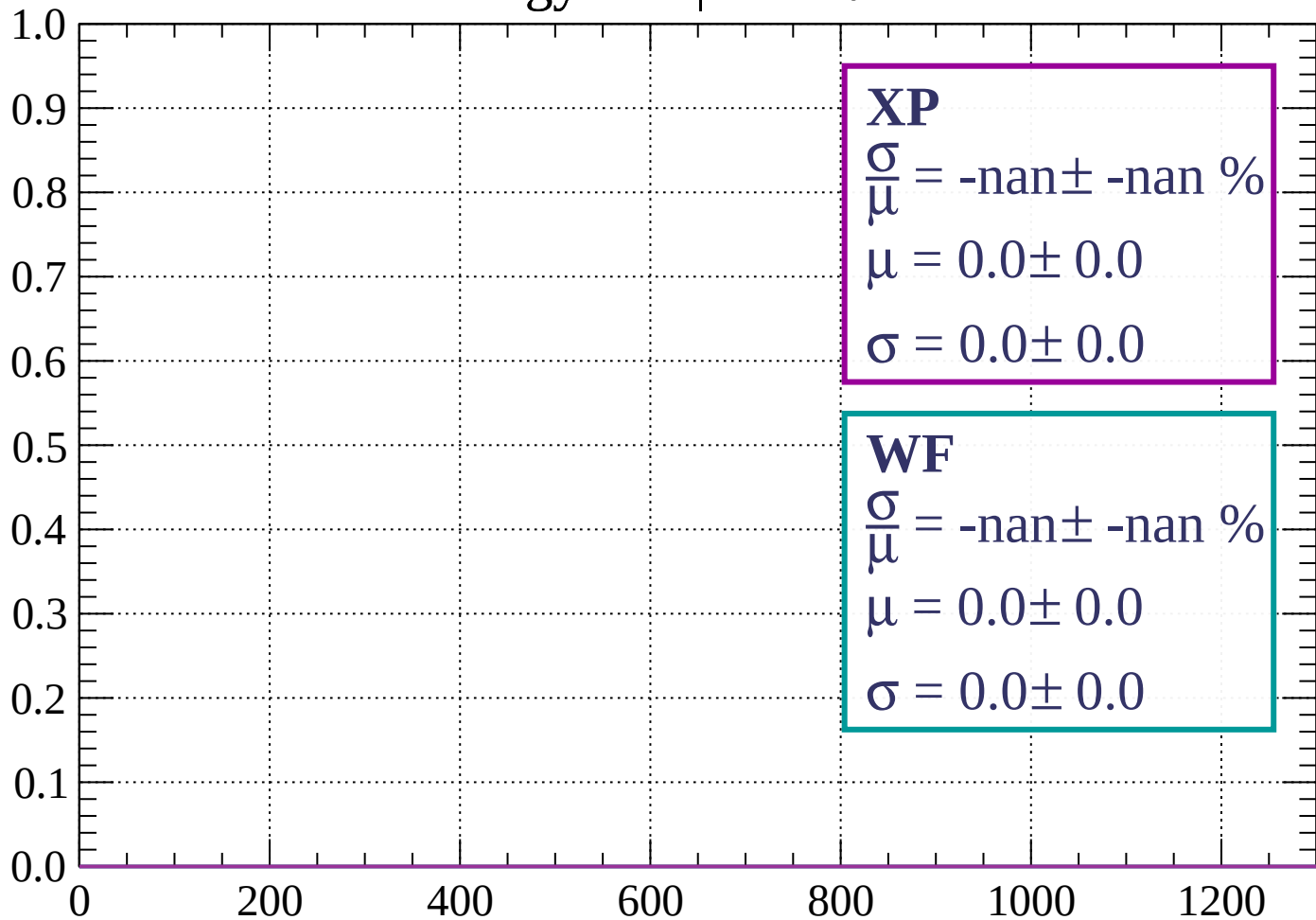
Count



dE/dx (ADC counts/cm)

Energy loss | $78 < \theta < 81$

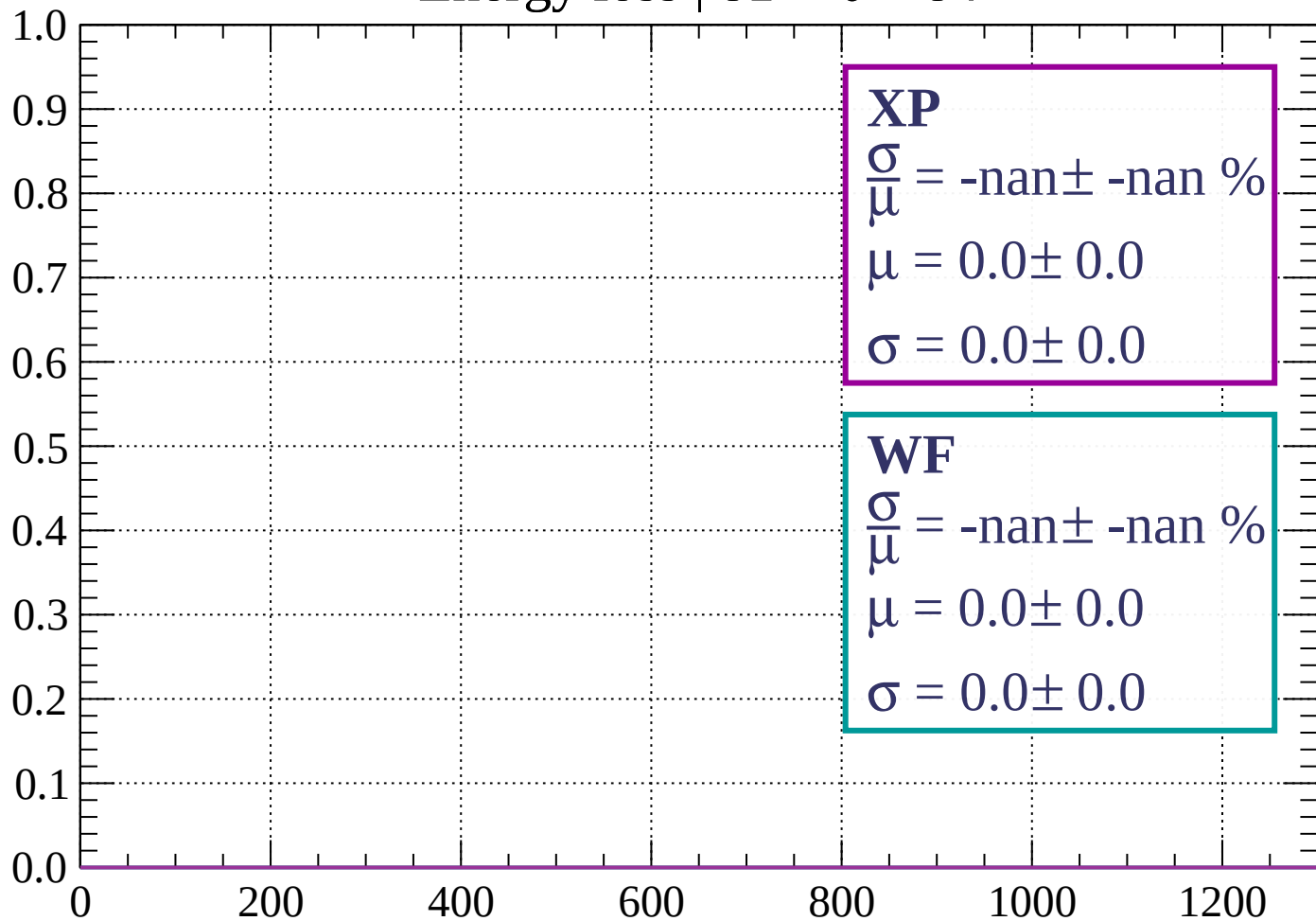
Count



dE/dx (ADC counts/cm)

Energy loss | $81 < \theta < 84$

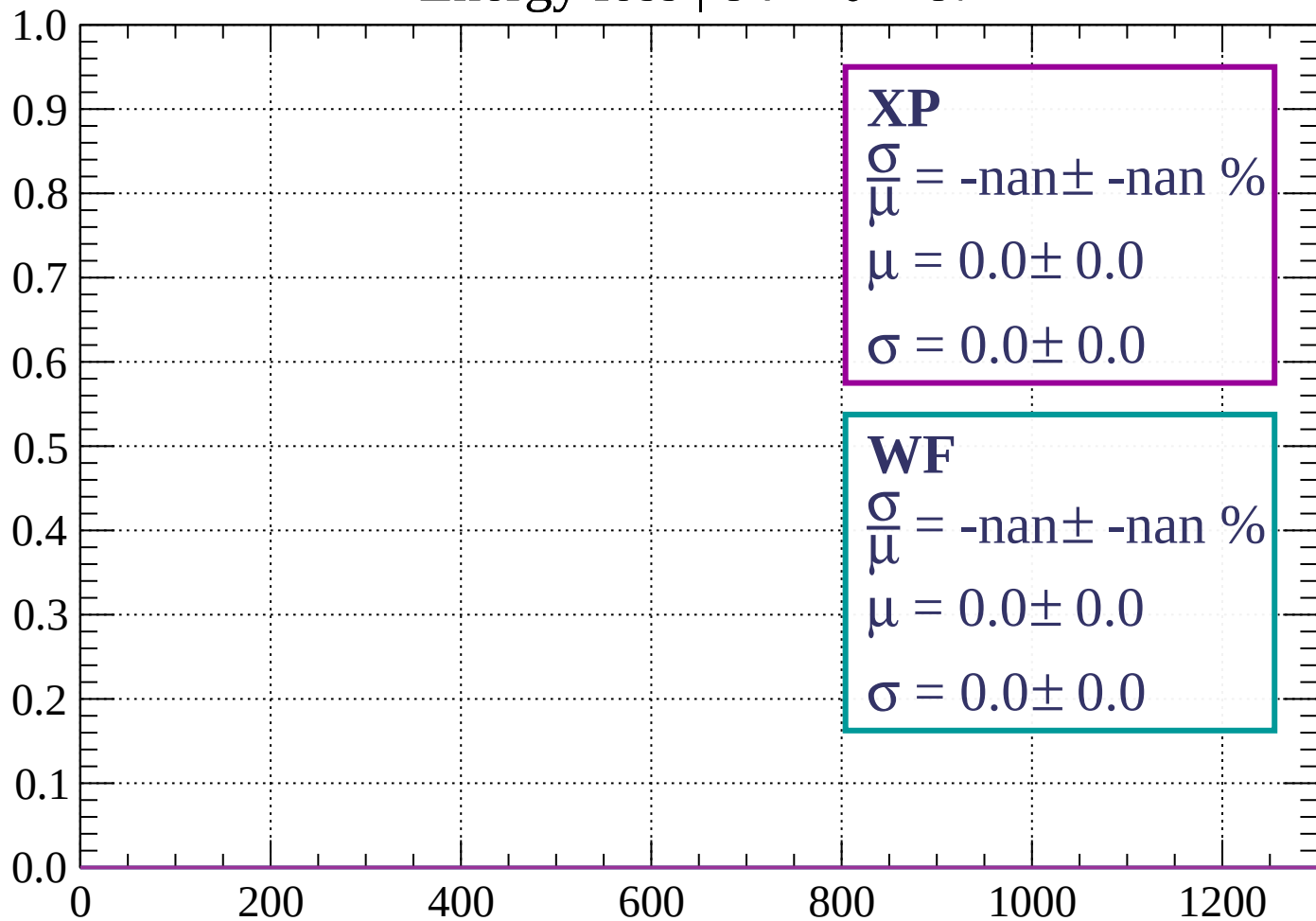
Count



dE/dx (ADC counts/cm)

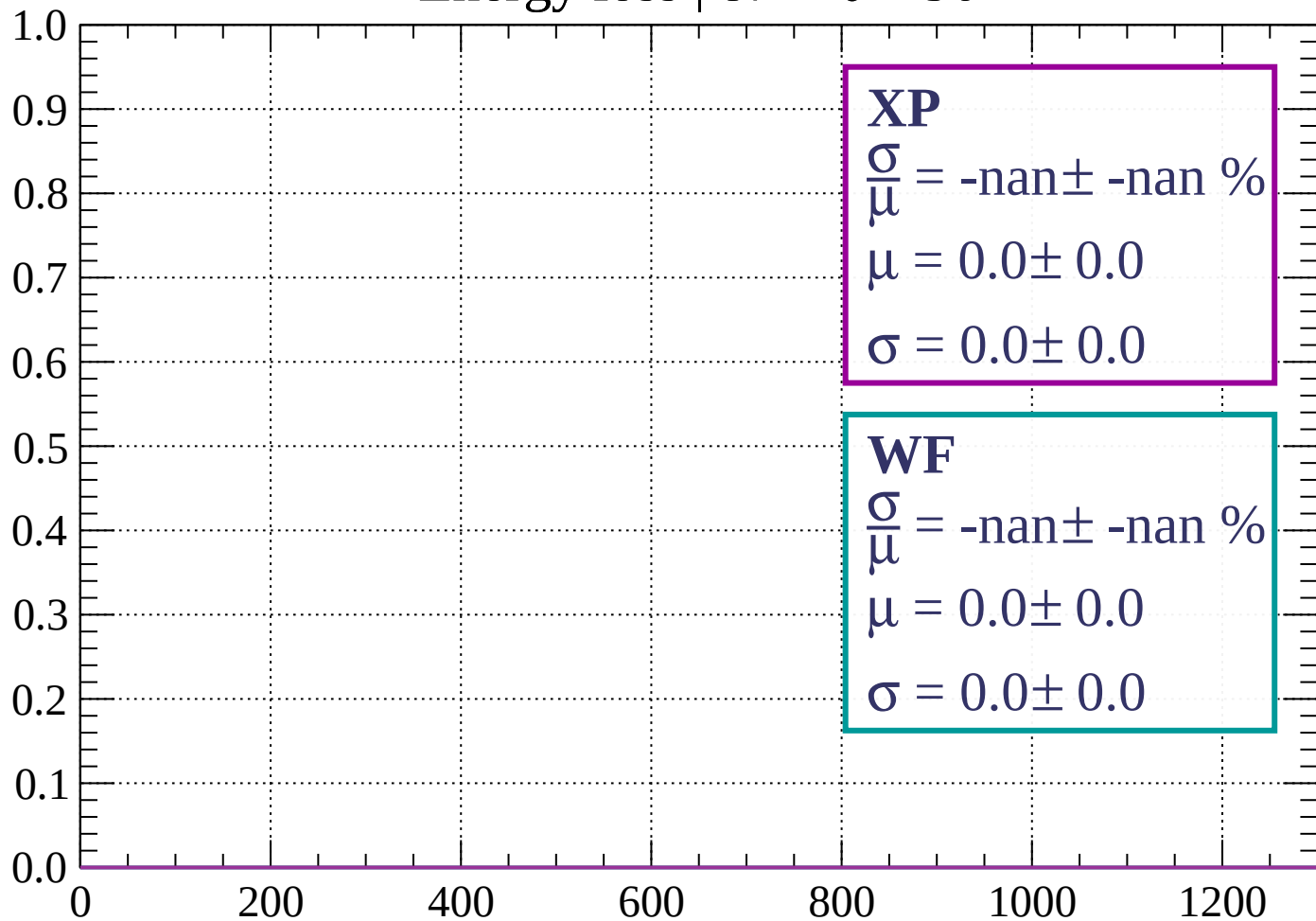
Energy loss | $84 < \theta < 87$

Count

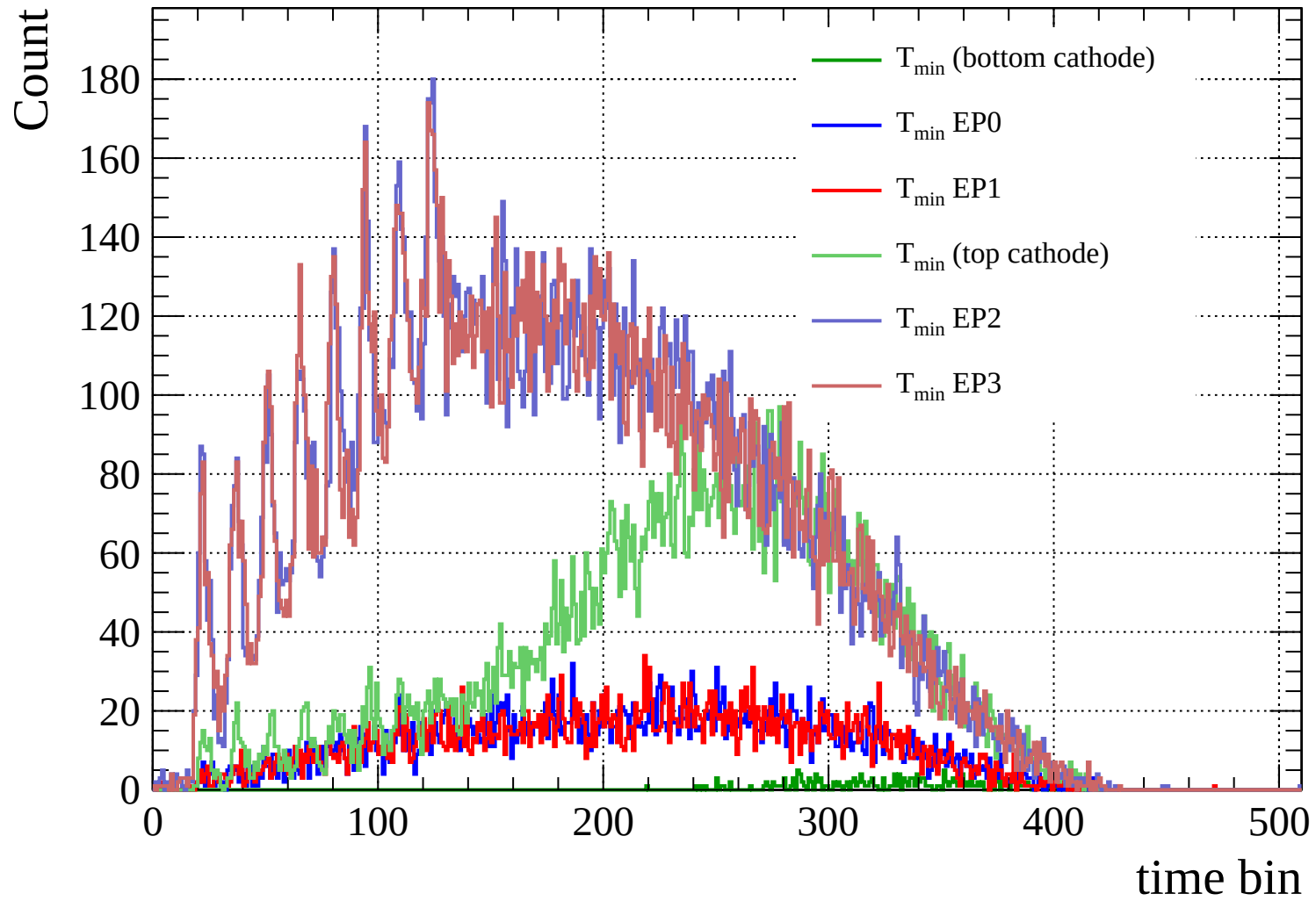


Energy loss | $87 < \theta < 90$

Count



dE/dx (ADC counts/cm)



Count

